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पोर्ट भवन
शुरजी वल्लभदास मार्ग
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CE. CF/226/CTP/154

01 JUN 2021

To,

Ministry of Environment, Forest and Climate Change
Indira Paryavaran Bhavan, Jor Bagh Aliganj,
New Delhi – 110 003

Subject: Compliance submissions in response to, 32th EAC meeting (Infra- 2, 2nd July 2018) for "Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour"

Reference: F.No.10-62/2016-IA.III], IA/MH/MIS/62185/2017

Dear Sir

With reference to above subject, MbPT herewith furnishing Compliance in response to, 32th EAC meeting as sought.

Submission of reply has been delay due to delay in obtaining of requisite clarifications/ recommendations from the respective offices due to lockdown due to Covid19 pandemic.

The compliance of the recommendations are hereby attached as per the below table of contents.

We request to the esteemed Ministry for condoning the delay in submission of the documents, and to re-consider our case for hearing before the EAC, and oblige.

Thanking you

Yours faithfully

(N.G.Mirajkar)
Chief Engineer(I/C)

Compliance of Minutes of the 32nd meeting of expert appraisal committee (infra-2) held on 2-4 July 2018.

1. **Compliance: latest recommendation of SCZMA.**

Answer:

- Proposal for proposed project was submitted to MCZMA on November 2019.
- Project was considered in 146th (4th September 2020) and 154th (15th April 2021) MCZMA meeting.
- MCZMA issue recommendation letter on 4th May 2021 (copy enclosed)

MAHARASHTRA COASTAL ZONE MANAGEMENT AUTHORITY

Tel. No. : 2202 9388
E-mail : dir1.mcz-mh@nic.in
Website: https://mczma.gov.in/

No. CRZ 2019/CR 217/TC 4
Office of the -
Maharashtra Coastal Zone Management Authority,
Environment & Climate Change Department,
15th Floor, New Administrative Building,
Mantralaya, Mumbai- 400 032
Date: 04th May, 2021

To,
Director (IA-III),
Coastal Zone Regulation, Government of India,
Ministry of Environment, Forests & Climate Change,
Indira Paryavaran bhavan, Jor Bagh Road,
New Delhi - 110 003.

Subject: Proposed construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour by Mumbai Port Trust (MbPT)

The Maharashtra Coastal Zone Management Authority in its 146th & 154th meeting held on 04th September, 2020 and 15th & 20th April, 2021 deliberated the proposal of construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour.

2. The Authority noted that, the project is an expansion of existing Mumbai Port Trust for improvement of Traffic at Mumbai Port. MbPT has proposed to develop offshore container Terminal (OCT) at Indira Dock, Mumbai Port. It was decided to execute project in two phases.

3. The Authority noted that, the proposal was earlier deliberated in 146th meeting of MCZMA held on 04.09.2020. The Authority noted that as per the Minutes of EAC, New Delhi, the Public hearing is exempted since it was already conducted in the year 2005. However, Queries/ Comments raised during earlier public hearing need to be documented in the form of Management Plan. It was noted that the Rapid EIA report submitted by the PP needs to add the chapter on the above said TOR point of the EAC, New Delhi. Accordingly, the matter was deferred for want of the Revised EIA report for the project which would have the chapter on Queries/ Comments raised during earlier public hearing need to be documented in the form of Management Plan. MbPT should also submit the CRZ map in 1:4000 scale superimposing the completed and remaining works. The MbPT vide letter dated 12.10.2020 submitted revised EIA report, CRZ map in 1:4000 scale.

4. The Authority noted that the revised EIA report has a chapter on details of public hearing and mitigation plan. One of the issues raised during public hearing was pertaining to mangroves and flamingoes. Reply of the MbPT states that no land is being reclaimed for the project. We are just filling up the old dock basins. About mangroves, this would not at all affected. The Mumbai Port has spent the dumping area is about 25 km away from the mangroves. The Mumbai port has developed and protected mangroves at sewri. Previously about 2000 flamingoes were there but of today there are about 14000 flamingoes. This project will no way affect habitat. Nearly about 15 Acres of area as sanctuary has been

replanted. MbPT spent Rs. 50 Lacs to develop this area as sanctuary. The mangrove situated between Sewri Wadala will be in no way affected.

5. The Authority noted the status of construction works at MbPT, which is as follows:

a) Completed works:
   i. 2 berths and approach trestle
   ii. Capital dredging for berth pocket
   iii. Filling of P & V Dock
   iv. Rail Container Depot
   v. Approach channel and Turning circle dredging
   vi. Berth Packet

b) Pending works:
   i. Procurement of Cargo handling equipment
   ii. Container yard (Reclamation work completed. Work of development of yard mainly metalling WBM and wearing coat balance)
   iii. Capital dredging in the approach channel and turning circle to the required depth

6. The Authority discussed the impact of the pending works such as work of development of container yard, capital dredging in approach channel etc. on the coastal environment and mangroves. The MbPT officials submitted that no mangroves would be disturbed while completing the pending works. No land is being reclaimed for the project. MbPT is just filling up the old dock basins.

7. The Authority noted that as per approved CZMP of Greater Mumbai of 2011, the site under reference falls in CRZ-II & IV areas.

8. The Authority noted that as per provisions of CRZ Notification, 2011, following is permissible:

a) As per para 3(c) of the CRZ Notification, 2011, maintenance or clearing of waterways, channels and ports, based on EIA studies;
b) As per para 4(i) (f) of CRZ Notification, 2011, Construction and operation for ports and harbours, jetties, wharves, quays, slipways, ship construction yards, breakwaters, groynes, erosion control measures are permissible activities.

9. The Authority felt that the project could be recommended to MoEF&CC, New Delhi from CRZ point of view so that pending works be completed by the MbPT.

10. In the light of above, after detailed discussion and deliberation, the Authority decided to recommend the proposal to MoEF&CC, New Delhi subject to strict compliance of following conditions:

Specific Conditions:

i. The proposed pending works should be carried out strictly as per the provisions of CRZ Notification, 2011 as amended from time to time.
ii. Project proponent should implement Environment Management plan for the project effectively and efficiently during construction and operational phase of the project to ensure that coastal environment is protected.
iii. MbPT to ensure no spillage of oil from ships/ vessels in the sea. MbPT should ensure proper Oil Spill Management Plan as well as barricading/isolation along with

danger/inflammable sign to be provided on the existing chemical berths. During operation phase, accidental spill of chemical/oil should be collected and disposed-off through authorized recyclers.

iv. The dredged material to be disposed-off at the designated locations to avoid any adverse impact on marine water quality.

v. Debris generated during the project activity should not be dumped in CRZ area. It should be processed scientifically at a designated place.

vi. PP to ensure that best industrial practices should be followed for fire safety measures and for conservation of coastal environment

vii. All other required permissions from different statutory authorities should be obtained prior to commencement of work.

General Conditions:

i. In case of any change in project profile, the project would require fresh appraisal by the MCZMA.

ii. The MCZMA reserves the right to revoke this recommendation, if the conditions stipulated are not complied with to the satisfaction of the MCZMA or Environment Department.

iii. The MCZMA or any other competent authority, concerned planning authority may stipulate any additional conditions subsequently, if deemed necessary, for environmental protection, which shall be complied with.

iv. The clearance accorded to the project under this notification shall be valid for a period of seven years from the date of issue of such clearance: Provided that the construction activities shall commence within a period of five years from the date of the issue of clearance and the construction be completed and the operations be commenced within seven years from the date of issue of such clearance.

v. The recommendation from CRZ point of view is being issued without prejudice to the action initiated under EP Act or any court case pending in the court of law and it does not mean that project proponent has not violated any environmental laws in the past and whatever decision under EP Act or of the Hon'ble court will be binding on the project proponent. Hence this recommendation does not give immunity to the project proponent in the case filed against him, if any or action initiated under EP Act.

11. Agenda item & minutes of the meeting is available on the website of MCZMA i.e. http://mczma.gov.in.

(Narendra Toke)
Director, Environment & MS, MCZMA

Copy for information to:

1. **PS (Environment) & Chairperson, (MCZMA),** Environment & CC Department, Room No. 217 (Annex), Mantralaya, Mumbai -32.
2. **Member Secretary, Maharashtra Pollution Control Board,** Kalpataru Point, 3rd and 4th floor, Road No. 8, Sion Cir, opp. PVR Theater, Mumbai -400022
3. **District Collector Mumbai City,** Old Custom House, Shahid Bhagat Singh Rd, Marg, Fort, Mumbai 400001
4. **Chairman, Mumbai Port Trust (MbPT),** 20, Shoorji Vallabhdas Rd, Kala Ghoda, Fort, Mumbai 400001 - -It is requested to submit application on Parivesh Portal of MoEF&CC.
5. **Select File-TC 4**

2. Compliance: Submit copy of consent to operate (CTO) for present project.

Answer:

- MbPT has applied for CTO on 10th December 2018,
- MPCB reply that, consent to operate will be considered after EC is amended to use the cargo other than containers (copy enclosed).
- So MbPT will applied for the same after receiving Environment and CRZ clearance

MAHARASHTRA POLLUTION CONTROL BOARD

Minutes of 8thConsent Appraisal Committee Meeting of 2018-2019 (1st sitting) held on 10.12.2018 at 2:00 p.m. in the chamber of Hon'ble Chairman, 3rd Floor, Maharashtra Pollution Control Board, Kalpataru Point, Sion, Mumbai, Maharashtra.

…….

Following members of the Consent Appraisal Committee were present:

1.	Shri Sudhir Shrivastava, Chairperson, MPCB, Mumbai.	Chairman
2.	Shri. P.P. Nandusekhar Technical Advisor, MIDC.	Member
3.	Shri. P. K. Mirashe Assistant Secretary (Technical), MPC Board, Mumbai.	Member Convener

Following Officer of MPCB were present for the meeting:

1.	Shri. Y. B. Sontakke, JD(WPC), MPCB, Mumbai.	Special Invitee
2.	Shri. N.N.Gurav RO(HQ), MPCB, Mumbai.	Special Invitee

Additional Chief Secretary, Home (Transport) Dept., Mumbai, Member Secretary, MPCB and Representative of NEERI could not attend the meeting. Leave of absence was granted to them.

Chairperson of the committee welcomed the members of the committee and allowed proceeding of the meeting to start.

The Minutes of 5th, 6th and 7th Consent Appraisal Committee Meeting of 2018-2019 (2st sitting) held on 27.11.2018,28.11.2018, 04.12.2018, 07.12.2018,10.12.2018 and 28.11.2018 respectively (Booklet No. 9,11,11 (Part),10, 11 (Part), and Booklet no.12) circulated vide letter no. BO/CAC-CAELL/TB/B-181217FTS-0160 Dated 17.12.2018 were confirmed.

The meeting thereafter deliberated on the fresh agenda items [Booklet No. 14 (Part-I)] placed before the committee and following decisions were taken:

Sr. No	Application Unique Number	Industry name and Address	Decision on grant of consent	Consent granted up to	Remarks / Discussion	
	Booklet No. 14 (Part-I)					
	Consent to Establish/Operate/Renewal					
1	MPCB-CONSENT-0000005220	Bharat Forge Ltd., Sr. No-72/76, Mundhwa, Tal. Haveli, Dist. Pune.	APPROVED Renewal of Consent with increase in C.I.	Up to 30.06.2021.	It was decided to grant Consent to 1st Operate (UAN No. 0000017921) for Phosphating activity with renewal of existing Consent (UAN No. 0000005220) and amalgamate by clubbing two applications by imposing the following conditions; 1. Industry shall extend the BG of Rs. 5 lakh obtained towards O&M of pollution control system.	
2	MPCB-CONSENT-0000017100	Sanvijay Rolling & Engineering Limited, Unit- B-2, Plot No B-203 to B-206, MIDC Industrial Area Butibori, Hingana, Nagpur.	APPROVED Consent to Establish for expansion	Commissioning of the unit or 5 years whichever is earlier	It was decided to grant Consent to Establish for proposed expansion to manufacture Blooms & Structural Steel i.e. Angles, Beams, Girders, Rounds, Rods, Circles, Channels etc.- 1,00,000 MT/A without manufacturing of Steel Billets by imposing following conditions; 1. Industry shall not manufacture Steel Billets in proposed expansion. 2. Industry shall submit the undertaking on Rs. 100 stamp paper stating that industry will not increase the production of Steel Billets/ blooms & Structural Steel- 2,00,000 MT/Annum from the existing unit at any point of time and submit the BG of Rs. 5 lakh towards compliance of the same. 3. Industry shall amend the existing EC within 03 months by incorporating expansion activity of rolling Mill. 4. Industry shall submit the BG of Rs. 5 lakh towards compliance of consent to establish conditions.	
3	MPCB-CONSENT-0000032035	M/s Karanja Terminals & Logistics Pvt. Ltd., Karanja	NOT APPROVED	----	It was decided to keep this case in abeyance and decided to call PP for personal hearing for presentation about propose plan of operation of port along with pollution control systems provided by the PP.	

Sr. No	Application Unique Number	Industry name and Address	Decision on grant of consent	Consent granted up to	Remarks / Discussion	
		Creek, Tal. Uran, Dist. Raigad.	Consent to 1st Operate (part)			
4	MPCB-CONSENT-0000033847	Mumbai Port Trust1873- Units Includes Alexandra (Indira Dock) (21 berths) -1914,PirPau berth -1922, In Sea of Mumbai Port Limit	APPROVED Renewal of Consent to Operate	31.12.2020	Committee noted that, Earlier Board has granted consents to operate for Residential Colonies located at Colaba, Mazgaon, Wadi Bander, Wadala, Cotton Green, Worli, Sewree, Govandi Etc. Pir Pau jetty, Alexandra (Indira Dock),Victoria Dock, Between Pir Pau and Jawahardeep, Princes Dock, Sassoon Dock and Ferry Wharf on 01.08.2015 & for operation of 2nd Liquid Chemicals/Specialized grades of POL product berth of Pir Pau by Mumbai Port Trust on 20.06.2016 which were valid till 30.09.2017 and 31.12.2017 respectively. Now PP has applied for renewal of consent to operate and also for consent to operate for 2 offshore container berths (OCT) for which consent to establish was issued on 17.01.2006 for construction of offshore container berths and development of container terminal in Mumbai Harbor. The earlier Environmental Clearance to Offshore berths (OCT) was given for handling of containers. PP has applied for renewal and modification of environmental clearance to handle containers, steel RO RO, Containers and any other clean cargo with increase in additional capacity 5.0 Million Tonnes. The amendment in environmental clearance is not issued till date and among all compliances, the PP has been asked to obtain consent to operate for present (current) project. As a part of earlier consent to operate PP was required to augment STP capacity from 250 CMD to 5796 CMD of sewage within 2 years. The PP is yet to take concrete steps towards providing additional capacity of STP. PP has also not indicated quantum of solid /Hazardous waste at the facilities and the pollution control measures adopted.	

Sr. No	Application Unique Number	Industry name and Address	Decision on grant of consent	Consent granted up to	Remarks / Discussion	
					In view of above, it was decided to renewed the consent by forfeiture of Bank Guarantee of Rs.10.0 lakhs and top up with double amount of Bank Guarantee i.e. Rs.20 Lakhs towards compliances of consent conditions. The consent to operate for Offshore container berths (OCT) will be considered After EC is amended and compliances of Environmental Clearance conditions.	
5	MPCB-CONSENT-0000043449	Wadhwa Group Holdings Pvt. Ltd. (Vishwaroop Infotech Pvt. Ltd.), Plot No. 34, 35 & 38, Sector 30 A, Vashi, Navi Mumbai.	APPROVED Renewal of Consent	Up to 31/01/2023.	It was decided to grant renewal of Consent to Operate for IT & ITES activity on total plot area 15,517.55 sq.mtrs. for total construction BUA 23,276.62 sq.mtrs. by imposing the following conditions; 1. PP shall extend the BG of Rs. 10 lakh towards O&M of pollution control system. 2. PP shall install online monitoring system to the STP and shall be connected to MPCB Server. 3. PP shall achieve the treated domestic effluent standard for the parameter BOD- 10 mg/lit. 4. The treated domestic effluent shall be 60% recycled for secondary purposes such as toilet, flushing, air conditioning, cooling tower make up, fire-fighting etc. and remaining shall be utilized on land for gardening and connected to the sewerage system provided by local body. 5. PP shall operate organic waste digester along with composting facility and compost obtained shall be used as manure in their own garden/ plantation. Consent shall be issued after Obtaining additional Consent fees for increased in C.I. and five terms consent fee for period up to 31.01.2023.	
6	MPCB-CONSENT-0000048993	Prasol Chemicals Pvt. Ltd., S. No. 8, 13, 15, 16, 25, 75, Village Honad, Tal. Khalapur, Dist. Raigad.	APPROVED Consent to Establish for Modernization of existing plant	Up to commissioning of the unit or 5 years whichever is earlier	It was decided to grant Consent to establish for Modernization of existing plant by imposing the following condition; 1. The industry shall not undertake production more than existing consented quantity or shall not manufacture new products till obtaining EC & C to E. 2. The industry shall 100% recycle/ reuse treated effluent into process, cooling tower and for utility purposes so as to achieve ZLD.	

3. **Compliance: Submit air pollution mitigation plan.**

Answer:

Port activities can potentially cause deterioration of air quality in the surrounding areas due to multifarious activities during construction and operational phases which will be addressed through a properly structured Air Pollution Mitigation Plan. The important components of the Air Pollution Mitigation Plan designed for the proposed project are as follows:

- Establish the baseline air quality.
- Identify potential sources of air pollution during construction and operational phases of the project.
- Strategize for eliminating or minimizing pollution through proper action and management.
- Conduct periodic monitoring and compare the results with the baseline air quality followed by corrective measures if warranted.
- Set adequate implementation mechanism for effectiveness of Air Pollution Mitigation Plan

Baseline Environment - Ambient Air Quality

Baseline status of the prevailing air quality within the study area was established through monitoring at 8 locations listed below:

Sr. No	Station no.	Station name	Distance (km)	Direction
1.	AAQ 1	Project Site, MbPT Shed no:13B	1.35	NW
2.	AAQ 2	Near Cosmos Builder, Malabar Hill	5.10	NW
3.	AAQ 3	Near Entry Gate, At Kohinoor Square, Dadar	9.76	N
4.	AAQ 4	At Shanti Kutir Apartment, Near Marine Plaza Hotel, Marine Drive, Mumbai	3.27	W
5.	AAQ 5	At Willson House, Colaba, Mumbai	4.23	SW
6.	AAQ 6	Near MbPT Hospital, Wadala	7.38	S
7.	AAQ 7	Near Pir Pau Fire Station	7.06	NE
8.	AAQ 8	Near Dr.AmbedakarBhavan	1.43	E

The parameters monitored included PM_{10}, $PM_{2.5}$, SO_2, NOx and CO. All the ambient air quality parameters were within the National Ambient Air Quality Standard, 2009.

Potential sources of air pollution and Air Pollution Mitigation Plan.

The proposed project involves the completion of the expansion project namely construction of two offshore container berths and development of container terminal within the designated port area. The construction of two berths, Y-shaped approach trestle, dredging in berth pockets, filling of Victoria and Princess Docks for development of yard has been completed. The remaining constructions will be completed and new facilities proposed due to changes in cargo profile will be added in the proposed expansion plan. Thus, most of the marine structures and land-based

constructions have been completed. Hence, the potential impact on air environment during the construction phase would be marginal.

The expansion project when operational will be handling steel cargo, automobiles and containers. Evidently, the loading /unloading operations at berths would not generate significant air contaminants. Major emissions would be limited to the wind-blown dust due to the movement of vehicles and topside equipment at berths as well from exhaust from vehicles, machinery and ships.

The probable emissions to environment during the construction and operational phases of the proposed expansion are identified and accordingly adequate mitigation measures are compiled in the following table:

S.No.	Sources of Air Pollution	Pollutants	Mitigation plan
Construction Phase			
1	• Civil works for the revised cargo configuration. • Installation of cargo-handling equipment. • Construction material handling (Loading and un-loading of raw materials). • Transportation of material and machinery. • Operation of construction equipment.	Generation of air pollutants like NO_2, SO_2, HC, CO, PM, VOCs	• Transportation vehicles will be properly and timely maintained, serviced regularly and inspected to control emissions. • Static and mobile diesel-powered equipment / generator sets will be used only during power exigency. Generators, machineries and vehicles will be serviced and maintained regularly. • It will be mandatory to use low sulphur diesel in transport vehicles, equipment and machinery where applicable. • Movement of bulk materials will be preferred during non-peak hours and regulated during peak hours. • Mobile equipment such as intermittently used machines and transport vehicles will be either switched off or throttled down to a minimum when not in use. • Environmental awareness program / training will be imparted to the key contractors' personnel and compliances will be enforced through in-house monitoring.
		Generation of Particulate Matter (dust generation)	• Automated sprinkler system will be installed at reclaimed areas of Victoria and Princess Docks earmarked for development of cargo storage yards during construction of yards and other civil works.

			• Speed control of 20 km/h within the port premises as per the policy of MbPT will be maintained through proper traffic management to reduce traffic-related dust emissions. • Dust masks will be provided to workers to minimize their exposure to dust. • Since most of the port area is concreted / asphalted, dust generation will be relatively low. Nevertheless, there areas will be regularly swept as per the ongoing practice and collected dust will be adequately disposed in the designated areas in the port premises. • It will be ensured that loading, transfer, and discharge of construction materials take place with a minimum height of fall, and shielded against wind. Dust suppression spray unit will be deployed where necessary. • Vehicles carrying construction materials will be covered with tarpaulin to avoid dust generation during their movement. • Vehicles having Government license and certificate for transportation of construction will be commissioned. • Dry dusty materials such as cement will be stored in sealed containers. • It will be ensured that contractors employed for site preparation and civil works carry out proper storage and handling of spoil, sub-soils, top-soils etc to minimize the risk of windblown material and dust. • Procedures and work instruction for all activities that may generate fugitive dust emissions will be created aimed at minimizing dust. • All loading and unloading activities will be carried out as close as possible to the storage facilities.
Operation Phase			
2	• Emission from ships, and	Generation of air	• *The Annex VI of MARPOL Convention deals with the “Regulations for the Prevention of

	onboard machinery. • Increase in the number of ships/traffic. • Vehicular emissions from transportation within the port. • Project operations such as loading, unloading of ships. • Ship fuelling activities and ancillary activities. • Emissions from DG sets.	pollutants like NO_2, SO_2, HC, CO, PM, VOCs	Air Pollution from Ships. It will be ensured that ships visiting the berths are compliant to Annex VI. • Transportation vehicles will be properly and timely maintained and serviced regularly to control emission of air pollutants and that they have valid PUC. • Static and mobile diesel-powered generator sets will be used only during power exigencies. • Diesel driven generators, machineries, vehicles and equipment will be serviced and maintained regularly. • Validity of certificates of all the vehicles entering the port area will be regularly checked. • Heavy vehicles older than 10 years will not be permitted to ply within the port. Use of electric cars inside the port area will be encouraged where feasible.
		Generation of Particulate Matter (dust generation) Resuspension of dust (due to wind and vehicular movement over the road surface)	• Regular cleaning of paved and unpaved roads and removal of the accumulated dirt from roadside will be routinely done. • Regular maintenance of unpaved shoulder on paved road will be practiced. • Paving of access roads will be undertaken wherever necessary. • Periphery and access roads of the cargo storage yard will be planted with fast growing trees of indigenous species. • Mobile sprinkling trucks will be deployed to reduce dust emissions from the roads. • Ambient air quality will be monitored at selected sites (which are in close proximity to the operations area) to ensure air quality meets the norms. \

** The Annex VI of MARPOL Convention deals with the "Regulations for the Prevention of Air Pollution from Ships". It sets limits on NOx and SOx emissions from ship exhausts, and prohibits deliberate emissions of ozone depleting substances. It aims at progressive reduction in emissions of SO_x, NO_x and particulate matter and the introduction of emission control areas (ECAs) to reduce emissions of those air pollutants further in designated sea areas. Under the revised MARPOL Annex VI, the sulphur limit would be reduced from current 3.50% to 0.50%, effective from 1*

January 2020. The limits applicable in ECAs for SO_x and particulate matter were reduced to 0.10%, from 1 January 2015. Progressive reductions in NO_x emissions from marine diesel engines installed on ships are also included. It also prohibits the incineration on board ships of certain products, such as contaminated packaging materials and polychlorinated biphenyls (PCBs). The vessels are required to comply with the regulations and should have the Prevention Certificate.

Many works and some support services will be outsourced during the construction and operational phases. Hence, appropriate contractor's agreement to manage fugitive emissions will be necessary and will from a part of the agreement while issuing the work order.

Environment Management Cell

The MbPT has an Environment Management Cell responsible for the management of the environment. Among its several other function the Cell will be responsible for compliance of the norms given under the EPA rules (including air environment) through the following actions:

- Develop and implement air environment management plan based on the framework in the EIA.
- Strictly follow the HSE policy of the MbPT.
- Install air monitoring equipment for the proposed expansion if required, after analysing the existing monitoring grid.
- Engage third party MOEFCC / NABET accredited consultant for air monitoring if necessary for measuring parameters listed in the EIA (Table ---) as per the schedule mentioned therein.
- Ensure, technical members of the Environment Management Cell Members must meet regularly to assess the air monitoring programme as designed and critically analyze the data collected periodically.
- Identify non-compliances to required Regulatory Environmental Norms and inform the management with recommendations to revert air quality to the set norms.
- Assure regulatory compliance with all relevant rules and regulations.
- Ensure regular operation and maintenance of pollution control devices.
- Prepare Contractor's Agreement and ensure its proper implementation.
- Maintain documents of good environmental practices and applicable environmental laws as ready reference.
- Maintain environmental related records.
- Coordinate with regulatory agencies, external consultants and monitoring laboratories.
- Maintain log of complaints and the action taken.

4. Compliance: Submit latest status of PIL 140 of 2014.

Answer: Yes, PIL has been filed against the Filling of Victoria and Princes Docks

Name of the Court	Bombay High Court
Case No	PIL no 149 of 2014/ 31 of 2015

PIL no. 140 of 2014 was changed to PIL 31 of 2015. The PIL was discharged on 09.07.2019 (copy enclosed).

8942

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IN THE HIGH COURT OF JUDICATURE AT BOMBAY
ORDINARY ORIGINAL CIVIL JURISDICTION

PUBLIC INTEREST LITIGATION NO. 31 OF 2015

1. Mr. Sayyed Abdul Nabi & Anr. .. Petitioners

Vs.

1. Union of India & Ors. .. Respondents

Mr. A. N. Nesarikar I/by R. N. Kachave for petitioners.
Mr. N. R. Prajapati for respondent No.1.
Mr. Abhay L. Patki, Addl. Govt. Pleader for respondent Nos.4 to 8 & 10-State.
Ms. K. H. Kastkar for respondent No.9 – MCGM.
Mr. Ajai Fernandes a/w Ms. Sneha Pandey I/by Motiwalla & Co. for respondent No.11.

CORAM: PRADEEP NANDRAJOG, CJ. &
N. M. JAMDAR, J.

JULY 09, 2019.

P.C.

1. The Petitioners had earlier filed Writ Petition No. 617 of 2013 pleading identical facts and making same prayers. The said Writ Petition was disposed of vide order dated 19 September 2013, which reads as under :-

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" *By the administrative directions of the Hon'ble the Chief Justice dated 25 July 2013 the hearing of the Petition has been assigned to this Bench.*

2] The two Petitioners before the Court in these proceedings under Article 226 of the Constitution have stated that they carry on the business of transportation and are contractors in respect of earth moving machinery and dumper operating trucks carrying debris, excavated soil etc. The petitioners claim to be office bearers of Mega City Debris Truck Welfare Association.

3] In pursuance of an Environmental Clearance which has been granted on 15 June 2006 by the Union Ministry of Environment and Forest (MOEF) the Mumbai Port Trust is carrying on certain work at the Prince's & Victoria Docks. MOEF modified the conditions of its clearance on 9 November 2006. The contract was awarded to the Seventh Respondent on 1 April 2009.

4] The grievance of the Petitioners is that the Seventh Respondent to whom the contract has been awarded was committing a breach of the environmental clearance by utilising unlicensed debris taken from building sites. The Collector Mumbai City had after issuing a notice to show cause imposed a penalty of Rs.1.68 Crores on 3 December 2012 on the Seventh Respondent. The relief which has been sought in the Petition

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is a direction to the competent authorities to stop trucks carrying debris of construction material from entering into the limits of the port area for the purpose of dumping into the sea.

5] An affidavit in Reply has been fled by the Chief Engineer denying the averments contained in the Petition. It has been stated that in pursuance of the Environmental Clearance that was granted on 15 June 2006 as modified on 9 November 2006 the construction of the berth has already been completed; dredging is about 75% complete and the work of filling was about 80% complete. It has been stated that the project is to be completed in two phases; the first phase for a capacity of 0.8 Million TE Units envisages a berth length of 700 meters, while in the second phase there would be an additional capacity of 0.4 Million TE Units and an additional berth of 350 meters. On behalf of the Port Trust it has been submitted that having regard to the modified clearance which has been granted by the MOEF on 9 November 2006, it has been envisaged that surplus dredged soil shall be dumped at the designated site.

6] For the purpose of these proceedings, this Court has not, it must be clarified, entered upon the question as to whether there is due compliance with the terms of the environmental clearance. MOEF is at liberty to enforce and verify compliance of the conditions of its clearance. The Petition, the learned Counsel appearing for the Petitioners clarifies is not filed

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in the public interest. The two Petitioners are contractors and their grievance was in regard to the conduct of the Seventh Respondent. Learned Senior Counsel appearing on behalf of the Port Trust has informed the court on instructions that the contract of the Seventh Respondent has been terminated on 27 June 2013 and that the Port Trust would now be required to issue fresh tenders for appointing a suitable contractor for carrying out the work of the project. Similarly the action which has been taken by the Collector against the Seventh Respondent does not fall for consideration and the Collector is at liberty to enforce the order which has already been passed. We see no reason to entertain the Petition in view of the fact that the contract of the Seventh Respondent has been terminated. We also clarify that this order will not come in the way of MOEF ensuring due compliance of the terms of its environmental clearance and from taking action if it is found that the conditions of the environmental clearance have not been observed. The Petition is accordingly dismissed. There shall be no order as to costs."

2. Instant petition has been filed as a Public Interest Litigation and we dismiss the same for the reasons same issues were raised by the petitioners in Writ Petition No. 617 of 2013. The order dated 19 September 2013 disposing of Writ Petition No. 617 of 2013 shows that the Mumbai Port Trust was carrying

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on works at Prince's and Victoria Docks resulting in soil being excavated and debris generated. A private contractor, impleaded as respondent No.7 in the said Writ Petition to whom contract was awarded to remove the debris and excavated soil, was named as the culprit on the plea that he was violating the environmental clearances. It be noted that after carrying out dredging activities, the debris had to be used for reclamation of sea at existing Prince's and Victoria Docks. The further grievance appears to be that the contractor started dumping hazardous bio-waste instead of dredging sand from the designated areas.

3. Suffice it to state the order dated 19 September 2013 left it open for the Competent Authority to decide whether there was compliance with the environmental clearances. Orders passed by the Collector were permitted to be taken to their logical conclusion.

4. The Public Interest Petition is accordingly dismissed.

N. M. JAMDAR, J. CHIEF JUSTICE

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CERTIFIED TO BE A TRUE COPY
this 11th day of July 20 19

For Registrar (O.S.) / Prothonotary
and Senior Master, High Court, Bombay

Applied On : 10/07/19

Section Writer / Comparer :

Examined By :

No. Of Pages : 5

Further Charges : Paid On:

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11 JUL 2019

Ready On :

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5. **Compliance: Submit details of quantity to be dredged with drilling and blasting and the alternate technology to be used avoiding blasting and the alternate technology to be used avoiding blasting as agreed.**

Answers:

Details of dredging:

Capital dredging of 13 million m3 of soft material and 0.3 million m3 of rock was approved vide Prior Environmental Clearance out of which 6.6 million m3 including rock is to be dredged. Thus, about half of dredging has been completed and the balance will be taken-up during the present proposal. The details of dredging completed and balance remaining are as follows:

Description	Status	Remarks
Capital dredging in berth pocket (-) 16 m (CD)	Completed in front of 2 berths	Will be taken at a later stage in front of the third berth
Capital dredging in approach channel and Turning Circle (-) 13.5 m (CD)	Partially completed up to levels varying from (-) 8 (CD) to (-) 13.5 m (CD)	Due to change in cargo dredging only up to (-) 11 m (CD) is Required
Rock dredging (0.3 million m3)	To be undertaken	Proposed to be completed in the present project.

Disposal of dredged material

The sediment in the area demarcated for deepening is largely composed of silt and clay with some fine sand and is not suitable for reclamation making it inevitable to dump it at a properly identified marine site. After hydrodynamic modelling and other studies in the past, CWPRS has recommended a site in the Arabian Sea at 20 m depth contour. The material dredged has been dumped at this site and the same procedure will be followed for the remaining dredged material. An annual maintenance dredging volume of 0.90 million m3 will also be dumped at this site.

Rock dredging and alternate technology envisaged

The rock formation at Mumbai mainly consists of basalt. The reddish brown residual soil is the weathering product of underlying basalt. These are hard to very hard in strength with Unconfined Compressive Strength (UCS) varying from 266±98 (from where taken it should be 39 t0 45 MPA MPA) and hence cannot be dredged using soft methods such as water jet or cutter head dredgers making blasting inevitable. The weathered rock will be removed by cutter section dredger and hard rock by controlled blasting.

Ground vibration and underwater shock waves are two of the major impacts from underwater blasting. The ground vibrations can potentially cause damage to structures; while excessive underwater shock wave pressure, apart from causing damage to nearby structures, marine life may be impacted. Special precautions are taken since underwater blasting is frequently conducted in the vicinity of structures such as berths, breakwaters, port buildings

etc. The magnitude of these vibrations is determined by factors such as quantity of explosive charge, characteristics of the rock and distance from the blast. Through controlled blasting these adverse effects can be effectively controlled and considerably minimized.

Controlled blasting generally adopts safe vibration level by developing appropriate Rock Blasting and Dredging Plan that takes into consideration attenuation relations with respect to propagation characteristics of blast vibrations, calculating safe explosive weights for different distances, design of blasting patterns and monitoring of blast vibrations during actual operations that enables modification in remaining blasting runs, if required. MbPT will contract known experts in the field such as CWPRS to examine the issue, provide their expertise while tendering the works and also critically evaluate the probable bids. The selected contractor will be bound to provide detailed plan for underwater controlled blasting and dredging which will be again examined by the expert and an agreed plan will be the included in the contract. The expert will be associated with the project in the supervisory capacity until the controlled blasting is completed.

From their previous experience of controlled blasting of rock and dredging MbPT is confident that the Elephanta Caves, a World Heritage Site located more than -- km from the site proposed for controlled blasting will not have any impact of ground vibrations and underwater shock waves. Underwater drilling and blasting of hard rock was successfully carried out for the construction of Liquid Chemical Berth within the Harbour and the safety of various structures was established by monitoring of blast vibrations on selected structures during actual blasting. Seismographs set-up at Elephanta Caves which was 2.6 km from the blast site did not record any vibrations due to blasting. The Elephanta Caves are about 7.9 km from the project site, hence, will not be affected.

If recommended by experts the possibility of using an Air Bubble Curtain system will be examined and deployed if feasible. Air Bubble Curtain system an air bubble curtain is created by using heavily perforated hoses around the construction site as a ring and are driven into the seabed. The hose ends are connected to superchargers on ship deck and pump compressed air into the hoses. The air escapes through the holes according to a well-defined pattern. The result is a curtain of millions of small bubbles that rise to the water surface, forming a kind of whirlpool around the construction site. The air bubbles change the density of the water and thus break down the sound waves thereby reducing the impacts of blast waves.

Underwater blasting operation

Generally drilling rigs mounted on pontoon or barge are stationed at the predetermined site to be dredged, drilling towers are positioned over the specified drill hole locations and drilling commences. In the Over Burden Drilling – the commonly used method, a casing pipe is driven up to the rock through the overburden, the inner drill rods are inserted through the casing pipe and the hole is drilled to the required depth. Usually a number of holes are drilled in a line. The barge is shifted to the next position for drilling and another row of holes are drilled. The operation is repeated until required number of holes are completed as planned. Each

hole is loaded with pre-decided number of cartridges of explosives and primed with nonelectric shock tube initiation system provided with delay detonators. Detonation is carried out through a lead-in line connected to an ignition device.

6. **Compliance: Details of provision of CETP/STP for the treatment of oily water, wash off, sewage and ships rejects, including solid waste treatment with capacities**

Answer:

Ship-generated Waste

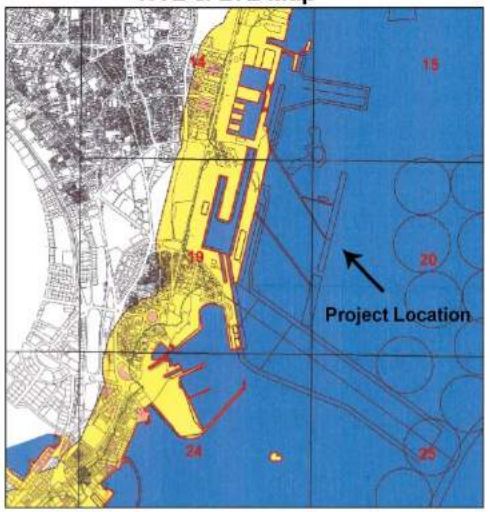
Major wastes generated by ships include garbage, solid waste, sewage, oily ballast and bilge water. According to MARPOL release of oily wastes directly to marine environment by ships is prohibited. MARPOL Annex V seeks to eliminate and reduce the amount of garbage being discharged into the sea from ships. Under MARPOL Annex V, garbage includes all kinds of food, domestic and operational waste, all plastics, cargo residues, incinerator ashes, cooking oil, fishing gear, and animal carcasses generated during the normal operation of the ship are liable to be disposed of continuously or periodically. The effectiveness of ships to comply with the discharge requirements of MARPOL depends largely upon the availability of adequate port reception facilities, especially within special areas. Hence, MARPOL Annex V also obliges Governments to ensure the provision of adequate reception facilities at ports and terminals for the reception of garbage without causing undue delay to ships, and according to the needs of the ships using them.

To comply with MARPOL Annex V requirement, MbPT has its own hazardous container cargo complex wherein hazardous cargo is stored as per the requirement under the supervision of qualified officer to handle such cargo. The quantum of hazardous cargo in containers which will be handled at the project is very insignificant and this will be shifted directly to the hazardous cargo complex.

7. Compliance: Submit a map as per C½a t 2018 superimposed with earlier CRZ map.
Answer: Superimposed map is attached below

HTL & LTL Map superimposed on CZMP Map

HTL & LTL Map



CZMP Map

8. **Compliance: Submit form-2 (Application for prior Environmental clearance) as per MoEFcc's OM No.22-8/2018-IA-III dated 20th April, 2018.**

Answer: Form 2 is attached below.

Form 2

APPLICATION FOR PRIOR ENVIRONMENTAL CLEARANCE

1.	**Details of Project**			
	a.	Name of the Project (s)		**Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour**
	b.	Name of the Company / Organization		**Mumbai Port Trust**
	c.	Registered Address		**Port House, S. V. Marg, Ballard Estate, Mumbai - 400001**
	d.	Legal Status of the Company		**Government**
	e.	Joint Venture (Yes/No)		**NA**
		If Yes,		**NA**
		(i) No. of JV Partners *(Multiple Entries Allowed)*		**NA**
2.	**Address for the correspondence**			
	a.	Name of the applicant		**N.G . Mirajkar**
	b.	Designation (Owner / Partner / CEO)		**Chief Engineers**
	c.	Address		**Mumbai Port Trust Port House, S. V. Marg, Ballard Estate, Mumbai**
	d.	Pin code		**400001**
	e.	e-mail		**chiefengineer@mumbaiport.gov.in**
	f.	Telephone No.		**022-65564031**
	g.	Fax No.		**022-2261 6804 / 3469**
3.	**Category of the Project/ Activity as per Schedule of EIA Notification, 2006**			
	a.	Project /Activity l(a)(i) / l(a)(ii) / l(b) / l(c) / l(d) / 1(e)/2(a)/ 2(b) / 3(a)/ 3(b) / 4(a) / 4(b)(i)/ 4(b) (ii) / 4(c) / 4(d) / 4(e) / 4(t) / 5(a)/ 5(b) /5(c) /5 (d) / 5(e) / 5(t) / 5(g) / 5(h) / 5(i) /5*(j)* / 6(a) / 6(b) / 7(a) / 7(b)/ 7 (c) / 7 (d) / 7 (d)/ 7 (e) / 7 (f) / 7 (g) /7(h) / 7 (i) / 8 (a) / 8 (b)]		**7(e)**
	B	Category (A/B1/B2)		A
		If B1 or B2		NA
		Reason for application at Central Level / State level (in case of B2 projects)		NA
		If Others		NA
	c.	Please Specify		NA
	d.	EAC concerned (only for category A Projects only) (Coal Mining / Non-coal Mining / Thermal / River Valley & Hydro / Industry- I / Industry-II / Infrastructure-I /Infrastructure-II /Nuclear & Defence /CRZ)		Infrastructure II

	e.	New /Expansion /Modernization / One Time Capacity expansion (only for Coal Mining) / Expansion under Para 7(ii) /Modernization under Para 7(ii) /Change of Product Mix under Para 7(ii))		Expansion
4.	**Location of the Project**			
	a.	Plot /Survey /Khasra No		--
	b.	Village		**NA**
	c.	Tehsil		**Mumbai**
	d.	District		**Mumbai**
	e.	State		**Maharashtra**
	f.	Pin Code		**400001**
	g.	Bounded Latitudes (North)		**18°56'37.35"N**
		From		
		To		
	h.	Bounded Longitudes (East)		**75°51'13.88"E**
		From		
		To		
	i.	Survey of India Topo Sheet No.		Topo sheet showing project location along with 10 km study area is attached (E43G13)
	j.	Upload Topo Sheet File *(Upload pdf only)*		**Annexure 1**
	k.	Maximum Elevation Above Means Sea Level (AMSL)		**5.5 m**
	l.	Upload (kml) File *(Upload kml only)*		**--**
	m.	Distance of Nearest HFL from the project boundary within the study area		
	n.	Seismic Zone (Zone: 1 / 2 /3 / 4 / 5)		**3**
5.	**Whether project is executed in multiple States (Yes /No)?**			**NO**
		If Yes		**NA**
	a.	Number of States in which Project will be Executed (e.g. *1,2,3,4,5,6)*		**NA**
	b.	Main State of the Project		**Maharashtra**
	c.	Other State *(Multiple Entries Allowed)* If the project to be executed, does not belong to any state, then state category could be selected as 'Other')		**NA**

State	**District**	**Tehsil**	**Village**
Maharashtra	**Mumbai**	**Mumbai**	**Mumbai**

6.	**Details of Terms of Reference (ToR)**			
	a.	Whether ToR is mandatory for submitting		**Yes**
		If Yes		
	b.	Date of issue of ToR / Standard ToR		**12.07.2017**
	c.	MoEF&CC / SEIAA File No.		**File no. 10-10-2017-IA-III**
	d.	Upload ToR letter *(PDF* only)		**Annexure 2**
7.	**Details of Public Consultation**			
	a.	Whether the Project Exempted from Public Hearing		**Yes**
		(Yes/No)?		
		If yes,		

		Reason		**The PH for the same project was conducted in 2005. Hence the proponent made the request for exemption which was accepted**
	b.	Supporting Document *(upload pdf only)*		**27.05.2017**
	c.	Whether details of Public Hearing available (Yes/No)?		**Yes** **Public Hearing for this Project Conducted on 29.09.2005**
		If No,		**NA**
		Reason thereof		**NA**
	d.	Supporting Document *(upload pdf only)*		**NA**
		If Yes,		-
	e.	Date of Advertisement of Public Hearing		-
	f.	Copy of advertisement in English (Upload PDF only)		-
	g	Whether Public hearing was presided over by an officer of the rank of Additional District Magistrate or above (Yes/No)?		**Yes**
		If yes		
	h.	Designation of Presiding Officer (District Magistrate / District Collector / Deputy Commissioner /others - please specify)		**District Collector**
	i.	Copy of duly signed Proceedings of Public Hearing in English *(Upload pdf only)*		**Annexure 3**
	j.	Date of Public Hearing		**29.09.2005**
	k.	Venue of Public Hearing:		**Durbar Hall, Asiatic Society of Mumbai**
		Village		**Mumbai**
		Tehsil		**Mumbai**
		District		**Mumbai**
		State		**Maharashtra**
	l.	Distance of Public Hearing Venue from the Proposed Project (km)		**Within 1 km**
	m.	No. of people attended		**11**
	n.	If the multiple public hearings conducted		**No**
		Pl give the details of each PH as per (e) to (o) above		**NA**
8.	**Details of Project Configuration /Product *(Multiple Entries Allowed)***			
	a.	Whether the project is New (Yes/No?)		**No this is an expansion**
		If yes,		
	b.	Project Configuration		

		Plant/Equipment/Facility	**Configuration**	**Remarks if any**

		• 2 Nos. of 100 Ton Mobile Harbour Crane for Ship Shore Handling • 27 Nos. of 20 Ton Tractor-Trailors for Shore at Stack yard. • 7 Nos. of 30 Ton Mobile Cranes at Stack yard for Steel Cargo. • 2 Nos. of Reach Stacker at Stack yard for Containers.	--	--
	c.	Product			

Product / Activity	Quantity	Unit	Mode of transport / transmission of product
Steel , RO- RO , containers and any other clean cargo	--	--	--

9.	**If Expansion /Modernisation /One Time Capacity expansion (only for Coal Mining) / Expansion under Clause 7(ii) /Modernisation under Clause 7(ii) / Change of Product Mix under Clause 7(ii)))**				
	a.	Details of environmental clearance granted earlier			
		(i)	Date of issue of environmental clearance		**15.06.2006** **EC Extension 07.01.2014**
		(ii)	MoEFCC / SEIAA File Number		**10-18/2005-IA-III**
		(iii)	Upload EC Letter		**Annexure 4**
	b.	**Details of certified report on compliance of earlier environmental clearance conditions**			
		(i)	Details of Regional Office of MoEFCC /Zonal Office of CPCB / SPCB / UTPCC from which certified report on compliance of earlier environmental clearance conditions obtained		**Mumbai**
		(ii)	Letter No		**CE.CF.226(CTP)/ENV/1435 dated 09.02.2016**
		(iii)	Status of clearance Compliance		**--**
		(iv)	Certified report on compliance of earlier environmental conditions (Including Monitoring Report) *(Upload pdf only)*		**Annexure 5**
		(v)	Date of site visit		--
	c.	**Details of Consent to Operate**			
		(i)	Whether Consent to operate obtained (Yes/No)?		--
			If yes,		--
		(ii)	Upload Copies of all Consent to operate obtained since inception *Upload pdf onlv)*		--
		(iii)	*Date of issue*		--
		(iv)	Valid up to		--
		(v)	File No.		--
		(vi)	Application No.		--
		(vii)	Upload Copy of Consent to operate valid as on date *(Upload pdf only)*		--

		Details of Capacity Expansion *(Multiple Entries Allowed)*				
		Product / Activity	Quantity from	Quantity to	Unit	Mode of transport / transmission of product
		NA	**NA**	**NA**	**NA**	**N**
		Details of Configuration *(Multiple Entries Allowed)*				
		Plant/Equipment/Facility	Configuration from	Configuration to	Remarks if any	
		NA	**NA**	**NA**	**NA**	
10.	**Project Cost**					
	a.	Total Cost of the Project at current price level (in Lakhs)			129000 Lakhs	
	b.	Funds Allocated for Environment Management (Capital) (in Lakhs)			100 Lakhs	
	c.	Funds Allocated Towards ESC (Entrepreneur Social Responsibility) (in Lakhs)			--	
	d.	Funds Allocated for Environment Management Plan (EMP) (Recurring per Annum) (in Lakhs)			Recurring cost per Annum is Included in 100 Lakhs mentioned in point 10.b above.	
11.	**Whether project attracts the General Condition specified in the Schedule of EIA Notification (Yes/No)? [provide name of WL/CPA/ESA/Inter-state boundary / International boundary and distance from the project**				**No**	
		If Yes				
	a.	Protected Area Notified Under the Wild Life(Protection) Act, 1972			**No**	
	b.	Critically Polluted Areas as identified by the Central Pollution Control Board from Time to Time			**No**	
	c.	Notified Eco-Sensitive Areas			**No**	
	d.	Inter-State Boundaries and International Boundaries			**No**	
12.	**Whether projects attract the Specific Condition specified in the Schedule of EIA Notification (Yes/No)?**				**No**	
		If Yes				
		If any Industrial Estate / Complex / Export processing Zones / Special Economic Zones / Biotech Parks / Leather Complex with homogeneous type of industries such as Items 4(d), 4(f), 5(e), 5(f), or those Industrial estates with pre-defined set of activities (not necessarily homogeneous, obtains prior environmental clearance, individual industries including proposed industrial housing within such estates / complexes will not be required to take prior environmental clearance, so long as the Terms			**NA**	

		and Conditions for the industrial estate/complex are complied with (Such estates/complexes must have a clearly identified management with the legal responsibility of ensuring adherence to the Terms and Conditions of prior environmental clearance, who may be held responsible for violation of the same throughout the life of the complex/estate		
13	**Raw Material / Fuel Requirement *(Multiple Entries Allowed)***			**NA**
	a.	Details of Raw Material / Fuel Requirement		NA

Raw Material /Fuel	Quantity /annum	Unit	Source (in case of Import, please specify country and Name of the port from which Raw Material / Fuel is received)	Mode of Transport	Distance of source from project site (in km) (in case of import, distance from the port from which raw material / fuel is received	Type of linkage (Linkage /fuel supply /Agreement / e-auction / MoU/IOA/Captive Open Market / Others)
Construction of jetty is completed only yard work is balance, which is fraction of work. Hence no deviation from previously approved quantities anticipated i.e sand – 1,72,830 cu.m and Metal 2.98,480 cu.m						

		in case of expansion proposals, total requirement of raw material /fuel shall be given		
	b.	Upload copy of Linkage / Fuel Supply Agreement / e-auction / Memorandum of Understanding / Letter of Allocation / Captive source /others.		**NA**
14.	**Baseline Data (Air / Water / Noise / Soil / Ground water table/ Others)**			
	a.	Period of Base Line Data Collection		**November 2017 to February 2018**
		From (DD/MM/YYYY)		**08.11.2017**
		To (DD/MM/YYYY)		**26.02.2018**
	b.	Season (Summer /Pre-monsoon / Post-monsoon / Winter)		**Winter**
	c.	No. of Ambient Air Quality (AAQ) Monitoring Locations		**Please Refer to chapter 3 of the EIA report**
	d.	Details of AAQ Monitoring *(Multiple Entries Allowed)*		
	e.	No. of Ground Water Monitoring Locations *(Multiple Entries Allowed)*		
	f.	Details of Ground Water Monitoring		
	g.	No. of Surface Water Monitoring Locations		

	h.	Details of Ground Water Monitoring *(Multiple Entries Allowed)*		
	i.	No. of Ambient Noise Monitoring Locations		
	j.	Details of Noise Monitoring *(Multiple Entries Allowed)*		
	k.	No. of Soil Monitoring Locations *(Multiple Entries Allowed)*		
	l.	**Ground Water Table**		
	i.	Range of Water Table Pre-Monsoon Season (Meters Below Ground Level (m bgl)):		--
		From		--
		To		--
	ii.	Range of Water Table Post-Monsoon Season (Meters Below Ground Level (m bgl)):		--
		From		--
		To		--
	iii.	Whether Ground Water Intersection will be there (Yes / No)?		No
		If Yes,		
		(i) Upload Copy of Central Ground Water Authority Letter *(Upload pdf only)*		NA
		(ii) Letter No.		NA
		(iii) Date of issue		NA

15. **Details of Water Requirement (During Operation)** *(Multiple Entries Allowed)*

a. Details

Source	Quantity KLD	Method of Water withdrawal	Distance from Source	Mode of Transport
Mumbai municipal corporation	**3.38 (Domestic)**	**Other**	**--**	**Pipeline**

- **Source: Surface/Ground water/ sea/others**
- **Mode of Transportation: Pipeline/ Canal/ Others**
- **Method of water withdrawal: Barrage/Weir/Intake well/ Jack well/Tube well/ Open well/others**

	b.	Upload copy of permission from Competent Authority (Upload pdf only)	
	c.	Letter No.	MbPT has its own network of water supply system, water will be supplied from the same.
	d.	Date of issue	
	e.	Permitted quantity	
		f. Whether Desalination is proposed (Yes/ No)	**NA**
		If Yes,	**NA**
		(i) Desalination capacity (KLD)	**NA**
		(ii) Quality of Brine (KLD)	**NA**
		(iii) Mode of Disposal of brine	**NA**

16 **Wastewater Water Management (During Operation)**

Type/ Source	Quantity of Waste Water Generated (Kilo Litre per Day)	Treatment Capacity (Kilo Litre per Day)	Treatment Method	Mode of Disposal	Quantity of Treated Water Used in Recycling / Reuse (Kilo Litre per Day)	Quantity of Discharged Water (Kilo Litre per Day)
Domestic	2.25	NA	NA	Existing Sewer system	NA	NA

17 **Solid Waste Generation Management** *(Multiple Entries Allowed)*

Item	Quantity per Annum	Unit	Distance from Site	Mode of Transport	Mode of Disposal
Domestic waste (Operation Phase)	12,150	NA	NA	Collecting vehicles	Solid Waste generated will be disposed to MCGM

Item:- (Industrial waste, Solid waste, Fly ash, Bottom Ash, Hazardous waste (as per Hazardous and Other Waste Management Rules 2016), E waste, Bio-Medical waste, Construction & Demolition waste, Plastic Waste, Others)
Units :(Tons. Kilometer)
Mode of Disposal: (Treatment, Storage and Disposal Facility(TSDF), Authorized Re-cyclers Landfills, Sanitary Landfills, Others)

18 **Air Quality Impact Prediction** ***(Multiple Entries Allowed)***

Criteria Pollutant	Unit	Baseline Concentration	Minimum value	Incremental Concentration	Total GLC	Prescribed Standard

Air Quality impact and mitigation measures is given in EIA report

Parameter:- (PM10, PM, S02, NOx, Others parameters specific to the sector)
Unit:- (Microgram per Meter Cube, NA)

19 **Power Requirement**

a.	Quantity (Kilo Volt Amps (KVA))		5000 KVA (5 MVA)
b.	Source		MbPT gets the electric supply from TATA electric and has its own distribution network The power needed will be supplied from MbPT network.
c.	Upload Copy of Agreement (Upload pdf only)		N.A
d.	Standby Arrangement (Details of DG Sets)		N.A
e.	Stack Height (in m)		N.A

20 **Land Ownership Pattern (Prior to the project proposal) in Ha**

a.	Forest land		NA
b.	Private Land		NA

	c.	Government Land		No extra land is required, as stated in earlier clearance, the existing Princess and Victoria Docks areas which were used for handling cargo have been filled up. At present, only Victoria dock will be used for stacking of cargo such as steel, car, containers and any other clean cargo.
	d.	Revenue Land		NA
	e.	Other Land		NA
		Total land		--
21	**Present Land Use breakup in Ha**			
	a.	Agriculture area		NA
	b.	Waste/Barren area		NA
	c.	Grazing/Community Area		NA
	d.	Surface Water Bodies		NA
	e.	Settlements		NA
	f.	Industrial		NA
	g.	Forest		NA
	h.	Mangroves		NA
	i.	Marine Area		NA
	j.	Others (Specify)		22.25 Ha of land is developed partly by backfilling Existing Victoria & Princes Docks
		Total		22.25 Ha

22. Land Requirement for various activities (Multiple entries allowed) in Ha

Description Activity/Facility/Plants/ Others	Land Requirement	Remarks
	20.83 ha, which includes 0.67 Ha covered storage area and 0.63 Ha for Misc.	
	As per earlier EC back filling of Victoria and Prince's dock for an area of 22.5 Ha already completed.	
Green Belt	In addition, existing areas of MbPT already has adequate green belt cover.	
Total		

Activity/Facility/Plants/Others include: Main Plant, Township, Greenbelt, Ash pond, Quarry area, OB dump area, Safety zone, Tailing Pond, Landfill ,Water Reservoir, De-salination plant, Area for solid waste management , Built-up area, Others

23.	**Ecological & Environmental Sensitivity (Within 10 Km):- WLS-Wild Life Species; NPA Notified Protected Area; ESAs-Eco Sensitive Area ;ESZs-Eco Sensitive Zone)**	
	a.	Details of Ecological Sensitivity

		Details of Ecological Sensitivity	Name	Distance from the project (km)	Remarks
		NA	NA	NA	NA
		Details of Ecological Sensitivity: - Critically Polluted areas, WLS, NPA,ESA's, ESZ's, Corridors, Wildlife Corridors)			
	b.	Whether NBW recommendation is require (Yes/No)?		**NA**	
		If yes		**NA**	
		Upload NBWL recommendation in PDF		**NA**	
	c.	Details of Environmental Sensitivity		**NA**	
		Details of Ecological Sensitivity	Name	Distance from the project (km)	Remarks
		NA	**NA**	**NA**	**NA**
		Details of Environmental Sensitivity:- (Forest, Archeological Sites, Defense Installations. Others			
	d.	Whether NOC /Permission from the competent authority is required		**No**	
		If yes		**NA**	
		Upload NOC/Permission from the competent authority in		**NA**	
24.	**Forest Land**				
	1	Whether any Forest Land involved (Yes/No)?		**No**	
		If Yes			
	a.	Forests Clearance Status (In-Principle(Stage-I) Approval Obtained /Final : (Stage-IT)Approval Obtained/ Forest Clearance Under Process (Stage-I) 1		**NA**	
		If In-Principle (Stage-I) Approval Obtained,		**NA**	
		(i) MoEFCC fie number		**NA**	
		(ii) Date of In Principle (Stage-I) approval		**NA**	
		(iii) Area diverted		**NA**	
		(iv) Upload FC Letter (Upload pdf only and attach it as Annexure-FC		**NA**	
		If Final (Stage-11) Approval Obtained,		**NA**	
		(i) MoEFCC file number		**NA**	
		(ii) Date of Final Approval		**NA**	
		(iii) Date of In-Principle Approval		**NA**	
		(iv) Area diverted		**NA**	
		(v) Upload FC Letter ***(Upload pdf only and attach it as Annexure-***		**NA**	
		If Forest Clearance under process (Stage-I),		**NA**	
		(i) MoEFCC file number		**NA**	
		(ii) Area applied		**NA**	
		(iii) If Forest Clearance under process (Stage-II),		**NA**	
		(iv) MoEFCC file number		**NA**	
		(v) Area applied		**NA**	
	b.	Legal Status of Forest Land (reserved, Protected, Private,		**NA**	

		If Others,		**NA**
		Please Specify Others		**NA**
25	**Tree Cutting, if any**			
	a.	No. of Trees Cut for the Project (if Forestland not involved)		**Permission already obtained from Tree authority (MCGM) at the time earlier EC.**
	b.	Details of Tree Cutting and Planting of Trees (Uploadpdf		**NA**
26	**Land Acquisition Status**			
	a.	Acquired Land		**NA**
	b.	Land yet to be acquired		**NA**
	c.	Status of Land acquisition if not acquired		**NA**
27	**Rehabilitation and Resettlement (R&R)**			
	a.	No. of Villages		**NA**
	b.	No. of Households		**NA**
	c.	No. of PDFs (Project Displaced Families)		**NA**
	d.	No. of PAFs (Project Affected Families)		**NA**
	e.	Funds Allocated for R&R		**NA**
	f.	Status of R&R (Completed / In-progress / Yet to start)		**NA**
28	**Whether there is Presence of Schedule-I Species (Yes/No)?**			
		If yes,		**NA**
	a.	Details of Schedule-I Species		**NA**
	b.	Whether conservation plan for Schedule-I Species has been prepared		**NA**
		If Yes.		**NA**
		(i) Upload conservation plan (Upload only PDF)		**NA**
		(ii) Fund Provision made		**NA**
		(iii)Period of Implementation		**NA**
	c.	Whether conservation ~Plan for Schedule-I species has been approved by		**NA**
		(i) Upload copy of approval (Upload PDF Only)		**NA**
		(ii) Letter No.		**NA**
		(iii)Date of issue		**NA**
		(iv) Recommendations if any		**NA**
29	**Whether there is Presence of Water Bodies in Core Area (Yes/No)?**			
		If yes,		
	a.	Details of Water Bodies in Core Area		Project lies in Arabian sea and is in CRZ IV as per CRZ Notification 2006
	b.	Whether there is Diversion required (Yes/No)?		**NO**
		If yes,		
	c.	Details of diversion required		**NA**

	d.	Details of study conducted		**NA**
	e.	Whether permission has been obtained from competent authority		**No**
		(i) Upload copy of permission (Upload PDF Only)		**NA**
		(ii) Letter No.		**NA**
		(iii) Date of issue		**NA**
		(iv) Recommendations if any		**NA**
30	**Whether there is Presence of Water Bodies in Buffer Area (YesNo)?**			
		If Yes		
	a.	Details of Water Bodies in Buffet Area		**NA**
	b.	Direction of water Bodies in Buffer Area (North/ south / East / West		**NA**
	c.	Distance of Water Bodies in Buffer Area (Kilo meters)		**NA**
31	**Manpower Requirement**			
	a.	Permanent employment during construction (for 18		40
	b.	Permanent employment during operation (for 342		50
	c.	Temporary employment during construction (for		75
	d.	Temporary emp1oyment during operation (for 342		25
	e.	No, of working days		9000
	f.	Total Manpower		62,445 man days during construction 7,69,500 man days during operation phase
32	**Green Belt in Ha**			
	a.	In case of new projects		The proposed OCT is coming up on offshore area. Existing areas of MbPT already has adequate green belt cover.
	i.	Total area of Green Belt		
	ii.	Percentage of Total Project Area		
	iii.	No. of Plants to be Planted		
	iv.	Funds Allocated for Plantation		
	v.	Upload Green Belt Plan (Upload PDF only)		
	b.	In case of Expansion /modernization / change in product		NA

	i.	Description	Existing	Proposed	Total
		Total Area of Green Belt	**NA**	**NA**	**NA**
		Percentage of Total Project Area	**NA**	**NA**	**NA**
		No of Plants	**NA**	**NA**	**NA**
		Fund Allocated	**NA**	**NA**	**NA**
	ii.	Upload Green Belt Plan (Upload PDF Only)			**NA**
33	**Project Benefit** *(Multiple Entry Allowed)*				

		(Project benefits shall include environmental. social and others)			:	The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai Region and Maharashtra.
34	**Whether the Project / Activity attract the provisions of CRZ (Yes/No)?**					
	If yes,					
	1	**Project Details**				
		a.	CRZ Classification: (CRZ I (A), CRZ 1(B), CRZ 11, CRZ 111, CRZ IV (A), CRZ IV(B))			**CRZ II & IV**
		b.	Location type: (Non-Eroding Coast, Low and Medium Eroding Coast,			**Non-Eroding Coast**
		c.	Details of Mangroves Land Involved, If Any			**Nil**
		d.	Are of Mangroves Land (hectare)			**Nil**
		e.	EIA(Terrestrial) Studies: (Carried out, Not Carried Out)			**Carries Out**
			If Carried out,			
			1)	Summary Details of EIA (Terrestrial) Studies		**Covered in chapter 3 of EIA report**
			2)	Upload Recommendation made in EIA (Upload		
			3)	Period of Study from (EIA Terrestrial)		**Nov 17- Feb 18**
			4)	Period of Study to (EIA Terrestrial)		**Nov 17- Feb 18**
			If not Carried Out			**NA**
				Give Reason		**NA**
		f.	EIA(Marine) Studies: (Carried out, Not Carried Out)			**Carried Out**
			If Carried out,			
			1)	Summary Details of EIA (Terrestrial) Studies		**Covered in chapter 3 of EIA report**
			2)	Upload Recommendation made in EIA (Upload PDF Only)		
			3)	Period of Study from (EIA Terrestrial)		**Nov 17- Feb 18**
			4)	Period of Study to (EIA Terrestrial)		**Nov 17- Feb 18**
			If not Carried Out			**NA**
				Give Reason		**NA**
		g.	Disaster Management Plan/ national oil Spill Disaster Contingency			**Covered in EIA report**
	2	**Description of the Project Under Consideration**				
		a.	Type of the Project: (Resort Buildings/civic amenities, Coastal Roads/Roads on Stilt, Pipelines from Thermal Power Blow Down, Marine Disposal of Treated Effluent. Facility for Storage of Goods/Chemicals, Offshore Structures, Desalination Plants, Mining of Rare Earth/Atomic Minerals, Sewage Treatment Plants, Lighthouse, Wind Mills, Others)			**Offshore Structures**
			If Resort/Buildings/Civic amenities,			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Total Area Built-up Area (hectare)		**NA**

			3)	Height of Structure		**NA**
			4)	FSI Ratio		**NA**
			5)	The governing Town Planning Rules/Regulations		**NA**
			6)	Details of Provision of Car Parking Area		**NA**
			If Coastal Roads/Roads on Stilt			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Area of Land Reclamation		**NA**
			3)	Estimated Quantity of Muck/Earth for		**NA**
			4)	Carrying Capacity of Traffic		**NA**
			If Pipelines from Thermal Power Blow Down			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Length of Pipeline		**NA**
			3)	Length Traversing CRZ Area		**NA**
			4)	Depth of Excavation		**NA**
			5)	Width of Excavation		**NA**
			6)	Length of Pipeline from Seashore to Deep Sea		**NA**
			7)	Depth of outfall Point from Surface of Sea Water		**NA**
			8)	Temperature of Effluent above Ambient at		**NA**
			If Marine Disposal of Effluent			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Location of Intake/Outfall		**NA**
			3)	Depth of Outfall Point		**NA**
			4)	Length of Pipeline		**NA**
			5)	Length Traversing CRZ Area		**NA**
			6)	Depth of Excavation		**NA**
			7)	Width of Excavation		**NA**
			8)	Length of Pipeline from Seashore to Deep Sea		**NA**
			9)	Depth of outfall Point from Surface of Sea Water		**NA**
			10)	Depthof Water at Disposal Point		**NA**
			11)	Type of Disposal		**NA**
			If Facilityty for Storage of Goods/Chemicals			
			1)	Agency Name for Preparing CFZ Maps		**NA**
			2)	Name and Type of Chemical		**NA**
			3)	End use of the Chemical		**NA**
			4)	No. of Tanks for Storage		**NA**
			5)	Capacity of Tanks		**NA**
			If Offshore Structure			
			1)	Agency Name for Preparing CRZ Maps		**Institute of remote sensing Anna University**
			2)	Exploration & Development		**--**

			3)	Depth of Sea Bed		**Dredging has been partially completed from (-) 8m Cd to (-)13.5m CD in approach channel and turning circle.**
			4)	No. of Ring/Platform		**700x58 m** **1.2 KM trestle**
			5)	Details of Group Gathering Station		**NA**
			If Desalination Plant			
			1)	Agency Name for Preparing CFZ Maps		**NA**
			2)	Capacity of Desalination		**NA**
			3)	Total Brine Generation		**NA**
			4)	Temperature of Effluent above Ambient at		**NA**
			5)	Ambient Salinity		**NA**
			6)	Disposal Point		**NA**
			If mining of Rare Earth/ Atomic Minerals			
			1)	Agency Name for Preparing CFZ Maps		**NA**
			2)	Capacity of Mining		**NA**
			3)	Volume/Area to be Mined		**NA**
			4)	Type of Minerals to be Extracted		**NA**
			5)	End use of the Mineral		**NA**
			If Sewage Treatment Plant,			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Capacity		**NA**
			3)	Total Area of Construction		**NA**
			4)	Compliance of effluent parameters as laid down by		**NA**
			5)	Whether discharge is in sea water/creek?		**NA**
				Distance of Marine Outfall Point from shore/from the tidal river		**NA**
				Depth of Outfall Point from Sea Water Surface		**NA**
				Depth of Sea at Outfall Point		**NA**
			If lighthouse			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Total Area of Construction		**NA**
			3)	Height of Structure		**NA**
			If Wind Mill			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Capacity (MW)		**NA**
			3)	Transmission Lines: (Overhead, Underground)		**NA**
			4)	Diameter of Windmill		**NA**
			5)	Length of Blade		**NA**
			6)	Speed of Rotation		**NA**
			7)	Height of Structure		**NA**

			If Others			
			1)	Agency Name for Preparing CRZ Maps		**NA**
			2)	Please Specify with Salient Features		**NA**
			3)	Upload relevant documents (Upload pdf Only)		
	3.	**Distance of the Project (In Meter) from LTL/HTL to be Stated**				
		a.	Clause of CRZ Notification Under which the Project is a Permissible/Regulated Activity			**Project located in CRZ II & IV**
		b.	Whether CRZ Map Indicating HTL/LTL Demarcation in 1:4000 Scales Prepared? (Yes/No)			**Yes**
			If Yes,			
			1)	Distance of the Project (In Meter) from LTL/HTL to be Stated		**Project located in CRZ II & IV**
			2)	Upload Maps *(kml file)*		**Annexure 6**
			3)	Distance of the Project (In Meter) from LTL/HTL to be Stated		**Project located in CRZ II &IV** **Distance: 0 meters**
			4)	Upload Maps *(kml file)*		**Annexure 6**
		c.	Whether Project Layout Superimposed on CRZ Map 1:4000 Scales?:			**Yes**
			If Yes,			
			1)	Upload Maps *(kml file)*		**Annexure 6**
		d.	Whether CRZ Map 1:25000 Covering 7 km Radius Around Project Site Prepared? (Yes/No)			
			If Yes,			
			1)	Upload Maps *(kml file)*		**--**
		e.	**Whether CRZ Map Indicating CRZ –I,II,III and IV including Other Notified ESAs Prepared?: (Yes/No)**			**--**
			If Yes,			
			1)	Upload Maps *(kml file)*		--
		f.	**NOC from State Pollution Control Board Obtained : (Yes/No)**			**No**
			If Yes,			
			1)	Upload Copy of NOC (Upload pdf Only)		-
		g.	Details of Rain Water Harvesting System			**NA**
	4.	**Recommendation of State Coastal Zone Management Authority**				
		a.	Upload Copy of CZMP *(Upload pdf Only)*			**Annexure 7**
		b.	State the Condition Imposed			**Annexure 8**

		c.	Social and Environmental Issue and Mitigations Measures Suggested Including but not Limited to R&R, Water, Air, Hazardous Waste, Ecological aspects, etc(Brief Detail to be Provided)		--
35	**Sector Specific Details**				
I	**Whether the Proposal is mining of minerals (coal/non-coal) Project (Yes/No)?**				
		If Yes.			**No**
	1.	No. mineral to be mined (Multiple Entries Allowed)			**NA**
	2	Mine Capacity in ROM (Run of Mine)			**NA**
	3	Upload 500 meter Cluster Certificate from State Mines and Geology in case of minor minerals (Upload pdf Only)			**NA**
	4	**Mining**			
		a.	Approval Letter No.		**NA**
		b.	Date of Approval		**NA**
		c.	Upload Approved Letter (Upload pdf only)		**NA**
		d.	Approved by State Mines & Geology Department / Indian Bureau of Mines / Ministry of Coal / Ministry of Mines /		**NA**
		e	If Others		**NA**
			Please Specify		**NA**
		f.	Approved Mining Lease Area		**NA**
		g.	Approved Capacity		**NA**
	5	**Technical Details**			**NA**
		a.	Total Geological Reserve (Million Ton)		**NA**
		b.	Mineable Reserve (Million Ton)		**NA**
		c.	Extractable Reserve (million Ton)		**NA**
		d.	Percent of Extraction (%)		**NA**
		e.	Grade of Ore/Core/Mineral		**NA**
		f.	Stripping ration		**NA**
		g.	Category of Gaseousness (Only for Core Mining, Others may write		**NA**
		h.	Average Gradient (Degree)		**NA**
		i.	Maximum Thickness of Seams (meter) (Only for Coal Mining, Others		**NA**
		j.	Mining Method (Opencast / underground/ Mixed(Opencast +		**NA**
		k.	Life of Mines (years)		**NA**

	6	**Details of beneficiation (including**			**NA**
		a.	Whether it is proposed to install crusher within the mining leas area		**NA**
			If Yes,		**NA**
		b.	No. fo Crusher		**NA**
		c.	Details of Crusher (Multiple Entries Allow)		**NA**
		d.	Whether it is proposed to install beneficiation Plant/ Coal washery		**NA**
			If Yes,		**NA**
		e.	Beneficiation/washing Technology		**NA**
		f.	Capacity		**NA**
	7	**Details of Seams if applicable**			**NA**
		a.	No. of Seams		**NA**
		b.	Thickness of seams to be worked on		**NA**
		c.	Maximum Thickness of Seams (meter) (if not Applicable, may write NA)		**NA**
	8	**Details of Mining Lease**			**NA**
		a.	Details of Mining Lease		**NA**
		b.	Upload Letter of Intent (Upload pdf Only)		**NA**
		c.	Date of Execution of mining with reference number		**NA**
		d.	Validity of Mining Lease		**NA**
		e.	Upload copy of Executed Lease deed valid as on Date (Upload pdf		**NA**
		f.	Earlier Renewals (Multiple Entries Allowed)		**NA**
					NA
	9	OB (Over Burden) Management (only if mining Method :			**NA**
		a.	**Details of External Dumps**		**NA**
			i) No. of OB Dumps		**NA**
			ii) Total Area (in Hectare)		**NA**
			iii) Height)in meter)		**NA**
			iv) Quantity (in Million Cubic meter)		**NA**
			v) No. of year backfill up vi)		**NA**
		b.	**Details of Internal Dumps**		**NA**
			i) No. of Internal Dumps		**NA**
			ii) Total Area (in Hectare)		**NA**
			iii) Height (in meter)		**NA**
			iv) Quantity (in Million Cubic Meter) v)		**NA**
	10	**Details of Topsoil Management**			**NA**

		a.	Quantity of Topsoil excavated during the entire life of the mine (in		**NA**
		b.	Quantity of Topsoil proposed for utilization for reclamation during the		**NA**
		c.	Quantity of Topsoil proposed for utilization for Other activities during the entire life of the mine (in Millions Cubic		**NA**
	11	**Details of Mine Void (Only if Mining Method : Opencast)**			**NA**
		a.	Area (in Hectare)		**NA**
		b.	Depth (in Meter)		**NA**
		c.	Volume (in Milion Cubic Meter)		**NA**
	12	**Details of Quarry (Only if Mining Method :Opencast)**			**NA**
		a.	Final Void of (Hectare_)		**NA**
		b.	At a Depth of (meter which is proposed to be converted into a Water Body)		**NA**
	13	**Details of Transportation**			**NA**
		a.	In Pit/ Underground to Surface		**NA**
		b.	Surface to Siding/Loading		**NA**
		c.	Transportation / Conveyor Details		**NA**
	14	**Details of Land Usage (Pre-Mining)**			**NA**
	15	**Details of Transportation**			**NA**
		a.	In Pit/ Underground to Surface		**NA**
		b.	Surface to Siding/Loading		**NA**
		c.	Transportation / Conveyor Details		**NA**
	16	**Details of Land Usage (Pre-Mining)**			**NA**
	17	**Details of Land Usage (post- Mining)**			**NA**
	18	**Details of Reclamation (only if Mining Method: Opencast) Total Afforestation Plan Shall be Implemented Covering of Mining. This will include:**			**NA**
		a.	External OB Dump (in hectare		**NA**
		b.	Internal Dump (in hectare)		**NA**
		c.	Quarry (in hectare)		**NA**
		d.	Safety Zone (in hectare)		**NA**
		e.	Final Void of (hectare)		**NA**
		f.	At a Depth of (meter which is proposed to be converted into a Water		**NA**
		g.	Density of Tree Plantation per ha (in no.)		**NA**

		h.	Others in ha (such as Excavation area along ML Boundary, along Roads and Infrastructure, Embankment area and in		**NA**
		i.	Total afforestation plant (in hectare)		**NA**
	19	**Status of Progressive Mining Closure Plan (For Expansion Projects only)**			**NA**
		a.	Implementation of Various Activities as per Approved Progressive		**NA**
		b.	Any deviation from the Approved Progressive Mine		**NA**
		c.	Total Area Excavated (in Hectare)		**NA**
		d.	Total Area Backfilled after Excavation (in Hectare)		**NA**
		e.	Total Area Reclaimed (in hectare)		**NA**
	20	**Actual Coal/Ore Production via-a-visa sanctioned capacity Since inception *(Multiple Entries Allowed)***			**NA**
II	**Whether proposal is for Thermal Project(including captive power plant and Waste Heat Recovery Plant) (Yes/No)**				**NA**
		If Yes.			**NA**
	1	**Specification of the Plant**			**NA**
		a.	Technology proposed (Supercritical, Sub-Critical, CFBC. AFBC. PFBC.IGCC. Incineration, Pyrolysis, Gasification. Bio-		**NA**
		b.	Plant Load Factor (%)		**NA**
		c.	Station Heat Rate (Kcal/Kwh)		**NA**
		d.	Steam Rate /Flow rate		**NA**
		e.	Boiler Temperature		**NA**
		f.	Boiler Pressure		**NA**
		g.	Type of Stack (Single flue /Bi- flue / Tri- flue)		**NA**
		h.	No. of Stacks		**NA**
	2	**Details of fuel Linkage (please specify if multiple linkages are involved)**			**NA**
		a.	Type of Linkage (Linkage / Fuel Supply Agreement / e-auction/MoU/ LOA/Captive / Open Market/ Others)		**NA**
		b.	Quantity of Linkage Granted		**NA**
		c.	Date of Linkage		**NA**
		d.	Duration of Linkage		**NA**
	3	**Details of Transportation of Fuel**			**NA**
	4.	**Details of Fuel Characteristics**			**NA**
		a.	Gross Calorific Value (Kcal/ Kg)		**NA**
		b.	Ash Content (%)		**NA**
		c.	Sulphur Content (%)		**NA**
		d.	Moisture Content (%)		**NA**
		e.	Mercury (mg/kg)		**NA**
		f.	Fixed Carbon (%)		**NA**
		g.	Volatile Matter (%)		**NA**

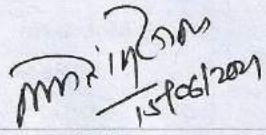
	5.	**Details of Cooling System**			**NA**
		a.	Type of Cooling System : Air Cooled / Water Cooled		**NA**
		b.	Type of Draft : Natural Draft/ Forced Draft		**NA**
		c.	Type of Air Circulation : Parallel floe/ Counter flow		**NA**
		d.	Cycles of Cooling (COC)		**NA**
		e.	Water requirement for cooling (m^3/day)		**NA**
		f.	Boiler blow down temperature		**NA**
III	**Whether proposal is for River Valley & Hydroelectric Project (Yes/ No)?**				**NA**
	If Yes,				**NA**
	1.	**Sub Sector:** (Multipurpose Project /Hydroelectric Project / Irrigation Project)			**NA**
	2.	Name of the River			**NA**
	3.	Whether Cumulative Impact Assessment ad Carrying Capacity Study of			**NA**
		a.	Status of Cumulative Impact Assessment ad Carrying Capacity Study		**NA**
		b.	Details of Cumulative Impact Assessment and carrying Capacity		**NA**
		c.	Upload relevant Document *(Upload pdf Only)*		**NA**
	4.	Type of Project: (Construction of Dam Barrage / Run of the River / Lift			**NA**
	5.	Dam Height (meter) if applicable			**NA**
	6.	Dam Length (meter) if applicable			**NA**
	7.	Total Submergence Area (hectare)			**NA**
	8.	Details of Gross Command Area (GCA) (hectare)			**NA**
	9.	Details of Irrigable Command Area (ICA) (hectare)			**NA**
	10.	Details of Culturable Command Area (CCA) (hectare)			**NA**
	11.	Total Command Area (hectare)			**NA**
	12.	Powerhouse Installed Capacity			**NA**
		a.	Type of Powerhouse : (Underground / Surface)		**NA**
		b.	Capacity (megawatts)		**NA**
		c.	No. of Tunnels		**NA**
		d.	No. of units		**NA**
		e.	Bank of River: (Left / Right / Both)		**NA**
	13.	**Generation of Electricity Annually**			**NA**
		a.	Quantity of Generation of Electricity Annually (in Giga Watt Hours		**NA**
		b.	Water availability at various percentage		**NA**
		c.	Machine Availability (%)		**NA**
		d.	Cost of Electricity Generation (in INR)		**NA**
		e.	Internal Rate of Return (%)		**NA**
		f.	Year of Achievement of Internal Rate of Return		**NA**

	14.	Catchment Area Treatment Plan (Upload PDF)			**NA**
	15	Muck Management Plan			**NA**
		a.	Upload Muck Management Plan in PDF		**NA**
		b.	Total Excavation Muck (Cubic meter)		**NA**
		c.	Utilized for Construction (Cubic meter)		**NA**
		d.	No. of Muck Disposal Area (Ha)		**NA**
		e.	Total Muck Disposal Area (Ha)		**NA**
	16	Fishery Development and Management Plan			**NA**
		a.	Fishery Development and Management Plan in PDF		**NA**
		b.	No. of Fingerling		**NA**
		c.	No. of Fish Species		**NA**
		d.	Name of the umbrella Fish Species (Scientific name)		**NA**
	17	Status of Environmental Flow			**NA**
	18.	Free Flow Stretch (km)			**NA**
	19.	Approval of Central Water Commission			**NA**
		a.	Approval Reference No		**NA**
		b.	Date of Approval		**NA**
		c.	Upload Copy of Approval *(Upload pdf Only)*		**NA**
	20	Approval of Central Electricity Authority			**NA**
		a.	Approval Reference No		**NA**
		b.	Date of Approval		**NA**
		c.	Upload Copy of Approval *(Upload pdf Only)*		**NA**
	21.	Details of Coffer Dam (upstream) if applicable			**NA**
		a.	Quantity of the Material (m^3)		**NA**
		b.	Length of the Dam (m)		**NA**
		c.	Quantity of the Muck to be Used (m^3)		**NA**
		d.	Quantity of borrow Material to be Used (m^3)		**NA**
		e.	Decommissioning Plan *(Upload pdf Only)*		**NA**
	22	Details of Coffer Dam (downstream) if applicable			**NA**
		a.	Quantity of the Material (m^3)		**NA**
		b.	Length of the Dam (m)		**NA**
		c.	Quantity of the Muck to be Used (m^3)		**NA**
		d.	Quantity of borrow Material to be Used (m^3)		**NA**
		e.	Decommissioning Plan *(Upload pdf Only)*		**NA**
IV	**Whether the proposal is infrastructure project (Yes/ No)**				**NA**
	If Yes?				
	1	**Details of Building Construction**			**NA**
		a.	Maximum Height of the Building (Meters)		**NA**
		b.	Total No. of Flats to be Build		**NA**
		c.	No, of Buildings		**NA**
		d.	Total Plot Area (sqm)		**NA**
		e.	Total Built up area (sqm)		**NA**
	2	**Foreshore Facilities And/or Marine Disposal**			**NA**

		a.	Whether Project Involves Foreshore Activities and/or marine Disposal (Yes/no)?		**NA**
		If Yes,			
		b.	Upload Shoreline Study (upload pdf Only)		**NA**
		c.	Type of Cargo		**Steel , RO RO, containers and any other clean cargo and occasionally passenger vessels.**
		d.	Quantity of Cargo		**Earlier EC granted for: 9.6 MTPA** **Now Proposed: 4.17 MTPA**
		e.	Control of Cargo		--
		f.	Dust Control Measures		**Since cargo is to be handled in RoRo, Steel, container and other clean cargo. Hence no dust generation is anticipated. However, regular cleaning will be done.**
		g.	Quantum		--
		h.	Quantity of Dredging (Cubic meter per Day)		**The capital dredging upto -16m CD in berth and upto -13.5 m CD at turning circle was approved in the earlier EC, part of which has already been done. Due to change in cargo profile, no additional dredging will be required, However, for balance dredging blasting will be done which was permitted in the earlier clearance.**
		i.	Type of Dredging (Capital/ Maintenance)		**Capital**
		j.	Disposal of Dredge Material		**DS3 AS per the earlier EC**
		k.	Details of Outfall Diffusers		**NA**
		m.	No. of Dilution Expected		**NA**
		n.	Distance at which the outlet will Reach Ambient		**NA**
		o.	Details of Monitoring at outfall		**NA**
		p.	Copy of NOC from PCB in case of Marine Disposal (Upload pdf		**NA**
	3.	**Rain Water Harvesting**			
		a.	No.of Storage		**NA**
		b.	Capacity		**NA**
		c.	No. of Recharge Pits		**NA**
		d.	Capacity		**NA**
	4.	**Parking**			
		a.	Details of 4- Wheeler/2- Wheeler parking		**NA**
	5.	**Energy Saving Measures**			

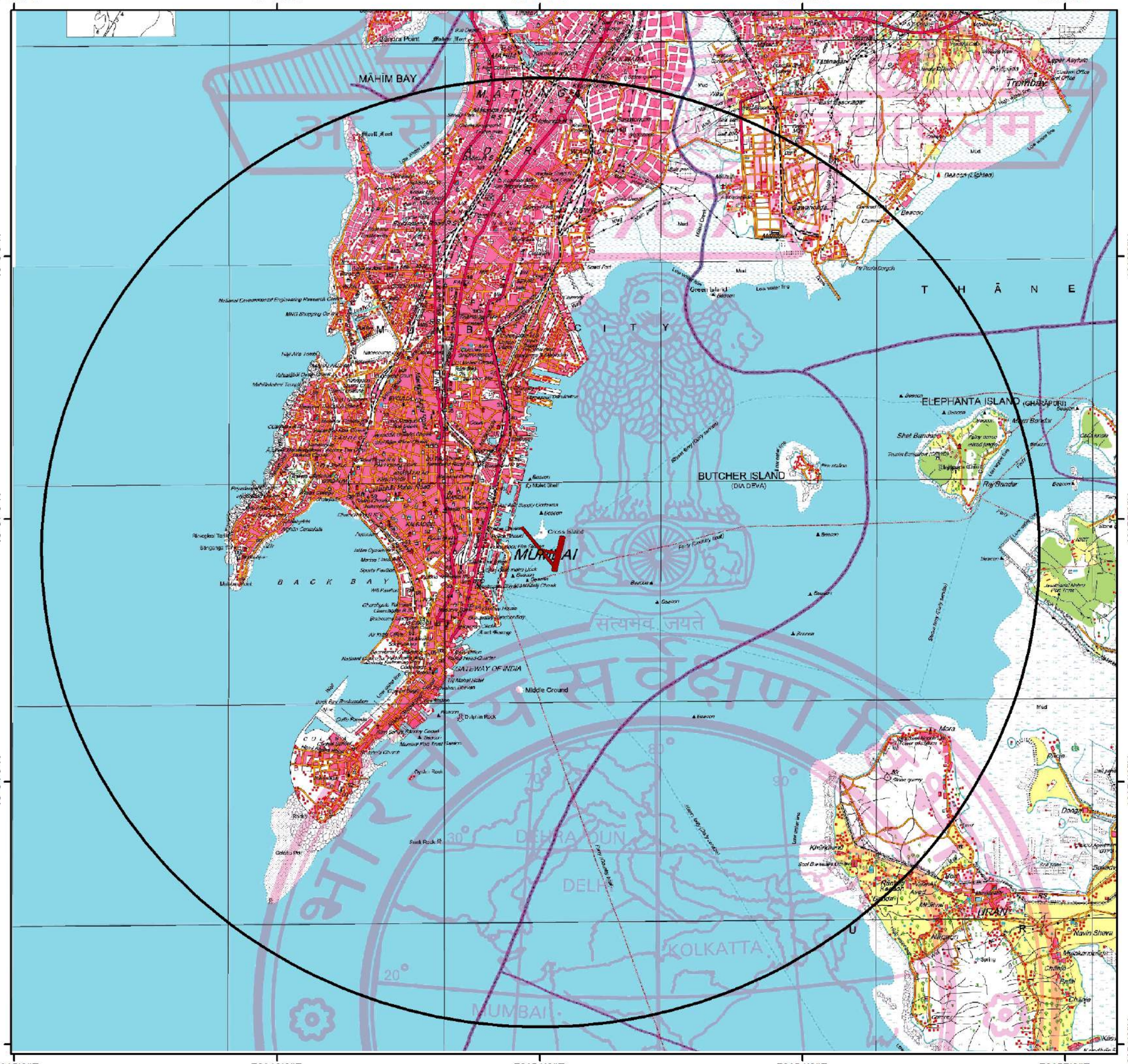
		a.	Source/Mode		**NA**
		b.	Percentage		**NA**
		c.	Quantity		**NA**
	6	**Other Details**			
		a.	Details of Impact on Water Bodies and Drainage patters of catchment area, if any		**NA**
		b.	Details of Traffic Density Impact Assessment and Modeling Study		**NA**
		c.	In case of Underground Tunnel Projects below the Forest land- Subsidence Impact Study report		**NA**
	7	**Type of Industries to be established with Industrial Estate as per their category A/B**			**NA**
	8	**Length of the alignment in case of highway projects**			**NA**
	9	**Details of Bridges / ROB/Inter changes, Flyovers, Vehicle Underpass and Pedestrian underpass(in case of Highway Projects)**			**NA**
		a.	Total no. of bridges		**NA**
		b.	Total No. of ROB		**NA**
		c.	Total No. of Interchanges		**NA**
		d.	Total No. of Flyovers		**NA**
		e.	Total No. of vehicle underpass		**NA**
		f.	Total No. of Pedestrian Underpass		**NA**
		g.	Details of Other utilities rail and road corridors		**NA**
36	**Details of Court Cases if any**				
	a.	Whether there is any Court Cases Pending against the Project and/or land in which the project is proposed to			**NA**
		If yes,			**NA**
	b.	Name of the Court *(District Court / High Court /*			**NA**
		If name of Court: (District Court, High Court,NGT,Tribunals)			**NA**
	c.	Name of the Sub-Court			**NA**
	d.	Case No.			**NA**
	e.	Orders/ Directions of the court,			**NA**
	f.	Case Details			**NA**
	g.	Upload Court Order if any *(Upload pdf Only)*			**NA**
37	**Details of direction issued under Environment (Protection) Act / Air (Prevention & Control of Pollution)) Act / Water (Prevention & Control of Pollution) Act**				
	a.	Whether any direction issued under Environment (Protection) Act / Air (Prevention & Control of Pollution)) Act / Water (Prevention & Control of Pollution) Act			**NA**
	b.	Details of directions issued under Environmental (Projection) Act / Air(Prevention & Control of Pollution)) Act / Water (Prevention & Control of Pollution) Act			**NA**
	c.	Upload Copy of direction issued under Environment (Protection) Act / Air (Prevention & Control of Pollution)) Act / Water (Prevention & Control of Pollution) Act			**NA**
	d.	Compliance of status of the directions			**NA**

38	**Details of EIA Consultant**			
	a.	Have you hired Consultant for preparing documents		**Yes**
		If No,		
		(i) Reason for not engaging the Consultant		--
		If yes,		
		(i) Accreditation No.		**NABET/EIA/1922/RA 0147**
		(ii) Name of The EIA Consultant		**Terracon Ecotech Pvt. Ltd.**
		(iii) Address		**202, Kingston, Tejpal Road, vile parle (E)**
		(iv) Mobile No.		--
		(v) Landline No		**+91.22.2086 3940/41/42**
		(vi) E-mail Id		--
		(vii) Category of Accreditation (Eligible for Category A/ Eligible for		**A**
		(viii) Sector of Accreditation		**9**
		(ix) Validity of Accreditation		**20.01.2022**
		(x) Upload Certificate of Accreditation cerfied by QCI/NABET		**Annexure 9**
39	**Documents to be attached**			
I	***If Project Type is New / Expansion / Modernization / one –time capacity expansion for coal mining:***			
	a.	Upload Copy of EIA/EMP Report		**Annexure 10**
	b.	Upload Copy of Risk Assessment Report		**Included in EIA report**
	c.	Upload Copy of Feasibility Report/ Detailed Project Report (DPR) / Detailed Engineering Report/ Detailed Conceptual Plan/ Approved mining Plan (in case of Mining Proposals) *(Upload pdf only)*		**Annexure 7 of EIA report**
	d.	Upload Copy of Final layout Plan *(Upload pdf only)*		**Annexure 11**
	e.	Upload Cover Letter (*Upload pdf only and attach it as Annexure- document of Cover letter)*		-
	f.	Upload a copy of documents in support of the competence/authority of the person making this application to make application on behalf of the User *Agency (Upload pdf only and attach it as Annexure-authorization)*		-
	g.	Upload copy of District Survey Report (for mining of minor minerals only)		**NA**
		Upload copy of Replenishment Study Report & Baseline Survey data		**NA**
	h.	Upload Addition File, if any *(Upload pdf Only)*		**NA**
II	***If project type is other than New / Expansion /Modernization / one-time capacity expansion for coal mining***			
	a.	Upload Copy of Feasibility Report/ Detailed Project Report (DPR) / Detailed Engineering Report/ Detailed Conceptual Plan/ Approved mining Plan (in case of Mining Proposals)		**NA**
	b.	Upload Copy of Final layout Plan *(Upload pdf only)*		**NA**

	c.	Upload Cover Letter (*Upload pdf only and attach it as Annexure- document of Cover letter)*		**NA**
	d.	Upload a copy of documents in support of the competence/authority of the person making this application to make application on behalf of the User		**NA**
	e.	Upload Addition File, if any *(Upload pdf Only)*		**NA**
	f.	Upload Updated Form I *(Upload pdf Only)*		**NA**
40	**Undertaking**			
	a	I hereby give undertaking that the data and information given in the application and enclosures are true to be best of my knowledge and belief and I am aware that if any part of the data and information found to be false or misleading at any stage,the project will be rejected and clearance given,if any to the project will be revoked at our risk and cost. In addition to above,I hereby give undertaking that no activity / construction / expansion has since been taken up		
		Signature		
	a	Name		**N. G. Mirajkar**
	b	Designation		**Chief Engineer**
	c	Company		**Mumbai Port Trust**
	e	Address		**Mumbai Port Trust Port House, S. V. Marg, Ballard Estate, Mumbai**

Annexure 1

Survey of India Topo Sheet



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Map Title

Map showing
Project Location and 10 km Study Area

Project Name

EIA and EMP for-
"Construction of two offshore construction berths and development of container terminal on BOT basis in Mumbai harbour"

Legend

Project site

10km Study Area

Scale

1:25000

Source

Survey of India
Toposheet no.-
E43A16 & E43G13

CONVENTIONAL SYMBOLS

Express highway with toll; with bridge; with distance stone
Roads, metalled; according to importance
Roads, double carriageway; according to importance
Unmetalled road. Cart-track. Path-track with pass. Foot-path
Streams: with track; in box; undefined. Canal
Dams: masonry or rock-filled; earthwork. Weir
River: dry; with water channel; with island & rocks. Tidal river
Submerged rocks. Shoal. Swamp. Reeds
Wells: lined; unlined. Tube-well. Spring. Tanks: perennial; dry
Embankments: road or rail; tank. Broken ground
Railways, broad gauge: double; single with station; under constn.
Railways, other gauges: double; single with distance stone; do.
Mineral line or tramway. Kiln. Cutting with arrow
Contours with sub-features. Rocky slopes. Cliffs
Sand features(1)flat, (2)sand-hills(permanent), (3)dunes(shifting)
Towns or Villages: inhabited; deserted. Fort
Huts: permanent; temporary. Tower. Antiquities
Temple. Chhatri. Church. Mosque. Idgah. Tomb. Graves
Lighthouse. Lightship. Buoys: lighted; unlighted. Anchorage
Mine. Wire on rails. Glass. Soda
Palms: palmyra; other. Plantain. Conifer. Bamboo. Other trees
Areas: cultivated; wooded. Surveyed tree
Boundary, international
,, state: demarcated; undemarcated
,, district; subdivision; tahsil or taluk; forest
Boundary pillars: surveyed; unfenced
Heights. Triangulated: station; point; approximate — △200 . 200 .200
Benchmark: geodetic; tertiary. Canal — BM 53-3 .BM 65-3
Post office. Telegraph office. Overhead tank
Rest house or Inspection bungalow. Circuit house. Police station
Camping ground. Forest: reserve; protected — R F P F
Spelled names: administrative; locality or tribal — KIKRI NAGA
Hospital. Dispensary. Veterinary Hospital / Dispensary
Aerodrome. Helipad. Tourist site
Power line with pylons surveyed; with poles unsurveyed

Annexure 2

ToR letter

C.E.-C.P.-226(CT D)/ENV
Annexure II 159

मुंबई पोर्ट ट्रस्ट
स्थापत्य अभियंत्रिकी विभाग
Mumbai Port Trust
Civil Engineering Deptt.
प्रकल्प ऑफिस/ Project Office
फाईल नं./File No. CE.CP.226/CT D/ENV
आ.रं.क्र./Inward No. 462
दिनांक/Date 20.07.2017

F. No. 10-10-2017-IA-III
Government of India
Ministry of Environment, Forest and Climate Change
(IA.III Section)

Indira Paryavaran Bhawan,
Jor Bagh Road, New Delhi - 3

Date: 11th July, 2017

21-7-17
DY. C.E. P-II
S.E. P-I 25/07/17
E.E. P/I
AE
20/7/17

To,

The Chief Engineer
M/s Mumbai Port Trust,
Mumbai Port Trust, Port House,
Shoorji Vallabhdas Marg, Ballard Estate,
Mumbai- 400 001 (Maharashtra)
Fax: 022- 22616804, 3469

Subject: Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour by Mumbai Port Trust - Terms of Reference - reg.

Sir,

This has reference to your online proposal No. IA/MH/MIS/62185/2017 dated 4th February, 2017, submitted to this Ministry for seeking Terms of Reference (ToR) in terms of the provisions of the Environment Impact Assessment (EIA) Notification, 2006 and Coastal Regulation Zone (CRZ) Notification, 2011, under the Environment (Protection) Act, 1986.

2. The proposal for grant of Terms of Reference (ToR) to the project 'Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour' promoted by Mumbai Port Trust was considered by the Expert Appraisal Committee (Infra-2) in its meeting held on 25th - 27th May, 2017.

3. The details of the project, as per the documents submitted by the project proponent, and also as informed during the above said meeting, are under:-

(i) Name of the proposal is "Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour" by Mumbai Port Trust. This project is an expansion of existing Mumbai Port Trust for improvement of traffic at Mumbai Port. The site was selected by the consultants (JICA) who prepared the master plan for development of Mumbai Port.

(ii) Ministry of Environment and Forest, Govt. of India had accorded the Environment Clearance for the subject project by letter No.10-18/2005-IA-III dated 15.6.2006 and modified the same vide corrigendum by letter No.10-18/2005-IA-III dated 09.11.2006. Thereafter, the validity has been extended upto 14.6.2016 vide letter No. 10-18/2005-IA-III dated 07.01.2014. Further 6 monthly reports were regularly sent to zonal office.

(iii) No extra land is required, as stated in earlier clearance, the existing Princess and Victoria Docks areas which were used for handling cargo other than containers have been filled up. At present, only Victoria docks will be used for stacking of cargo such as steel, car, containers and any other clean cargo.

(iv) Cost of the project is estimated at Rs. 1290 crores.

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(v) Court cases: PIL no 149 of 2014 later changed as PIL 31 of 2015 has been filed by the petitioner Shri Sayyed Abdul against the Filling of Prince's and Victoria Docks. The works of filling of Prince's and Victoria Docks have already been completed.

(vi) Employment potential: The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai region and Maharashtra.

(vii) Benefits of the project: The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai region and Maharashtra.

4. During deliberation in the meeting held on 25th-27th May, 2017, the project proponents made a request for an exemption from public hearing as the same was conducted in 2005. The Committee allowed the same and advised the project proponent to include a chapter on queries and comments raised by the participants during the earlier Public hearings suitably documented in the form of a management plan drawn to address to the expressed concerns.

5. The EAC, after detailed deliberations, recommended the project for grant of ToR as specified by the Ministry as Standard ToR in April, 2015 for the said project/activity and the following TOR in addition to Standard ToR for preparation of EIA-EMP report without public hearing. As per the recommendation of the EAC, the Ministry of Environment, Forest and Climate Change hereby accords ToR to the project 'Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour by Mumbai Port Trust for preparation of the Environmental Impact Assessment (EIA) Report and Environmental Management Plan (EMP) with the following additional conditions:

i. Importance and benefits of the project.

ii. Submit a copy of layout superimposed on the HTL/LTL map demarcated by an authorized agency on 1:4000 scale.

iii. Recommendation of the SCZMA.

iv. Stage – I forest clearance to be submitted.

v. Various Dock and shipbuilding facilities with capacities for existing and proposed project.

vi. Study the impact of dredging on the shore line.

vii. A detailed impact analysis of rock dredging.

viii. Study the impact of dredging and dumping on marine ecology and draw up a management plan through the NIO or any other institute specializing in marine ecology.

ix. A detailed analysis of the physico-chemical and biotic components in the highly turbid waters round the project site (as exhibited in the Google map shown during the presentation), compare it with the physico-chemical and biotic components in the adjacent clearer (blue) waters both in terms of baseline and impact assessment and draw up a management plan.

x. Details of Emission, effluents, solid waste and hazardous waste generation and their management in the existing and proposed facilities.

xi. The existing project should avail of and submit consent to operate from the State Pollution Control Board.

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xii. Requirement of water, power, with source of supply, status of approval, water balance diagram, man-power requirement (regular and contract).

xiii. Wastewater management plan.

xiv. Details of Environmental Monitoring Plan.

xv. The Project proponents will submit copy of the certified compliance report on the Environmental clearances issued earlier to the project.

xvi. To prepare a detailed biodiversity impact assessment report and management plan through the NIOS or any other institute of repute on marine, brackish water and fresh water ecology and biodiversity. The report shall study the impact on the rivers, estuary and the sea and include the intertidal biotopes, corals and coral communities, molluscs, sea grasses, sea weeds, subtidal habitats, fishes, other marine and aquatic micro, macro and mega flora and fauna including benthos, plankton, turtles , birds etc. as also the productivity. The data collection and impact assessment shall be as per standard survey methods.

xvii. The EIA report shall address to all representations and complaints received against the project including those received from the representations from the Conservation Action Plan.

xviii. The Study area should be 15 Kms round the project site for base level studies in order that the areas already polluted or under stress can also be identified for impacts and mitigation.

xix. The EIA would follow the orders of the respective courts and include a chapter in the EIA on the Court cases including those at the Supreme Court and the NGT.

xx. The Queries and comments raised by the participants during the earlier Public hearing held in 2005 will be suitably documented in the form of a management plan drawn up to address to the expressed concerns.

xxi. Disaster Management Plan for the above terminal.

xxii. Layout plan of existing and proposed Greenbelt.

xxiii. Status of court case pending against the project.

xxiv. A tabular chart with index for point wise compliance of above TORs.

General Guidelines

(i) The EIA document shall be printed on both sides, as for as possible.

(ii) All documents should be properly indexed, page numbered.

(iii) Period/date of data collection should be clearly indicated.

(iv) Authenticated English translation of all material provided in Regional languages.

(v) The letter/application for EC should quote the MoEF&CC File No. and also attach a copy of the letter prescribing the ToR.

(vi) The copy of the letter received from the Ministry on the ToR prescribed for the project should be attached as an annexure to the final EIA-EMP Report.

(vii) The final EIA-EMP report submitted to the Ministry must incorporate the issues mentioned in ToR. The index of the final EIA-EMP report, must indicate the specific chapter and page no. of the EIA-EMP Report where the

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specific ToR prescribed by the Ministry. Questionnaire related to the project (posted on MoEF&CC website) with all sections duly filled in shall also be submitted at the time of applying for EC.

(viii) Grant of ToR does not mean grant of EC.

(ix) The status of accreditation of the EIA consultant with NABET/QCI shall be specifically mentioned. The consultant shall certify that his accreditation is for the sector for which this EIA is prepared.

(x) On the front page of EIA/EMP reports, the name of the consultant/consultancy firm along with their complete details including their accreditation, if any shall be indicated. The consultant while submitting the EIA/EMP report shall give an undertaking to the effect that the prescribed ToRs (ToR proposed by the project proponent and additional ToR given by the MoEF&CC) have been complied with and the data submitted is factually correct (Refer MoEF&CC Office memorandum dated 4th August, 2009).

(xi) While submitting the EIA/EMP reports, the name of the experts associated with/involved in the preparation of these reports and the laboratories through which the samples have been got analysed should be stated in the report. It shall clearly be indicated whether these laboratories are approved under the Environment (Protection) Act, 1986 and the rules made there under (Please refer MoEF&CC Office Memorandum dated 4th August, 2009). The project leader of the EIA study shall also be mentioned.

(xii) All the ToR points as presented before the Expert Appraisal Committee (EAC) shall be covered.

6. The above ToR should be considered for the project 'Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour by Mumbai Port Trust, in addition to all the relevant information as per the 'Generic Structure of EIA' given in Appendix III and IIIA in the EIA Notification, 2006.

7. The project proponent shall submit the detailed final EIA/EMP prepared as per ToR to the Ministry for considering the proposal for environmental clearance within 3 years as per the MoEF&CC O.M. No.J-11013/41/2006-IA-II(I) (P) dated 08.10.2014.

8. The consultants involved in preparation of EIA/EMP report after accreditation with Quality Council of India/National Accreditation Board of Education and Training (QCI/NABET) would need to include a certificate in this regard in the EIA/EMP reports prepared by them and data provided by other Organization(s)/ Laboratories including their status of approvals etc. vide Notification of the MoEF&CC dated 19.07.2013.

9. The prescribed ToR would be valid for a period of three years for submission of the EIA/EMP Reports.

14/7/17

(Kushal Vashist)
Director

Copy to:

The Member Secretary, Maharashtha State Pollution Control Board Kalptaru Point, 3rd & 4th Floor, Sion Matunga Scheme, Road No 6, Opposite Cine, Sion Circle, Sion (E), Mumbai-400 022 E Mail- ms@mpcb.gov.in) Fax- 022 24023516

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Annexure 3

Copy of duly signed Proceedings of Public Hearing

Minutes of Environmental Public Hearing about proposed Project of Offshore container berths and development of container terminal on BOT basis by M/s. Mumbai Port Trust, at Mumbai Harbour. Held on 29/09/2005, at Durbar Hall, of the Asiatic Society of Mumbai-23.

The Environmental Public Hearing about proposed project of "Offshore container berths and development of container terminal on BOT basis of M/s. Mumbai Port Trust was held on 29/09/2005 under the Chairmanship of Dr. D.B.Boralkar , Member Secretary, Maharashtra Pollution Control Board, at Durbar Hall, of the Asiatic Society of Mumbai-23. Convener of the public hearing panel welcomed the Chairman, panel members and the public participants present for public hearing.

Convener of the Public Hearing panel explained purpose and necessity of public hearing as per the Notification No. S.O.318(E), dtd. 10th April, 1997, under the Environment (Protection) Act 1986. Convener also placed facts about the procedure following by MPCB as per Notification.

As per the notice published in newspapers, written objections were called in the office of Maharashtra Pollution Control Board at Regional Office, Mumbai, from the concerned persons. Also, copies of the executive summary each in English and Marathi about the proposed project were made available at Collector Office, Mumbai City, Department of Environment , Govt. of Maharashtra , Mumbai, Head Office of Maharashtra Pollution Control Board at Sion, Mumbai & Regional Office, Mumbai of Maharashtra Pollution Control Board at Shri Chhatrapati Shivaji Maharaj Mandai Municipal Bldg., 4th Floor, Mata Ramabai Ambedkar Road, Mumbai-400 001 to appraise the public about the proposed project.

Following are the panel bers for the public hearing.

1)	Dr. D.B.Boralkar, Member Secretary, Maharashtra Pollution Control Board.	-Chairman
2)	District Collector or his Nominee.	-Member
3)	Shri. Surendra Jadhav, Dy. Secretary Env. Dept., Govt of Maharashtra, Mumbai.	-Member
4)	Representative of Urban Development & Sanitation, Govt. of Maharashstra, Mumbai.	-Member
5)	Shri. Father Divkle Sr. Citizen.	Member

8) Smt. A.P.Remedios, Sr. Citizen -Member

7) Shri. D.D Mane, Sr. Citizen -Member

8) Smt. Kunoji Geetabai Chhe[illegible], Corporator, M.C.G.B. -Member

9) Shri. Dosti Puran Mohanlal, Corporator, M.C.G.B. -Member

10) Shri. Dhulia Vijaya Kantilal, Corporator, M.C.G.B -Member

11) Shri. B.D.Wadde, S.R.O. Mumbai- -Convener

Convener of the Public Hearing panel declared that in spite of appeal in the newspapers, the office of Maharashtra Pollution Control Board have not received any written objection/suggestion in the register kept in the Regional Office-Mumbai.

Thereafter, the project proponents proceedings commenced with brief presentation on salient features of their proposed project. They covered description of site, location, details of construction, facilities provided, impacts on environment and related points

Thereafter, Chairman asked the participants/citizens to register the objections/points if any they have Chairman and Members have also raised certain points about the said proposed project to which project representative has given explanation. Details of Questions & Answers raised during the public hearing are as follow:-

1) **Shri. Debu Goenka, (N.G.O.):**

Point No.1 :- Where is the CWPRS report?

Explanation:- The report is not enclosed and CWPRS have done siltation studies in the Mathematical Model. The Mathematical Model studies for the entire Mumbai harbour has been done. The study for the subject project has also been done and is a part of the report. The location and the orientation of the proposed berths have also been studied on the model and they have opined that the proposed berth is well aligned with the flow direction. They have also stated that since the proposed berth and its approaches to the shore are on piles the flow pattern is not affected. The quantum of siltation due to this project is estimated to be about 2 million Cu. M. Thus the extra siltation is hardly 10-15% more than what we are doing now. The dumping of material due to capital dredging will be done at location where there is good dispersion and will not affect the mangrove locality. The cement Manufacturing from the proposed Jetty, would be [illegible] km. He further explained the flow [illegible] siltation in the vicinity of the jetty [illegible] and the [illegible]

2) Smt. Neera Punj, (N.G.O.)

Point No.1:- She asked abo . road management, space management, supporting infrastructure, mainland infrastructure, proper transport system. Why are we investing this money for when there is no proper infrastructure available to the general public?

Explanation:- Mumbai Port Trust owns a very good network of road and rail. MbPT owns roads from Docks to Wadala dedicated for the Port Traffic. At present hardly 2000 trucks ply on these roads owing to Port Traffic. We have handled 6 lakh containers with the existing road system in 1997-98 without Truck Terminus road at that time . Now, with the Truck Terminus Road, in place and also proposed Anik Panjrapole Link Road for which construction has already started, the Port Traffic will directly join the National Highways bypassing the city roads. Also when the proposed berthes are commissioned in 2009, the traffic at these berths for the first 5 years will not be more than 6 Lakh TEUs. The existing roads within the Port estate will be widened wherever required for smooth movement of Traffic.

As regards Railways, MbPT owns its own rail network from Docks to Wadala (55 km track length and 10 km route length). Mumbai Port is also proposing dedicated additional railway line from Wadala to Kurla for which the feasibility studies have been done and the pro is feasible. The final estimate and location survey have been completed and C al Railway is going to submit final eport within a week's time. With this dedic d link from Wadala to Kurla and also the 5th and 6th line laid by the Central Railway from Thane to Kurla (work almost completed) and Thane to Kalyan (work commenced) which is dedicated for the through trains and goods trains , there will be smooth sail of cargo movement through rail system.

Point No.2:- If feasibility study is already done, then it should be part of this report Then why this Public Hearing without the report?

Explanation:- In the Year 1996 we have handled 6 Lakh TEUs. Now i.e. in the year 2004-2005 we have handled only about 2-3 Lakh TEUs. Thus we have spare capacity available. To enhance the capacity of handling cargo it is necessary to construct offshore container berth.

3) **Shri. Father Berkie, Member, Environmental Public Hearing**

Point No.1 :- The project is necessary, but since the container traffic is shifted to JNPT, what is the benefits of the project and what is the ratio of evacuation by road and rail ?

Explanation:- 6:40 Ratio is c idered in feasibility report.

4) **Shri. A.O.Remedios, Member, Environmental Public Hearing**

Point No.1 :- What about MTHL project? If the project is delayed how the evacuation will be problem. The offshore project will be constructed by international firms and will complete the works well in time. Whereas the Bandra Worli link is delayed by the Indian company. Thus the evacuation will be right on

Explanation: - Shri. Puran Doshi objected Mr. Remedios above point stating that the Indian companies are quite competent and comparable to any International firms. I have been in the standing committee MCGM for so long and I know the problems. The award of works to consultants is dictated by the World Bank and consultants are from abroad only. Don't blame Indian engineers and Companies alone for the delay.

Project proponent:- The evacuation of Port cargo is not at all dependent of the MTHL. What I want to say that it will only supplement the evacuation.

B) Shri. Debu Gonnka, (N.G.O.) :-

Point No.1:- You have assumed one-inch leakage. It may be more also at the site. What would be the follow up done by the Disaster Management Plan (DMP is minimum risk) while taking into consideration the minimum risk?

Explanation:- The Disaste[r] [M]anagement prepared by the Port is adequate to handle, the condition. Accident are [n]ot, leakages (one or two inch) happen due to unloading operation, fire fighting is able to handle such situation and maximum precautions are taken. The hazardous chemicals come in the form of containers. There will be hardly few container carrying cylinders. The existing DMP of the Port is adequate to handle any risk.

Point No. 2:- Both the road and the rail network is congested/saturated and now. Traffic handled is proposed to be handled is not clear in the report. What is the waiting time? JNPT was created to decongest the city. Now what is the necessity of this project?

Explanation:- JNPT was set up as a part of Mumbai Port and subsequently it was developed as a satellite Port. We are not setting up as a competition to JNPT. The container traffic is increasing at a phenomenal rate and almost 60% of the Country's traffic is handled by Mumbai Region and this Project is the requirement of the trade and in the interest of the Nation. JNPT has already awarded for the construction of 3rd terminal. They are conducting feasibility studies for the 4th container terminal. Even after that there will be a substantial gap which is to be serviced by another Port so why not Mumbai Port. [A]ny other point is regarding the hinterland services Mumbai city itself will account for [ ] of the container traffic, which can be well served by the Mumbai Port with the new [t]erminal. Mumbai Port contributes to Customs nearly 500-600 Crores worth goods which will be more than double if the project is envisaged. MbPT also contributes Rs. 350 Crores towads Octroi to MCGM, which will increase upto Rs. 800-900 Crores. Similarly Stamp Duty, Sales Tax will go up. We should gear up to handle huge container traffic projected in the next ten years and since 60% of Hinterland consumption is from the Mumbai Region, this project is necessary in the interest of Nation. Other thing is that it creates employment, other economical activities, labour contractors, service providers etc. These changes are positive aspects of this project. It is first of its kind in India and pride of Mumbai. Mumbai Port has studied all the issues comprehensively.

Point No.3:- He desired to know about the evacuation of cargo by roads (roads already being congested)

Explanation: - There should be no apprehension on this count. On the road front, I divide the connectivity into 2 parts, one is internal roads within the Mumbai Port Estate, where we are planning to widen the roads, construct flyovers etc. We are investing about Rs. 90 Crores for this.

The second is the road improvements being carried out by MMRDA. The Anik Panjrapole Link Road. This is a [illegible] link and the work has already begun. Mumbai Port is also sensitive to the co[illegible]station issue. We are funding Rs. 35 Crores to ensure that the project is completed well in time before the offshore berths are commissioned. The work would be completed in 2 years time. From Panjrapul MMRDA is planning to develop two stretches, which directly connects to Eastern Express Highway. Eastern Express Highway is 10 lane road. With the above projects taken up by MMRDA, the entire Port traffic will bypass the City roads. I would like to tell you that Mumbai Prot handled 6 Lakh TEUs in 1996-1997 and evacuation was not at all a problem at that time even without Wadala Truck Terminus Road. This traffic of 1.2 Million TEUs will come only after 10 years by which time even MTHL will be completed.

About Rail connectivity, members / citizen's should appreciate that his traffic will put minimum constraints to the general public. The Wadala Kurla Link will come and it is being monitored by the Prime Minister Office. The ministry has given high priority for this Rail Link. Mumbai Port is funding this project and I am sure that this would be completed before commissioning of the berths. Even though I am very much confident that this rail link would be completed, assuming that this will not complete in time, there should be no concern to MbPt as MbPT is capable of handling 6 lakhs TEUs with the existing link itself. We have handled without any difficult previously. After all the traffic of [illegible] million TEUs will be achieved in the 10th Year.

I would like to state that MMRDA has submitted a report for construction of East island freeway from Museum to Ghatkopar. Mumbai Port Trust has given clearance for this link for aligning in Mumbai Port estate. This link will further ease the traffic on City roads.

6) Shri. Father Berkle, Member, Environmental Public Hearing

Point No – 1: - What comes to the common man? P. D'mello Road is fully blocked / encroached by the Slums. What about their rehabilitation? What profit will MbPT make out of this project?

Explanation: - I share your sentiment fully. MbPT has constructed several roads, which are given to public, and P.D' Mello road is one of them. This road is maintained by MCGM. Slums have developed not because of MbPT. This matter is being taken up with BMC / Chief Secretary, Government of Maharashtra. In fact for the Wadala Kurla rail link, about 2540 hutments will have to be resettled. We have made provision for their rehabilitation. MbPT have become victim of migration and hence the proliferation of slums. MbPT will see to that no further slums will develop. It is only [illegible] roads to be taken [illegible] slums of encroachment.

7) Smt. Neera Puni, (N.G.O.) & Dr. Boralkar, Chairman.

Point No.-1: - Whether any heritage buildings are to be demolished for this Project? No mention has been made in the report.

Prepare list of heritage structures under Mumbai Port and state if any of these heritage structures, are affected by this project, if yes, list out the heritage structures.

Explanation: - There is no heritage building in the Project area. All we are demolishing is the old sheds and warehouses.

8) Shri. Debu Goenka, (N.G.O.).

Point No.-1: - Whether any other land will be reclaimed? Whether mangroves will be affected due to additional dredging?

Explanation: - No land is being are claimed for this project. We are just filling up the old dock basin. About Mangroves, I should say this would not at all affected. The dumping area is about 25 km away from Mangroves. Mumbai Port has spent to develop and protect Mangrove at Sewree. Previously about 2000 Flamingos were there but of today there are about 14000 flamingos. This project will in no way affect the habitat. Nearly about 15 acres of area have been replanted. We spent Rs. 50 Lakhs to develop the area as sanctuary. The mangroves situated between Sewree-Wadala will be in no way affected.

9) Shri. Father Berkle, Mer . Environmental Public Hearing

Point No – 1 :- What about passenger water transport from Ferry Wharf. Whether this is affected by the project.

Explanation :- Not affected. In fact MbPT has given clearance to Ferry Wharf Belapur Cruise Terminal to the Maharashtra Government. They have appointed consultants for finalizing the terms.

10) Shri. Debu Goenka, (N.G.O).

Point No-1 :- Trucks are parked on the roads. Wadala Truck Terminus is more than 10 years old No use is made of it.

Suggestion by Remedios: - I suggest that MbPT think of beautifying the MbPT Roads so that general public can use the same and income is earned by MbPT. MbPT should create health centre, gymnasium and swimming pool at the recreation area – Botanical Garden developed by them.

I should mention that Mumbai came up because of Mumbai Port Trust. There should be no objection for revival of the Port.

Suggestion by Dr. Bora . :- I agree with you. In fact the entire MbPT should be properly planned to impro the environment of their estate, which includes proper drainage system for the slum s etc. Proper environmental plan can be taken. There is an Institute for Environmental Planning at Ahmedabad, which will give all inputs to finalize the plan.

Road within the MbPT estate should be paved.

The point is noted by project proponent.

Explanation :- Sheds will be demolished and the same will be used as filling material after processing Further excavated material and demolished structures due to construction activity in an around Mumbai will also be used. This in one way will solve the dumping of debris generated out of construction activity in Mumbai.

11) Shri. Doshi Puran Mohanlal, Member , Evvironmental Public Hearning

Point No- 1 :- MbPT should give NOC for providing drinking water, toilet blocks etc for the slums in their areas. Municipality is ready to provide. What about the water supply to be given to the hutments?

Explanation :- Requested all the members who attended the meeting not to delay the project. He alap explained that these unauthorized hutments, which are post 1995, and are already having unauthorized water connections and MbPT is suffering at their cost. If water connection is given to them this will results in to huge proliferation of hutments and blame MbPT for the same and vice-versa. He again stated that already MbPT has constructed about 55 Sulabh Sanchalay mainly in Darukhana area.

1) **Suggestion by Father Berkie:-** MCGM and MbPT should co-ordinate with each other.

2) **Suggestion by Remedios:-** Stated that this query is related to MCGM.

3) **Suggestion by Doshi Puran :-** Stated that government has ordered that after 26th July torrential rains rehabilitate the slums outside the city limits.

Lastly, Chairman and Convener of the public hearing panel again suggested the participants/public to ask questions/points/objections/doubts if still they have. However, nobody raised any questions/points/objections/doubts about the said.

Finally, the Public Hearing was concluded with vote of thanks to all those who were present by Chairman and convener of the Public Hearing panel.

(D.B.Boralkar)
Chairman
Environmental Public Hearing Panel
Member Secretary
Maharashtra Pollution Control Board
Kalpataru Point, Sion (E),
Mumbai - 400 022.

Annexure 4

EC Letter

Date of public hearing → 29/9/2005
29 Sept, 2005

Annexure XIIIE

COPY OF ENVIORNMENTAL CLEARANCE

No.10-18/2005-IA-III
Government of India
Ministry of Environment and Forests
(IA-III Division)

Paryavaran Bhavan,
CGO Complex, Lodhi Road,
New Delhi – 110003.

Dated the 15th June, 2006

Sub: Construction of two offhshore container berths and development of Container terminal on build, Operate and Transfer (BOT) basis in Mumbai Harbour Environmental Clearance – regarding.

****

Reference is invited to your letter No.MCZMA/78, dated 9.5.2005 from Member Secretary, Maharashtra State Coastal Zone Management Authority, Government of Maharashtra regarding the above project. Subsequent letters dated No.CE.CF/226(CTP)/1429, dated 27th May, 2005, No.CE.CF/226(CTO)/2306, dated 30th June, 2005, No.CE.CF/226(CTP)/2962, dated 21st July, 2005, No.CE.CF.226(CTP)/6942 dated 29.12.2005, No.CE.CF.226(CTP)/7372, dated 16.1.2006, No.CE/CF.226(CTP)/8081, dated 14.2.2006, No.CE.CF.226(CTP)/8444, dated 27.2.2006 and No.CE.CF226(CTP)/1485, dated 25.5.2006 from Mumbai Port Trust has also been taken into account. No Objection Certificate from Maharashtra Pollution Control Board vide their letter No.BO/RO(P&P)/CC-151; dated 17.1.2006 has also been obtained.

The project pertains to construction of two offshore container berths (length 350 m, width 55 m per container berth), in the first stage and the third berth in second stage and development of container terminal on BOT basis in Mumbai Harbour. The project involves expansion of the existing Mumbai Port for improvement of container traffic at Mumbai Port. For the project 43 ha of land would be required which is a Government land. The length of two new container have been decided to be 700 mts for accommodating two 6000 TEU (approximately 65,000 DWT) size vessels with average dimensions of LOA 275 m, beam 42 m and fully laden draft 14.5 m. The container berth will accommodate on the deck across its width for rails, crane movement, space for movement of transport vehicles/trucks/equipment, space for power cables and service lines. Both the berths and trestle are constructed over piled formation only the berths and approached trestle falls within Coastal Regulation Zone area rest the other facilities are located outside the Coastal Regulation Zone area.

For the above project rail and road connectivity will be provided. With regard to railway a new railway container depot/siding for rail transportation of containers will be created in the space behind the proposed Victoria Dock container stackyard. It is required to provide full train length line to accommodate 45 wagons rake (700 m in length) near the traffic office.

With regard to road construction, an overpass at Malet road junction connecting the dock expressway and the link road is proposed. The flyover crossing mallet Road will free the port traffic (containers and other traffic) from the present heavy congestion. The arrangement of roads system around the Victoria Dock will need to be modified re-built and new roads be built for the stackyard ensuring that traffic circulation of the stackyard and Indira Dock do not interfere with each other.

The other facilities to be created of the above port project is as follows:-

- Terminal operational buildings/offices including office space for major port users – 1200 sq m.
- Maintenance building including repair workshop, and the central fire station – 4000 sq m.
- New substations for power supply and distribution – 200 sq m.
- Stores 0 150 sq m.
- Customs/security buildings, amenity buildings/ conveniences – 500 sq m.

The project requires dredging. Considering the tidal range at Mumbai Port is 5.2 m, it is proposed that the entrance channel and the harbour basin be dredged to -13.5 m CD. A 70 m wide it in front of the berth will be dredging to -15 m CD. These depths will permit entry and exit of 4,500 TEU size vessels to and from the harbour basin (at half tides) and the vessel to lie at the berths at all stages of the tide. The dumping site for the dredged material is to be located at geographical coordinates 18°55'00" (± 30") and 72°42'00" (± 30") is recommended as the final site for dumping the capital dredging material. The site is located about 8.5 km from the proposed channel where the water depth is abut 17.0 m below chart datum. The sedimentation model studies carried out for selected dumping site indicates that the deposition rates at the dumping site are negligibly small when compared to the prevailing depths at the dumping site. The dredged material is proposed to be dumped at DS-3 site.

The diameter of the turning circle will also be increased from 1.7 to 2 times the length of the largest vessel to be turn. The turning circle will be 350 mts. For creating the above facilities several structures are proposed to be demolished. It is indicated that the dredging operation will not have any impact on the mangroves near Sewree or Dharamtar/Rewas. An annual maintenance dredging quantity of 0.90 million cu mts will also be dumped at the existing dumping sites.

The proposal has been examined in the Ministry of Environment & Forests and environmental clearance to this project is hereby accorded under Environment Impact Assessment Notification, 1994 and Coastal Regulation Zone Notification, 1991 subject to effective implementation of the following conditions:

A. SPECIFIC CONDITIONS:

(i) All the conditions stipulated in the No Objection Certificate from Maharashtra State Pollution Control Board vide their letter No.BO/RO(P&P)/CC-151, dated 17.1.2006 should be strictly implemented.

(ii) The reclamation of the port area should be carried out with the dredged material; dredged material should not be dumped into the sea. No reclamation should be carried outside the port limits.

(iii) The coastal protection works should be carried out after detailed hydrodynamic modelling studies and it should be ensured that no erosion or accretion takes place in other areas due to the shore protection works.

(iv) Reclamation of 500 acres should be carried out only for port development. The height of the reclaimed area will be maintained above the maximum flood level.

(v) The wave tranquility study and the ship maneuvering studies carried out should be taken into account while operating the port.

(vi) The revetment should be designed after conducting flume studies and the section designed based on this study.

(vii) The project proponent should ensure that during construction and operation of the port there will be impact on the livelihood of the fisherman. The fishermen should be provided free access to carry out the fishing activity.

(viii) Al necessary precaution while undertaking construction and operation of the port should be taken keeping in view the bathymetric changes caused due to tsunami.

(ix) All development in the port should be carried out in accordance with the Coastal Regulation Zone Notification, 1991 and approved Coastal Zone Management Plan of Maharashtra.

(x) The ecological, terrestrial and biological studies should be carried out by recognised scientific institutions and report submitted to this Ministry every six months.

(xi) The project proponent should undertake a comprehensive hydrodynamic modelling study with regard to river diversion and submit the report to the Ministry within 6 months from the date of receipt of this letter. Further, the unit should comply with all the findings/recommendations of the study.

(xii) Construction labour camps should be located outside Coastal Regulation Zone areas and should be provided with adequate cooking and sanitation facilities.

B. GENERAL CONDITIONS:

(i) Development of the proposed channel should be undertaken meticulously conforming to the existing Central/local rules and regulations including Coastal Regulation Zone Notification, 1991 and its amendments. All the construction designs/drawings relating to the proposed development activities must have approvals of the concerned State Government Department/Agencies.

(ii) A well-equipped laboratory with suitable instruments to monitor the quality of air and water shall be set up as to ensure that the quality of ambient air and water conforms to the prescribed standards. The laboratory will also be equipped with qualified manpower including a marine biologist so that the marine water quality is regularly monitored in order to ensure that the marine life is not adversely affected as a result of implementation of the said project. The quality of ambient air and water shall be monitored periodically in all the seasons and the results should be properly maintained for inspection of the concerned pollution control agencies. The periodic monitoring reports at least once in 6 months must be send to this Ministry (Regional Office at Bhopal) and Pollution Control Committee.

(iii) Adequate provisions for infrastructure facilities such as water supply, fuel for cooking, sanitation etc. must be provided for the labourers during the construction period in order to avoid damage to the environment. Colonies for the labourers should not be located in Coastal Regulation Zone area. It should also be ensured that the construction workers do not cut trees including mangroves for fuel wood purpose.

(iv) To prevent discharge of sewage and other liquid wastes into the water bodies, adequate system for collection and treatment of the wastes must be provided. No sewage and other liquid wastes without treatment should be allowed to enter into the water bodies.

(v) Appropriate facility should be created for the collection of solid and liquid wastes generated by the barges/vessels and their safe treatment and disposal should be ensured to avoid possible contamination of the water bodies.

(vi) Necessary navigational aids such as channel markers should be provided to prevent accidents. Internationally recognized safety standards shall be applied in case of barge/vessel movements.

(vii) The project authorities should take appropriate community development and welfare measures for villagers in the vicinity of the project site, including drinking water facilities. A separate fund should be allocated for this purpose.

(viii) The quarrying material required for the construction purpose should be obtained only from the approved quarries/borrow areas. Adequate safeguard measures shall be taken to ensure that the overburden and rocks at the quarry site do not find their way into water bodies.

(ix) For employing unskilled, semi skilled and skilled workers for the project, preference should be given to local people.

(x) The recommendations made in the Environmental Management Plan and Disaster Management Plan, as contained in the Environmental Impact Assessment and Risk Analysis Reports of the project shall be effectively implemented.

(xi) A separate Environmental Management Cell with suitable qualified staff to carry out various environment should be set up under the charge of a Senior Executive who will report directly to the Chief Executive of the Company.

(xii) The project affected people, of any should be properly compensated and rehabilitated.

(xiii) The funds earmarked for environment protection measures should be maintained in a separate account and there should be no diversion of these funds for any other purpose. A year-wise expenditure on environmental safeguards should be reported to this Ministry.

(xiv) Full support should be extended to the officers of this Ministry's Regional Office at Bangalore and the officers of the Central and State Pollution Control Boards by the project proponent during this inspection for monitoring purposes, by furnishing full details and action plans including the action plans including the action taken reports in respect if mitigative measures and other environmental protection activities.

(xv) In case there is an intention of deviation or alternation in the project including the implementing agency, a fresh reference should be made to this Ministry for modification in the clearance conditions or imposition of new ones for ensuring environmental protection. The project proponents should be responsible for implementing the suggested safeguard measures.

(xvi) This Ministry reserves the right to revoke this clearance, if any of the conditions stipulated are not complied with to the satisfaction of this Ministry.

(xvii) This Ministry or any other competent authority may stipulate any additional conditions subsequently, if deemed necessary for environmental protection, which shall be complied with.

(xviii) The project proponent should advertise at least in two local newspapers widely circulated in the region around the project, one of which shall be in the vernacular language of the locality concerned informing that the project has been accorded environmental clearance and copies of clearance letters are available with the State Pollution Control Board and may also be seen at Website of the Ministry of Environment & Forests at http://www.envfornic.in. The advertisement should be made within 7 days from the date of issue of the clearance letter and a copy of the same should be forwarded to the Regional Office of this Ministry at Bhopal.

(xix) The project proponents should inform the Regional Office as well as the Ministry the date of financial closure and final approval of the project by the concerned authorities and the date of start of Land Development Work.

The above mentioned stipulations will be enforced among others under the Water (Prevention and Control of Pollution) Act, 1974, the Air (Prevention and Control of Pollution) Act, 1981, the Environment (Protection) Act, 1986, the Hazardous Chemicals (Manufacture, Storage and Import) Rules, 1989, the Coastal Regulation Zone Notification, 1991 and its subsequent amendments and the Public Liability Insurance Act, 1991 and the Rules made thereunder from time to time. The project proponents should also ensure that the proposal complies with the provisions of the approved Coastal Zone Management Plan of Maharashtra and Supreme Court's order dated 18th April, 1996 in the Writ Petition No.664 of 1993 to the extent the same are applicable to this proposal.

Sd/-
(A. Senthil Vel)
Additional Director

To

Shri Keshav Sundar,
Executive Engineer,
Mumbai Port Trust, Civil Engineering Department,
Port House, Shoorji Vallabhdas Marg,
Ballard Estate, Mumbai – 400 001.

Copy to:

1. The Chief Conservator of Forests (Central), Ministry of Environment & Forests, Regional Office, Western Region, "Kendriya Paryavaran Bhavan", Link Road No.3, Ravishankar Nagar, Bhopal –462016 (M.P.).
2. Principal Secretary, Environment Department, 15th Floor, New Administrative Building, Opposite Mantralaya, Madam Cama Road, Bombay – 400020.
3. The Chairman, Central Pollution Control Board, Parivesh Bhavan, CBD-cum-Office Complex, East Arjun Nagar, Delhi – 110032.
4. Member Secretary, Maharashtra Coastal Zone Management Authority, Maharashtra State Coastal Zone Management Authority, Maharashtra State Pollution Control Board, Kalpataru Point, 3rd and 4th Floor, Sion Matunga Scheme Road No.8, Opposite Cine Planet Cinema, Near Sion Circle, Sion (East), Mumbai – 400022.
5. DIG (SU), Regional Office Cell, Ministry of Environment & Forests, New Delhi.
6. Guard File.
7. Monitoring Cell.
8. Director (EI), Ministry of Environment & Forests, New Delhi.

Sd/-
(A. Senthil Vel)
Additional Director

No.10-18/2005-IA-III
Government of India
Ministry of Environment and Forests
(IA-III Division)

Paryavaran Bhavan,
CGO Complex, Lodhi Road,
New Delhi – 110003.

Dated the 9th November, 2006

CORRIGENDUM

Sub: **Construction of two offshore container berths and development of Container terminal on build, Operate and Transfer (BOT) basis in Mumbai Harbour Environmental Clearance – regarding.**

****

Reference is invited to this Ministry's clearance of even number dated 15th June, 2006 regarding the above subject. Condition Nos. (ii), (iv) and (ix) under Specific Conditions may be substituted as follows:

"(ii) The surplus dredged spoil shall be dumped at the designated site i.e., DS-3 as proposed by CWPRS.

(iv) Reclamation of 55 acres should be carried out only for port development. The height of the reclaimed area will be maintained above the maximum flood level.

(xi) "The location and orientation of the berth should be as per recommendations of the CWPRS studies."

The rest of the terms and conditions of the clearance letter remain unchanged.

(Dr. A. Senthil Vel)
Additional Director

To

Shri Keshav Sundar,
Executive Engineer,
Mumbai Port Trust, Civil Engineering Department,
Port House, Shoorji Vallabhdas Marg,
Ballard Estate, Mumbai – 400 001.

Copy to:

1. The Chief Conservator of Forests (Central), Ministry of Environment & Forests, Regional Office, Western Region, "Kendriya Paryavaran Bhavan", Link Road No.3, Ravishankar Nagar, Bhopal –462016 (M.P.).
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4. Member Secretary, Maharashtra Coastal Zone Management Authority, Maharashtra State Coastal Zone Management Authority, Maharashtra State Pollution Control Board, Kalpataru Point, 3rd and 4th Floor, Sion Matunga Scheme Road No.8, Opposite Cine Planet Cinema, Near Sion Circle, Sion (East), Mumbai – 400022.
5. DIG (SU), Regional Office Cell, Ministry of Environment & Forests, New Delhi.
6. Guard File.
7. Monitoring Cell.
8. Director (EI), Ministry of Environment & Forests, New Delhi.

(Dr. A. Senthil Vel)
Additional Director

141

No.10-18/2005-IA-III
Government of India
Ministry of Environment and Forests
(IA-III Division)

Paryavaran Bhavan,
CGO Complex, Lodhi Road,
New Delhi - 110003.

Dated the 4th October, 2007

CORRIGENDUM

...tion of two offshore container berths and development of Container terminal on ...erate and Transfer (BOT) basis in Mumbai Harbour Environmental Clearance –

****

...is invited to this Ministry's clearance of even number dated 15th June, 2006 and dated 9th ...regarding the above subject. The numbers "55m" appearing in the second line of para 2 of ...ted 15th June, 2006 shall be substituted with "65m" instead of "55m".

...of the terms and conditions of the clearance letter remain unchanged.

(Dr. A. Senthil Vel)
Additional Director

...tiav Sundar,
... Engineer,
... Port Trust, Civil Engineering Department,
...se, Shoorji Vallabhdas Marg,
...state, Mumbai – 400 001.

...nservator of Forests (Central), Ministry of Environment & Forests, Regional Office, ...tion, "Kendriya Paryavaran Bhavan", Link Road No.3, Ravishankar Nagar, Bhopal –462016

...ecretary, Environment Department, 15th Floor, New Administrative Building, Opposite ... Madam Cama Road, Bombay – 400020.

...dm, Central Pollution Control Board, Parivesh Bhavan, CBD-cum-Office Complex, East ... Delhi – 110032.

...retary, Maharashtra Coastal Zone Management Authority, Maharashtra State Coastal Zone ...nt Authority, Maharashtra State Pollution Control Board, Kalpataru Point, 3rd and 4th Floor, ...ga Scheme Road No.8, Opposite Cine Planet Cinema, Near Sion Circle, Sion (East), ...00022.

...egional Office Cell, Ministry of Environment & Forests, New Delhi.

... Cell.

..., Ministry of Environment & Forests, New Delhi.

(Dr. A. Senthil Vel)
Additional Director

(P) env 143

F.No.10-18/2005-IA.III
Government of India
Ministry of Environment & Forests
(IA Division)

Paryavaran Bhawan,
CGO Complex, Lodhi Road,
New Delhi - 110 003.

Dated: 7th January, 2014

CE 23/1/14
CFI/C
S.E.(P)I
E.E. OCT I / II 23/1
DY 23/1
21.01.2014

To,
Deputy Chief Engineer, Project I,
M/s Mumbai Port Trust,
Civil Engineering Department,
3rd Floor, Port Bhavan, S.V. Marg,
Ballard Estate, Mumbai – 400 001

Contact Person Details:
Shri G.M. Yerandekar,
Tel No. 022 6656 4032
Mob No. 9870997224

Subject: Extensions of validity of Clearance dated 09.11.2006 granted to M/s Mumbai Port Trust for construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour – Reg.

***

This has reference to your letters No: CE.CF.226(CTP)/ENV/173 dated 02.06.2011 and 19.06.2013 seeking extension of validity of Environmental Clearance granted vide letter no. F.No.10-18/2005-IA-III dated 15.06.2006 and two corrigendums dated 09.11.2006 and 04.10.2007. The issue was discussed by the EAC in its 127th meeting held on 28th -30th October, 2013 and committee recommended for the extension of validity of the clearance dated 15.06.2006 for a period of five years. Accordingly, the validity of the clearance is extended up to 14.06.2016.

All other conditions remain unchanged.

(Lalit kapur)
Director (IA-III)

Copy to:
(1) The Secretary, Department of Environment, Govt. of Maharashtra, Mantralaya, Mumbai – 400 032.
(2) The Chairman, CPCB, Parivesh Bhawan, CBD-cum-Office Complex, East Arjun Nagar, Delhi – 32.
(3) The Member Secretary, Maharashtra Pollution Control Board, Mumbai.
(4) The Chief Conservator of Forests, Ministry of Environment and Forests, Regional Office, Western Region, Kendriya Paryavaran Bhavan, Link Road No. 3, Ravishankar Nagar, Bhopal – 462016 (M.P.)
(5) Guard File.
(6) Monitoring Cell.

मुंबई पोर्ट ट्रस्ट
स्थापत्य अभियांत्रिकी विभाग
Mumbai Port Trust
Project Cell
आ. CE CF/226 (CTP)
Inward ENV/1755
दिनांक- 21.01.2014
Date

(Lalit Kapur)
Director (IA-III)

F env 143

F.No.10-18/2005-IA.III
Government of India
Ministry of Environment & Forests
(IA Division)

Paryavaran Bhawan,
CGO Complex, Lodhi Road,
New Delhi - 110 003.

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To,
Deputy Chief Engineer, Project I,
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All other conditions remain unchanged.

(Lalit kapur)
Director (IA-III)

Copy to:
(1) The Secretary, Department of Environment, Govt. of Maharashtra, Mantralaya, Mumbai – 400 032.
(2) The Chairman, CPCB, Parivesh Bhawan, CBD-cum-Office Complex, East Arjun Nagar, Delhi – 32.
(3) The Member Secretary, Maharashtra Pollution Control Board, Mumbai.
(4) The Chief Conservator of Forests, Ministry of Environment and Forests, Regional Office, Western Region, Kendriya Paryavaran Bhavan, Link Road No. 3, Ravishankar Nagar, Bhopal – 462016 (M.P.)
(5) Guard File.
(6) Monitoring Cell.

मुंबई पोर्ट ट्रस्ट
स्थापत्य विभाग
Mumbai Port Trust
Inward No. CE CF/226 (CTP)
Inward ENV/1755
दिनांक 21.01.2014
Date

(Lalit Kapur)
Director (IA-III)

Annexure 5

Certified report on compliance of earlier environmental conditions

MUMBAI PORT TRUST
Port House,
Shoorji Vallabhdas Marg,
Ballard Estate,
Mumbai - 400 001
Tel. No. 65564031
Telefax No.2261 6804 / 3469

मुंबई पोर्ट ट्रस्ट
MUMBAI PORT TRUST
स्थापत्य अभियांत्रिकि विभाग
CIVIL ENGINEERING DEPARTMENT

मुंबई पोर्ट ट्रस्ट
पोर्ट भवन, शूरजी वल्लभदास मार्ग,
बेलार्ड इस्टेट, मुंबई 400 001.
दूरध्वनी क्र. 65564031
फॅक्स क्र. 2261 6804 /3469

No.CE.CF.226(CTP)/ENV/1435

Date : 09 FEB 2016

The Director (S)
Government of India,
Ministry of Environment & Forests
Regional Office, Western Region,
Kendriya Paryavaran Bhavan,
Link Road No. 3, Ravi Shankar Nagar,
Bhopal – 462016 (M.P)
E-mail : ajaymehrotra13@gmail.com

BY REGD.A.D.
BY Email

Dear Sir,

(Kind Attn. : Dr. A. Mehrotra)

Sub : Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour – Environmental Clearance - **Compliance Report ending December 2015.**

Ref : 1. Your letter No.10-18/2005-IA-III dated 15.6.2006 dated 09.11.2006 and 07.01.2014.
2. This office letter No.CE.CF.226(CTP)/ENV/103 dated 02.06.2011, 434 dated 03.12.2011,534 dated 11.1.2012,41dated 13.04.2012,223 dated 18.06.2012,304 dated 21.07.2012,409 dated 4.9.2012,564 dated 30.10.2012,683 dated 18.12.2012,744 dated 15.1.2013 and 817 dated 8.2.2013 and 515 dated 29.7.2014 and 14 .1.2015

.....

The compliance report (ending December 2015) is enclosed herewith as required.

It is to state that the civil works have been completed. The capital dredging contract has been terminated from 27.06.2013 and there is no dredging work since then, The reports will be forwarded once the dredging works recommences.

Yours faithfully,

(N.G.Mirajkar)
Dy. Chief Engineer, Project I

DA : As above.

c.c. to the Secretary, Government of India, Ministry of Environment & Forests,Paryavaran Bhavan, CGO complex, Lodhi Road, New Delhi – 110 003.
(Attn. : Shri Lalit Kapoor, Addl. Director, Infrastructure Projects) **Fax No.011-2436 8536**

c.c. to the Maharashtra Pollution Control Board, Kalpataru Point, 3rd & 4th floor,Sion Matunga Scheme, Road No.8, Opp. Cine Planet Cinema, Near Sion Circle, Sion (E), Mumbai 400 022.

MUMBAI PORT TRUST
Port House,
Shoorji Vallabhdas Marg,
Ballard Estate,
Mumbai - 400 001
Tel. No. 65564031
Telefax No.2261 6804 / 3469

मुंबई पोर्ट ट्रस्ट
MUMBAI PORT TRUST
स्थापत्य अभियांत्रिकि विभाग
CIVIL ENGINEERING DEPARTMENT

मुंबई पोर्ट ट्रस्ट
पोर्ट भवन, शूरजी वल्लभदास मार्ग,
बेलार्ड इस्टेट, मुंबई 400 001.
दूरध्वनी क्र. 65564031
फॅक्स क्र. 2261 6804 /3469

209
CE

No.CE.CF.226(CTP)/ENV/1435

Date : 09 FEB 2016

The Director (S)
Government of India,
Ministry of Environment & Forests
Regional Office, Western Region,
Kendriya Paryavaran Bhavan,
Link Road No. 3, Ravi Shankar Nagar,
Bhopal – 462016 (M.P)
E-mail : ajaymehrotra13@gmail.com

BY REGD.A.D.
BY Email

Dear Sir,

(Kind Attn. : Dr. A. Mehrotra)

Sub : Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour – Environmental Clearance - **Compliance Report ending December 2015.**

Ref : 1. Your letter No.10-18/2005-IA-III dated 15.6.2006 dated 09.11.2006 and 07.01.2014.
2. This office letter No.CE.CF.226(CTP)/ENV/103 dated 02.06.2011, 434 dated 03.12.2011,534 dated 11.1.2012,41dated 13.04.2012,223 dated 18.06.2012,304 dated 21.07.2012,409 dated 4.9.2012,564 dated 30.10.2012,683 dated 18.12.2012,744 dated 15.1.2013 and 817 dated 8.2.2013 and 515 dated 29.7.2014 and 14 .1.2015

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Yours faithfully,

CE

(N.G.Mirajkar)
Dy. Chief Engineer, Project I

SE PI
EE OCT

DA : As above.

c.c. to the Secretary, Government of India, Ministry of Environment & Forests,Paryavaran Bhavan, CGO complex, Lodhi Road, New Delhi – 110 003.
(Attn. : Shri Lalit Kapoor, Addl. Director, Infrastructure Projects) **Fax No.011-2436 8536**

c.c. to the Maharashtra Pollution Control Board, Kalpataru Point, 3rd & 4th floor,Sion Matunga Scheme, Road No.8, Opp. Cine Planet Cinema, Near Sion Circle, Sion (E), Mumbai 400 022.

MUMBAI PORT TRUST
CIVIL ENGINEERING DEPARTMENT

CE.CF.226(CTP)/ENV/

Sub : Construction of two offshore container berths and development of Container Terminal on BOT basis in Mumbai Harbour.

Compliance Status upto 31.12.2015

Sr. No	Conditions stipulated	MbPT's Compliance status
	A. Specific Conditions	
i)	All the conditions stipulated in the No Objection Certificate from Maharashtra State Pollution Control Board vide their letter No. BO /RO (P&P) / CC-151, dated 17.1.2006 should be strictly implemented.	Noted. Will be complied with. The HSE plan submitted by M/s. Indira Container Terminal Pvt. Ltd. (ICTPL) was forwarded vide this office letter dated 23.07.2009 to MoEF. The EMP for capital dredging work prepared by NIT, Suratkal was forwarded by this office till the dredging work was in progress till January 2013. the work is terminated on 27.6.2013. • The last EMP of January 2013 was forwarded to your office vide this office No.CE.CF.226(CTP)/ENV/ 817dated 8.2.2013 • The EPC contractor's Monthly report on Ambient air quality, noise level measurements have been submitted to your office in 1st week of every month till 16.11.2013. • The civil works has been completed on 31.12.2013 and therefore no reports are generated.
ii)	The reclamation of 55 acres should be carried out only for Port development. The height of reclaimed area will be maintained above the maximum flood level.	Yet, it is complied with.
iii)	The coastal protection works should be carried out after detailed hydrodynamic modeling studies and it should be ensured that no erosion or accretion takes place in other areas due to the shore protection works.	The project scope does not envisage any coastal protection works. As such no erosion or accretion would take place.
iv)	Reclamation of 500 acres should be carried out only for port development. The height of the reclaimed area will be maintained above the maximum flood level.	This has been modified, the reclamation is 55 acres and not 500 acres. Will be complied with.
v)	The wave tranquility study and the ship maneuvering studies carried out should be taken into account while operating the port.	Will be complied with.

Sr. No	Conditions stipulated	MbPT's Compliance status
vi)	The revetment should be designed after conducting flume studies and the section designed based on this study.	This has been taken care in detailed design by the BOT operator M/s. Indira Container Terminal Pvt. Ltd. (ICTPL)
vii)	The project proponent should ensure that during construction and operation of the port there will be no impact on the livelihood of the fishermen. The fishermen should be provided free access to carry out the fishing activity.	Yes. There is no fishing activity in the project area.
viii)	All necessary precaution while undertaking construction and operation of the port should be taken keeping in view the bathymetric changes caused due to tsunami.	Yes. Will be ensured.
ix)	All development in the port should be carried out in accordance with the Coastal Regulation Zone Notification, 1991 and approved Coastal Zone Management Plan of Maharashtra.	Yes. Will be complied with.
x)	The ecological, terrestrial and biological studies should be carried out by recognized scientific institutions and report submitted to this Ministry every six months.	The studies not yet over by National Institute of Technology, Suratkal. The 1st set of sample collected in Aug.2009. Monthly reports till December 2010 have already been forwarded every month. the dredging work is terminated on 27.6.2013. The reports will be forwarded once the balance Capital Dredging work is awarded.
xi)	The project proponent should undertake a comprehensive hydrodynamic modeling study with regard to river diversion and submit the report to the Ministry within 6 months from the date of receipt of this letter. Further, the unit should comply with all the findings/ recommendations of the study.	There is no river diversion involved in the Project. The hydrodynamic modeling studies have been already carried out by CWPRS, Pune. There is no impact on the tidal flow due to the Project. The report was forwarded vide this office letter No.CE.CF.226(CTP)/1429 dated 27.05.2015.
xii)	Construction labour camps should be located outside Coastal Regulation Zone areas and should be provided with adequate cooking and sanitation facilities.	The construction labour camps is not within Coastal Regulation Zone.
	B. General Conditions	
i)	Development of the proposed channel should be undertaken meticulously conforming to the existing Central/local rules and regulations including Coastal Regulation Zone Notification, 1991 and its amendments. All the construction designs/drawings relating to the proposed development activities must have approvals of the concerned State Government Department/Agencies.	The necessary statutory approvals will be obtained.

Sr. No	Conditions stipulated	MbPT's Compliance status
ii)	A well-equipped laboratory with suitable instruments to monitor the quality of air and water shall be set up as to ensure that the quality of ambient air and water conforms to the prescribed standards. The laboratory will also be equipped with qualified manpower including a marine biologist so that the marine water quality is regularly monitored in order to ensure that the marine life is not adversely affected as a result of implementation of the said project. The quality of ambient air and water shall be monitored periodically in all the seasons and the results should be properly maintained for inspection of the concerned pollution control agencies. The periodic monitoring reports at least once in 6 months must be send to this Ministry (Regional Office at Bhopal) and Pollution Control Committee.	Mumbai Port has testing laboratory. The Air & Water quality is being regularly monitored every month and monthly reports are sent to Regional Office, Bhopal.
iii)	Adequate provisions for infrastructure facilities such as water supply, fuel for cooking, sanitation etc. must be provided for the labourers during the construction period in order to avoid damage to the environment. Colonies for the labourers should not be located in Coastal Regulation Zone area. It should also be ensured that the construction workers do not cut tress including mangroves for fuel wood purpose.	Yes.
iv)	To prevent discharge of sewage and other liquid wastes into the water bodies, adequate system for collection and treatment of the wastes must be provided. No sewage and other liquid wastes without treatment should be allowed to enter into the water bodies.	Yes.
v)	Appropriate facility should be created for the collection of solid and liquid wastes generated by the barges/vessels and their safe treatment and disposal should be ensured to avoid possible contamination of the water bodies.	Presently most of the vessels visiting the Port are provided with Incinerator on Board for burning waste oil/used oil. No ship is allowed to drain off any waste oil, used oil, dirty ballast in the Harbour water and the same is either burned by the incinerator on board of the concerned ship or disposed through private firms registered with Central Control Pollution Board.
vi)	Necessary navigational aids such as channel markers should be provided to prevent accidents. Internationally recognized safety standards shall be applied in case of barge/vessel movements.	All necessary navigational aids is already provided as per the international standards for safety of the barges/vessels.

Sr. No	Conditions stipulated	MbPT's Compliance status
vii)	The project authorities should take appropriate community development and welfare measures for villagers in the vicinity of the project site, including drinking water facilities. A separate fund should be allocated for this purpose.	Not Applicable.
viii)	The quarrying material required for the construction purpose should be obtained only from the approved quarries/borrow areas. Adequate safeguard measures shall be taken to ensure that the overburden and rocks at the quarry site do not find their way into water bodies.	Yes.
ix)	For employing unskilled, semi skilled and skilled workers for the project, preference should be given to local people.	This is a BOT project and the construction contractor will employ the unskilled, semi-skilled and skilled workers available locally.
x)	The recommendations made in the Environmental Management Plan and Disaster Management Plan as contained in the Environment Impact Assessment and Risk Analysis Reports of the project shall be effectively implemented.	Yes.
xi)	A separate Environmental Management Cell with suitable qualified staff to carry out various environment should be set up under the charge of a Senior Executive who will report directly to the Chief Executive of the Company.	Trained staff has been employed by M/s. Indira Container Terminal Pvt. Ltd. (ICTPL) for civil work and reports are submitted to the MoEF. The Environmental Cell of NIT, Suratkal was monitoring and the reports were being submitted to MoEF regularly till Jan. 2013. Thereafter is no work is done.
xii)	The project affected people, of any should be properly compensation and rehabilitated.	This project is within the Mumbai Port Trust area and no resettlement and rehabilitation is involved in the project.
xiii)	The funds earmarked for environment protection measures should be maintained in a separate account and there should be no diversion of these funds for any other purpose. A year-wise expenditure on environmental safeguards should be reported to this Ministry.	Yes.
xiv)	Full support should be extended to the officers of this Ministry's Regional Office at Bangalore and the officers of the Central and State Pollution Control Boards by the project proponent during this inspection for monitoring purposes, by furnishing full details and action	Yes.

Sr. No	Conditions stipulated	MbPT's Compliance status
	plans including the action plans including the action taken reports in respect if mitigative measures and other environmental protection activities.	
xv)	In case there is an intention of deviation or alternation in the project including the implementing agency, a fresh reference should be made to this Ministry for modification in the clearance conditions or imposition of new ones for ensuring environmental protection. The project proponents should be responsible for implementing the suggested safeguard measures.	Yes.
xvi)	This Ministry reserves the right to revoke this clearance, if any of the conditions stipulated are not complied with to the satisfaction of this Ministry.	Noted.
xvii)	This Ministry or any other competent authority may stipulate any additional conditions subsequently, if deemed necessary for environmental protection, which shall be complied with.	Yes.
xviii)	The project proponent should advertise at least in two local newspapers widely circulated in the region around the project, one of which shall be in the vernacular language of the locality concerned informing that the project has been accorded environmental clearance and copies of clearance letters are available with the State Pollution Control Board and may also be seen at Website of the Ministry of Environment & Forests at http://www.envfornic.in. The advertisement should be made within 7 days from the date of issue of the clearance letter and a copy of the same should be forwarded to the Regional Office of this Ministry at Bhopal.	This has already been complied.
xix)	The project proponents should inform the Regional Office as well as Ministry the date of financial closure and final approval of the project by the concerned authorities and the date of start of Land Development Work.	Government vide letter No.PD-11011/2/99/BPT dated 19.11.2007 conveyed sanction to the project estimates for MbPT component of work. The BOT operator M/s. Indira Container Terminal Pvt. Ltd. have achieved financial closure on 14.11.2008. The BOT operator has also finalized EPC contractor for construction on 02.12.2008. The construction work has commenced from October 2009.

215

PRESENT STATUS Works by BOT Operator, M/s. ICTPL.

- Berthing structure (700mx58m) with approach jetty completed.

PRESENT STATUS Works by Mumbai Port Trust.

- The contract for capital dredging and filling up of Prince's and Victoria Docks was awarded to M/s.Jaisu Shipping Co. Pvt. Ltd on 1.4.2009 with a completion period 24 months.
- The capital dredging and filling up of Prince's & Victoria Dock contract has been terminated on 27.06.2013.
- Contracts for balance filling work and dock closure work awarded on 04.04.2014.
- Depth achieved at the time of termination

- At berth pocket – (-)16m CD (achieved)
- At turning circle & approach channel – (-) 11m CD as against (-13.5 m CD).
- The present depth due to siltation is At berth pocket – (-)10m CD.
- Filling of Prince's Dock is complete.
- Filling of Victoria Dock is in progress.
- Laying of railway track for RCD is in progress.

Notes :-

1. During the 127th Meeting held on 28.10.2013, the Expert Appraisal Committee on CRZ Infrastructure & Miscellaneous Projects and New Construction and Industrial Estates Projects held at New Delhi, the validity of the environment clearance (No. 10-18/2005-IA-III dated 9.11.2006) issued to the project has been extended for five years.

2. Vide letter dated 07.01.2014, the extension to validity of environmental clearance for a period of 5 years upto 14.06.2016 has been communicated by the MoEF.

3. The scheduled date of commissioning of the project is January 2016.

4. The Dock Closure works at Victoira Dock and Prince's Dock gate No. 1 are in progress, which have been awarded on 04.04.2014 and works are in progress.

5. M/s. SBI Capital has been appointed for revival of the project by MbPT. M/s. KPMG has been appointed as consultants for feasibility assessment for handling containerised and non- containrised clean cargo by M/s. ICTPL.(BOT Operator)

Sd/-

(N.G.Mirajkar)
Dy. Chief Engineer, Projects I

Sr. No	Conditions stipulated	MbPT's Compliance status
		presently the project is stalled after completion of civil works at jetty & berth.

PRESENT STATUS Works by BOT Operator, M/s. ICTPL.

- Berthing structure (700mx58m) with approach jetty completed.

PRESENT STATUS Works by Mumbai Port Trust.

- The contract for capital dredging and filling up of Prince's and Victoria Docks was awarded to M/s.Jaisu Shipping Co. Pvt. Ltd on 1.4.2009 with a completion period 24 months.
- The capital dredging and filling up of Prince's & Victoria Dock contract has been terminated on 27.06.2013.
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- Depth achieved at the time of termination
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- Laying of railway track for RCD is in progress.

Notes :-

1. During the 127th Meeting held on 28.10.2013, the Expert Appraisal Committee on CRZ Infrastructure & Miscellaneous Projects and New Construction and Industrial Estates Projects held at New Delhi, the validity of the environment clearance (No. 10-18/2005-IA-III dated 9.11.2006) issued to the project has been extended for five years.

2. Vide letter dated 07.01.2014, the extension to validity of environmental clearance for a period of 5 years upto 14.06.2016 has been communicated by the MoEF.

3. The revised date of commissioning of the project is April 2018.

4. Based on the directions given during the meeting taken by the Principal Secretary to the Prime Minister on 17.08.2015, MbPT appointed M/s. SBI Capital as a Transaction Advisor.

O/C

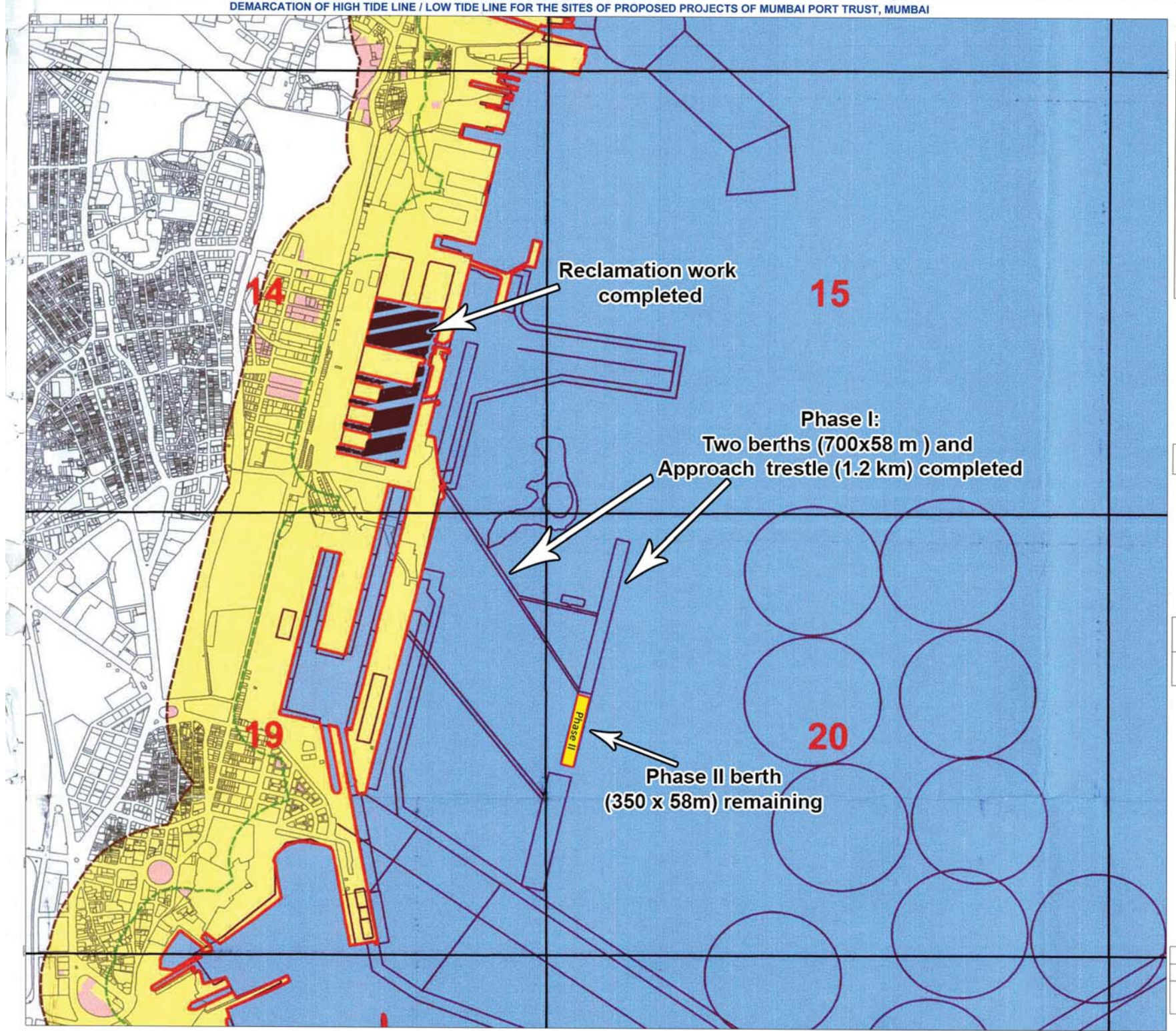
(N.G.Mirajkar)
Dy. Chief Engineer, Projects I

Dy CE PI
SE PI 15/2/16
EE DC I
15/2/2016

Annexure 6

CRZ Map Indicating HTL/LTL Demarcation in 1:4000 Scales

DEMARCATION OF HIGH TIDE LINE / LOW TIDE LINE FOR THE SITES OF PROPOSED PROJECTS OF MUMBAI PORT TRUST, MUMBAI



LEGEND

- LOW TIDE LINE (LTL)
- HIGH TIDE LINE (HTL)
- 100m / CREEK WIDTH FROM HTL
- 200m FROM HTL
- 500m FROM HTL
- MANGROVES (CRZ - IA)
- 50m BUFFER FROM MANGROVES (CRZ - IA)
- MUD FLAT (CRZ - IA)
- CRZ - II
- CRZ - III
- CRZ - IVA
- CRZ - IVB
- PROPERTY BOUNDARY
- HTL REFERENCE POINTS

SOURCE : CLIENT

- PROJECT SITE

PROPOSED PROJECTS

1. FIFTH OIL BERTH
2. TRANSIT SHED AT 16/17 ID
3. TRANSIT SHED AT 4 ID
4. MULTIPURPOSE CARGO BERTH
5. DEVELOPMENT OF BERTHING FACILITIES FOR OFFSHORE SUPPLY VESSELS
6. NEW CRUISE TERMINAL AT GATEWAY OF INDIA

N

SCALE 1 : 25,000

0 312.5 625 1,250 1,875 2,500 M

PREPARED BY
INSTITUTE OF REMOTE SENSING
ANNA UNIVERSITY
CHENNAI - 600 025

FOR
MUMBAI PORT TRUST
PORT HOUSE, SHOORJI VALLABHDAS MARG
BALLORD ESTATE, MUMBAI - 400 001

PREPARED BY	
VERIFIED BY	
APPROVED BY	Prof. S.S. Ramakrishnan, Lt.(Mrs.) M.Sc.(Tech.), Director, Institute of Remote Sensing, Anna University, Chennai-600 025.

Annexure 7

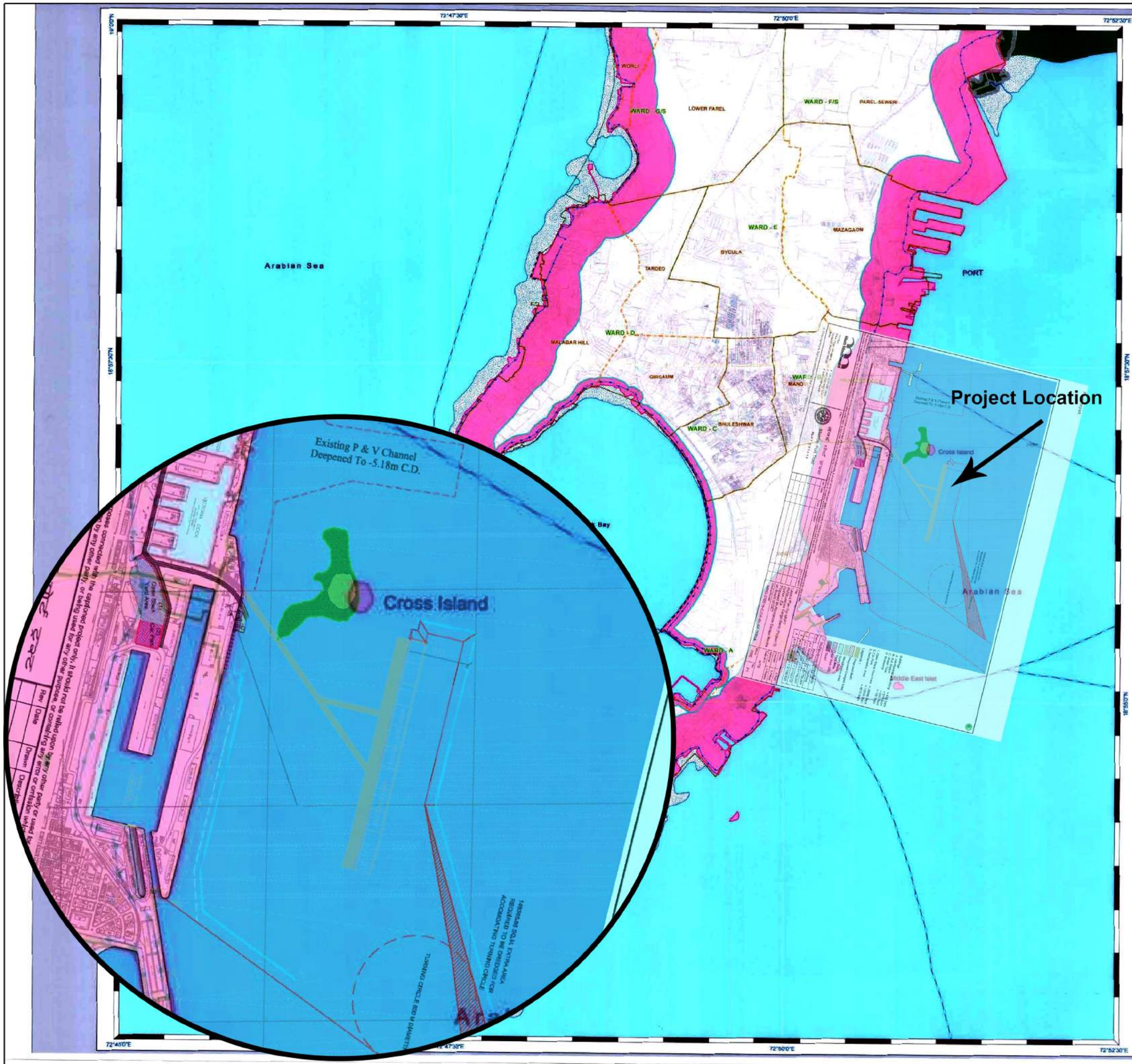
Copy of CZMP

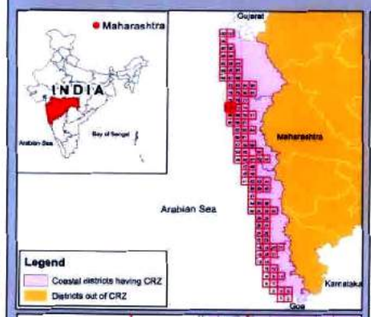
COASTAL ZONE MANAGEMENT PLAN
MAHARASHTRA

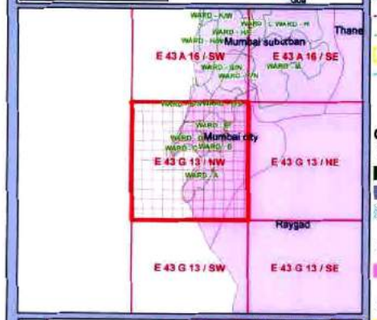
Sheet No: E 43 G 13 / NW

Projection :- UTM Datum :- WGS 1984 Map No. :MH 72

0 1,000 2,000
Meters
1:25,000







Legend

- Road
- Ferry
- Railway Line
- High Tide Line (HTL)
- Low Tide Line (LTL)
- Bund
- Seawall
- Ward Boundary
- Village Boundary
- 22 Survey Plots

CRZ Lines & Boundary

- Hazard Line
- 100 m Line in CRZ III Area
- 200 m CRZ Line - NDZ
- CVCA
- CRZ Boundary (300 m Line, 100 m for Day, 100 m or width of the creek whichever is less along the tidal influenced water bodies)

CRZ CATEGORY

CRZ - I

- CRZ - IA
- 50 m Mangrove Buffer Zone (CRZ - IA)
- CRZ - IB

CRZ - II

- CRZ - II

CRZ - III

- No Development Zone
- 200 to 500 m from HTL

CRZ - IV

- CRZ - IVA
- CRZ - IVB

DATA SOURCE

i) National Centre for Sustainable Coastal Management
1) HTL, 2) CRZ - IA, 3) CVCA

ii) Survey of India
1) Hazard Line

iii) Maharashtra Coastal Zone Management Authority
1) Administrative Bouncaries

iv) Institute of Remote Sensing
1) CRZ Categories and Lines (except CRZ - IA)
2) Infrastructure facilities such as Road, Ferry, etc

ABBREVIATIONS

CRZ : Coastal Regulation Zone
NDZ : No Development Zone
CVCA : Critically Vulnerable Coastal Area

Mapped During 2017-18

PREPARED AS PER COASTAL REGULATION ZONE NOTIFICATION 2011

Scrutinized by	*Certified by*	*Approved by*
Technical Scrutiny Committee National Centre for Sustainable Coastal Management	DIRECTOR National Centre for Sustainable Coastal Management Ministry of Environment, Forest & Climate Change	National Coastal Zone Management Authority Ministry of Environment, Forest & Climate Change Government of India

Prepared by

Institute of Remote Sensing
Anna University
Chennai - 600 025

State Government of Maharashtra

Annexure 8

Recommendation of State Coastal Zone Management Authority

MAHARASHTRA COASTAL ZONE MANAGEMENT AUTHORITY

Tel. No. : 2202 9388
E-mail : dir1.mcz-mh@nic.in
Website: https://mczma.gov.in/

No. CRZ 2019/CR 217/TC 4
Office of the -
Maharashtra Coastal Zone Management Authority,
Environment & Climate Change Department,
15th Floor, New Administrative Building,
Mantralaya, Mumbai- 400 032
Date: 04th May, 2021

To,
Director (IA-III),
Coastal Zone Regulation, Government of India,
Ministry of Environment, Forests & Climate Change,
Indira Paryavaran bhavan, Jor Bagh Road,
New Delhi - 110 003.

Subject: Proposed construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour by Mumbai Port Trust (MbPT)

The Maharashtra Coastal Zone Management Authority in its 146th & 154th meeting held on 04th September, 2020 and 15th & 20th April, 2021 deliberated the proposal of construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour.

2. The Authority noted that, the project is an expansion of existing Mumbai Port Trust for improvement of Traffic at Mumbai Port. MbPT has proposed to develop offshore container Terminal (OCT) at Indira Dock, Mumbai Port. It was decided to execute project in two phases.

3. The Authority noted that, the proposal was earlier deliberated in 146th meeting of MCZMA held on 04.09.2020. The Authority noted that as per the Minutes of EAC, New Delhi, the Public hearing is exempted since it was already conducted in the year 2005. However, Queries/ Comments raised during earlier public hearing need to be documented in the form of Management Plan. It was noted that the Rapid EIA report submitted by the PP needs to add the chapter on the above said TOR point of the EAC, New Delhi. Accordingly, the matter was deferred for want of the Revised EIA report for the project which would have the chapter on Queries/ Comments raised during earlier public hearing need to be documented in the form of Management Plan. MbPT should also submit the CRZ map in 1:4000 scale superimposing the completed and remaining works. The MbPT vide letter dated 12.10.2020 submitted revised EIA report, CRZ map in 1:4000 scale.

4. The Authority noted that the revised EIA report has a chapter on details of public hearing and mitigation plan. One of the issues raised during public hearing was pertaining to mangroves and flamingoes. Reply of the MbPT states that no land is being reclaimed for the project. We are just filling up the old dock basins. About mangroves, this would not at all affected. The Mumbai Port has spent the dumping area is about 25 km away from the mangroves. The Mumbai port has developed and protected mangroves at sewri. Previously about 2000 flamingoes were there but of today there are about 14000 flamingoes. This project will no way affect habitat. Nearly about 15 Acres of area as sanctuary has been

replanted. MbPT spent Rs. 50 Lacs to develop this area as sanctuary. The mangrove situated between Sewri Wadala will be in no way affected.

5. The Authority noted the status of construction works at MbPT, which is as follows:

a) Completed works:
   i. 2 berths and approach trestle
   ii. Capital dredging for berth pocket
   iii. Filling of P & V Dock
   iv. Rail Container Depot
   v. Approach channel and Turning circle dredging
   vi. Berth Packet

b) Pending works:
   i. Procurement of Cargo handling equipment
   ii. Container yard (Reclamation work completed. Work of development of yard mainly metalling WBM and wearing coat balance)
   iii. Capital dredging in the approach channel and turning circle to the required depth

6. The Authority discussed the impact of the pending works such as work of development of container yard, capital dredging in approach channel etc. on the coastal environment and mangroves. The MbPT officials submitted that no mangroves would be disturbed while completing the pending works. No land is being reclaimed for the project. MbPT is just filling up the old dock basins.

7. The Authority noted that as per approved CZMP of Greater Mumbai of 2011, the site under reference falls in CRZ-II & IV areas.

8. The Authority noted that as per provisions of CRZ Notification, 2011, following is permissible:

a) As per para 3(c) of the CRZ Notification, 2011, maintenance or clearing of waterways, channels and ports, based on EIA studies;
b) As per para 4(i) (f) of CRZ Notification, 2011, Construction and operation for ports and harbours, jetties, wharves, quays, slipways, ship construction yards, breakwaters, groynes, erosion control measures are permissible activities.

9. The Authority felt that the project could be recommended to MoEF&CC, New Delhi from CRZ point of view so that pending works be completed by the MbPT.

10. In the light of above, after detailed discussion and deliberation, the Authority decided to recommend the proposal to MoEF&CC, New Delhi subject to strict compliance of following conditions:

Specific Conditions:

i. The proposed pending works should be carried out strictly as per the provisions of CRZ Notification, 2011 as amended from time to time.
ii. Project proponent should implement Environment Management plan for the project effectively and efficiently during construction and operational phase of the project to ensure that coastal environment is protected.
iii. MbPT to ensure no spillage of oil from ships/ vessels in the sea. MbPT should ensure proper Oil Spill Management Plan as well as barricading/isolation along with

danger/inflammable sign to be provided on the existing chemical berths. During operation phase, accidental spill of chemical/oil should be collected and disposed-off through authorized recyclers.

iv. The dredged material to be disposed-off at the designated locations to avoid any adverse impact on marine water quality.

v. Debris generated during the project activity should not be dumped in CRZ area. It should be processed scientifically at a designated place.

vi. PP to ensure that best industrial practices should be followed for fire safety measures and for conservation of coastal environment

vii. All other required permissions from different statutory authorities should be obtained prior to commencement of work.

General Conditions:

i. In case of any change in project profile, the project would require fresh appraisal by the MCZMA.

ii. The MCZMA reserves the right to revoke this recommendation, if the conditions stipulated are not complied with to the satisfaction of the MCZMA or Environment Department.

iii. The MCZMA or any other competent authority, concerned planning authority may stipulate any additional conditions subsequently, if deemed necessary, for environmental protection, which shall be complied with.

iv. The clearance accorded to the project under this notification shall be valid for a period of seven years from the date of issue of such clearance: Provided that the construction activities shall commence within a period of five years from the date of the issue of clearance and the construction be completed and the operations be commenced within seven years from the date of issue of such clearance.

v. The recommendation from CRZ point of view is being issued without prejudice to the action initiated under EP Act or any court case pending in the court of law and it does not mean that project proponent has not violated any environmental laws in the past and whatever decision under EP Act or of the Hon'ble court will be binding on the project proponent. Hence this recommendation does not give immunity to the project proponent in the case filed against him, if any or action initiated under EP Act.

11. Agenda item & minutes of the meeting is available on the website of MCZMA i.e. http://mczma.gov.in.

(Narendra Toke)
Director, Environment & MS, MCZMA

Copy for information to:

1. **PS (Environment) & Chairperson, (MCZMA),** Environment & CC Department, Room No. 217 (Annex), Mantralaya, Mumbai -32.
2. **Member Secretary, Maharashtra Pollution Control Board,** Kalpataru Point, 3rd and 4th floor, Road No. 8, Sion Cir, opp. PVR Theater, Mumbai -400022
3. **District Collector Mumbai City,** Old Custom House, Shahid Bhagat Singh Rd, Marg, Fort, Mumbai 400001
4. **Chairman, Mumbai Port Trust (MbPT),** 20, Shoorji Vallabhdas Rd, Kala Ghoda, Fort, Mumbai 400001 - -It is requested to submit application on Parivesh Portal of MoEF&CC.
5. **Select File-TC 4**

Annexure 9

Certificate of Accreditation cerfied by QCI/NABET

Quality Council of India

National Accreditation Board for Education & Training

CERTIFICATE OF ACCREDITATION

Terracon Ecotech Pvt. Ltd

202, Kingston, Tejpal Road, Vile Parle (E), Mumbai 400057

Accredited as Category - A organization under the QCI-NABET Scheme for Accreditation of EIA Consultant Organizations: Version 3 for preparing EIA-EMP reports in the following Sectors:

Sl. No.	Sector Description	Sector (as per)		Cat.
		NABET	MoEFCC	
1	Mining of minerals including opencast/ underground mining	1	1 (a) (i)	A
2	Off shore and on shore oil and gas exploration, development and production	2	1 (b)	A
3	River valley projects	3	1 (c)	A
4	Metallurgical industries (ferrous & non ferrous)	8	3 (a)	A
5	Oil & gas transportation pipeline (crude and refinery/ petrochemical products), passing through national parks/ sanctuaries/coral reefs /ecologically sensitive Areas including LNG terminal	27	6 (a)	A
6	Isolated storage & handling of hazardous chemicals (As per threshold planning quantity indicated in column 3 of Schedule 2 & 3 of MSIHC Rules 1989 amended 2000)	28	6 (b)	B
7	Ports, harbours, break waters and dredging	33	7 (e)	A
8	Common municipal solid waste management facility (CMSWMF)	37	7(i)	B
9	Building and Construction projects	38	8 (a)	B
10	Townships and Area development projects	39	8 (b)	B

Note: Names of approved EIA Coordinators and Functional Area Experts are mentioned in RA AC minutes dated Oct 11, 2019 posted on QCI-NABET website.

The Accreditation shall remain in force subject to continued compliance to the terms and conditions mentioned in QCI-NABET's letter of accreditation bearing no. QCI/NABET/ENV/ACO/19/1188 dated Dec 31, 2019. The accreditation needs to be renewed before the expiry date by Terracon Ecotech Pvt. Ltd, Mumbai following due process of assessment.

Sr. Director, NABET
Dated: December 31, 2019

Certificate No.
NABET/ EIA/1922/ RA 0147

Valid till
Jan 20, 2022

For the updated List of Accredited EIA Consultant Organizations with approved Sectors please refer to QCI-NABET website.

Annexure 10

EIA report

ENVIRONMENTAL IMPACT ASSESSMENT REPORT

FOR

OFFSHORE CONTAINER TERMINAL (OCT) AT INDIRA DOCK, MUMBAI PORT

BY

MUMBAI PORT TRUST

CONTENTS

COMPLIANCE FOR TOR....6
CHAPTER 01: INTRODUCTION....10
1.1 PURPOSE OF THE EIA STUDY....10
1.2 ABOUT MUMBAI PORT TRUST....11
1.3 ABOUT THE PROJECT....11
1.4 SEQUENCE OF EVENTS FOR ENVIRONMENTAL CLERANCE....13
1.5 PRESENT STATUS OF THE PROJECT....13
1.6 PROPOSAL FOR REVIVAL OF THE PROJECT....14
1.7 TERMS OF REFERENCE....14
1.8 SCOPE OF WORK....14
1.9 APPROACH AND METHODOLOGY....15
1.9.1 Baseline study....15
1.9.2 Project Impact Assessment....15
1.9.3 Formulation of Mitigative Measures....16
1.9.4 Environmental Management Plan....16
1.9.5 Generic Structure of Environmental Impact Assessment Report....16
CHAPTER 02 - DESCRIPTION OF THE PROJECT....19
2.1 INTRODUCTION....19
2.2 PROJECT DESCRIPTION....19
2.3 PRESENT STATUS OF THE PROJECT....20
2.3.1 Existing Civil Works....22
2.3.2 Site Photographs....23
2.4 NEED OF THE PROJECT....24
2.5 PROJECT DETAILS....24
2.6 PROJECT LOCATION....27
2.7 CRZ STATUS OF THE PROJECT....27
2.8 LAND REQUIREMENT....27
2.9 CONSTRUCTION WORK....28
2.9.1 Demolition Work....28
2.9.2 Dredging....28
2.10 UTILITY REQUIREMENT....28
2.10.1 Construction Material....28
2.10.2 Power Requirement....28
2.10.3 Man Power Requirement....28
2.10.4 Water Requirement....28
2.10.5 Waste Managements....28
2.11 TRAFFIC ANALYSIS....28
2.11.1 PLANNING PARAMETERS....30
2.12 CAPITAL COST AND IMPLEMENTATION SCHEDULE....30

2.12.1 Operation And Maintenance Cost....31
2.12.2 Revenue Earnining....31
2.11.3 Financial Viabiliy Analysis....32
2.13 Court Cases....32
CHAPTER 03 - BASELINE ENVIRONMENTAL STUDY....33
3.1 SCOPE OF WORK....33
3.2 BASELINE ENVIRONMENTAL STUDY....34
3.2.1 Land Environment....36
3.2.1.1 Topography & Geology....36
3.2.2 Air Environment....38
3.2.2.1 Meteorology of the Study Area....38
3.2.2.2 Ambient Air Quality....42
3.2.3 Water Environment....45
3.2.3.1 Coastal Hydrography Of The Study Area....45
3.2.3.2 Marin Water Quality....46
3.2.3.3 Sediment Quality....47
3.2.4 Noise Environment....48
3.2.4.1 Applicable Standards & Classification....48
3.2.4.2 Status of Noise Environment in the Project Area....49
3.2.4.3 Ambient Noise Levels....49
3.2.5 Socio-Economic Environment....51
3.2.5.1 Objective....52
3.2.5.2 Baseline Data....52
3.2.5.3 Settlement Pattern and Demography....52
3.2.5.4 Distribution of population....52
3.2.5.5 Sex Ration....53
3.2.5.6 Literacy levels....53
3.3 TRAFFIC STUDY....53
3.4 ECOLOGY & BIODIVERSITY....73
3.4.1 Marine Ecology....73
3.4.1.1 Water Quality Biotic....73
3.4.2 Terrestial Flora And Fauna....79
CHAPTER 04 - ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATION MEASURES....97
4.1 INTRODUCTION....97
4.2 IMPACTS FROM THE PROJECT LOCATION....98
4.3 IMPORTANT ATTRIBUTES DURING CONSTRUCTION PHASE....99

4.4 IMPACTS DURING CONSTRUCTION PHASE 100
4.5 IMPACTS DURING OPERATION PHASE 106
4.6 IMPACTS ON QUALITY OF LIFE 112
4.7 EVALUATION OF IMPACTS 122
CHAPTER 05 - ENVIRONMENTAL MONITORING PROGRAMME 124
5.1 THE NEED 124
5.2 AREAS OF CONCERN 126
5.3 WATER QUALITY 126
5.4 SOIL QUALITY 127
5.5 AMBIENT AIR QUALITY 128
5.6 NOISE LEVELS 129
5.7 SUMMARY OF ENVIRONMENTAL MONITORING PROGRAMME 129
CHAPTER 06 - DISASTER MANAGEMENT PLAN 130
6.1 CONCEPT OF RISK ASSESSMENT 130
6.2 LEAKS AND SPILLAGES 130
6.3 RISK ANALYSIS STUDY 133
6.4 ONSITE EMERGENCY PLAN 133
6.5 LIFE SAVING APPLIANCES AND ARRANGEMENTS 135
6.6 OCCUPATIONAL HEALTH AND SAFETY 136
6.7DISASTER MANAGEMENT PLAN (DMP) 136
6.8 EMERGENCY PLAN FOR BERTHS AND VESSEL 138
6.9 IDENTIFICATION OF MAJOR HAZARDS 139
6.10 EMERGENCY RESPONSE MEASURES FOR NATURAL HAZARDS 140
6.11 IDENTIFICATION AND ASSESSMENT OF HAZARDS 144
CHAPTER 07 - PROJECT BENEFITS 146
CHAPTER 08 - ENVIRONMENTAL MANAGEMENT PLAN 147
8.1 INTRODUCTION 147
8.2 SUMMARY OF CRITICAL IMPACTS/ ISSUES 149
8.3. IDENTIFICATION OF IMPLEMENTING AUTHORITY 149
8.4 IMPLEMENTATION OF RECOMMENDED MITIGATION MEASURES 150
8.5 MONITORING PLAN 150
8.6 ENVIRONMENTAL TRAINING 151
8.7 BUDGETS FOR ENVIRONMENTAL MANAGEMENT PLAN 151
8.8 DETAILS OF MANAGEMENT PLANS 158
CHAPTER 09 - EXECUTIVE SUMMARY 160
9.1 INTRODUCTION 160
9.2 TRAFFIC ANALYSIS 160

9.3 PLANNING PARAMETERS 161
9.4 PROJECT DETAILS 162
9.5 CAPITAL COST AND IMPLEMENTATION SCHEDULE 165
9.6 BASELINE ENVIRONMENT 166
CHAPTER 10 - CONSULTANTS ENGAGED 168

List of Tables

Table 1 Comparative statement existing & proposed project 21
Table 2 Baseline Environmental Monitoring Locations 34
Table 3 - Average mean temperature for the period 2002-2013 39
Table 4 -Average monthly rainfall for the years 2002-2013 40
Table 5-The predominant wind direction for the years 1971-2000 at at 03 UTC 41
Table 6 - The predominant wind direction for the years 1971-2000 at 12 UTC 41
Table 7 Ambient Air Quality Standards (National) – India 42
Table 8 - Ambient Air Quality Of Study Area 43
Table 9 - Marine water and sediment quality monitoring location 46
Table 10 - Marine Water Monitoirng results 46
Table 11 – Sediment Quality 47
Table 12 Permissible Noise Level (CPCB Standards) 48
Table 13 - Ambient Noise Levels in Study Area 50
Table 14 - Distribution of Population 52
Table 15 - Literacy Rate 53
Table 16 Enumeration of Phytoplankton in Multipurpose cargo berth 76
Table 17 Enumeration of Zooplankton at Multipurpose Cargo Berth 77
Table 18 Benthic fauna recorded at multipurpose cargo berth 79
Table 19 - List of all sampling points for flora and fauna survey 80
Table 20 - Environmental Impacts and Mitigation Measures 113
Table 21 Environmental Impacts and Mitigation Measures 119
Table 22 - List of Possible Environment Impacts due to Proposed Project 123
Table 23 - Project Start-Up Checklists 125
Table 24 - The Summary of Environmental Monitoring Program for Implementation during Project Construction and Operation Phase 129
Table 25 - Summary of the Disasters Preparedness Plans 145
Table 26 -Critical Impacts / Issues 149
Table 27 - Reporting of The Major- Parameters And Responsible Organization- Construction Phase 152

Table 28 Reporting of The Major- Parameters And Responsible Organization- Operation Phase 155

List of Figures

Figure 1 Exisitng Civil Works..... 22
Figure 2 Primary approach with forked extension..... 23
Figure 3 Sub-station; Fire station and Office buildings..... 23
Figure 4 – Google Image showing project site..... 27
Figure 5 Environmental Monitoring Locations..... 36
Figure 6 - Average Mean Temperature year 2002-2013..... 39
Figure 7 - Average monthly rainfall for year 2002-2013..... 40
Figure 8 -Windrose for the year 1971-2000..... 42
Figure 9 - Ambient Air Quality of Study Area..... 44
Figure 10 - Marine water and sediment quality monitoring locations 46
Figure 11 - Ambient Noise Levels at Study Area..... 51
Figure 12 - Image showing sampling points in 10 km radius of proposed Heliport site 80
Figure 13 -Relative Frequency of trees in study area 82
Figure 14 Relative Density of Trees in Study area..... 83
Figure 15 - Relative Abundance of Trees in Stusy Area..... 83
Figure 16 - Shanon Index of Trees in Study Area..... 84
Figure 17 - Relative Frequency of Shrubs in Study Area..... 84
Figure 18 – Relative Density of Shrubs in Study Area..... 85
Figure 19 - Relative Abundance of Shrubs in Study Area..... 85
Figure 20 - Shanon Index of Shrubs..... 86
Figure 21 - Relative Frequency of Herbs in Study Area..... 86
Figure 22 - Relative Density of Herbs in Study Area..... 87
Figure 23 – Relative Abundance of Herbs in Study Area..... 87
Figure 24 -Shanon Index of Herbs in Study Area 88
Figure 25 – Relative Frequencies of Birds..... 91
Figure 26 – Relative Densities of Birds..... 91
Figure 27 – Relative Abundance of Birds 92
Figure 28 - Shannon Wiener Index values of all the points with respect to bird diversity..... 93
Figure 29 -Relative Frequencies of Butterflies..... 94
Figure 30 - Relative Densities of Butterflies..... 94
Figure 31 – Relative abundances of butterflies..... 95
Figure 32 – Shanon Wiener Index of all points with respect to Butterfly diversity..... 95
Figure 33 - sampling points with maximum bird and butterfly diversity 96

COMPLIANCE FOR TOR		
i.	Importance and benefits of the project.	The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai region and Maharashtra.
ii.	Submit a copy of layout superimposed on the HTL/LTL map demarcated by an authorized agency on 1:4000 scale.	CRZP Map enclosed as **Annexure 1**
iii.	Recommendation of the SCZMA.	CRZ Clearance dated 09.05.2005 enclosed as **Annexure 2**
iv.	Stage – I forest clearance to be submitted.	--
v.	Various Dock and shipbuilding facilities with capacities for existing and proposed project.	Please refer Table 1 of EIA
vi.	Study the impact of dredging on the shore line.	Impacts of dredging on various environmental components are studied and given in Chapter 4 of EIA
vii.	A detailed impact analysis of rock dredging.	Impacts of dredging on various environmental components are studied and given in Chapter 4 of EIA
viii.	Study the impact of dredging and dumping on marine ecology and draw up a management plan through the NIO or any other institute specializing in marine ecology.	Environmental Management Monitoring Plan & Green Plan for Mumbai Port was carried out by NIO, Nagpur in 2016
ix.	A detailed analysis of the physico-chemical and biotic components in the highly turbid waters round the	Please refer Table 10 Marine Water Monitoring Results of EIA

	project site (as exhibited in the Google map shown during the presentation), compare it with the physico- chemical and biotic components in the adjacent clearer (blue) waters both in terms of baseline and impact assessment and draw up a management plan.	
x.	Details of Emission, effluents, solid waste and hazardous waste generation and their management in the existing and proposed facilities.	Please refer to point 4.5.2 for Emission details, refer point no. 4.5.4 for effluent details & refer to 8.8.2 for Solid & Hazardous Waste
xi.	The existing project should avail of and submit consent to operate from the State Pollution Control Board.	Consent to operate will be obtained from MPCB
xii	Requirement of water, power, with source of supply, status of approval, water balance diagram, man-power requirement (regular and contract)	No significant change from earlier scope of work envisaged.
xiii.	Wastewater management plan.	Please refer to 8.8.1 Chapter 8 of EIA
xiv.	Details of Environmental Monitoring Plan.	Please refer to Chapter 5 of the EIA
xv.	The Project proponents will submit copy of the certified compliance report on the Environmental clearances issued earlier to the Project.	Not Applicable
xvi.	To prepare a detailed biodiversity impact assessment report and management plan through the NIOS or any other institute of repute on marine, brackish water and fresh	Environmental Management Monitoring Plan & Green Plan for Mumbai Port was carried out by NIO, Nagpur in 2016

	water ecology and biodiversity. The report shall study the impact on the rivers, estuary and the sea and include the intertidal biotopes, corals and coral communities, molluscs, sea grasses, sea weeds, subtidal habitats, fishes, other marine and aquatic micro, macro and mega flora and fauna including benthos, plankton, turtles , birds etc. as also the productivity. The data collection and impact assessment shall be as per standard survey methods.	
xvii.	The EIA report shall address to all representations and complaints received against the project including those received from the representations from the Conservation Action Plan.	--
xviii	The Study area should be 15 Kms round the project site for base level studies in order that the areas already polluted or under stress can also be identified for impacts and mitigation.	Yes the area of 15 kms around project site is considered for baseline environmental studies. There are no areas already polluted or under stress water bodies the study area.
xix.	The EIA would follow the orders of the respective courts and include a chapter in the EIA on the Court cases including those at the Supreme Court and the NGT.	Please refer point 2.11 of Chapter 2 of EIA
xx.	The Queries and comments raised by the participants during the earlier Public hearing held in 2005 will be	Reply to the public hearing MOM are enclosed as **Annexure 3**

	suitably documented in the form of a management plan drawn up to address to the expressed concerns.	
xxi.	Disaster Management Plan for the above terminal.	Please refer Chapter 6 of the EIA
xxii.	Layout plan of existing and proposed Greenbelt	Project Layout (Google Image) enclosed as **Annexure 4**
xxiii.	Status of court case pending against the project.	Yes, PIL no. 149 of 2014 has been filed against the filling of Victoria & Princess Dock. By the the petitioner Shri Sayyed Abdul Nabi & other have filed an rejoinder to MbPT's a reply. It is pertinent to note that i) the petitioner has filed an PIL against the filling work of Princes and Victoria Docks which has already been completed i) the same petitioner had filed the same case earlier, but as a writ Petition no 617 of 2013 which has been disposed off by the Hogh court Bench comprising of Juctice Dr. D.Y.Chandrachud and Juctice MN.S. Sonak JJ
xxiv.	A tabular chart with index for point wise compliance of above TORs.	--

CHAPTER 01: INTRODUCTION

1.1 PURPOSE OF THE EIA STUDY

Environmental Impact Assessment (EIA) of a project ensures accountability of all the environmental impacts of the various project activities right from the stages of project initiation. The study incorporates the various environmental issues into planning and design stages of the project. It further guarantees the initiation of the various steps for minimization of the identified project impacts and assures a careful consideration of the different project alternatives. An exhaustive EIA process is inclusive of the various steps as described below:

- Screening
- Scoping and consideration of alternatives
- Baseline data collection
- Impact prediction
- Assessment of alternatives, delineation of mitigation measures and environmental impact statement
- Environmental Management Plan
- Decision – making
- Monitoring the clearance conditions

The Rapid EIA of the proposed project was undertaken to achieve the following goals:

- Identification of the various project activities and their potential impacts on the environment.
- Generation of a comprehensive information database for the project planning team on the nature of environmental risks posed by the project activities, such as human health effect, habitat loss, pollution levels and change in land use pattern among other issues.
- Careful consideration of the different alternatives for the project activities that may have serious consequences on the environment.
- Improving the overall decision-making process and ensuring that project options under consideration, are environmentally sound and sustainable.

The key objectives of the study are as follows:

- Including the viable environmental options into micro planning of the project
- Providing mitigation measures as may be required for the successful implementation of the overall project.
- Providing an Environment Management Plan for the site, considering them likely environmental issues and mitigative action plans in the near future.
- Providing a Disaster Management Plan for making sound arrangements for emergency preparedness at the time of natural or man-made disasters.

1.2 ABOUT MUMBAI PORT TRUST

Mumbai Port has long been the principal gateway to India and has played a pivotal role in the development of the national economy, trade & commerce and prosperity of Mumbai city in particular. The port has achieved this position through continuous endeavor to serve the changing needs of maritime trade. Though traditionally designed to handle general cargo, over the years, the port has adapted to changing shipping trends and cargo packaging from break bulk to unitisation/palletisation and containerisation. Besides, it has also developed specialised berths for handling POL and chemicals. For decades, Mumbai Port was India‘s premier port. Even today, with the development of other ports, it caters to 10% of the country‘s sea-borne trade handled by Major Ports of the country in terms of volume. It caters about 19% of POL Traffic handled by Major Ports.

Having weathered and survived many a changes in maritime trade in its long history, Mumbai Port is today facing challenges posed by competition from adjoining ports and private ports, changing traffic patterns, inherent physical constraints and continuing labour intensive operations, etc. However, Mumbai Port is taking various measures to render cost effective and quality services to the trade.

1.3 ABOUT THE PROJECT

During 2006, Mumbai Port invited global tenders for development of an offshore container terminal on DBFOT basis. Based on this tendering process, the licence

was awarded to Indira Container Terminal Private Limited (ICTPL), a special purpose vehicle promoted by Gammon India Ltd. & Gammon Infrastructure Projects Ltd, collectively called the Gammon Group (74%) and Dragados SPL, Spain (26%). The license was for the development and operation of OCT for a period of 30 years including construction period of 3 years at gross revenue share of 35.064%. The License also requires the company to operate and maintain the existing container operations of at the Ballard Pier Station (BPS) for the first five years of the License or 2 years from commencement of operations, whichever is earlier. The license was awarded during December 2007.

As per the concession agreement, the project was to be executed in two phases. During the first phase, the minimum port facilities will be as follows:

Berth length of not less than 700 m and width of 58 m with adequate equipment such as ship to shore handling cranes, yard handling equipment, rail container yard handling facilities, reefer plug points and sufficient container stacking yard etc., commensurate with the traffic. The berth is expected to be designed to receive vessels of capacity up to 6000 TEUs

A Y-Shape approach trestle to connect the berths with the onshore terminal facilities. For this an 18 m wide road will split into two 12 m wide roads reaching the berth and thus forming a Y-shape. Approximate length of the trestle will be 1000m.

Minimum equipment to be commissioned at the berth:

- Four QGC capable of handling 6000 TEU design vessels at the time of commissioning of berths and two more after achieving throughput of 600,000 TEUs in an operating year
- 12 RTGs at the time of commissioning of berths and six more after achieving throughput of 600,000 TEUs in an Operating year
- Two RMGs for rail container depot at the time of commissioning

Correspondingly, the Mumbai Port was to fill up of Princess and Victoria Docks and with the surrounding areas to create a stackyard space of 35 hectares; provide

railway connectivity to create a railway container depot with 3 sidings; dredge the approach channel, turning basin and berth to handle container vessels of 13.5 m draft with tidal advantage.

The project was designed to handle 1.2 million TEU of container traffic during the first phase. During the second phase, the berth length would be increased by 350 m, thereby increasing the handling capacity of the terminal to 2.0 million TEU.

1.4 SEQUENCE OF EVENTS FOR ENVIRONMENTAL CLERANCE

1. CRZ clearance obtained from MCZMA on 09.05.2005
2. Public Hearing for this Project Conducted 29.09.2005
3. MOEF granted Environmental Clearance for 5 years 15.06.2006
4. MOEF modified the EC 09.11.2006
5. MOEF further Extended the Validity for 5 Years upto 14.06.2016 on 07.01.2014
6. MOEF denied further Extention & directed to apply afresh for TOR stage 21.04.2016
7. Applied for fresh ToR on 04.02.2017
8. Proposal was considered in 16th EAC (Infra 2) MoEF&CC on 01.05.2017
9. Terms of reference granted to project on 27.05.2017 by the EAC (Infra 2) MoEF&CC in its 18th meeting

1.5 PRESENT STATUS OF THE PROJECT

Though the commercial operations at the offshore terminal should have been commissioned by December 2010, the project got delayed due to various reasons beyond the control of both the Concessionaire and the Licensee. The berths with the approach could be completed only by September 2013. During this period, the container traffic through Mumbai Port also did not grow as expected due to the influence of JNPT and it stagnated between 40,000 to 50,000 TEU only.

As of now, part of Project facilities like offshore jetty, approach trestle, and yard has been physically completed, however, the completion of the balance mechanical and other facilities of the Project have not been taken up. As an interim measure to ensure utilization of assets created, MbPT decided to permit handling of RoRo at OCT for one year.

1.6 PROPOSAL FOR REVIVAL OF THE PROJECT

MbPT, desirous to revive the OCT Project, evaluated various options including the structure adopted by Kamarajar Port, whereat, Kamarajar Port had proposed to rebid the existing iron ore terminal to include coal cargo based on highest revenue share, with Right of First Refusal (ROFR) to the existing concessionaire and assessment of the existing iron ore terminal assets at Replacement cost. The same model is being followed by Kandla Port for Berth 13 and Berth 15. Accordingly, MbPT is contemplating to revise the cargo to be handled at OCT to include no-containerised cargo like steel and RoRo.

1.7 TERMS OF REFERENCE

The Terms of Referecnce was reccomonded for project of Construction of Two Offshore Container Berths and development of Container Terminal on BOT basis in Mumbai Harbour' by Expert Appraisal Committee (Infra 2), Ministry of Environment, Forest & Climate Change in its 25th Meeting held on 27.05.2017.

1.8 SCOPE OF WORK

As per the ToR the study area for the EIA pof 15 km radius around project site was defined. The primary focus of the study is restricted to the project boundaries, an area within radii of 5 km around project site. the secondary data was collected from 15 km radius around project site. The EIA study is primarily based on collection of baseline information and assessment of short-term as well as long-term impacts. In order to fulfill these objectives a three-phased approach to the study was followed:

Phase I : Data Collection and Description of the Baseline Environment

Phase II : Assessment and Evaluation of potential Environmental Impact of the project

Phase III : Development of recommendations including an EMP and Disaster Management Plan (DMP) for improving the environmental features of the project on a long term.

The three phases of the Environmental Impact Assessment study includes the various activities/tasks for exhaustive coverage of the various issues and concerns

for the project site. The description of each task has been done in the following sections in accordance with the Environmental Impact Assessment guidelines of Ministry of Environment & Forests, Government of India.

1.9 APPROACH AND METHODOLOGY

The general approach followed for carrying out the EIA for the project is summarized in the enclosed flow chart on the following page.

1.9.1 Baseline study

Baseline study comprised generation of primary data and collection of secondary data. Primary survey including environmental monitoring was conducted for the project area to gather information on the following environmental attributes.

- Air quality in respect of SPM, NOx, SO2, CO and RSPM
- Noise Levels
- Water Quality of key physico-chemical parameters
- Soil Quality
- Ecology and Bio- diversity

Secondary data was collected on the following aspects to supplement the primary data in order to assess the baseline environmental setup. The data was gathered on

- Geological and physiological characteristics
- Meteorological data – rainfall, humidity, temperature, wind speed and wind direction
- Landuse
- Flora and Fauna (Ecology & Bio- Diversity)

1.9.2 Project Impact Assessment

Impacts of the project on environment both during construction phase and operational phase was assessed against the baseline information. Baseline information along with predicted air quality and noise levels data were compared with the National Standards stipulated by regulatory agencies like CPCB and MoEF. Qualitative assessment of the impacts of increased noise level, water and soil

pollution due to project on public health, animal and surrounding vegetation has been assessed.

1.9.3 Formulation of Mitigative Measures

The standard guidelines of MoEF and other regulatory agencies were considered to suggest different / alternative mitigative measures. Trade –off and multi –alternative analysis was carried out arrive at the most appropriate measures to minimize the negative impacts of the project.

1.9.4 Environmental Management Plan

Environmental Management Plan (EMP) is prepared after identifying, predicting and evaluating the significant impacts on each component of the environment with a view to maximize the economic benefits from the project. Post-project Environmental Monitoring program is also detailed in the report.

1.9.5 Generic Structure of Environmental Impact Assessment Report

In terms of the EIA notification of the MOEF dated 14 September 2006, the generic structure of the EIA document shall be as under:

Chapter no.	Chapter Title	Description
Chapter 1	**Introduction**	This chapter contains the general information on the port sector, major sources of environmental impact in respect of port projects and details of the environmental clearance process.
Chapter 2	**Project Description**	This chapter should cover the description of the project, such as the type of project, need for the project, project location, project layout, cargo handling methods, utilities and services, and the project implementation schedule, estimated cost of development etc
Chapter 3	**Description of Environment**	This chapter designs the heart of the manual, since it is in this chapter that the proponent shall give the comprehensive data on the existing and additional

Chapter no.	Chapter Title	Description
		data collected by him on the baseline environmental data in the study area as well as in the surrounding area that is likely to be affected by the proposed activity.
Chapter 4	**Anticipated Environmental Impact and Mitigation Measures**	This chapter should cover the anticipated impact on the environment and mitigation measures. The method of assessment of impact including studies carried out, modeling techniques adopted to assess the impact where pertinent should be elaborated in this chapter. It should give the details of the impact on the baseline parameters, both during the construction and operational phases and the mitigation measures to be implemented by the proponent.
Chapter 5	**Environmental Monitoring Programme**	This chapter should cover the planned Environmental Monitoring Program. It should include the technical aspects of monitoring the effectiveness of mitigation measures
Chapter 6	**Additional Studies**	This chapter should cover the details of the additional studies, if any, required in addition to those specified in the TOR and which are necessary to cater to more specific issues applicable to the particular project. These studies may be suggested either by the proponent itself or the regulatory authority.
Chapter 7	**Project Benefits**	This chapter should cover the benefits accruing to the locality, neighbourhood, region and nation as a whole. It should bring out details of benefits by way of improvements in the physical infrastructure, social infrastructure, employment potential and other tangible benefits.

Chapter no.	Chapter Title	Description
Chapter 8	**Environmental Management Plan**	This chapter should comprehensively present the Environmental Management Plan (EMP), which includes the administrative and technical setup, summary matrix of EMP, the cost involved to implement the EMP, both during the construction and operational phases.
Chapter 9	**Summary and Conclusions**	This chapter makes the summary of the full EIA report condensed to ten A-4 size pages at the maximum. It should provide the overall justification for implementation of the project and should explain how the adverse effects are proposed to be mitigated
Chapter 10	**Disclosure of Consultants Engaged**	This chapter should include the names of the consultants engaged with their brief resume and nature of consultancy rendered.

proposed project received ToR on 12th July 2017 and proposal was considered in 32nd EAC meeting Infra – 2, 2-4 July 2018, however some of the project details has been revised and defer from EIA report submitted during application. Project details changed /updated are given as Annexure 8 of EIA report.

CHAPTER 02 - DESCRIPTION OF THE PROJECT

2.1 INTRODUCTION

MbPT, desirous to revive the OCT Project, evaluated various options including the structure adopted by Kamarajar Port, whereat, Kamarajar Port had proposed to rebid the existing iron ore terminal to include coal cargo based on highest revenue share, with Right of First Refusal (ROFR) to the existing concessionaire and assessment of the existing iron ore terminal assets at Replacement cost. Accordingly, MbPT is contemplating to revise the cargo to be handled at OCT to include no-containerised cargo like steel and RoRo.

2.2 PROJECT DESCRIPTION

Proposed project obtained MCZMA clearance on 09.05.2005 and public hearing for this Project was conducted on 29.09.2005. MoM of the Hearing are attached in **Annexure 3** and Action/Mitigation plan for concerns raised in public hearing is attached in **Annexure 5**. In 2006 Mumbai Port invited global tenders for development of an offshore container terminal on DBFOT basis. Based on this tendering process, the licence was awarded to Indira Container Terminal Private Limited (ICTPL), a special purpose vehicle promoted by Gammon India Ltd. & Gammon Infrastructure Projects Ltd, collectively called the Gammon Group (74%) and Dragados SPL, Spain (26%). The license was for the development and operation of OCT for a period of 30 years including construction period of 3 years at gross revenue share of 35.064%. The License also requires the company to operate and maintain the existing container operations of at the Ballard Pier Station (BPS) for the first five years of the License or 2 years from commencement of operations, whichever is earlier. The license was awarded during December 2007.

As per the concession agreement, the project was to be executed in two phases. During the first phase, the minimum port facilities will be as follows:

Berth length of not less than 700 m and width of 58 m with adequate equipment such as ship to shore handling cranes, yard handling equipment, rail container yard handling facilities, reefer plug points and sufficient container stacking yard etc., commensurate with the traffic.The berth is expected to be designed to receive vessels of capacity up to 6000 TEUs

A Y-Shape approach trestle to connect the berths with the onshore terminal facilities. For this an 18 m wide road will split into two 12 m wide roads reaching the berth and thus forming a Y-shape. Approximate length of the trestle will be 1000m.

Minimum equipment to be commissioned at the berth:

- Four QGC capable of handling 6000 TEU design vessels at the time of commissioning of berths and two more after achieving throughput of 600,000 TEUs in an operating year
- 12 RTGs at the time of commissioning of berths and six more after achieving throughput of 600,000 TEUs in an Operating year
- Two RMGs for rail container depot at the time of commissioning

Correspondingly, the Mumbai Port was to fill up of Princess and Victoria Docks and with the surrounding areas to create a stackyard space of 35 hectares; provide railway connectivity to create a railway container depot with 3 sidings; dredge the approach channel, turning basin and berth to handle container vessels of 13.5 m draft with tidal advantage.

The project was designed to handle 1.2 million TEU of container traffic during the first phase. During the second phase, the berth length would be increased by 350 m, thereby increasing the handling capacity of the terminal to 2.0 million TEU.

2.3 PRESENT STATUS OF THE PROJECT

Though the commercial operations at the offshore terminal should have been commissioned by December 2010, the project got delayed due to various reasons beyond the control of both the Concessionaire and the Licensee. The berths with the approach could be completed only by September 2013. During this period, the container traffic through Mumbai Port also did not grow as expected due to the influence of JNPT and it stagnated between 40,000 to 50,000 TEU only. As of now, part of Project facilities like offshore jetty, approach trestle, and yard has been physically completed, however, the completion of the balance mechanical and other facilities of the Project have not been taken up. The detailed status of the project is as under:

Completed Work

- 2 berths and Approach Trestle Completed
- Capital Dredging for berth pocket
- Filling of P&V Dock
- Rail container Depot
- Approach Channel and Turning Circle dredging (-)8mCD to (–)13.5m CD) (Target (–) 13.5 CD)
- Berth Pcket (-)15.0 CD to (-)16.0 CD

Pending Work

- Procurement of Cargo handling Equipment
- Container Yard not developed. (Reclamation work completed. The work of development of yard mainly Metelling WBM, and wearing coat balance)
- Capital Dredging in the approach channel and turning circle to the required Depths.

Table 1 Comparative statement existing & proposed project

Sr. No.	Description	As approved by MoEF	Now Proposed
1.	Cargo profile	New cargo(Containers only)	What is already been handled at the port will be shifted to OCT. Mainly RO-RO, Steel, any other clean cargo and occasionally passenger vessels.
2.	Capacity	9.6 MTPA in Phase I	4.17 MTPA in Phase I
3.	Area of stackyard	45 ha	20.83 ha
4.	Cargo	New cargo	Existing cargo handled at MbPT will be shifted to OCT.

5.	Berth and Trestle	700x58 m 1.2 KM trestle	Same The work has been completed
6.	Capital Dredging	12 million cum soil 3 lakh cum rock (as calculated earlier)	Only 6.3 Million cum soil 30,000 Cum rock remaining.
7.	Equipment	6 quay gantry cranes 18 RTGs 2 RMCGS	• 2 Nos. of 100 Ton Mobile Harbour Crane for Ship Shore Handling • 27 Nos. of 20 Ton Tractor-Trailors for Shore at Stack yard. • 7 Nos. of 30 Ton Mobile Cranes at Stack yard for Steel Cargo. • 2 Nos. of Reach Stacker at Stack yard for Containers.

2.3.1 Existing Civil Works

The Licensee, ICTPL have constructed the jetty (700 m x 58 m) ; primary approach trestle (612 m x 18 m) and two forked extensions (518 m x 12m & 374 m x 12 m). These are completed fully and are operational. The exisitng civil works are shown in the figure 1 below.

Figure 1 Exisitng Civil Works

2.3.2 Site Photographs

Figure 2 Primary approach with forked extension

In addition they have constructed on the rear of the berth a sub-station, a fire- fighting pump room and a jetty office. However, these are incomplete and are to be equipped with the necessary machinery.

Figure 3 Sub-station; Fire station and Office buildings

Mumbai Port, on their part, have filled up both Princess Dock and Victoria Dock and levelled the whole area. In addition, Mumbai Port has also developed a dedicated railway yard with 3 sidings each of 740 m length on the western side of P&V Docks as shown in Figure 6 hereunder. This railway yard is connected to MbPT railway network. Out of the 3 sidings, one is likely to be retained by the port for its own usage. Presently, the connectivity to the national network is available only around midnight and for a very limited time because of the suburbum traffic congestion Hence the port has undertaken to lay a dedicated line (1 up + 1 down) from

Wadala to Kurla. This is 4.41 km line which, when completed, is expected to facilitate smooth evacuation of MbPT cargo by rail. This project is being executed by Central Railway and MMRDA for the port. It is expected that this project will be completed within 2 years. Meanwhile, MbPT has also facilitated receipt of export passenger car units by rail instead of by road by trailers reducing the trffiac congestion on roads.

2.4 NEED OF THE PROJECT

The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai region and Maharashtra.

2.5 PROJECT DETAILS

The following facilities have been constructed by the Present Licensee, ICTPL, which are completed and are operational.

- Jetty (700 m x 58 m)
- Primary approach trestle (612 m x 18 m)
- Two forked extensions (518 m x 12m & 374 m x 12 m).

In addition they have constructed on the rear of the berth a sub-station, a fire- fighting pump room and a jetty office. However, these are incomplete and are to be equipped with the necessary machinery.

2.5.1 New Civil works to be executed for the revised cargo configuration

The stackyard development:

Steel	:	26,600 sq.m of area of which
		6,650 sq.m will be covered and
		19,950 sq.m will be open area
RoRo	:	135,600 sq.m of paved area
Cotainers	:	20,160 sq.m of parking yard

For stacking steel and containers in the open a heavy duty pavement has to be prepared while for RoRo a yard developed with paver blocks will be required.

Outof the total area of 18.23 Ha required, 14.65 Ha will be made available at Victoria Dock and the balance made available at Frere Basin or eny other

In addition to the above, 1.19 Ha of railway yard area for handling steel cargo, 2552 sqm for Fire water platform including substation and an area of 0.69 Ha for misc. operational buildings is required to be given. Hence the total area required to be given is 2,08,260 sq m or say 20.83 Ha besides the Berth area and the trestle.

2.5.2 New Mechanical equipment required for the revised cargo configuration

2 Nos. of 100 Ton Mobile Harbour Crane for Ship Shore Handling will be mounted on the crane rail tracks already provided at the berth.
27 Nos. of 20 Ton Tractor-Trailors for Shore at Stack yard.
7 Nos. of 30 Ton Mobile Cranes at Stack yard for Steel Cargo.
2 Nos. of Reach Stacker at Stack yard for Containers.

2.5.3 New Mechanical Equipment

Cargo handling equipment are needed for steel. For RoRo berth, the automobile units will be directly driven into the carrier through ramps by drivers. Containers will be handled through ships‘ own gear. If gearless vessels or barges are to be handled, the cranes that handle steel could be used by replacing hooks with spreaders.

For steel equipment are needed for ship-shore transfer, transfer from berth to the stackyard, handling at the yard and also for loading onto the railway wagons.

2.5.4 Ship-shore transfer

For ship-shore transfer, it is suggested to equip one berth with two Harbour Mobile Cranes for efficient handling of cargo. However, these HMCs should be mounted on the rail tracks laid for the container quayside gantry cranes. The deck slab will not be able to take the point loads from the crane leg pads. This crane rail track is of 30 m gauge. Hence the HMCs should be tailer made as rail mounted with this rail span. The capacity should be to handle minimum 30 Te at 50 m radius. Market enquiry has already been made and a typical crane with technical details has been

presented in the earlier section. For the other berth, the cargo transfer will be through ships‘ own gear as is being done now.

2.5.5 Berth to Yard transfer:

It has been indicated earlier that the productivity of the HMC equipped berth will be 510 TPH while the ships‘ gear operated berth will have a productivity of 405 TPH. It has also been mentioned that the unit weight of steel coils will be 20 Te while that of the steel flats will be 10 Te.

For transfer of cargo between the quay and the storage, 10 truck-trailers will be required, with a distribution of 3 trailers on the each way, 2 trailers on the quay side and 2 trailers at the storage site for achieving the required productivity. Assuming 20 Te capacity trailer/trucks being used, the cycle time for each will be about 35 minutes: 5 minutes each at berth and at yard, 15 minutes travel time from berth to yard and 10 minutes travel time back to berth from the yard. Each trailer should be fitted with twist locks and other container securing arrangement as the same Truck-Trailers will be used for haulage for both steel and containers. Based on these, the berth will need 15 trailers/trucks and the other berth will need 12 trailers/trucks.

2.5.6 Handling equipment at the steel stack yard

When both the steel berths are being occupied and operated, 27 trailers/trucks will be arriving at the yard per hour. In order to unload the cargo, a minimum of 5 numbers of 30 Te mobile cranes will be required.. When the berths are not occupied or not in operation, these cranes will be used for loading the cargo onto evacuating trucks/trailers.

2.5.7 Handling equipment at the container yard

In addition to the tractor trailers, the containers yard will require two reachstacker and receipt/ delivery operations.

2.5.8 Handling equipment at the railway yard

For loading the cargo into railway wagons at the railway yard, at least 2 numbers of 30 Te mobile cranes will be required.

2.6 PROJECT LOCATION

The projet is located at Indira Docks, Mumbai Port Trust, Maharashtra. The Google Earth Image showing location of project is given in Figure 4 below:

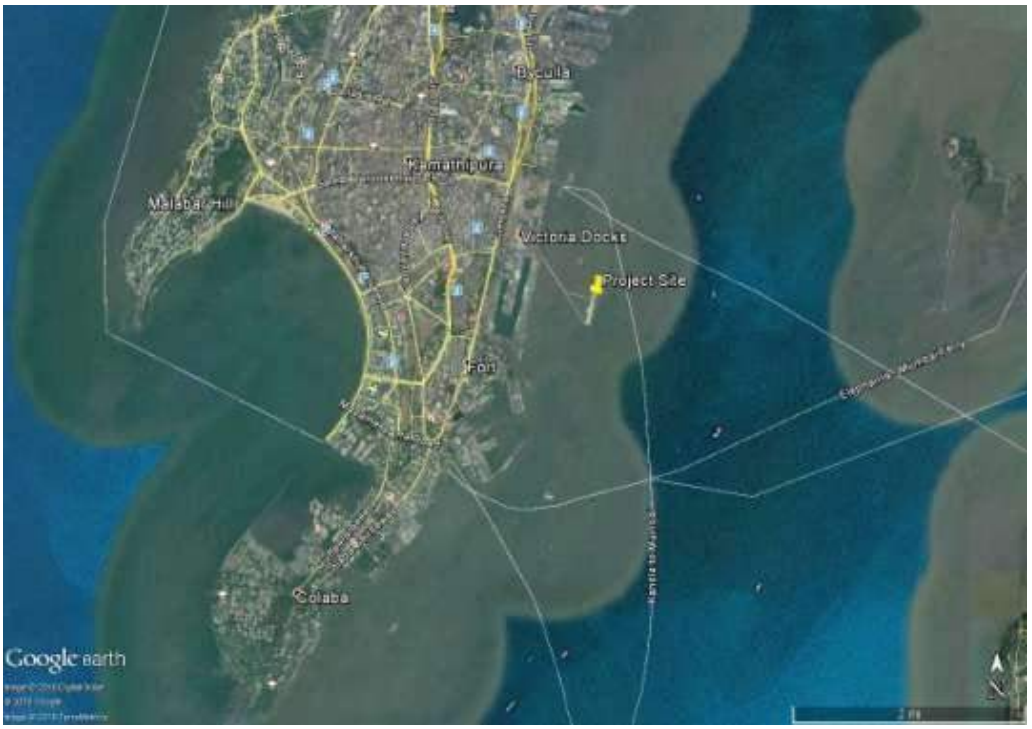


Figure 4 – Google Image showing project site

2.7 CRZ STATUS OF THE PROJECT

CRZ Clearance was obtained from Maharashtra Coastal Zone Management Authority on 09.05.2005. Copy of the CRZ Clearance enclosed as **Annexure – 2**,

Copy of project layout superimposed on CZMP is enclosed in **Annexure 1** and Map demarcating HTL/LTL line for proposed site prepared by Institute of remote sensing Anna University Chennai is attached in **Annexure 6**.

2.8 LAND REQUIREMENT

- Additional land area of 55 acres already created by filling up of P&V Docks, as per earlier clearance.
- No extra land is required, as stated in earlier clearance, the existing Princess and Victoria Docks areas which were used for handling cargo other than containers have been filled up. At present, only Victoria docks will be used for stacking of cargo such as steel, car & containers

2.9 CONSTRUCTION WORK

As per earlier clearance granted berth of 700x58m has already been constructed. Further, 3rd berth of 350mx65m will be constructed after Traffic from these berths

reaches a certain level. Development of Yard in filled up area of Victoria Dock only is pending. Further construction of office, gate complex, workshop, canteen, sub- station, etc. is envisaged. Further for handling of steel cargo a covered shed of 6650 sq.m. will be constructed.

2.9.1 Demolition Work

All the existing structures affecting the project have been already demolished. No change from earlier scope of work.

2.9.2 Dredging

The capital dredging upto -16 m CD in berth and upto -13.5 m CD at turning circle was approved in the earlier EC, part of which has already been done. Due to change in cargo profile, no additional dredging will be required , However, for balance dredging blasting will be done which was permitted in the earlier clearance.

2.10 UTILITY REQUIREMENT

2.10.1 Construction Material

Construction of Jetty completed only yard work is balance, which is fraction of work. As such no deveiations from previously approved quantities ancticipated i.e. Sand – 1,72,830 cum. & Metal 2,98,480 cu.m.

2.10.2 Power Requirement

Power requirement is about 5 MVA. No change from earlier scope of work.

2.10.3 Man Power Requirment

Manpower requirement in construction phase is estimated to be 115 and in operation phase is 78.

2.10.4 Water Requirement

Total water requirement for domestic use is 5.17 KLD in construction phase and 3.78 KLD in operation phase.

2.10.5 Waste Management

Total sewage water generated 3.45 KLD in construction phase and 2.25 KLD operation phase. Total solid water generated 51.75 Kg/day in construction phase and 33.7Kg/day in operation phase.

2.11 TRAFFIC ANALYSIS

Based on the past traffic trends in Mumbai Port, steel and automobile are identified as the potential growth cargoes for OCT. Accordingly, the future prospects for these cargoes were analysed The assessment based on the present trends in the domestic

and global scenario. The Study shows that in both steel and automobile cargo segments there will be appreciable growth in the near term, moderate growth in the medium term and subdued growth in the far term. Container traffic appears to have very limited scope for expansion. It may therefore be prudent to wait and watch for a few years how the traffic in the above cargoes behaves. Thereafter, the Port may take the decision for the construction of the third berth considering upon the behaviour of the cargo profile and the performance of the licensee. The expected future traffic of Steel, Ro-RO and containers for the whole of Mumbai Port is as follows:

Traffic Projections for Steel (Million Tonnes)

Year	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Mn Tons	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

Base: actual traffic of MbPT in 2015-16 @ 5.985 million tons

Traffic Projections for Automobiles (Lakh PCU units)

Year	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Lakh Units	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

Traffic Projection for Containers

The scope for container are limited to about 40000 to 50000 TEUs assuming the entire present container traffic coming to Mumbai Port will be continued in future also. Considering the above projections, the expected traffic at the proposed OCT is as follows.

Year	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Steel Cargo											
Mn Tons	2.50	2.50*	2.63*	2.76*	2.89*	3.03*	3.03*	3.03*	3.03*	3.03*	3.03*
Automobiles											
Lakh Units	2.29	2.63	3.03	3.5	4.01	4.40*	4.40*	4.40*	4.40*	4.40*	4.40*
Containers											
Lakh TEU's	0.20	0.22	0.24	0.27	0.29	0.32	0.39	0.53*	0.53*	0.53*	0.53*

* Capped at Optimal Capacity

2.11.1 PLANNING PARAMETERS

The existing OCT jetty structure has an overall length of 700M designed to handle two container vessels upto 6,000 TEU capacity each ie. 75,000 DWT; 300 m LOA; 40 m Beam and 12.0 m Draft. With the revised traffic configuration i.e. with RoRo, Steel and containers through smaller ships, the design ship would be Panamax size general cargo vessels upto 75,000 DWT. Based on the past performance, the following conclusions are made.

Steel : The maximum ship size could be Panamax vessels of 225 m LOA but with draft limited to 12 m only. The beam will be 32 m.

RoRo: The largest pure car carrier of 8,000 unit capacity is only 30,000 DWT with 232 m LOA; 32 m Beam and 9.50 m draft.

Container vessels: With the present container traffic at around 50,000 TEUs and which is not expected to have any dramatic increase in future, the size of the veseels will be much smaller, Hence, it should be possible to berth 3 ships concurrently at berth.

Considering the size of the vessels of Ro-Ro and Steel calling Mumbai Port, besides Ro-Ro vessel all the time, it is consider to possible to handle 2 steel vessels part of time and one steel vessel all the time.

2.12 CAPITAL COST AND IMPLEMENTATION SCHEDULE

The total capital cost of the project as a replacement cost of the existing assets is estimated at Rs. 549.31 crores by M/s SBI Caps. An amount of Rs. 166.36 crores has been considered towards additional cap-ex for development of stack yards etc and procurement of cargo handling cranes without which Steel cargo cannot be handled. Thus total cost of the project was estimated at Rs. 715.67 crores. The entire project is expected to be completed in 18 months' time from the date of grant of concession to the operator. Due to delay in project revised project cost is estimated to be 1290 Crores.

2.12.1 Operation And Maintenance Cost

The annual operation and maintenance cost of the Project based on the TAMP Guidelines 2008 is estimated at Rs. 112.79 Crores.

2.12.2 Revenue Earnining

The estimated annual revenue based on provisional tariff assessed as per the upfront tariff guidelines 2008 / Tariff orders is given below The estimated annual revenue from the project is given below:

Sr. no.	Particulars	2017-18
1.	**Estimated Throughput**	
	(a) RoRo (in lakh Per Car Units p.a)	4.40
	(b) Steel (in Lakh tons p.a.)	30.06
	(c) Containers (TEUs)	52560
2.	Handling RateRate (Rs)	
	(a) Automobiles (Per PCU)	978.45
	(b) Steel(Per ton)	311.57
	(c)Container (PerTEU)	4334.42
	(d) Stevedoring on RoRo	250.00
3.	**Annual Revenue Requirement (ARR) (Rs. In lakhs)**	**17049.14**
	(a) RoRo	4305.18
	(b) Steel	9365.79
	(c)Conatiner	2278.17
	(d) Stevedoring on RoRo	1100.00
4.	**Estimated GRT (Lakh GRT hours)**	**4293.61**
5.	**Berth hire (Rs./ GRT hour)**	**1.42**
6.	**Revenue on Berth hire (Rs. In lakhs)**	**6096.93**
7.	**Total Estimated Income (Rs. lakhs)**	**23146.07**

2.11.3 Financial Viabiliy Analysis

The pre tax project IRR from the private operator‘s perspective is as follows:

Sr. No.	Pre-Tax Project IRR	IRR (%)	NPV @ 12% (in Rs. Cr)
1.	Base case	12.01%	0.45
2.	Capital Cost up by 10%	10.92%	-59.43
3.	Revenue down by 10%	9.45%	-125.91
4.	Annual O&M Cost up by 10%	10.70%	-65.98
5.	Combined effect of Sl. no. 2, 3 & 4	7.18%	-252.22

However, the viability of the project will further improve, if the operator achieves the productivity norms and become eligible for 15% increase in tariff over the notified tariff.

2.13 Court Cases

PIL no. 149 of 2014 has been filed against the filling of Victoria & Princess Dock. By the the petitioner Shri Sayyed Abdul Nabi & other have filed an rejoinder to MbPT‘s a reply.

It is pertinent to note that

iii) the petitioner has filed an PIL against the filling work of Princes and Victoria Docks which has already been completed the same petitioner had filed the same case earlier, but as a writ Petition no 617 of 2013 which has been disposed off by the Hogh court Bench comprising of Juctice Dr. D.Y.Chandrachud and Juctice MN.S. Sonak JJ

CHAPTER 03 - BASELINE ENVIRONMENTAL STUDY

3.1 SCOPE OF WORK

This section of the report gives description of the existing Environmental Studies within the project area, which constitutes the baseline for the study. Natural conditions are often critical when designing and constructing infrastructure works. The assessment of baseline studies of the appropriate environmental parameters, which may be affected by the project implementation, is a pre-requisite for any Environmental Impact Assessment (EIA) study.

In order to investigate likely impacts due to commissioning of proposed project, the consultants, M/s Fine Envirotech Engineers carried out estimation of impacts based on data generated, secondary data as well as literature studies.

Monitoring surveys of the study area (project area) has been carried out in one season. Field monitoring for meteorological conditions, ambient air quality, water quality, noise quality, etc. has been carried out in one season, which constitutes major portion of the baseline environmental studies. In addition to these important parameters, certain aspects like land use, socio-economic studies, geo-technical investigations etc. are covered during the study period. This information is based on secondary information sources and constitutes remaining part of the baseline environmental studies.

It is brought to the notice of the authorities that, since the project is of the construction of bridge the impact on existing baseline of environmental parameter will be very restricted and of temporary in nature. These are further controlled and minimized by adopting various mitigative measures. Even during operational phase the impact on environmental settings will be negligible and will be controlled by adopting proper environment management plan (EMP). These aspects have been studied in depth with reference to the proposed project and baseline data has been presented in this chapter. These details have been given in the following sections.

EIA is often mandatory requirement for planning of infrastructure and marine structures. The EIA determines the environmental consequences of the project prior to construction, assessment of environmental impact due to construction, its impact on existing baseline environmental parameters and also importantly on land use and socio-economic parameters. The entire data has been collected through actual physical surveys and observations, literature surveys, interaction with locals, government agencies and departments. This chapter describes the baseline environment settings in the area and will throw light on its effect on day-to-day environment.

3.2 BASELINE ENVIRONMENTAL STUDY

In order to understand the baseline environmental status of the study area monitoring of various parameters viz. air, water, soil, noise etc. were undertaken. The locations selected for monitoring of Ambient Air Quality, Noise Levels and Water Quality are represented in Table 2 & Figure 5 below.

Table 2 Baseline Environmental Monitoring Locations

Sr. no	Station no.	Station name	Distance (km)	Direction
Ambient Air Quality				
1.	AAQ 1	Project Site, MbPT Shed no:13B	1.35	NW
2.	AAQ 2	Near Cosmos Builder, Malabar Hill	5.10	NW
3.	AAQ 3	Near Entry Gate, At Kohinoor Square, Dadar	9.76	N
4.	AAQ 4	At Shanti Kutir Apartment, Near Marine Plaza Hotel, Marine Drive, Mumbai	3.27	W
5.	AAQ 5	At Willson House, Colaba, Mumbai	4.23	SW
6.	AAQ 6	Near MbPT Hospital, Wadala	7.38	S
7.	AAQ 7	Near Pir Pau Fire Station	7.06	NE
8.	AAQ 8	Near Dr. Ambedakar Bhavan	1.43	E
Noise Level Monitoring Locations				
1.	NQ 1	Project Site, MbPT Shed no:13B	1.35	NW

2.	NQ 2	Near Cosmos Builder, Malabar Hill	5.10	NW
3.	NQ 3	Near Entry Gate, At Kohinoor Square, Dadar	9.76	N
4.	NQ 4	At Shanti Kutir Apt, Nr Marine Plaza Hotel, Marine Drive	3.27	W
5.	NQ 5	At Willson House, Colaba, Mumbai	4.23	SW
6.	NQ 6	Hughes Dry Dock in Indira Dock	1.23	W
7.	NQ 7	Orange Gate - Pricess Dock (OGPD)	1.92	NW
8.	NQ 8	Lakadi Bandar, Darukhana Ship Breaking Yard	3.19	N
Soil Quality Monitoring Locations				
1.	S1	At St. Joseph Church, Colaba	5.45	SW
2.	S2	At Dadhi Seth Agyari Garden	3.02	NW
3.	S3	Near Front Gate, Hanging Garden	5.37	NW
4.	S4	Opp. To Water Tank Mukesh Mill Compound	3.99	SW
Surface Water Quality Monitoring Locations				
1.	SW1	Water Tank at Kashivishwnath Temple, Kalbadevi	2.97	NW
2.	SW2	Water Tank at Banganaga water tank, Walkeshwar Temple, Malabar Hill	6.26	W
Ground Water Quality Monitoring Locations				
1.	GW1	Well Water Opposite Metro Cinema	2.44	W
2.	GW2	Open Well water at Dadi Seth Agyari Kalbadevi	3.02	NW

Figure 5 Environmental Monitoring Locations

3.2.1 Land Environment

3.2.1.1 Topography & Geology

Mumbai consists of two distinct regions: Mumbai City district and Mumbai Suburban district, which form two separate revenue districts of Maharashtra. The city district region is also commonly referred to as the Island City or South Mumbai. The total area of Mumbai is 603.4 km2. Of this, the island city spans 67.79 km2, while the suburban district spans 370 km2, together accounting for 437.71 km2 under the administration of Brihan Mumbai Municipal Corporation (BMC). The remaining area belongs to Defence, Mumbai Port Trust, Atomic Energy Commission and Borivali National Park, which are out of the jurisdiction of the BMC. Mumbai lies at the mouth of the Ulhas River on the western coast of India, in the coastal region known as the Konkan. It sits on Salsette Island, partially shared with the Thane district. Mumbai is bounded by the Arabian Sea to the west. Many parts of the city lie just above sea level, with elevations ranging from 10 m to 15 m; the city has an average elevation of 14 m. Northern Mumbai is hilly, and the highest point in the city is 450 m at Salsette in the Powai-Kanheri ranges. Sanjay Gandhi National Park (Borivali National Park) is

located partly in the Mumbai suburban district, and partly in the Thane district, and it extends over an area of 103.09 km2

The original seven islands of Bombay consisted of 22 hills. Most of them were razed to fill in the shallows to connect the islands. The hills still standing today are:

- Malabar Hill — the highest point in the city area
- Cumbala Hill
- Antop Hill
- Sewri Hill
- Gilbert Hill
- Worli Hill
- Pali Hill
- Mazgaon Hill
- Sion Hill
- Mahakali Hill
- Golanji Hill
- Pulshachi Dongri
- Salamati Hill

There are three hill ranges with the city limits. The Ghatkopar Hills are present near the station of Ghatkopar. The hill range runs parallel to the Central Railway track and is inhabited by slums. During the monsoon season, landslides are common. The Trombay Hills occupy a large portion of Trombayon the eastern part of the city. The highest hill is about 302 metres above sea level. The Powai Hills are present north of the city. The Borivali National Park occupies most of the region. The Vihar and Tulsi Lakes are present within the hills. The highest point of the metropolis, at 450 metres, is located in this region. There are three lakes in the city. The Vihar Lake and the Tulsi Lake are present within the National Park and supply part of the city's drinking water. The Powai Lake is immediately south of these two. Back Bay is the largest bay in the city. The coastline of Back Bay is an inverted C-shaped region 4 kilometres in length, and Marine Drive is located along this stretch. North of Marine Drive is Worli Bay. In the centre of the bay is the tomb of Haji Ali, a 13th century Muslim saint. The bay perimeter is about two kilometres in length. Mahim Bay is the

second largest bay in the city. The Mithi River empties into the Mahim Creek which drains into the bay. The border between the city and its suburbs bisects the bay. To the north lies Bandra and to the south, Mahim. Mumbai has numerous creeks with close to 71 km2 of creeks and mangroves along its coastline. The Vasai Creek to the north and Thane Creek to the east separates Salsette Island from the mainland. Within the city the Malad (or Marve) Creek and the Gorai (or Manori) Creek inundate the suburban region. The Mahim Creek forms the border between the two districts. On the eastern front, a large creek—the Thane Creek—is present. There are also the Mahul Creek and the Mahim Creek.

3.2.2 Air Environment

3.2.2.1 Meteorology of the Study Area

Meteorology is an important parameter in the Environment Impact Assessment process. The main parameters of meteorology are temperature, rainfall, relative humidity, wind speed and wind direction. In order to assess the impact on existing ambient environment due to the project, it is necessary to have baseline status of ambient meteorological parameters.

The 10 year meteorological data was procured for the Colaba station which is nearest to the site, from the National Data Centre, Indian Meteorological Department (IMD, Pune).

Temperature

The mean temperature data for the years 2002-2013 is given in Table 3 and the graph showing the averages for the mean temperature are graphically presented in Figure 6.

Table 3 - Average mean temperature for the period 2002-2013

AVERAGE TEMPERATURE				
Month	**Mean Maximum temperature in °C**	**Mean Minimum temperature in °C**	**Highest maximum temperature in °C**	**Lowest minimum temperature in °C**
	MMAX	**MMIN**	**HMAX**	**LMIN**
JAN	30.1	19.3	34.7	15.7
FEB	30.4	20.4	35.2	16.7
MAR	31.8	23.0	36.1	20.2
APR	33.1	25.4	35.0	23.5
MAY	34.1	27.4	35.7	25.3
JUN	32.2	26.5	35.3	23.3
JUL	30.0	25.6	32.1	23.2
AUG	29.8	25.2	31.9	23.4
SEP	30.5	25.1	32.7	23.2
OCT	33.3	24.9	36.4	22.9
NOV	33.5	23.4	36.0	20.8
DEC	32.2	21.2	35.5	17.8

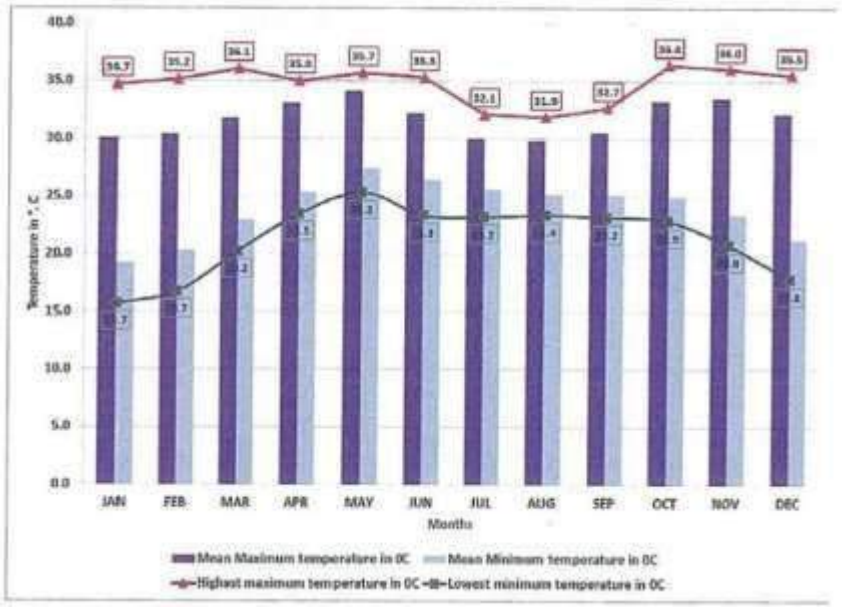


Figure 6 - Average Mean Temperature year 2002-2013

Rainfall

The rainfall data for the years 2002-2013 is given in the Table 4. The per cent rainy days was calculated from the no. of observations and the total no of rainy days as illustrated in Figure 7.

Table 4 -Average monthly rainfall for the years 2002-2013

Month	Total Rainfall in the month in mm	% Rainy days
JAN	0.1	0
FEB	0.5	0.36
MAR	2.2	0.65
APR	0.0	0.00
MAY	16.3	0.97
JUN	561.7	51.09
JUL	665.4	65.70
AUG	464.9	64.19
SEP	331.1	51.56
OCT	69.4	9.03
NOV	15.0	1.67
JEC	0.1	0.00

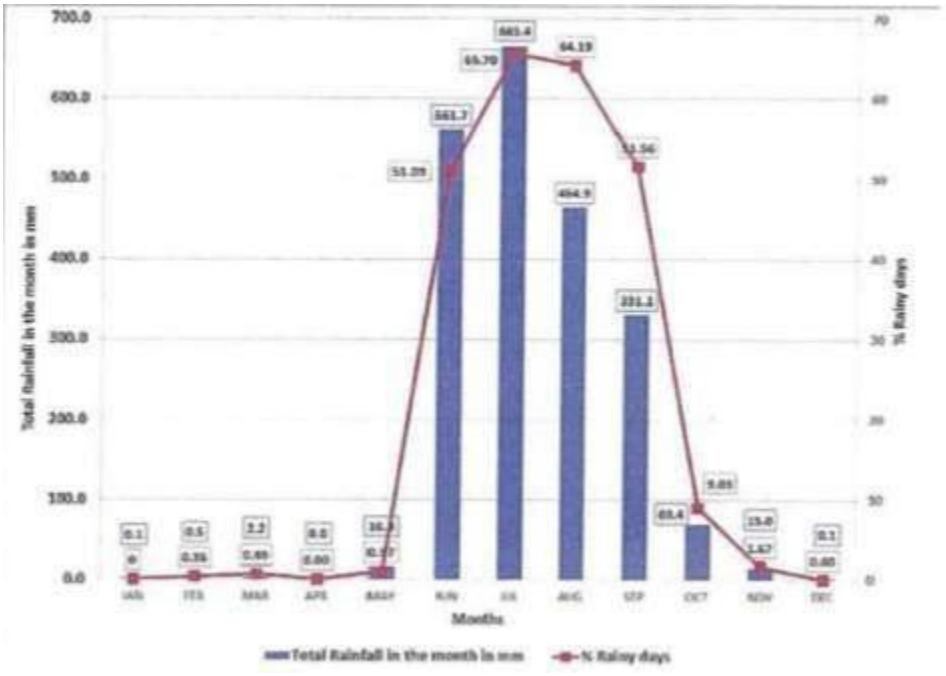


Figure 7 - Average monthly rainfall for year 2002-2013

Wind

The wind speed and wind direction data for the years 1971-2000 was obtained from National Data Centre (IMD Pune). The predominant wind direction and wind speed at 03 UTC (Coordinated Universal Time) and 12 UTC has been given in the Table 5 and Table 6 respectively. The windrose for the same has been given in Figure 8.

Table 5-The predominant wind direction for the years 1971-2000 at at 03 UTC

03 UTC		
MONTH	**PREDOMINANT WIND DIRECTION FOR THE YEARS 1971-2000**	**WIND SPEED (M/S)**
JAN	NE	1.5-4.5
FEB	NE	1.5-4.5
MAR	NE	1.5-4.5
APR	NNW,SE	1.5-4.5
MAY	NW	1.5-4.5
JUN	WSW	9.0-11.0
JUL	W	11.0-14.0
AUG	W	7.0-9.0
SEP	W,SE	7.0-9.0
OCT	ENE,E	4.5-7.0
NOV	ENE	1.5-4.5
DEC	ENE	4.5-7.0
ANNUAL	NE	1.5-4.5

Table 6 - The predominant wind direction for the years 1971-2000 at 12 UTC

12 UTC		
MONTH	**PREDOMINANT WIND DIRECTION FOR THE YEARS 1971-2000**	**WIND SPEED (m/s)**
JAN	NW,NNW	4.5-7.0
FEB	NW	4.5-7.0
MAR	NW	7.0-9.0
APR	NW	7.0-9.0
MAY	WNW	4.5-7.0
JUN	WSW	7.0-9.0
JUL	W	7.0-9.0
AUG	W	7.0-9.0
SEP	WNW	7.0-9.0
OCT	NW	1.5-4.5
NOV	NW	1.5-4.5
DEC	NW	4.5-7.0
ANNUAL	NW	4.5~7.0

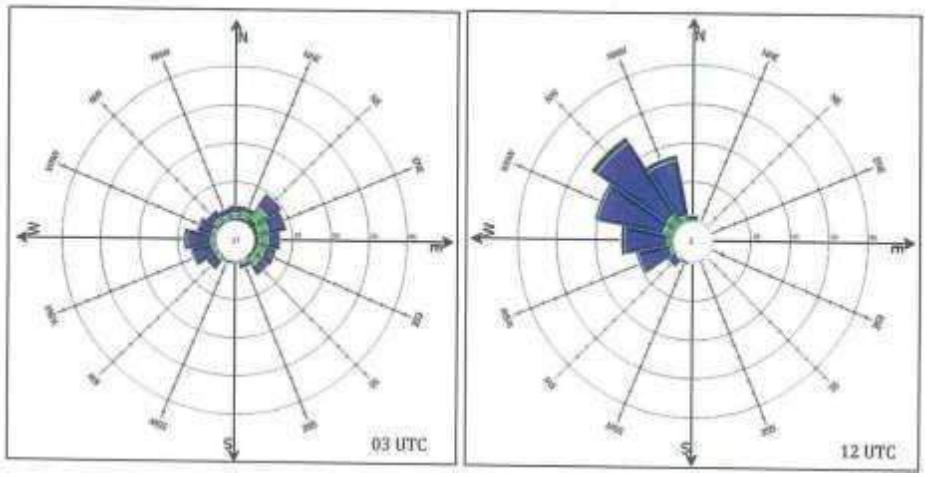


Figure 8 -Windrose for the year 1971-2000

3.2.2.2 Ambient Air Quality

In order to access the baseline air quality of the study area, ambient air quality monitoring was undertaken at 8 locations. The parameters for AAQM include Particulate Matter (PM10 AND PM2.5), Oxides of Sulphur (SO2), Oxides of Nitrogen (NOx), Carbon Monoxide (CO). All the parameters were monitored on 24 hourly basis except for CO, which was monitoring on 8 hourly basis.

The details of location are given in Table 2 and Fig. 5 above. The air quality criteria as per Nation anal Ambient Air Quality Standards are given in Table 7. The results of monitoring are represented in Table 8 & Figure 9.

Table 7 Ambient Air Quality Standards (National) – India

Pollutant	Time Weighted Average	Concentration in Ambient Air		
		Industrial, Residential, Rural And Other Areas	Ecologically Sensitive Area (Notified by Central Govt.)	Methods of Measurement
Sulphur Dioxide (SO2) (μg/m3)	Annual * 24 hours**	50 80	20 80	Improved West & Gaeke method Ultraviolet Fluorescence
Nitrogen	Annual *	40	30	Modified Jacob &

Pollutant	Time Weighted Average	Concentration in Ambient Air		
		Industrial, Residential, Rural And Other Areas	Ecologically Sensitive Area (Notified by Central Govt.)	Methods of Measurement
Dioxide (NO2) (μg/m3)	24 hours**	80	80	Hochheiser method. (Na – Arsenite) Chemiluminescence
Particulate Matter (Size less than 10 μm) or PM10 μg/m3	Annual * 24 hours**	60 100	60 100	Gravimetric TOEM Beta Attenuation
Particulate Matter (Size less than 2.5 μm) or PM2.5 μg/m3	Annual * 24 hours**	40 60	40 60	Gravimetric TOEM Beta Attenuation
Carbon Monoxide (CO) (mg/m3)	8 hours** 1 hour	2 4	2 4	Non dispersive infra red (NDIR) Spectroscopy

Table 8 - Ambient Air Quality Of Study Area

Observations	**Parameters**				
	Particulate Matter 10 (PM10)	**Particulate Matter 2.5 (PM2.5)**	**Oxides of Sulphur (SO2)**	**Oxides of Nitrogen (NOX)**	**Carbon monoxide (CO)**
	μg/m3	**μg/m3**	**μg/m3**	**μg/m3**	**mg/m3**
Project Site, MbPT Shed no:13B	89.7	40.5	20.7	37.8	1.0
Near Cosmos Builder, Malabar	85.3	35.7	23.7	40.2	1.2

Hill					
Near Entry Gate, At Kohinoor Square, Dadar	88.6	45.1	22.1	45.3	1.8
At Shanti Kutir Apt, Nr Marine Plaza Hotel, Marine Drive	87.5	48.0	20.7	47.1	1.8
At Willson House, Colaba, Mumbai	96.2	29.7	19.3	32.7	1.7
Near MbPT Hospital, Wadala	76.3	26.8	9.0	20	1.03
Near Pir Pau Fire Station	75.6	30.4	15.2	28.6	1.10
Near Dr. Ambedakar Bhavan	87.3	45.0	14.8	29.5	1.60

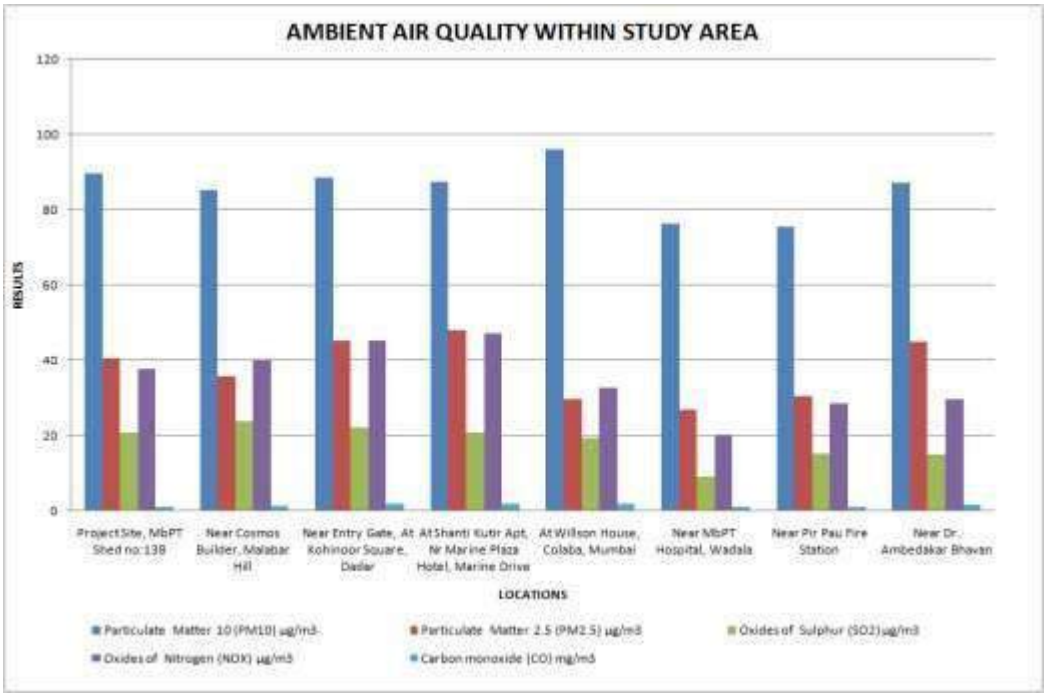


Figure 9 - Ambient Air Quality of Study Area

Conclusion

All the Ambinet Air quality parameters were forund within National Ambient Air Quality Standard, 2009.

3.2.3 Water Environment

3.2.3.1 Coastal Hydrography Of The Study Area

Tides

The tidal variation in the Mumbai harbour is mainly semi-diurnal but with an appreciable diurnal element which produces unequal tides on most days. The various tide levels with respect to Chart Datum at the Mumbai Port are as follows:

1.	Chart Datum (CD)	0.00 m CD
2.	Mean Low Water Springs (MLWS)	+0.76 m CD
3.	Mean Low Water Neap (MLWN)	+1.85 m CD
4.	Mean Sea Level (MSL)	+2.51 m CD
5.	Lowest High Water	+2.80 m CD
6.	Highest Low Water	+2.74 m CD
7.	Mean High Water Neap (MHWN)	+3.30 m CD
8.	Mean High Water Springs (MHWS)	+4.42 m CD
9.	Highest High Water Level (Iune 1924) (HHWL)	+5.38 m CD
10.	Lowest recorded LW	-0.44 m CD

Statistical inferences from the above are:

All high tides exceed +2.7 m CD

About 5% of all high tides would be less than +3.2 m CD

About 5% of lower high tides would be less than +2.8 m CD

Waves

The significant waves entering the harbour are the long period as well waves generated by deep sea storms. These mainly arise just before and during the monsoon and their direction of approach is normally from South-West. Whatever the wave-front orientation outside the harbour, the waves running up the harbour tend to be refracted in to a constant pattern. Waves are also substantially attenuated by the time they reach the berthing facilities in the harbour

Currents

The currents in Mumbai harbour are almost entirely due to tidal ebb and flood flows but in the SW monsoon during periods of heavy rainfall the run-off from the rivers is

sufficient to alter the flow pattern appreciably. The normal maximum currents inside the harbour are about 2 to 3 knots although 4 knots can sometimes occur on the ebb during the wet weather

3.2.3.2 Marin Water Quality

The surface and bottom marine water monitoring was carried out at 2 stations. The distance and direction of these stations from the project site are given in Table 9. The marine water monitoring locations are shown in the Figure 4.10. Niskin sampler was used for sampling the bottom water. The results are given in the Table 10.

Table 9 - Marine water and sediment quality monitoring location

S.N	Locations	Aerial Distance (Km)	Direction
1	Station 1	0.11	S
2	Station 2	0.56	NW
3	Station 3	1.6	NE
4	Station 4	2.5	SE
5	Station 5	2.61	E

Figure 10 - Marine water and sediment quality monitoring locations

Table 10 - Marine Water Monitoirng results

Parameters	Unit	Buitcher Island Jetty Check Post	Butcher Island Landing Jetty -3
Colour	Hazen	BDL[DL=5]	BDL[DL=5]
Odour	--	Agreeable	Agreeable
Turbidity	NTU	6.1	5.8
pH	--	7.3	7.7
TSS	mg/l	184	228
DO	mg/l	3.8	5.8
BOD	mg/l	BDL[DL=2]	BDL[DL=2]
O&G	mg/l	BDL[DL=10]	BDL[DL=10]
Ammonia	µmol/l	1.3	2.4
Nitrate	µmol/l	12.2	15.7
Nitrite	µmol/l	0.3	0.3
Phosphate	µmol/l	2.3	2.8
Phenol	µmol/l	BDL[DL=10]	BDL[DL=10]
Petroleum Hydrocarbon	µg/l	10.8	4.8
Polyaromatic Hydrocarbon	µg/l	6.9	2.5
Mercury	mg/l	0.009	0.006
Cadmium	mg/l	BDL[DL=0.015]	BDL[DL=0.015]
Lead	mg/l	BDL[DL=0.01]	BDL[DL=0.01]
F.Coli	MPN/100ml	34	22

3.2.3.3 Sediment Quality

To know the present sediment quality near the project site, 2 monitoring stations were selected within 10km (aerial distance) from the project location. The sediment sampling was done using a grab sampler. The results for the sediment analysis are given in the Table 11.

Table 11 – Sediment Quality

Parameters	Unit	Buitcher Island Jetty Check Post	Butcher Island Landing Jetty -3
Texture	--	Sandy Clay	Clay
Total Organic Carbon	%	0.7	1.2
Total Phosphorus	mg/kg	268	301
Total Petroleum Hydocarbon	mg/kg	5.6	5.9
Polyaromatic Hydrocarbon	mg/kg	2.5	1.9
Cadmium	mg/kg	BDL[DL=2]	BDL[DL=2]
Chromium	mg/kg	105	74
Cobalt	mg/kg	23	24
Copper	mg/kg	72	70
Iron	mg/kg	68835	757584
Lead	mg/kg	21	43
Mercury	mg/kg	BDL[DL=2]	BDL[DL=2]
Nickel	mg/kg	58	50
Zinc	mg/kg	1147	1517

3.2.4 Noise Environment

Sound is a common part of everyday life that its presence, benefits and threats are often overlooked. It can be a source of entertainment or annoyance. It has been observed that a quiet sound can be annoying if it grates on the ear, and a loud noise can be ignored if it fits in and is unobtrusive. Unpleasant sounds are classed as noise.

3.2.4.1 Applicable Standards & Classification

The Ministry of Environment & Forests has prescribed the noise level standards for the residential and commercial areas during the day as well as night time. It is important to note that the noise levels during the day are more relaxed as compared to night levels because of the importance of undisturbed sleep and relaxation of the entire body during the nights. The prescribed MoEF standards are given in Table 12 below.

Table 12 Permissible Noise Level (CPCB Standards)

Area	Category of Area	Permissible Limit	
		Leq Day time	Leq Night time
A	Industrial Area	75	70
B	Commercial Area	65	55
C	Residential Area	55	45
D	Silence Zone	50	40

Note – 1 Day time is reckoned in between 6.a m and 10 p.m.

Note – 2 Nighttime is reckoned in between 10 p.m. and 6 p.m.

Silence zone is defined as an area comprising not less than 100 meters around hospitals, educational institutions and courts. The silence zones are zones, which are declared as such by the competent authority.

Mixed categories of areas may be declared as one of the four above-mentioned categories by the competent authority.

3.2.4.2 Status of Noise Environment in the Project Area

Considering the various effects of noise, it is important to identify the primary cause of noise in the ambient environment in the project area. With the present development and project activities, it is understood that the construction phase of the project would generate more noise with the various construction and maintenance activities. It is important, therefore, to identify the various causes of noise within the project area. Some of these causes are described below:

- Transport noise
- Social Noise
- Construction Noise

During construction phase there will be a slight increase in noise levels within the plot area, which will not affected to the nearby residential area. Adequate measures will be taken to mitigate such effect.

3.2.4.3 Ambient Noise Levels

The main objective of noise monitoring in the study area is to establish the baseline noise levels, and assess the impact of the total noise that is expected to be

generated by the proposed activities. Considering the site topography, proposed activities during construction and operation phase, noise monitoring has been undertaken for 24 hours at 8 locations. The noise monitoring locations are represented in Table 2 & Figure 5 above. The noise levels in study area are represented in Table 13 & Figure 11 below.

Table 13 - Ambient Noise Levels in Study Area

Location	Leq Day	Leq Night
Project Site, MbPT Shed no:13B	61.4	47.9
Near Cosmos Builder, Malabar Hill	56.8	48.3
Near Entry Gate, At Kohinoor Square, Dadar	60.5	47.5
At Shanti Kutir Apt, Nr Marine Plaza Hotel, Marine Drive	62.4	45.3
At Willson House, Colaba, Mumbai	63.7	46.8
Hughes Dry Dock in Indira Dock	65.2	58.9
Orange Gate - Pricess Dock (OGPD)	68.5	66.1
Lakadi Bandar, Darukhana Ship Breaking Yard	66.3	64.8

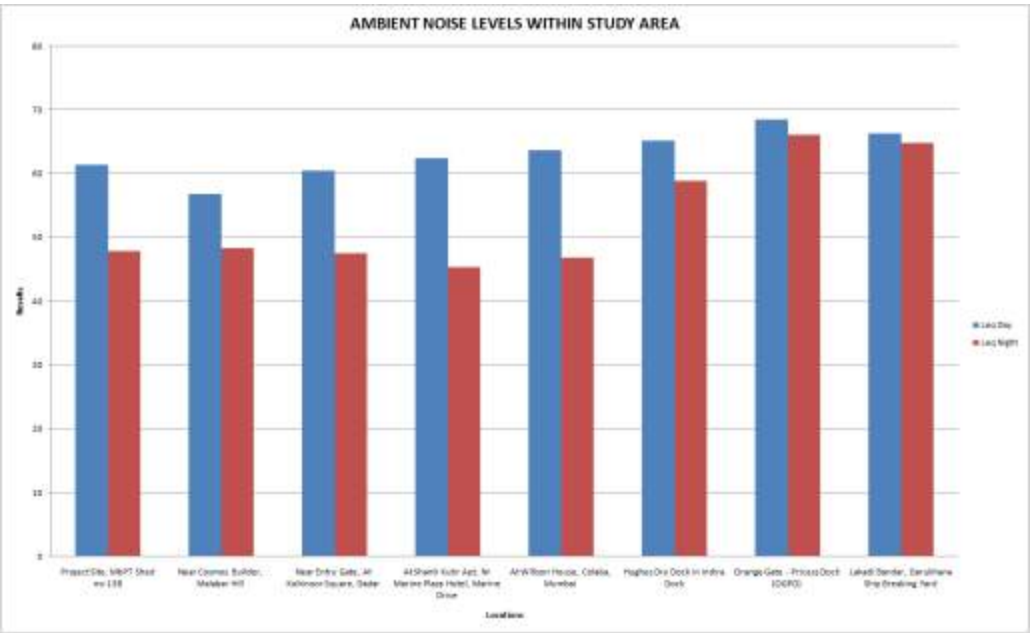


Figure 11 - Ambient Noise Levels at Study Area

3.2.5 Socio-Economic Environment

Socio-economic environment in the vicinity of any ensuing project is affected by the mode of change that is likely to occur due to the beneficial or adverse effects arising out of the project activity. The impact of such change is dependent upon likely job opportunities economic output immigration strain on the existing basic amenities and overall impact on the quality of environment which may impair the health status of the people living in that geographical area. The projection of such assessment calls for collection of the baseline and background information about the socio-economic and demographic environment of the surrounding areas of the proposed site.

The socio-economic profile of the study area based on Census of India data is classified into following points:

- Population and dwelling units
- Social profile
- Education
- Health and medical infrastructure
- Communication facilities

3.2.5.1 Objective

Socio-economic factor plays an important role in any development, as it may have positive or beneficial impact. Therefore it is necessary to check with socio-economic impact due to the proposed project. The various factors dealing with socio-economic study included direct and indirect employment during construction and operation phases, health and hygiene etc. The local body / authority for the region have adequately and wide spread facilities provided to the region.

3.2.5.2 Baseline Data

The methodology adopted for the study is based on the review of secondary data, which mainly comprises 2011 Census records, as these are more comprehensive and authentic. The sociological aspects studied include human settlements, demographic and other socio economic aspects and infrastructural facilities available in the study area. The salient features of the demographic and socio-economic details are described in the following sections.

3.2.5.3 Settlement Pattern and Demography

The socioeconomic condition was studies for Mumbai City.

3.2.5.4 Distribution of population

As per 2011 census, Mumbai city consists of a total population of 30, 85, 411 residing in 6, 74, 339 residential households. This indicates an average household size of 5 persons per household. The distribution of settlements and population in the Mumbai city is shown in Table 14.

Table 14 - Distribution of Population

Sr. No.	Particulars	Study Area
1	No. of Households	674339
2	Total Population	3085411
3	Male population	1684608
4	Female population	1400803

Source: 2011 Census Handbook

3.2.5.5 Sex Ration

The average national sex ratio in India is 832 females per 1000 males as per latest reports of Census 2011 Directories.

3.2.5.6 Literacy levels

The analysis of the literacy levels reveals a medium literacy rate in the study area. The area experienced a literacy rate of 89.2% in 2011. If this compared only for the people of above the age group of 5 years, i.e. the school going age people, this slightly increase the literacy rate. The distribution of literates and literacy rates in the study area is given in Table 15 below:

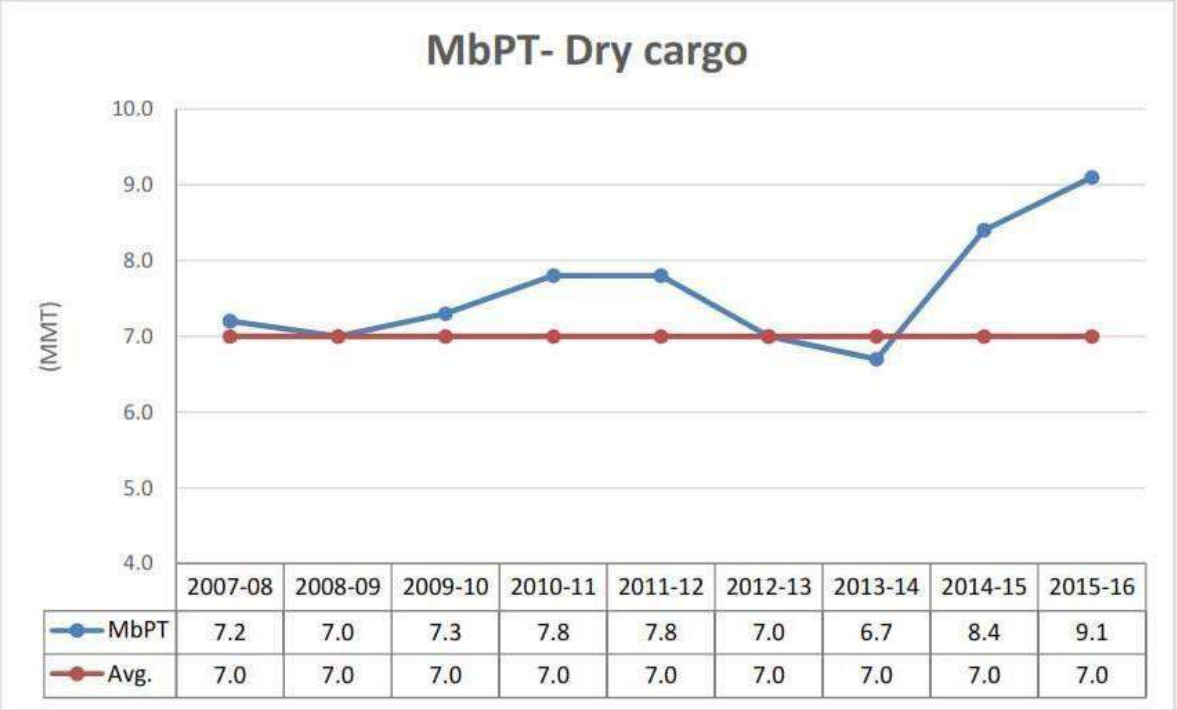
Table 15 - Literacy Rate

Sr. No.	Particulars	Percentage
1	Literacy rate	89.2 %
2	Male Literacy rate	91.48 %
3	Female Literacy rate	86.45 %

This indicates that there is a need for sociological development in the region. The demographic characteristics i.e. the emerging trend in population growth, its characteristics, spatial distribution are sure to have profound influence on the structure and size of the city. It is therefore essential ad interesting to study the dynamics of population growth and structures of city. As population is a key parameter for planning of infrastructure and other activities, a thorough analysis as discussed above shall help city to take a analytical and holistic approach towards urban management.

3.3 TRAFFIC STUDY

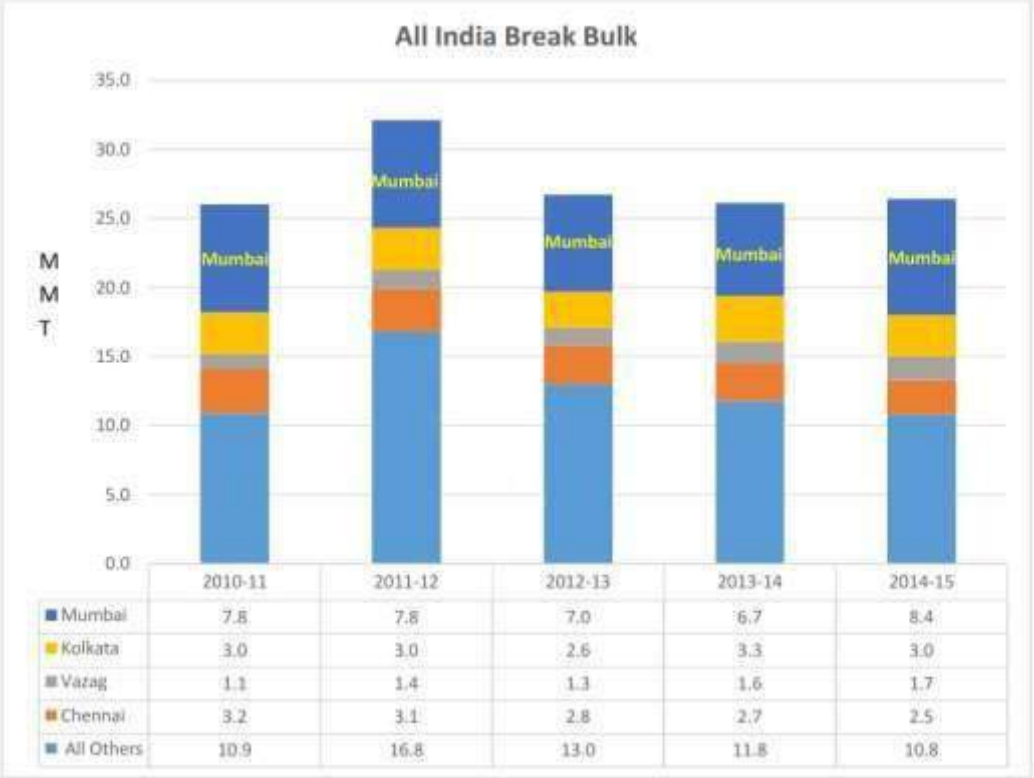
Mumbai Port‘s dry cargo traffic (excluding stream-handled cargo) had been steady at around 7 million tons per annum for 7 years, from 2007-2008 to 2013-14. It spluttered to about 8.5 million tons in 2014-15 and over 9 million tons in 2015-16. Both the non-reducing and increasing performances are significant given the Port‘s limitation in handling deep-drafted and large vessels, which have been coming in increasing numbers in the last 15 years due to availability of abundant cargo and the economies of scale it brings.



	2007-08	2008-09	2009-10	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16
MbPT	7.2	7.0	7.3	7.8	7.8	7.0	6.7	8.4	9.1
Avg.	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0

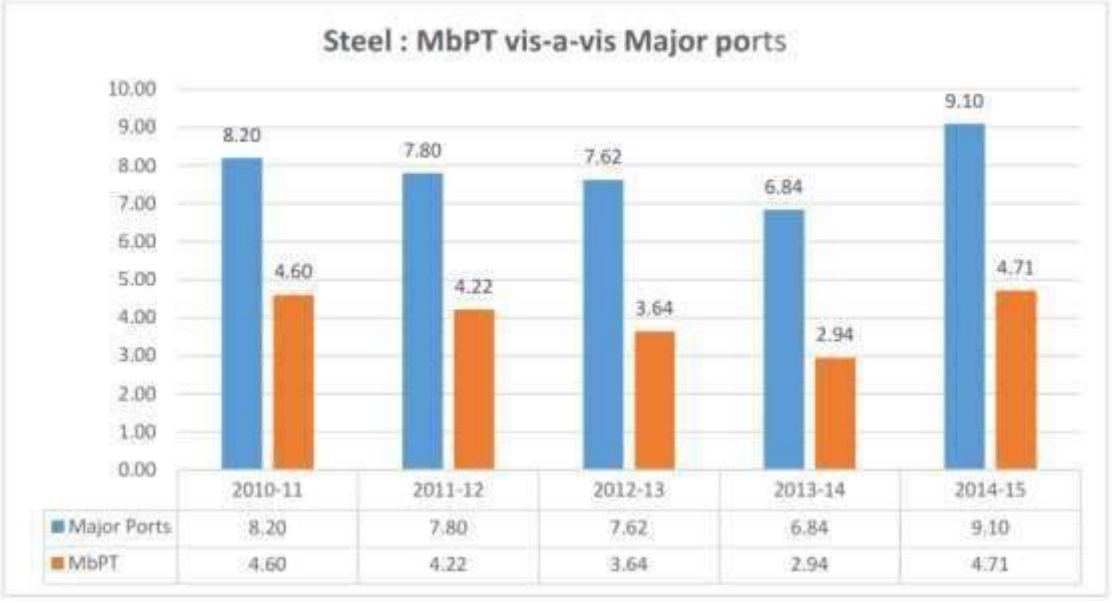
The reason for the Port's undiminished volume appears tolie in the kind of dry cargo it handles Break-bulk and automobiles are the two dry cargo items the Port predominantly handles. A good proportion (about 60%) of break-bulk cargo still comes to the Port in modest parcels. They are carried on small or medium-size vessels (handysize/ handymax) of 150 metre to 205 length requiring 9.0 to 10.5 metre draft max. The Port handles smaller, lighter vessels with no difficulty. The medium-size vessels are a struggle as the Port has only one berth to cater to vessels beyond 9.5metre draft and 190 metre length.

The above may mean one of two things. One, there is ample break bulk cargo for Mumbai Port and because of its limitation most of them are coming in smaller, lighter vessels. Or, some of the medium-size vessels and all the big vessels of the Supramax/ Panamax type are slipping through the fingers of the Port. Either way, it indicates the Port's position and potential in break bulk cargo. For many years now, Mumbai Port has been number one in break bulk traffic among the 12 major ports, handling 25-30% of the total traffic.



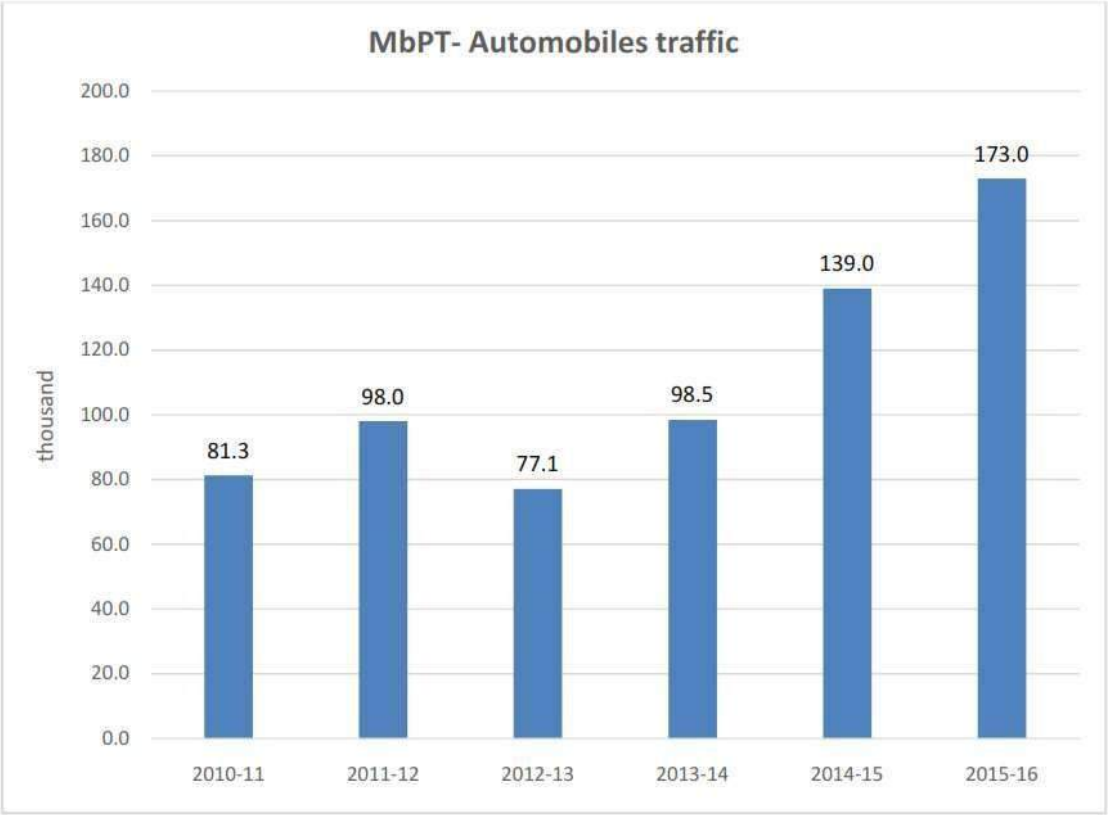
	2010-11	2011-12	2012-13	2013-14	2014-15
Mumbai	7.8	7.8	7.0	6.7	8.4
Kolkata	3.0	3.0	2.6	3.3	3.0
Vazag	1.1	1.4	1.3	1.6	1.7
Chennai	3.2	3.1	2.8	2.7	2.5
All Others	10.9	16.8	13.0	11.8	10.8

Another notable feature is steel is the major break bulk item the Port handles. Steel makes up more than 50% of its break bulk volume. In steel throughput also, Mumbai Port occupies a preeminent position, handling more than 40% of the total steel traffic of the 12 major ports. In 2013-14 its share was 43% and in 2014-15, 52%.



	2010-11	2011-12	2012-13	2013-14	2014-15
Major Ports	8.20	7.80	7.62	6.84	9.10
MbPT	4.60	4.22	3.64	2.94	4.71

One more dry cargo the Port is handling in increasing quantity is automobiles. In 2015-16 it handled about 1.7 lakh vehicles. From April to September'16 it has handled above one lakh units and it is poised to clock more than 2 lakh units by the end of the financial year 2016-17. In automobile traffic, Mumbai Port now is number 1 in western region and number 3 at all India level. Soon it may become the 2nd largest among all the Indian ports including non-major ports.



These performances have come about despite the Port's handicap in two vital attributes: draft and berth size. Had the port been better endowed it would have attracted much more break bulk cargo and automobile as it has the geographical advantage and a rich and growing hinterland for its key cargoes, namely steel and automobiles.

That is where OCT (the terminal may be referred to as Offshore Cargo Terminal or ODT as it is now intended to handle all dry cargo and not just container) can make a difference. With 14.5metre depth at its berth pockets and a liner quay of 700 metre,

extendable to another 350 metre, it is exactly what the doctor ordered for Mumbai Port‘s growing steel and automobile traffic.

Thus, from the above perspective, steel and automobiles have been identified as the most potential cargoes for OCT. Accordingly, their prospects for OCT have been analyzed and traffic projections thereof worked out, as detailed under.

Steel

State of the Indian steel industry

India is the third largest producer of crude steel and the largest producer of sponge iron or DRI (direct reduced iron) in the world. It accounts for 2% of India‘s GDP and 6% of industrial production.

It is one of the first core sectors de-licensed and de-regulated under the 1991 liberalization policy initiative. Since then, its capacity and production have grown in the range of CAGR 5 to 8%. The capacity increased from 20 million tons in 1991-92 to 110 million tons in 2014-15, production from 14.34 million tons to 91.6 million tons and consumption from 44.33 to 77.09million tons.

India‘s steel consumption currently is quite low, considering its population size of 1.29 billion. But the consumption is set to surge significantly over the coming 20 years. Upcoming big-ticket projects in infrastructure development, expansion and improvement, boom in construction and housing industry, rising activity in engineering and manufacturing sectors and an increasing appetite for personal consumption products such as automobiles and white goods are expected to drive the demand for steel to unprecedented levels.

Steel import-export scenario

Import prospects

Steel import has been on a steady rise in the past 5 years, reaching a peak in 2015-16. Primarily, 2 factors contribute to the increasing import. One: domestic price being higher than imports. Two: non-availability locally of some products like high quality, and special grade steel required by automotive and engineering goods industry. At present, they are not manufactured in the country.

Steel import is likely to increase tremendously for another reason: demand exceeding production. Per capita consumption of steel in India today is 65 kg compared with 510 kg in China and the world average at 235 kg. It is expected that India will in the coming two decades catch up at least with the world average. When that happens, there is a question mark whether domestic production will be ready for it.

An Exim Bank paper on India‘s prospects for steel export says two problems beset the steel industry in ramping up its production. One is that its expansion projects are moving slowly or stuck. A plethora of factors including high cost of funding, land acquisition delays, environmental issues, non-availability of reliable raw materials supplier and others have been identified as the chokers of capacity augmentation. Another problem is underutilization of capacity by about 25%. This level of underutilization may be tolerable and even prudent in the present climate of low consumption. But not when the demand shoots up in the foreseeable future.

Signs of falling production and increasing consumption were already seen in 2015-16. In that fiscal, production came down to 91.12 tons from the previous year‘s 91.46 million tons, a 1.1% drop. In the same period demand went up by 4.3%, from 76.96 million tons to 80.27 tons. Not coincidentally, in the same year import reached a newhigh of 11.71 million tons, a sharp increase of 25.6% over the previous year.

Another major challenge the steel industry faces is dumping of steel on India byChina and Russia, to a lesser extent. On average, Chinese steel is cheaper by 15% than domestic steel and Russian by about 9%. So long as China‘s downturn continues and Russia‘s economy, battered by fall in oil prices, remains weak they will keep tweaking their currencies, which will further cheapen their prices.

The Government has taken a series of measures in the recent past to stem the flow of foreign steel into the country. These include anti-dumping duty, ad valorem safeguard tax and MIP or minimum import price. These are interim steps. They may not continue forever. Both MIP and safeguard duty have been challenged by a host of countries under WTO rules. Anti-dumping duty is the only

one sustainable but it takes a long time to implement. Thus, these protective measures may work for a limited time. They are not a permanent solution for stanching the import.

The Hon'ble Finance Minister in a recent seminar on steel industry said as much. He remarked that the Government could do the hand-holding to the steel sector only for a limited extent and period. The industry has to stand on its own by staying competitivein world market.

In the backdrop of the above scenarios and trends the outlook for steel import has been examined and the key points bulleted as below:

Positives for import:

Steel derives its demand from 3 broad sources of consumption: public, private and personal, roughly at45%, 35% and 20% respectively. Public consumption includes improvement, expansion and development of infrastructure like roads, rails, bridges and ports. As per the Working group on steel for the 12th five-year plan, spending on infrastructure will in the coming years go up to a trillion dollars. In the Union budget of 2016-17, Government has allocated an expenditure of Rs. 280000 crores for road and rail development and improvement. Other Government initiatives include housing for all by 2022, power for all by 2019, 100 smart cities by 2022 and Atal Mission for Rejuvenation of Urban Transportation or AMRUT.

Private demand comprises manufacturing, engineering services, housing and other construction activities. Manufacturing sector is projected to grow from current 8 to 12%. Urban population is expected to increase to 600 million by 2030 from the present 400 million. All these will escalate the demand for steel appreciably.

On the personal front, due to rise in standard of living and disposable incomes, a steep increase is anticipated in the demand for automobiles and personal goods like television, washing machine, air conditioner and others, which all will need massive quantities of steel.

As a result of these voracious demands, the country‘s steel consumption is expected to reach about 300 million tons by 2035, from the current 80 million tons.
Meanwhile, China‘s steel juggernaut will roll on. Today China has an over capacity of 300 million tons. It will do all it can to dump them on other countries including India. At their price, it will be hard for us to resist it.

South Korea and Japan are among the top steel producing countries in the world. We have free trade agreements with both. This means, duty-free or substantially reduced tariff on their steel imports. We have already seen the effect of the above factors: import on a continuous climb in the last 6 years, except for a drop in 2013- 14.

	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16
MT	6.66	6.86	7.93	5.45	9.32	11.71

Negatives for import

- If the Government could hold on to the protective measures, at least in the near-term the import will come down
- If the local industry could control its cost and bring domestic prices on par with or below imports cost, it might retard import to an extent
- Once again, China factor: if China‘s economy revives and they go on an infrastructure bingeagain, they may not have enough steel to spare to others.
- These are all big ‗ifs‘.

On balance for import

Steel import started falling from June-July 2016 as the Government‘s punitive measures took hold. However, import looked up again in Oct 2016.

The Government‘s interventions are likely to continue well into 2017. So, the imports may keep falling till around the end of calendar year2017.Nevertheless, it will not reduce drastically. It may bottom out at around 9 million tons.

The reduction in import will most likely to be a temporary phenomenon. The demand is certainly set to surge sooner or later to about 300 million tons by 2035 from the current 80 million tons, as already indicated. When that happens, the import will override the restrictions or the Government may dilute or dissolve the curbs.

The present steel production looksfarmore than the consumption. But it is not. In 2015-16 the production was about 91 million tons, consumption 80 million tons and exportabout 4 million tons. Thus, the gap between production and consumption even at today‘s low demand is not much. And when the demand shoots up the gap will widen.

In sum, regardless of the present reduction in import,caused by the curbs, the long-term prospects for steel import appears to be strong. It may scale to about 25-30 million tons around 2030-35

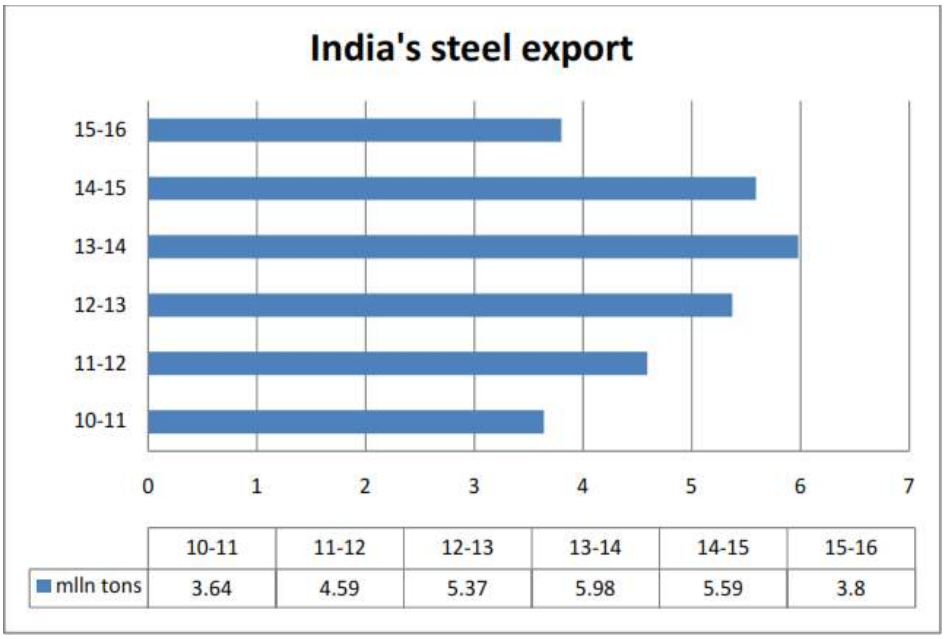
Export prospects

- India started exporting steel in 1964. Its real pace picked up after the liberalization in 1991.

- In 2013-14its export turned around when it briefly became a net exporter.

- Since then our export has been falling. It plunged to 3.8 million tons in 2015- 16 against 5.59 tons in 2014-15, a precipitous fall of 32%.

- Even at the best of times our export was around 7 million tons. It is a fraction of the 110 million tons China is exporting.



Our export market is well diversified with America leading the pack. America is our major steel buyer, taking almost 80% of our export. Today, it is the only market that has a strong demand outlook. Its steel use increased 6.7% increase last year. Robust growth in its automotive industry and energy sectors, and recovery in non- housing (their houses are wooden) construction – all point to a forthcoming massive steel demand in America. This augurs well for our steel export.

Besides America, the rest of the world too is expected to recover from the present economic sluggishness, set off by China‘s downturn. With China‘s recovery expected by the end of 2017, global demand, too, for steel is expected to pick up. This offers hope for our exports.

Positives for steel export

Like the shipping industry, the steel industry is cyclical and fiercely competitive. For the last few years, demand for steel and prices of steel in international market were bearish. This has struck a blow to our export. Last year‘s export was particularly bad at 3.8 million tons. However, the analysts are predicting the cycle is turning and resurgence expected around 2018. If they are right prices, may perk up, which will give a fillip to our export

In his victory speech, Donald J Trump said infrastructure is his top priority. This is not surprising. America's infrastructure is fraying. Its roads, rail, subways and airports are cracking and crumbling. To repair and rebuild them America will need a humungous infusion of steel. And India may be a major source for their import with our long history of selling steel to them.

It is expected that the Chinese economy will revive by the end of 2017 and it will revert to its infrastructure splurging days of the 2000s and early 2010s. If and when that turnaround happens, China might become a net importer of steel. This will be another opportunity for our export.

On the domestic front, the Government has set a production target of 300 million tons by 2025. The steel industry is skeptical about it. However, if it does materialise, the industry may be able to offer its surplus for export at a competitive price.

Negatives for export

China and Russia, to a lesser extent, remain a threat to our exports, with their predatory prices.

America produces less steel than it needs. A few years ago, we took their place as the world's third largest producer of steel. The expected developmental boom in America's various sectors will further scale up its demand. Despite it, they have been talking of antidumping action against India, a regular and reliable supplier. Anti-dumping action is not easy to enforce, however. It requires investigation and proof that the unloading country is selling its products at prices less than the domestic prices. This may be the case with China, not with India. Yet, there is a lurking danger.

If domestic demand rises, as anticipated, and international prices continue to remain weak there may be no incentive for export.

Assessment of overall steel traffic

Import prospects certainly look bright, notwithstanding the recently invoked counter measures. Over the long haul, demand is set for a huge spurt due to factors already

mentioned. It is not certain production will be able to match it. Import is therefore most likely to climb.

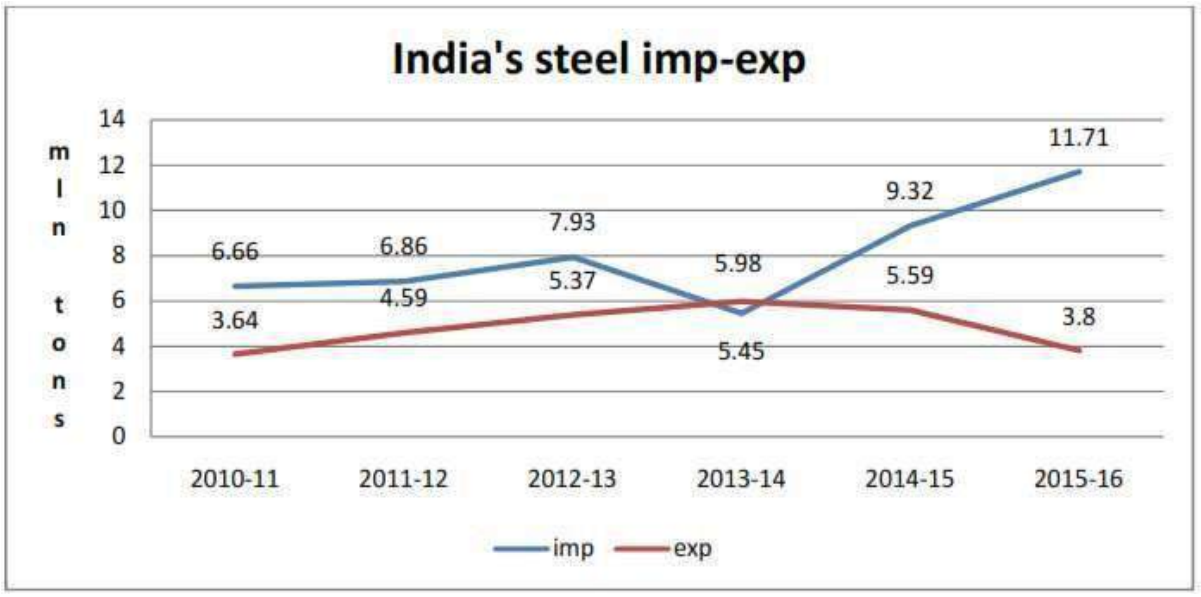
Export future does not look that cut and dried. It mostly depends on global factors. If major economies like Europe and China revive, that may push up demand and prices, giving some traction to our export.

Export future does not look that cut and dried. It mostly depends on global factors. If major economies like Europe and China revive, that may push up demand and prices, giving some traction to our export.

If the anticipated domestic demand does not materialize and production goes up, the surplus will go for export.

In any case, the overall steel traffic will keep going up. If import falls export may pick up the slack or vice versa. It is interesting to note import and export seem to be inversely proportional. In the last 6 years whenever import fell export went up or vice versa, as the following graph shows.

Last year, import constituted about 14% of the total steel consumed in India. At that rate, even if the consumption by 2035 increases to 250 million tons, if not 300 million tons astargeted, import alone will be around 35 million tons. At an optimistic scenario, export may grow to about 15 million tons by the same period.



Mumbai Port's place in steel traffic

Mumbai Port leads major ports in steel traffic. In 2014-15,it handled 4.7 million tons at a share of 43% and in 2015-16, 6.0 million tons at a share of 52%.The Port will continue its dominance as it caters to a rich and growing hinterland of both producers and consumers of steel.

The other ports in Mumbai region are Dharamtar and Dighi. They pose no threat to Mumbai Port. Dighi has a maximum draft of 8.5 meters and Dharamtar is a lighterage port. In any case, there will be more than enough steel cargo in the region that the ports do not have to poach each other's cargo.

Drivers for MbPT's steel traffic

Mumbai Port's proximity to both major consumers and producers of steel and its good connectivity by road and rail and its commercial capital status give it a clear advantage in steel import and export.

Steel consumers in the region include automobile majors like Tata, Mahindra, Bajaj, Volkswagen and GM, white goods producers like Godrej, Voltas, Whirlpool, Blue Star, construction giants like L&T, Afcons and HCC, and engineering goods manufacturers such as Thermax, Sandvik and Atlas Copco

Out of the top 10 steel manufacturers 5 have their plants at proximate distances from Mumbai Port

- No 3: JSW Steel at Vasind, Kamleshwar, Tarapur and Dolvi
- No 6:Bhushan Steel at Khopoli
- No 7: Esssar Steel at Pune
- No 8: Ferro Alloys at Nagpur
- No 9: Mahindra Ugine atKhopoli

Besides the above, Posco and Uttam Galva have their plants at Pune and Khopoli for processing HR coil to CR coil for export

Uttam Galva is a regular shipper of steel through Mumbai Port. They export about 2.5 lakh tons CR coils per year to US. They were also importing 30000 tons of HR Coils a month before the curbs. As per their representatives, if the Port could handle bigger vessels, like Panamax, their volume will increase.

Maharashtra Seamless is another user of the Port for their pipes export. They now import about 7000 tons of billets for making pipes, which they export in equal quantity. Their projection is their import and export will increase to 10000 tons each per month.

Posco exports to US about 25000 tons of coil every month. Their projection is 1 million tons a year starting next year

Octroi was an irritant for using Mumbai Port. With the imminent introduction of GST, octroi will go. This may divert some more steel traffic to Mumbai Port

Steel traffic projection for Mumbai Port

For the past 6 years, Mumbai Port, on average, has been handling each year about 35% of the country‘s total steel traffic (including import and export). Based on this trend and the outlook visualized for the country‘s total steel import-export for the next 20-25 years, the forecast for Mumbai Port is:

The Port's overall steel traffic including OCT's will grow at CAGR 5% until about 2025 and dip to CAGR 3% over the next 5 years, up to 2030. It is likely to drop to CAGR 2% over the next 10 years up to 2040 and then peter out to 1% by 2045 and plateau or retard from then on.

The projection for OCT is done with its capacity limitation in view. Projection for entire Mumbai Port

Base: actual traffic of MbPT in 2015-16 @ 5.985 million tons

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Mn Tons	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

Projection exclusively for OCT

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Lakh Units	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

*capacity saturation of existing berth

Automobiles

State of the Indian Automobile industry

Liberated in 1991 from the cocoon of Ambassador-Premier era, Indian automotive industry has grown the fastest and fiercest, clocking a spectacular growth of ~ 10% in the last ten years. It is expected to grow even faster in the coming decade at a CGAR of ~ 15%

The industry contributes 7.1 to the GDP and accounts for 4.3% of country's exports. In the past decade, 2005-2016, it attracted $ 15 billion in FDI and created 19 million jobs.

Indian automotive industry is the largest in 3-wheelers, 2nd largest in 2-wheelers, 6th largest in passenger vehicles and largest in commercial vehicles overall and 2nd largest in buses.

Drawn by the size of the market and the growing income level of the populace, almost every international automobile major has set up shop in the country. What is even more striking is most of them have made India as one of the manufacturing hubs for their export markets. As of 2015-16, 35% of the small cars sold in the world were made in India.

What pulls international auto makers to India to set up their production facilities for exports is availability of inexpensive labour, a large pool of blue-collared workers with previously acquired skills in auto industry, highly educated technical staff with expertise in design and engineering and a well-established spare parts network.

Indian automobile industry's export performance and prospects

During the decade 2005-15, overall automobile exports grew at a CGAR of 17.79%. Passenger vehicles grew at 7.8% and commercial vehicles at 15.1%. The growth in passenger vehicle segment was subdued in 2015-16 at ~ 5.2 % but the commercial vehicle segment clocked an impressive rise of 18.3% in that period.

Under the National Electric Mobility Mission Plan 2020 (MEMMP 2020), the Government has set an ambitious production target of 10 million passenger vehicles during 2015-20 at a CGAR of 25.6%.The industry expects the production of PVs will be at least 70% of the target. Presently, 20% of the cars manufactured are exported. At that proportion, by 2020 export will increase to 1.4 million units, from the present 0.65 million units.

As regards commercial vehicles, the export is expected to rise to about 3 lakh units by 2020, @ 25% CGAR

Positives for India's light vehicle (PV+CV) export

There is a growing appetite for small cars (hatchbacks and compact sedans) from across developing and emerging economies including Latin American and some European countries. Most Indian car manufacturers are concentrating on this segment in order to tap into those fertile markets.

Automobile is one market where we have no competition from China. China is two places behind us in automobile export. Because of quality issues China is likely to remain a laggard in auto export.

After Latin America, Africa maybe a big market. Its potential has not been tapped so far. Our competitive price in small car segment will be a USP to them.

Negatives for auto export

Thailand is a distinct threat to our passenger vehicle exports. They are the 15th largest exporter with a share of 1.4%. We are at the 20th position with a 0.8% share.

Thailand is on par with us in cost, quality and productivity but better in logistics and supply chain efficiency. From India Suzuki has shifted its Celereio production to Thailand and Ford its Fiesta plant.

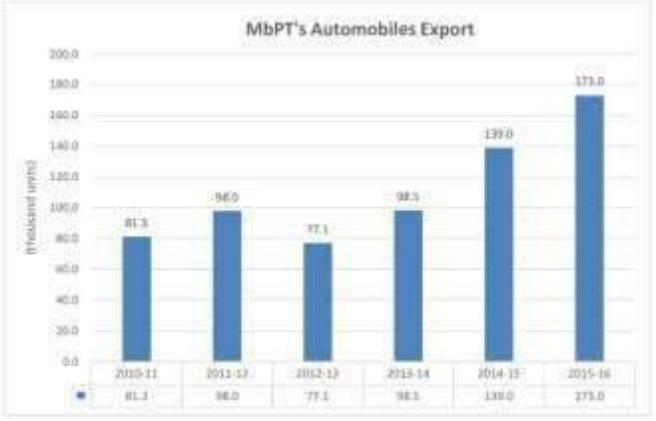
Latin American and European markets may saturate in about 20 years. Unless we find alternative markets in Africa export may stagnate or fall.

Overall outlook for India's automobile export

From all indications, it appears the country's light vehicle (PV and CV) export will boom in the near term (up to 2025), grow moderately in the medium term (up to 2035) and subside in the far term (after 2035)

Mumbai Port's place in automobile export

Mumbai Port's auto export traffic has been on the risesince the past 6 years except for a slide in 2012-13.Today, Mumbai Port is at the third position in passenger vehicles and commercial vehicles export with a share of 27%.



Growth drivers for Mumbai Port's auto export

Mumbai Port is close to many indigenous and international automobile production facilities including that of Tata, Mahindra, GM, Volkswagen and Mercedes
General Motors will be exportingout of Mumbai Port 70000 small cars (Beat)in 2016-17 and about 81000 units next year.

GM expects the volume to substantially increase in the coming years.They have plans to invest $ 1 billion in India to boost exports

Jaguar Land Rover plans to manufacture SUVs in Pune for local and overseas consumption

Chrysler will invest Rs 3500 crores in Maharashtra for manufacturing its highend Grand Cherokee model

Tata Motors' representative stated they are presently shipping 16K to 20K commercial vehicles and they expect the volume to increase by 15% a year

A representative of Mahindra and Mahindra, another commercial vehicle shipper from Mumbai Port, informed that in 16-17 they will be exporting about 36000 units and they expect the volume to grow by 10% every year
It is expected that some volume of small cars exported to Europe from Thailand may divert to Mumbai Port due to shorter transit time from Mumbai. 22days is the transit time from Mumbai to European destinations whereas it is 28 days from Bangkok.

As a policy, Mumbai Port does not handle dangerous cargo. Hence, an accident like the one that happened in Tianjin Port in China will not occur in Mumbai Port. On 12 August 2015, hazardous chemicals, Sodium nitrate and Potassium nitrate (IMCO class 5.1), stored in a warehouse near Tianjin Port caused huge explosion in which nearly 170 people and more than 10000 vehicles including about 8000 cars perished. This will never happen to the cars handled in Mumbai Port.

Estimated automobile export traffic for Mumbai Port and OCT

Mumbai Port's auto export traffic has been continuously on the rise in the last four years. Besides Tata and Mahindra, who are the steadfast user of Mumbai Port, new players have come in including General Motors and Volkswagen.

Signs are their exports will take off in a big way. This is evident from the auto traffic handled by the Port in the first half of the current fiscal. From April to September 2016 it has handled 1lakh units as against 1.7 lakh units in the whole of last year, 2015-16. Many more players are setting up shop in the Maharashtra cluster, including Jaguar, Chrysler and BMW.

Mumbai Port currently handles about 27 % of country's total light vehicle exports. This proportion will certainly rise with the entry of new players and the expected increase in volumes of existing OEMs.

In specific terms, Mumbai Port's automobile traffic will grow at 15% CAGR in the near term until 2020 and then come down to about 7% for the next 5 years, until 2025. Thereafter as our main export markets, South America and Europe, start maturing and other markets, mainly in Africa, open up, the growth may be at a sedate pace of CGAR 5% up to 2040. A subdued growth of 3% CGARcan be expected then on.

Automobile export projection

(for thewhole of MbPT)

Year	**16/17**	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Lakh Units	1.99	2.29	2.63	3.03	3.5	4.03	4.64	5.34	6.15	8.62	14.02	16.28

Projection exclusively for OCT

Year	**16/17**	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Mn Tons	1.99	2.29	2.63	3.03	3.5	4.01	4.40*	4.40*	4.40*	4.40*	4.40*	4.40*

*capacity saturation

Container Traffic

OCT appears to have limited scope for container traffic due to the following reasons: Mumbai Port on an average has been handling only 40000 teus per annum for the last 10 years

These are mostly JNPT-bound or JNPT-origin containers moved on barge or by road. These are the only containers handled in the last 10 years. These containers carry import goods for local (Mumbai city‘s) consumption or export consignments shipped from city-based units. They may remain more or less at the same level.

Feeder vessels up to the 2nd generation, which are light and small, could havecome to Mumbai Port. They have not been coming.

The present capacity of JNPT is 4.1 million teus. The 4th terminal is expected to go on stream in another 2 years. The capacity of JNPT will then reach nearly 9 million teus. This leaves little room for others in the region. In fact, JNPT will become a threat even to Mundra.

Without quayside gantry cranes no mainline container vessel will call at the terminal In view of above, the scope of OCT for container may be limited to about 40000 to 50000 assuming the entire present container traffic coming to Mumbai Port will shift to OCT. On this basis the capacity of OCT for container traffic is calculated as below The berth occupancy for the automobile berth of OCT has been assumed at 50%. So, there is a slack of 20% occupancy is available at the auto berth. This slack can be used for handling the above limited container traffic. On that basis, the capacity for container will be as follows:

No. of moves per hour per crane	: 12
No. of cranes at a time	: 2
Ship output per hour:	: 24 moves
Average no. of working hours per day	: 20
Ship output per day	: 480 moves
Proportion of 40‘ to 20‘	: 50:50
So, ship output per day	: 720 teus

Berth days available for container ships at auto berth: 365×0.2= 73 days

So, container capacity of OCT : 73×720= 52560 teus

Conclusion on Traffic Analysis

Based on the past traffic trends in Mumbai Port, steel and automobile were identified as the potential growth cargoes for OCT. Accordingly, the future prospect for these cargoes for OCT was analyzed. The study shows that in both steel and automobile cargo segments there will be appreciable growth in the near term, moderate growth in the medium term and subdued growth in the far term. Container traffic appears to have very limited scope for expansion.

The above assessment is based on the present trends in the domestic and global scenario. It may therefore be prudent to wait and watch for a few years how the traffic in the above cargoes behaves. Thereafter, the Port and the prospective concessionaire, with mutual understanding, may take a call on the construction of the third berth.

3.4 ECOLOGY & BIODIVERSITY

3.4.1 Marine Ecology

3.4.1.1 Water Quality Biotic

In view of the need for conservation of environmental quality and biodiversity, study of biological environment is one of the most important components for ecological assessment. Ecological system shows inter relationship between biotic and abiotic components including dependence, competition and mutualism. Biotic component comprises of both plant and animal communities, which interact not only within and between them but also with the abiotic components viz., physical and chemical components of the environment.

Generally biological communities are the indicators of climatic conditions, dependent on environmental condition and resource of its distribution and survival. It may change if there is alteration in the environmental variables like temperature, humidity, rainfall, soil characteristics, topography etc., which are responsible for maintaining the homeostasis of the environment. The species of flora and fauna in the environment are organized into natural communities with mutual dependencies and

show various responses and sensitivities to anthropogenic influences. The changes in biotic community are studied in the pattern of distribution, abundance and diversity.

Plankton

Plankton is an important component of an aquatic ecosystem, which responds to ecosystem alterations rather rapidly. It is due to the fact that planktonic organisms, which react to different types of water pollution, play a key role in turnover of organic matter and energy through the ecosystem. This reaction is very rapid because of relatively short lifetime and high reproduction rates of these organisms. Since the phytoplankton plays a key role of primary producer in aquatic environment, it is the first component in the trophic tier to be affected by pollution. Phytoplankton can grow rapidly and form massive blooms that can be regulated by environmental factors such as nutrients, availability of light and biotic interaction with grazers. Phytoplankton are generally microscopic and passive drifters with the currents.

Among phytoplankton, diatoms are a highly diverse and an abundant group of phytoplankton in the aquatic environment. They are responsible for about 25% of global primary productivity and play a central role in the biogeochemical cycling of important nutrients such as carbon, nitrogen and silica. Most of the N is bound inorganic compounds and its importance to phytoplankton bloom formation is acknowledged. Phytoplankton bloom can have negative influence on the water quality of water bodies. Phytoplankton blooms decrease light penetration through the water column and can depress primary productivity.

It may have diminished ecosystem integrity and the abundance and sustainability of living resources (e.g. fish and shrimp). Moreover, when bloom organisms die at the end of their life cycle, there can be oxygen depletion which can be serious in enclosed or esmi-enclosed water bodies.

Alike phytoplankton, zooplankton also form a very important group in the aquatic ecosystem and acts as the primary consumer. The zooplankton serves as the natural food source for many aquatic organisms, including fishes. Freshwater zooplankton show considerable variety comprising of members of almost every

group from protozoato chordate. Depending on seasons and environmental conditions, the plankton community shows pronounced variation in its character and composition.

Enumeration and Indices

The inhabitants of a particular ecosystem serve as biological indices and reflect the environmental conditions that are required for their optimum growth and survival. In view of this, studies were carried out towards distribution, diversity and other ecological aspects of phytoplankton and zooplankton from different sampling locations of multipurpose cargo berth site. Phytoplankton's were enumerated from unfiltered water samples. Desired volumes of the waters were· filtered through plankton net to represent all the available groups of zooplankton. The samples were fixed immediately with 5 % buffered formalin. The parameters studied were numerical count of individual species, groups and indices as described here under.

Shannon Wiener Diversity Index

A biological community, whose composition is influenced by environmental conditions and availability of required resources, may undergo changes with respect to their number and types of species and their populations, depending upon changes in factors influencing their composition. For e.g. organic wastes eliminate the sensitive organisms and provide food for survival of tolerant forms. By natural purification, water quality improves and subsequently diversity increases. Thus, the nature and type of biological species present in particular environment is dependent on various chemical characteristics such as pH, conductivity, alkalinity, nutrients, DO etc. It is also dependent upon the nature of water body that is, whether the aquatic ecosystem is characterized by standing water (lentic) or flowing water (lotic). Additionally it is influenced by physical parameters like temperature, turbidity, colour, light penetration and flow of water. Usually diversity increases with decrease of pollution and vice versa. Shannon Wiener Diversity Index (d) is a measure of diversity which takes into account the total count and individual count in water sample and is expressed as

$$\text{Shannon Wiener Diversity Index: SWI} = -\sum \frac{ni}{N} \log_2 \frac{ni}{N}$$

Where,

SWI = Shannon Wiener Diversity Index

N = Total number of individuals of species in a sample

n = number of individuals of species in a sample

A widely accepted ecological concept is that community with large number of species i.e. with high diversity will have stability and thus have the capability to resist adverse environment influences to certain extent. "The Shannon Wiener index" values in the range of 3 and above are generally considered to represent healthy conditions of water. The values between 1 and 3 are believed to indicate semi and poor productivity respectively.

Phytoplankton

Count: Phytoplankton counts recorded at different sampling stations are presented in Table 4.10. Total lowest and highest algal population varied between 290 and 510 algal cells ml-r at station W4 and W3. The phytoplankton population comprised of 4 major groups, namely Bacillariophyceae, Chlorophyceae, Cyanophyceae and Chrysophyceae. Bacillariophyceae dominated all the samples. The most common genera found were Oscillatoria, Fraqillariy Anabaena, Uothrix, Closterium, Cosmarium, Navicula' and Gyrosiqma.

Shannon - Wiener Diversity Index (SWI): During the survey, the SWI values varied between 1.00 and 2.10 at stations 3 and 10. The values suggest low to medium impact of pollution or adverse factors,

Table 16 Enumeration of Phytoplankton in Multipurpose cargo berth

S.N	Sampling station	Phyto-plankton, [No/ml]	Percentage Composition of Algal Groups				SWI
			Bacillario-phyceae	Chloro-phyceae	Cyano-phyceae	Chryso-phyceae	
Multipurpose cargo berth							
1.	Wl	480	50	10	20	20	1.26
2.	W2	370	60	20	10	10	1.85
3.	W3	510	45	25	20	10	1.00
4.	W4	290	50	20	10	20	1.62

5.	W5	340	60	20	20	-	1.70

Ranges of Shannon Wiener Diversity Index (SWI):

<I: Indicate maximum impact of pollution or adverse factor 1-<3: Indicate medium impact of pollution or adverse factor 3 & above: Indicate lowest or minimum impact of pollution

Zooplankton

Zooplankton counts were, recorded at different sampling stations. Since large volume of water was to be filtered through plankton net, middle and bottom samples could not be collected. Density of zooplankton in marine water varied between 210 and 350 N/m3. Copepoda was the dominant group followed by Cladocera. Maximum count was observed at station W4 which probably accounts for less number of phytoplankton in the respective station. The results are given in the Table 4.11.

Shannon- Wiener Diversity Index (SWI): The SWI, observed to vary from 1.56 to 2.21 at stations 1 & 5 respectively indicated low to medium load of organic pollution or adverse factors for zooplankton growth. Generally "Shannon Wiener index" values between 1 and 3 are considered to indicate moderate productivity of the water body, while the values above 3 may represent no or minimum impact of pollution.

Table 17 Enumeration of Zooplankton at Multipurpose Cargo Berth

S.N	Sampling station	Zooplankton, [No/m3]	Percentage Composition of Zooplankton Groups			SWI
			Copepoda	Cladocera	Rotifera	
Multipurpose cargo berth						
1.	WI	210	50	20	30	1.56
2.	W2	315	60	20	20	1.89
3.	W3	280	50	30	20	1.64
4.	W4	350	40	30	30	1.82
5.	W5	315	75	15	10	2.21

Ranges of Shannon Wiener Diversity Index (SWI):

<1 : Indicate maximum impact of pollution or adverse factor. 1-<3 : Indicate medium impact of pollution or adverse factor. 3 & above : Indicate lowest or minimum impact of pollution

Benthos

The organisms which inhabit the bottom of aquatic body are called benthos. Many of them are sessile while; some creep over or burrow in mud. The quality and quantity animals found at the bottom is not only related to the nature of substrata but also to depth, the kind and the quality of aquatic plants present in such environment and food availability. Their number and distribution also depend upon physico-chemical properties of water, community structure and other factors.

The bottom mud was collected from various sampling points by Van veen grab sampler having the mouth area 0.02 m-, The sediment was sieved through 500 micron mesh size sieve and the organisms retained the sieve were preserved immediately with 5% buffered formalin. Subsequently, all the macrobenthic specimens were identified to the lowest possible level under a stereoscopic microscope. All unidentified specimens are referred to by their generic/family names and were considered in single taxonomic category.

A total of three macrobenthic groups were obtained from the remaining sediment samples. Foraminifera were the most abundant group. Species like Triloculina and Flori/us were observed. Forams are abundant all over the ocean. They either live on the sea bottom (benthic) or float in the upper water column (planktonic). The size of the foraminiferal test typically ranges from 0.05mm to 0.5mm although some forams may be as large as several centimetres with a recorded maximum of 18cm in diameter. They not only provide surface for respiration but also perform feeding, locomotion, test building, metabolite release, adhering, etc. Foods of the foraminifera are variable and include dissolved free amino acids, bacteria, unicellular algae, and even metazoans, such as copepods. Foraminifera group was followed by Gastropods. Among the Gastropods, the dominant species were Litiopa sp. and Morula sp. The highest count was 250 No/m- in sampling point 4 (Table 4.12) The SWI values vary from 1.40 to 2.15 which indicate medium impact of pollution on these communities.

Table 18 Benthic fauna recorded at multipurpose cargo berth

Sample No.	Sampling Station	Macro benthos No/m2	Percentage composition of Macrobenthos			SWI
			Foraminifera	Gastropods	Polychaeta	
Multipurpose cargo berth						
1.	W1	210	50	20	30	1.5
2.	W2	180	60	25	15	1.85
3.	W3	215	45	25	30	1.96
4.	W4	250	50	30	20	2.15
5.	W5	170	65	15	20	1.4

Ranges of Shannon Wiener Diversity Index (SWI):

<1: Indicate maximum impact of pollution or adverse factor.

1- <3: Indicate medium impact of pollution or adverse factor.

3 & above: Indicate lowest or minimum impact of pollution or adverse factor

Observations in Sewri Mudflats

The Sewri open mudflats are major intertidal area in the Thane creek. The flats cover an intertidal segment of 10 km long and 3 km wide, dominated by lush green mangroves. The flora and fauna of the Sewri Mudflats were also investigated during the present study to generate baseline information on its ecological status

3.4.2 Terrestial Flora And Fauna

Sampling locations

The sampling locations were randomly chosen within a 10 km radius of the proposed project site. This included many green regions like Jijamata Udyaan and Sewri Mudflats. It also included many gardens like Sagar Upvan (Mumbai Port Trust Garden) and the hanging gardens. Table 19 lists all the sampling points along with their point name and geographical locations. Figure 12 represents all the sampling points in a 10 km radius of the proposed project site.

Table 19 - List of all sampling points for flora and fauna survey

Point	Name	Latitude	Longitude
A	Sagar Upvan 1	18.91126	72.82372
B	Sagar Upvan 2	18.90946	72.82377
C	RC Church	18.89999	72.81388
D	Horniman Circle	18.93217	72.83505
E	Indira Dock 1	18.93796	72.83928
F	Indira Dock 2	18.94356	72.8392
G	Marine Drive	18.92603	72.81861
H	Five Gardens	19.01993	72.85375
I	Sewri 1	18.99705	72.86037
J	Sewri 2	18.99578	72.86137
K	Pare I	19.00491	72.84038
L	Nehru Science Centre 1	18.99132	72.81978
M	Nehru Science Centre 2	18.99081	72.81751
N	Hanging Garden 1	18.95686	72.80497
0	Hanging Garden 2	18.95459	72.80481
P	Walkeshwar	18.94496	72.79516
Q	Rani Baug 1	18.97977	72.83728
R	Rani Baug 2	18.97758	72.83596
S	Gol Deool	18.95815	72.82826

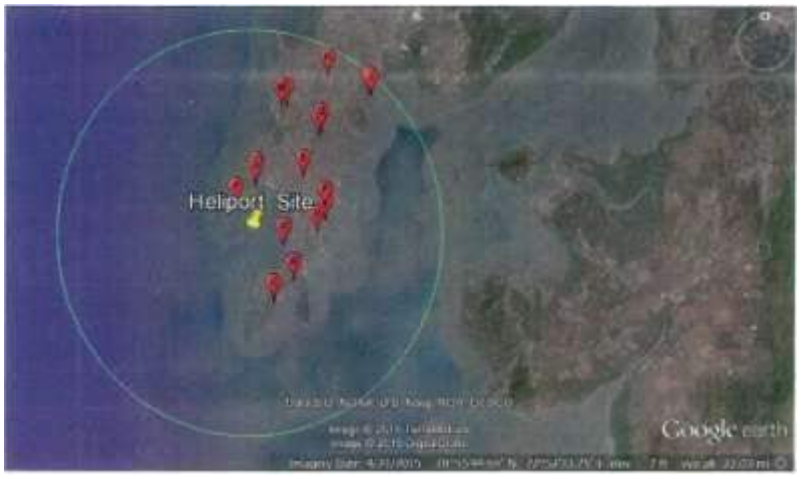


Figure 12 - Image showing sampling points in 10 km radius of proposed Heliport site

3.4.2.1 FLORA SURVEY

A total of 19 points within 10 km radius of the proposed project area were surveyed to document the flora diversity. These points had mostly urban and garden habitats wherein the flora was primarily planted by humans. Therefore, no naturally occurring flora was observed. The diversity of trees, herbs and shrubs were more or less similar in all the points. During the entire flora survey, 42 trees, 15 shrubs and 34 herbs were observed

Flora Observations in Urban Habitats & Gardens

From the secondary literature review, it was found that one tree species, Saraca asoca, is listed as "Vulnerable" by the International Union for Conservation of Nature (mCN) Red List. Other than that, all the other trees, herbs and shrubs are listed as "Least Concern"

The ground cover in most sampling points was poorly developed. Trees were more dominant in most sampling points, with very few herbs and shrubs. Only the sampling points in gardens had a well-developed ground cover of herbs and shrubs. The points that were closer to the sea showed a dominance of the tree Samudraphal. Most of the species observed were ornamental plants and less native, naturally occurring plants.

Flora Observations in Sewree mudflats

The Sewri Mudflats formed a different habitat from the other sampling points. It consisted of plants that are found in wetland type of habitats. Mangroves a single tree group covered most of the Sewree mudflats that was sampled; Avicennia marina being the dominant species. Therefore, the plant diversity in Sewree was very poor.

Flora Analysis

During the flora survey of the study area, besides the number of plants seen, the number of individuals of each plant was also documented. Using this data, the distribution patterns of the plants in the study area were determined. The relative frequency, relative density and relative abundance of trees, shrubs and herbs were calculated. It is difficult to compare the diversity of Sewri Mudflats with the rest of the sampling points as it consists of a habitat entirely different from the rest. Hence, the Sewri Mudflats were excluded from the analysis. Along with the distribution patterns, the species composition of the sampling points also needs to be determined. Thus, to understand the species composition of the project area, the Shannon Wiener Index was calculated

Trees

Distribution Patterns

During the survey, Pimpal was found to be the most frequently occurring tree in all the sampling points. Pimpal is an easily growing tree species that does not need a lot of maintenance. Its seeds are also very easily dispersed and grow in any habitat. Followed by Pimpal, the second most frequent tree species is Amba (Mango). A graph representing all the relative frequencies of all the tree species is given in Figure 13.

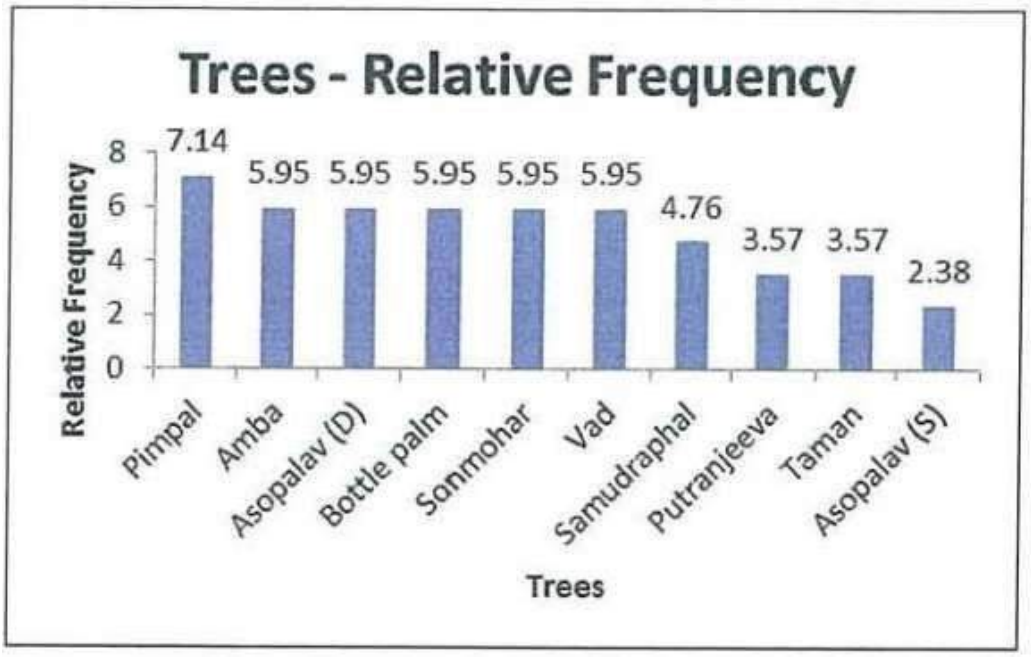


Figure 13 -Relative Frequency of trees in study area

The relative densities calculated showed that Asupalav was the most densely populated tree in the entire project area. This is because Asupalav was present in maximum points. They were planted in large numbers in all of the gardens as well. The second most densely populated tree species was Samudraphal which was present in many points close to the sea. The relative densities of all the tree species in the project area are represented in Figure 14.

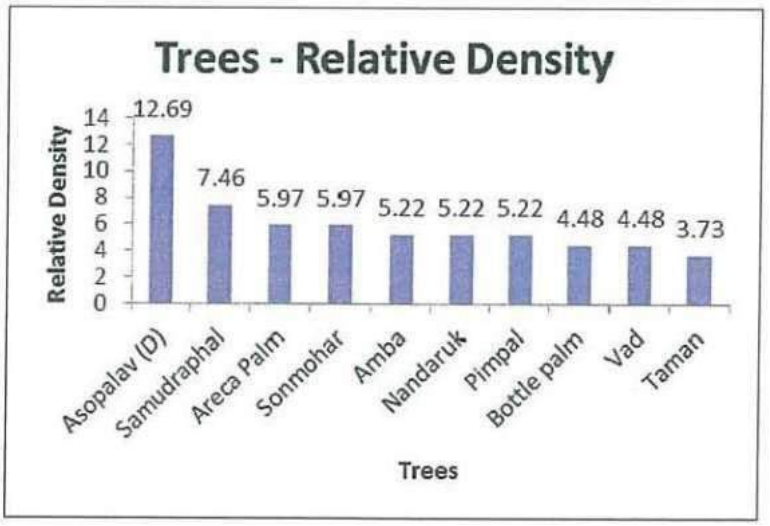


Figure 14 Relative Density of Trees in Study area

The most abundant tree species found in all sampling plots was Nandruk (Ficus benjamina). This is an ornamental tree which is planted in many gardens, thus it was present in all points. The second most abundant tree species was Areca Palm. This tree was also present in large numbers in many gardens. The relative abundance of all the tree species in the project area are represented in Figure 15.

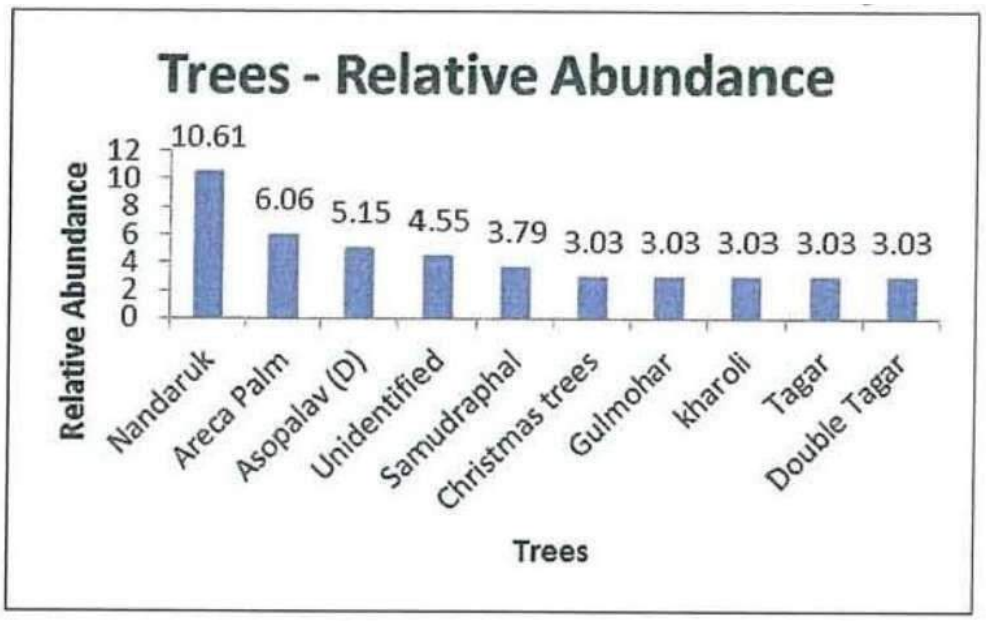


Figure 15 - Relative Abundance of Trees in Stusy Area

Species composition - Diversity Indices

To determine the most diverse region in all the sampling plots, the Shannon Wiener diversity index was calculated. From the calculations, it can be concluded that the sampling points Indira Dock 1 and Walkeshwar are the most diverse regions. This means that the two points are not only rich in diversity but also species abundance. The Shannon Index of Trees in the study area is given in Figure 16.

Figure 16 - Shanon Index of Trees in Study Area

Shrubs

Distribution Patterns

Good luck plant (Cordyline [ruticosa) was the most frequently occurring plant species in the entire project area. The second most frequently distributed plant species was Lantana. Lantana is an invasive species that is very fast growing and grows in all habitats. It is planted in many places as an ornamental shrub. The frequency distribution of all the shrubs is represented in Figure 17..

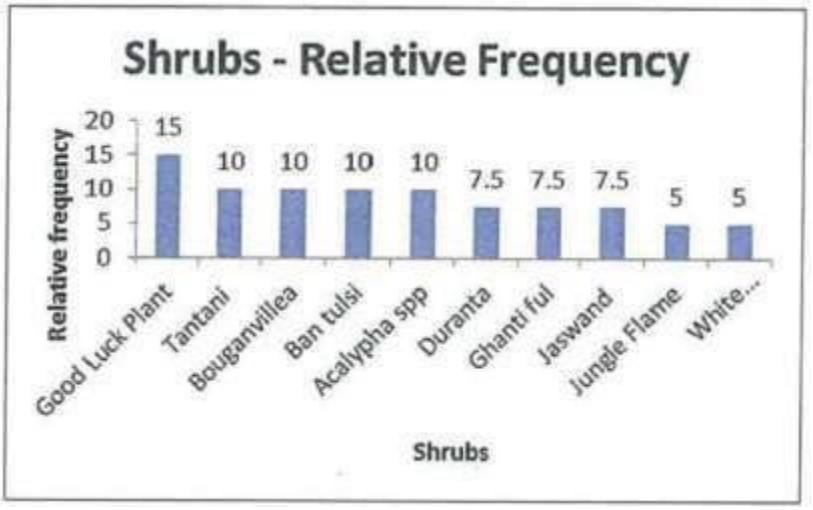


Figure 17 - Relative Frequency of Shrubs in Study Area

The most densely populated plant throughout the survey was Acalypha spp. It was present in large numbers in all the points. The second most densely populated plant species was Lantana, followed by Cordyline fruticosa. The density distribution of all the species found in the study area is represented in Figure 18.

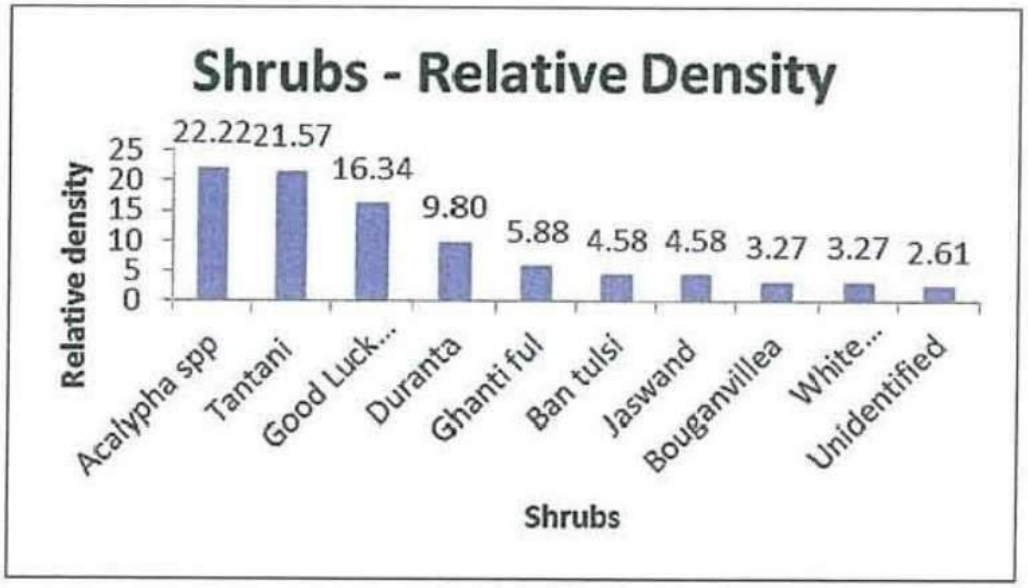


Figure 18 – Relative Density of Shrubs in Study Area

The most abundant plant species in the entire study area was Acalypha spp as it was present in almost all the sampling points. The second most abundant plant species was Lantana, followed by Duranta. Both these shrubs are planted in many sampling points. Figure 19 represents the abundance of all species in the areas they were found in.

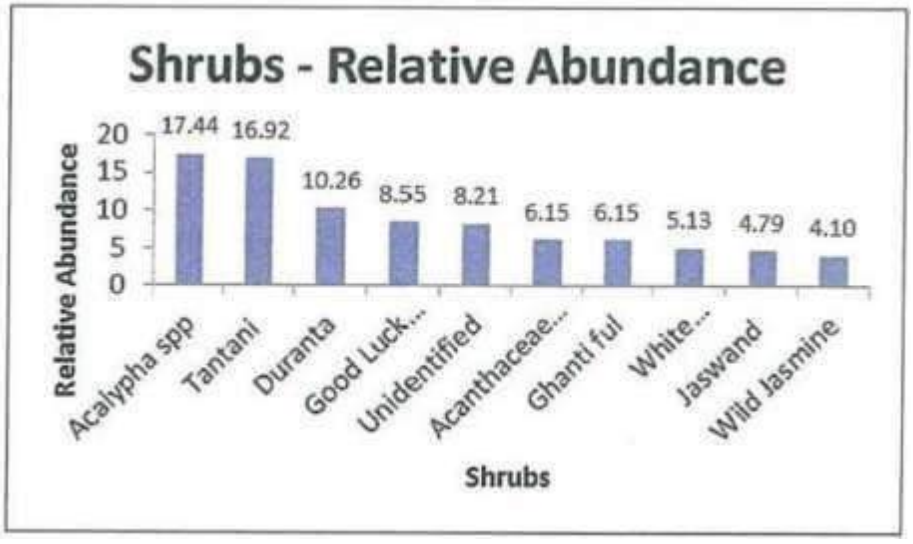


Figure 19 - Relative Abundance of Shrubs in Study Area

Species Composition - Diversity Indices

The most diverse region in terms of shrubs diversity was Nehru Science Centre 1. This point showed a high diversity of shrubs. It also showed high species abundance. The second most diverse region was Hornimon Circle. Many sampling points showed no shrub diversity at all. The Shannon Index for shrubs is shown in Figure 20.

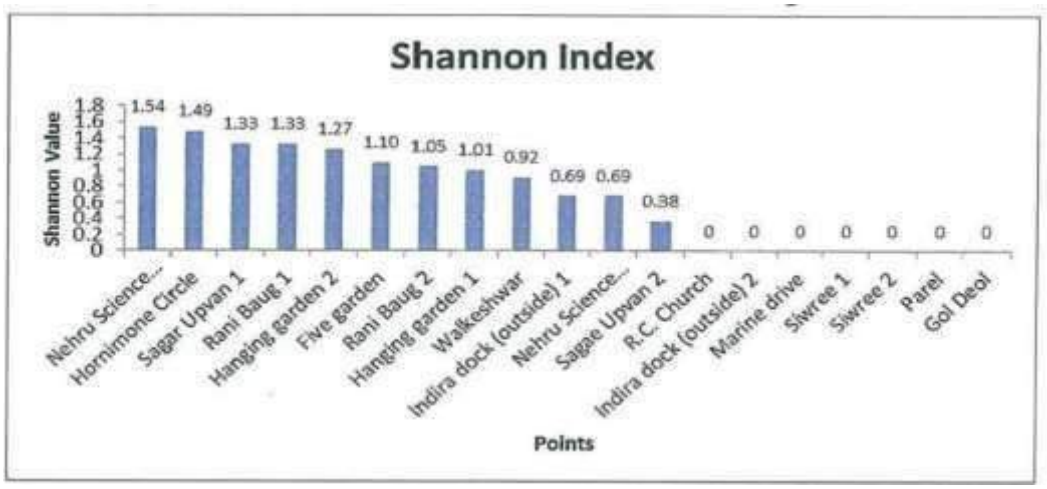


Figure 20 - Shanon Index of Shrubs

Herbs

Kanchari (Alternanthera sessilis) was the most frequently distributed herb in the entire project area. It was present in all the points surveyed. The second most frequently distributed plant species was Durva. The frequency distribution of all the herbs is represented in Figure 21.

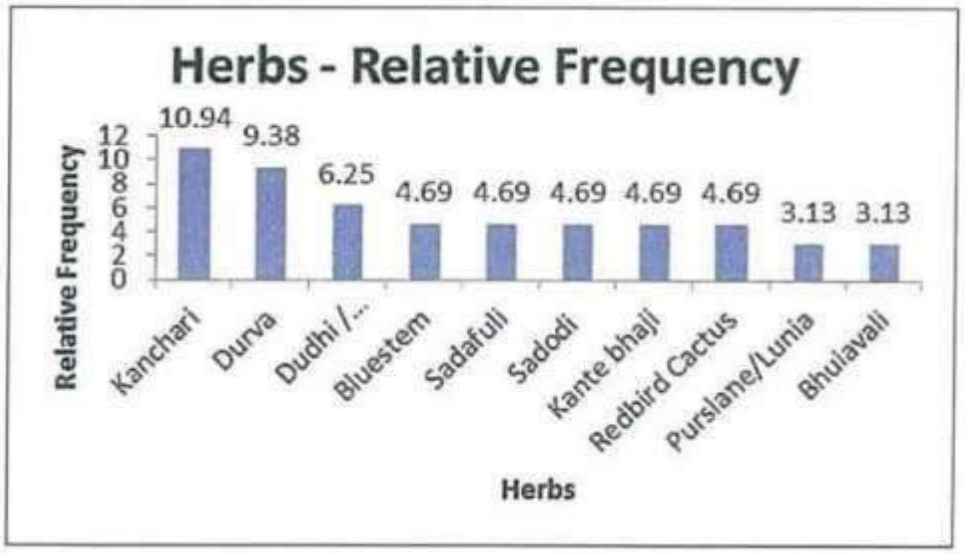


Figure 21 - Relative Frequency of Herbs in Study Area

The most densely populated plant species throughout the survey was Sadodi (Vernonia cinerea). The second most densely populated plant species was Kanchari, followed by Redbird Cactus (Pedilanthus tithymaioides). The density distribution of all the species found in the study area is represented in the following Figure 22.

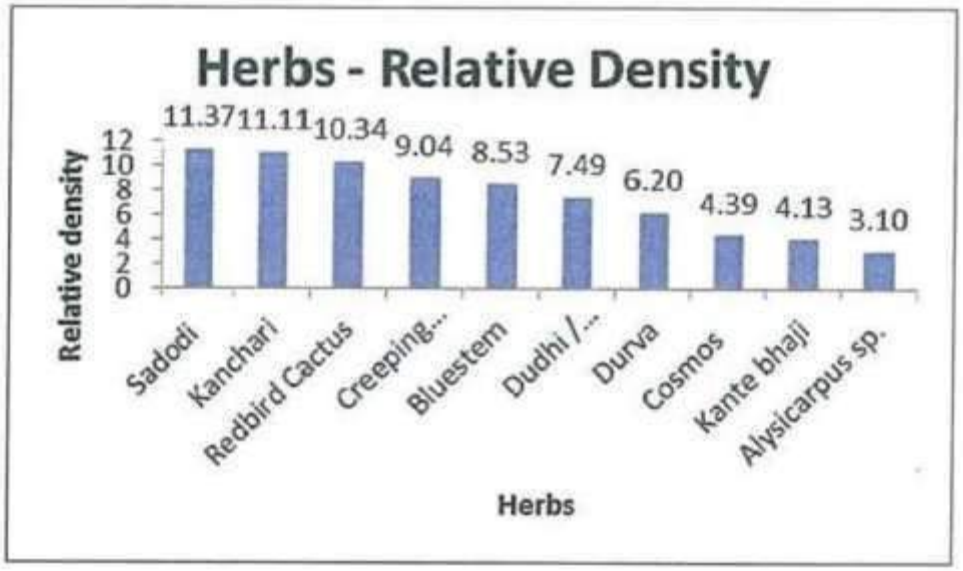


Figure 22 - Relative Density of Herbs in Study Area

The most abundant plant species in the entire study area was Creeping Oxalis (Oxalis corniculata). It was common in almost all of the sampling points. The second most abundant plant species was Cosmos. This herb was present in large numbers in one of the gardens. Figure 23 represents the abundance of all species in the areas they were found in.

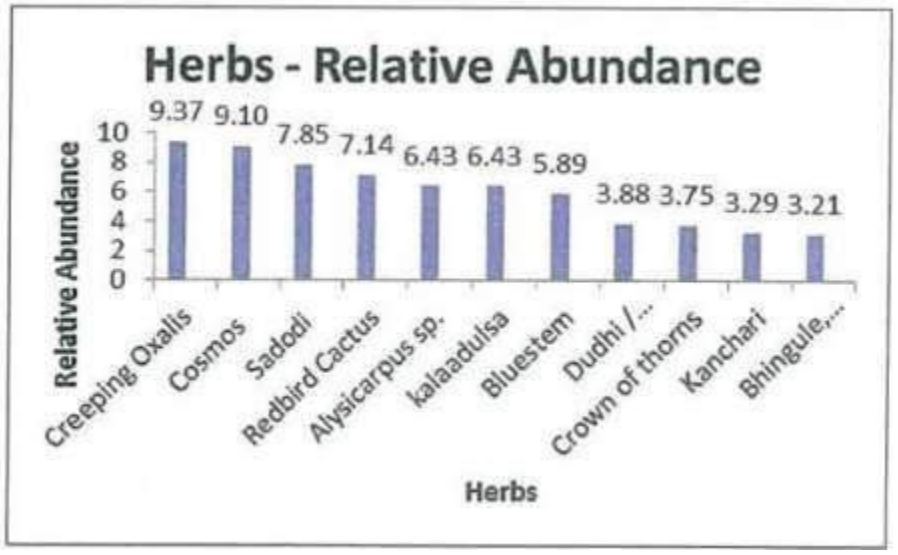


Figure 23 – Relative Abundance of Herbs in Study Area

Species Composition - Diversity Indices

From the Shannon Wiener Index calculations, Five Gardens showed the highest herb diversity in terms of species composition and species abundance. The second most diverse areas were the hanging gardens sampling points. Many sampling plots showed zero herb diversity. These plots were mostly areas in the city. The Shannon Index of herbs in the study area is shown in Figure 24.

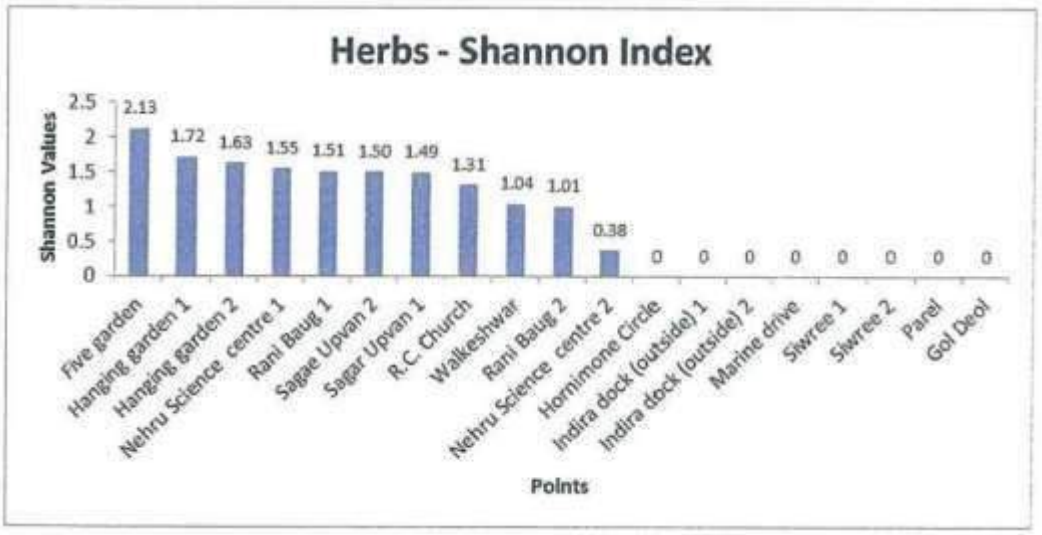


Figure 24 -Shanon Index of Herbs in Study Area

3.4.2.2 Fauna Survey

The sampling method used for fauna survey was by random sampling. Certain points within 10 km radius of the proposed project area were randomly chosen as the sampling points. For birds, point count method was followed, where observations of bird diversity were made at a single point. For butterflies, reptiles and mammals, opportunistic observations were taken.

Fauna Observations

A total of 19 points within 10 km radius of the proposed project site were surveyed to document the fauna diversity present in the area. Except the Sewri Mudflats, all the points had more or less the same habitats. The points were mostly gardens or other urban habitats. Hence, the fauna diversity observed during the survey was more or less the same. However, the fauna diversity in Sewri Mudflats was different from the diversity in the rest of the points. A total of 25 birds, 19 butterflies, 2 mammals and 1 reptile species were observed in the entire survey.

Fauna observations in urban habitats

The International Union for Conservation of Nature (IUCN) Red List of Threatened Species lists all the species documented in the survey as "Least Concern". The Wildlife Protection Act lists them under "Schedule IV", except House Crow. The House Crow is listed under "Schedule V". There were no migratory species observed in the points with urban habitats.

Fauna observations in Sewri Mudflats

In the year 2004, the Sewri Mudflats were designated as an Important Bird Area (IBA) by The Bombay Natural History Society (BNHS) and the Indian Bird Conservation Network (IBCN). An Important Bird Area is a globally recognized habitat for conservation of birds. However, the Sewri Mudflats are unprotected. The Sewri Mudflats offer typical wetland habitat for water birds. It is home to many migratory bird species that spend their winters in these wetlands from the months of October to May.

A total of 12 bird species were observed during the survey of the Sewri Mudflats. Out of the total 25 species found in the entire study area, 7 species were documented in the Sewri Mudflats alone. More than 100 individuals of Lesser Flamingo were observed during the field survey, along with many Painted Storks, Intermediate Egrets, Little Egrets and Grey Heron. From these 7 bird species, three are listed as Near Threatened according to the IUCN Red List of Threatened Species. These species are namely, the Lesser Flamingo, the Black-headed Ibis and the Painted Stork. Three species, namely, Lesser Flamingo, Grey Heron and Gull-billed Tern are known migratory species that visit the Sewri Mudflats from different parts of India. The Lesser Flamingo is also listed under Appendix 2 by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). The rest of the bird species are listed as Least Concern by IUCN Red List and Schedule IV by the Wildlife Protection Act (WPA).

A total of 19 species of butterflies were documented from all the 19 points surveyed during the fauna survey. Out of these 19 species, only one species was recorded from the Sewri Mudflats, namely, Small Salmon Arab (Colotis amata). The Small

Salmon Arab is found mostly in coastal regions, hence, in the project area, it was restricted to the Sewri Mudflats.

Fauna Analysis

Apart from the species name and number, the number of individuals of each species was also recorded during the fauna diversity survey. This data is useful in determining the distribution patterns of the species in the study area. To determine this, the relative frequency, relative density and relative abundance of all the species of birds and butterflies were calculated. As the habitats and diversity in Sewri Mudflats and the rest of the points differ greatly, the calculations were made for all points except Sewri Mudflats. Additionally, the Shannon Wiener Index was also calculated for each survey point.

Birds

Distribution Patterns

From the total 16 bird species observed in the 18 points of survey, the House Crow was the most frequent species in all the points. It is the most habituated bird species found in the urban habitats. The second most frequent species was the Common Myna, followed by House Sparrow and Blue Rock Pigeon. The relative frequency distribution of all species is given in Figure 25. The House Crow was also the most densely populated bird species in the entire project area, as it was present in large numbers everywhere. The second most densely populated bird species was Blue Rock Pigeon, followed by House Sparrow and Common Myna. The relative density distribution of all the bird species is given in Figure 26.

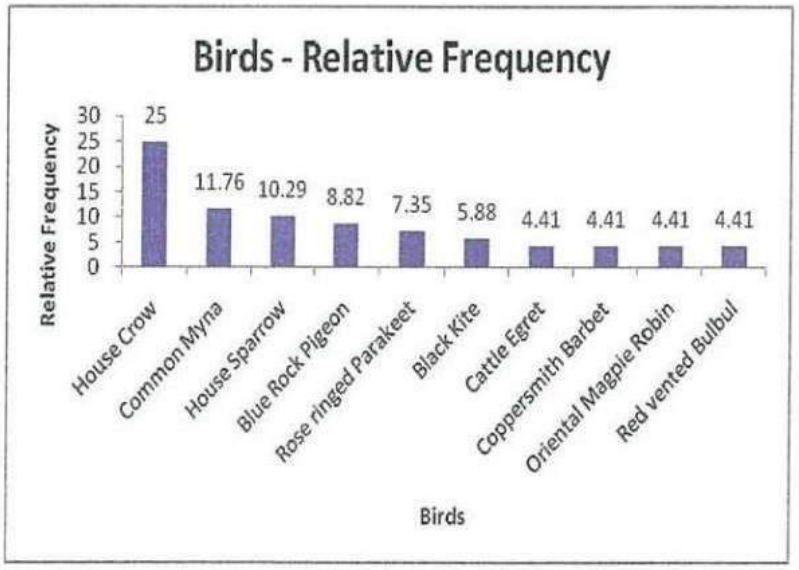


Figure 25 – Relative Frequencies of Birds

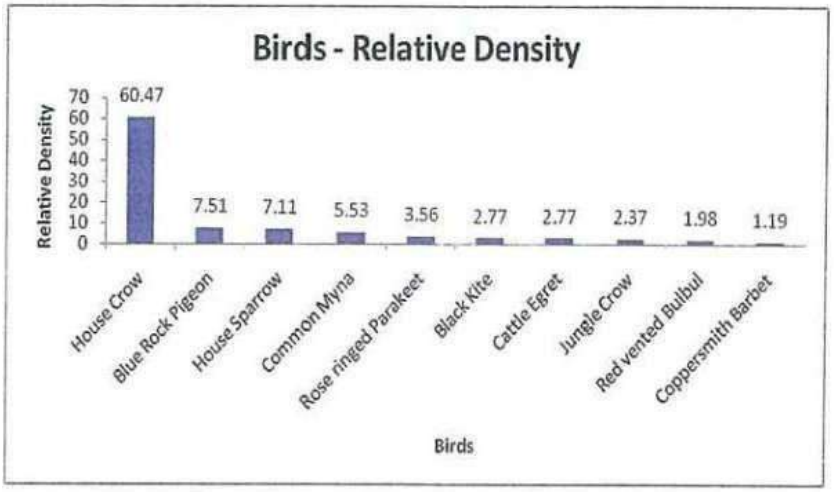


Figure 26 – Relative Densities of Birds

Apart from the most frequent and densely populated, House Crow is also the most abundant species in all of the 18 survey points. It was present in all the survey points in large numbers. The second most abundant species was the Jungle Crow, followed by Blue Rock Pigeon and House Sparrow. Figure 27 represents the relative abundances of all birds.

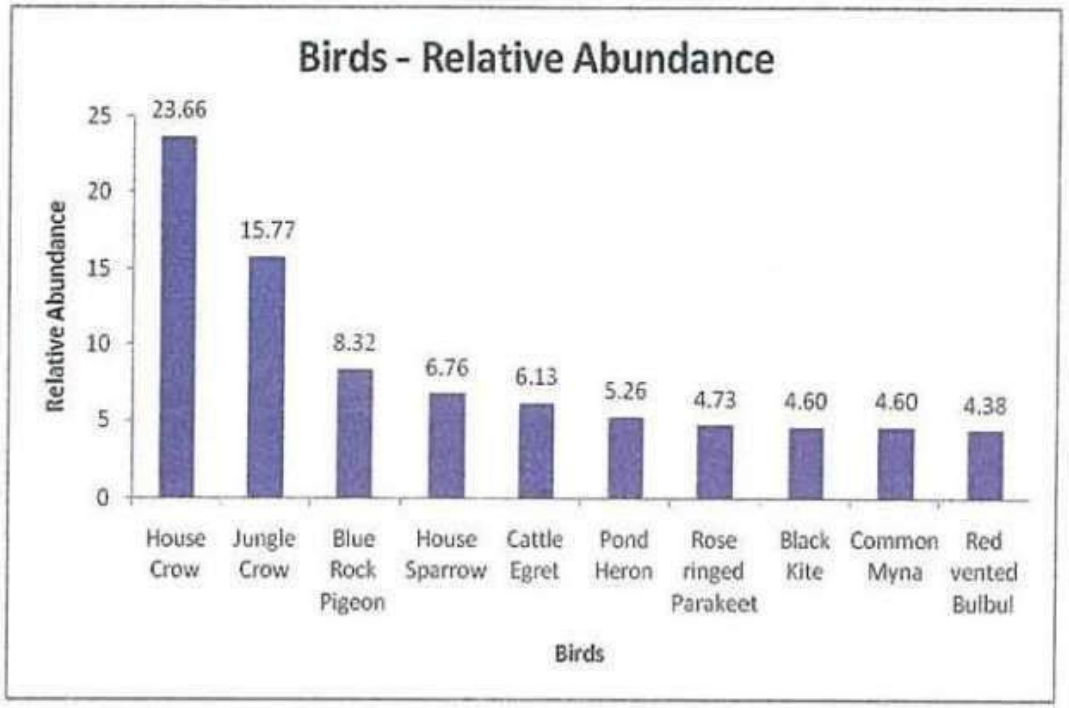


Figure 27 – Relative Abundance of Birds

Species Composition - Diversity Indices

A diversity index is a measure of species diversity in a community. In simple terms, a diversity index reflects upon how many different types of species are found in a given area and how they are distributed in the area. Thus, diversity indices help in determining the species composition of an area. The diversity indices can also be used to compare multiple areas to find out which areas are poor in species diversity and distribution. In our study, the Shannon Wiener Index of diversity was calculated from the data collected during the field survey and the species list is attached as Annexure XIII.

The Shannon Wiener Diversity Index takes into account both species diversity and species abundance, thus indicating the distribution of all species in an area.

According to the Shannon indices obtained from the calculations, Horniman Circle 1 is the most diverse region from all the points that were surveyed. Horniman Circle was a garden area which showed the presence of birds like Coppersmith Barbet, Common Tailor Bird, Red-vented Bulbul, Asian Koel, etc. The second most diverse region in the study area was Walkeshwar 1. This point was typically in an urban

habitat. The Shannon Wiener values of all the points surveyed are represented in Figure 28.

Figure 28 - Shannon Wiener Index values of all the points with respect to bird diversity

BUTTERFLIES

Distribution Patterns

Out of the 18 butterflies documented in all 18 points, the two most frequently observed butterfly species were Common Emigrant and Tailed Jay. Common Emigrant is found very commonly in cities, especially places that have lawns. As the study area consisted of many gardens, the frequency of Common Emigrant occurring in these areas is very high. Tailed Jay is also a relatively common butterfly species in the city. The second most frequent butterfly species in the entire project area was the Common Jay, followed by Common Mormon and Tawny Coster. The relative frequency of all the butterfly species documented in the study area is represented in Figure 29.

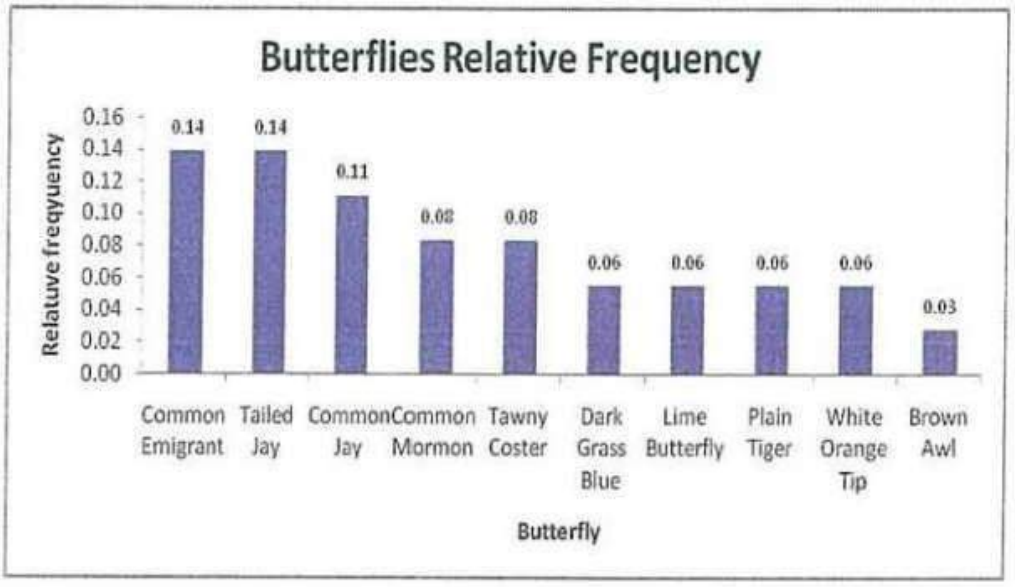


Figure 29 -Relative Frequencies of Butterflies

The most densely populated butterfly species in the entire study area was the Tailed Jay. The second most densely populated butterfly was Common Jay, followed by Common Emigrant and Tawny Coster. The relative density of all species is represented in Figure 30.

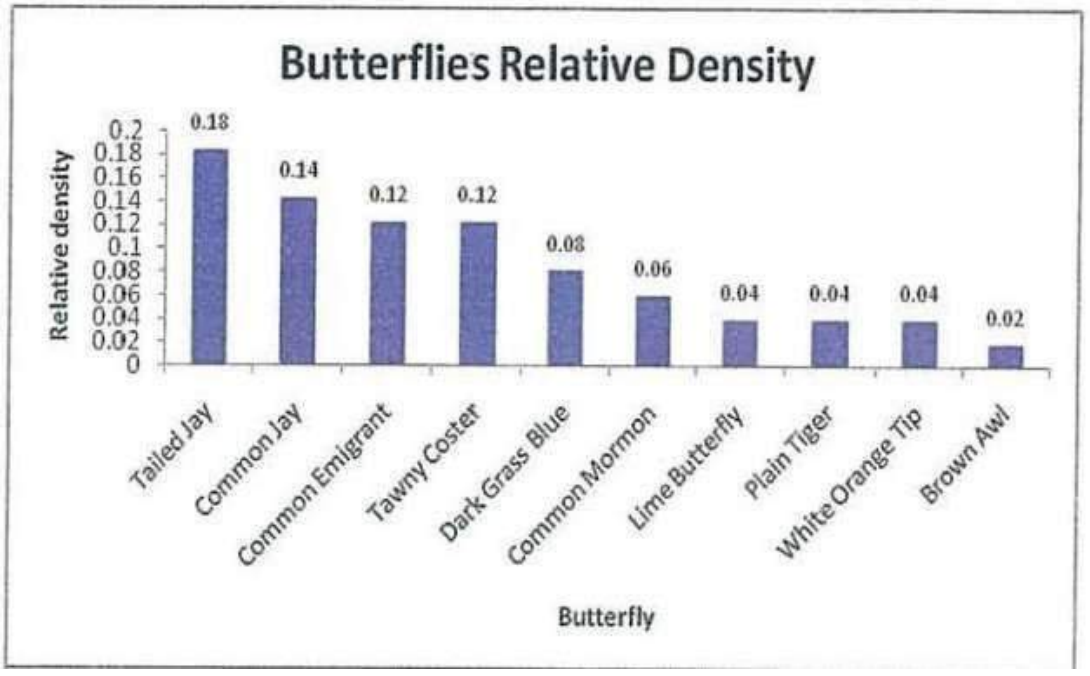


Figure 30 - Relative Densities of Butterflies

Dark Grass Blue was abundantly present in survey points with good lawn cover. Tawny Coster was also present in large numbers in various survey points. The relative abundance of all species is represented in Figure 31.

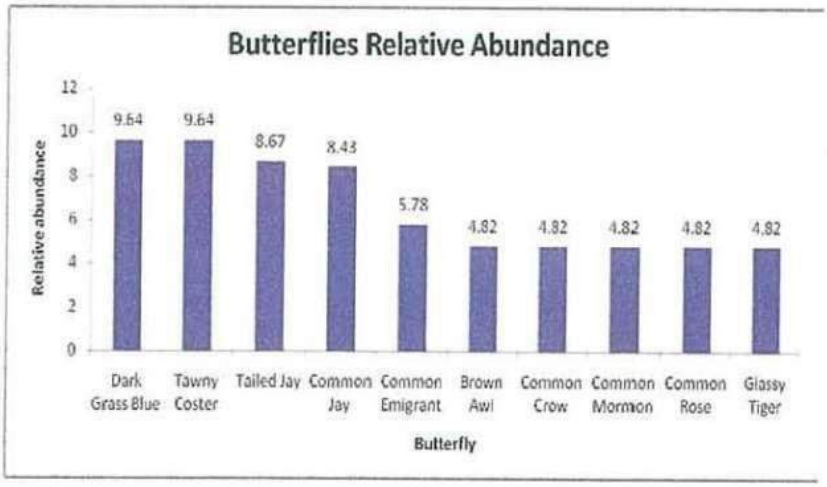


Figure 31 – Relative abundances of butterflies

Species Composition - Diversity Indices

The Shannon Wiener Index of the all the survey points were calculated to determine the most diverse and least diverse region in the study area. The most diverse region in the entire study area was the point Hanging Garden 1. The Hanging Garden has a lot of plants that butterflies use for nectar. Thus, the number of butterfly species observed in this area was very high. Apart from the number of species, the number of individuals of species in this region was also high. Nehru Science Centre 1 was the second most diverse region in the study area. Figure 32 shows identification of butterflies.

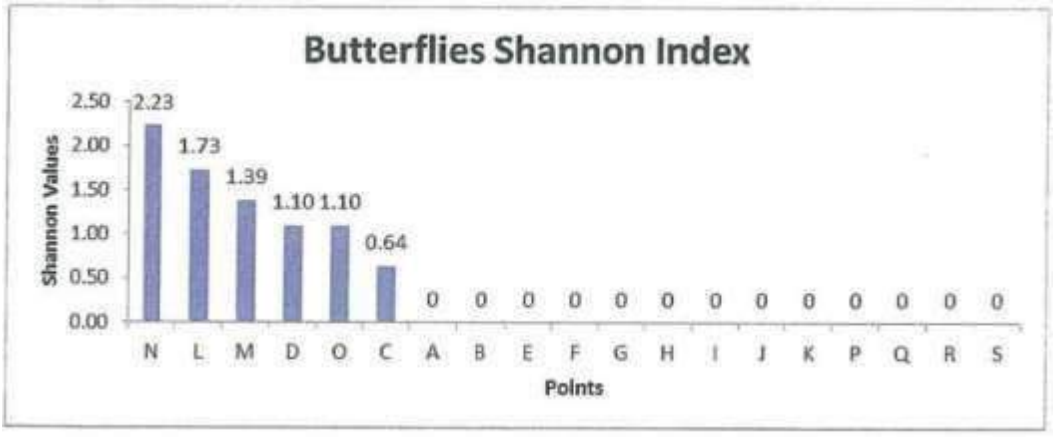


Figure 32 – Shanon Wiener Index of all points with respect to Butterfly diversity

Mammals and Herpetofauna

In mammals, two species were recorded, namely the Indian Palm Squirrel (Funambulus palmarum) and Indian Flying Fox (Pteropus giganteus). The Indian Palm Squirrel was present in many of the survey points in good numbers. A roosting colony of Indian Flying Fox was observed in Jijamata Udyan.

Only one species of reptile was documented in the entire field survey, i.e., the Indian Garden Lizard (Calotes versicolor). It was found in very few regions of the study area.

All the mammal and reptiles species documented in the study area are listed as Least Concern in the IUCN Red List of Threatened Species. The Indian Flying Fox is listed under Schedule V by Wildlife Protection Act and Appendix 2 by CITES. The Indian Garden Lizard is also listed under Appendix 2 by CITES. Figure 33 shows sampling points with maximum bird and butterfly diversity.

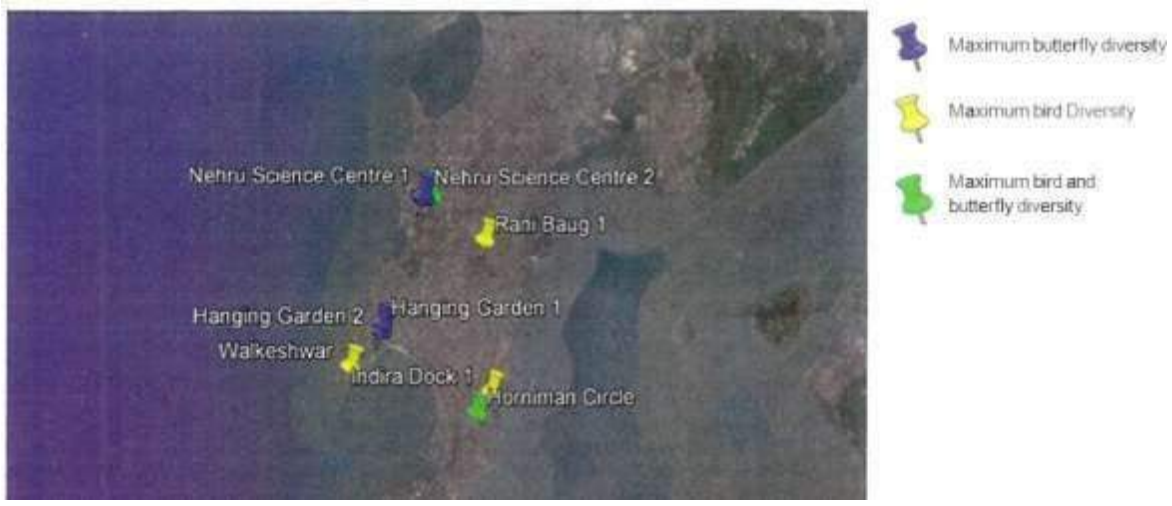


Figure 33 - sampling points with maximum bird and butterfly diversity

CHAPTER 04 - ANTICIPATED ENVIRONMENTAL IMPACTS & MITIGATION MEASURES

4.1 INTRODUCTION

This chapter deals with the assessment of project impacts on environment. Mitigative measures are suggested to minimize the likely negative impacts. An environmental management plan is also suggested along with an estimate of environmental costs as an input for evaluation the economic feasibility of the project.

The project will have impacts of varying magnitude on different environmental components. These impacts could be categorized as-

Primary impacts, i.e. impacts which occur as a direct result of the project activities
Secondary and tertiary impacts, i.e. impacts that occur as a result of primary impacts.

Impacts could occur during the construction phase as well as during the operational phase. Impacts during these phases are discussed separately in this chapter.

Significant Environmental Impacts and Mitigative Measures
In view of the above study, we will sub divide the key environmental factors into 3 groups:-

- In respect of existing status
- In respect of construction phase and
- In respect of operation phase

The type and magnitude of the impacts is entirely site specific. In order to logically analyze situation on a probable scale, following standards have been devised for the project under consideration to quantify the impact:

0 = No Impact
- 1 = Negligible
- 2 = Mild
- 3 = Moderate

- 4 = Significant
- 5 = Severe

Utility of the above noted qualitative scale is that it can be used as a method to approximately indicate varying order of caution while dealing with different stretches of the project.

Each of the negative impacts on the environment requires consideration of mitigative measures. Some of these measures require judicious application of road engineering design and construction methodology while others require special techniques. An attempt has been made to indicate the required mitigative measures for each type of identified negative impact.

4.2 IMPACTS FROM THE PROJECT LOCATION

Potential Impact due to Port Location

- Construction related activities may involve changes in land terrain like dredging, reclamation of areas for creation of huge extant of port infrastructure like operational areas, storages, roads, railway lines etc.
- Coastline changes like erosion or accretion may be expected due to the establishment of berths/terminals.
- Transportation of quantities of construction material for the berths, operational, administrative and welfare buildings, land filling/development, formation of storage and stacking yards etc.,
- During operations phase of the port may result in excessive use of existing public infrastructure like roads, railways and in-land waterways etc., resulting in congestion and early ageing etc.
- Public utilities such as water supply, drainage, electrical power etc. may also get undue demand.

Mitigation measures

- Careful site selection and port design should be carried out, focusing on the possibility of limiting the extent of land area requirement.

- Coast line changes like erosion or accretion due to interference to littoral process, likely to have serious consequences are to be assessed by proper techniques sometimes involving mathematical/physical model studies to ascertain remedial measures such as shore protection works, sand by passing etc.
- Drainages, roads and rail connectivity should suitably designed and implemented to overcome affects of changes in land terrain.
- Existing network of road and rail infrastructure will be improved by suitable expansion including development of bye passes as part of project to avoid congestion of existing road rail net works and resulting inconvenience to the public.
- Utilities like water supply, surface drainage including storm water, sewage treatment/disposal, waste management/disposal, and electrical power supply shall be augmented or developed as the case may be part of the port project.

4.3 IMPORTANT ATTRIBUTES DURING CONSTRUCTION PHASE

The major activities during the construction phase of the project will be:

- Transportation of construction material and wastes.
- Construction activities for the proposed project.
- Dredging & Reclamation activities
- Construction of berthing facility, ship parking facility.
- Ship lifts stations development.
- Storage of materials and equipments.
- Construction of support infrastructure such as offices, toilets, etc.
- Construction of internal roads, and other infrastructure.
- Operational construction equipment.
- Runoff from construction debris and sanitary wastes
- Disposal of liquid and solid wastes generated during construction.

Impacts of these activities on various components of the environment are discussed in the following subsections. It must be noted that most of the impacts during the construction phase will be temporary, and will not be felt at the end of the

construction phase. However, proper mitigative measures are provided for the minimization of the impacts caused by these activities which we shall see further in detail.

POSITIVE IMPACTS

Some of the potential positive impacts expected from the project are:

- Significant improvement in coastal trade along with the development of nearby ports.
- Development of ancillary facilities along with the ship/ barge resting facilities.
- Opportunities for locals for better employment.
- No additional burden on existing water and power supply.
- Commercial development of the locality

ADVERSE IMPACTS

Assessment has been made of the adverse impacts due to the project. These could be during construction and / or operation stages of the project. The details of the measures to be adopted including mitigative measures have been mentioned in this chapter.

4.4 IMPACTS DURING CONSTRUCTION PHASE

Air Quality

During construction phase the loading un-loading of raw-materials, dredging, reclamation, demolisiton of exsitng structure, plying of construction vehicles, operation of construction equipments & other construction activities can have impact ambient air quality on account of emissions of dust generation, vehicular exhaust/emissions et. Impact also includes accidental leakage of emissions, exposures, fumes, odors, hazardous airborne emissions vehicles, construction. However this will be a temporary and short duration, reversible impact. Moreover, proper mitigative measures shall be followed during the construction phase.

Baseline ambient air quality was monitored at 8 locations within study area. The results are presented in the Chapter 3.0. The air quality of the area seems to be within the standard limits.

Mitigation measures

Mitigation measures to be proposed during the construction stage include suitable dust suppression techniques.

- Vehicles will not be loaded above the height of the side and tailboards. Vehicular emission will be kept to minimum by avoiding unnecessary engine running time.
- Restriction of vehicle movements and vehicle speeds to reduce dust emissions.
- General housekeeping practices and regular maintenance of equipment and machinery will be done.
- Sealing of unsealed roads, exposed surfaces, active construction areas and stockpiles (and water sprinkling if required).
- No construction material except piles and other support structures will be permitted into waterway during construction phase.
- Wind sheltering with help of barriers shall be installed during the stock piling activity. The raw-material storage shall be temporarily covered as far as possible with adequate covering material such as plastic, tarps etc.
- Green belt development as well as institutional arrangements proposed with other agencies for effective implementation of environmental measures, applicable environmental standards and compliance.
- This vegetation cover will also act as a barrier for any penetration of air quality and odor in the nearby area.
- Approach roads will be covered with green belt on both the sides to avoid any air quality problems to the nearby residents.
- Road Furniture/Signboards will be put along the approach roads and at project Site requesting motorists to avoid idling or/and stoppage of the vehicles at non-designated places.

Noise Quality

Once the construction phase is over noise quality will be further improved and will be limited only at certain points within project area. There will not be any noise pollution on the approach roads due to decongestion. The approach road is of high quality and broad, which will not have impact on increase in noise levels.

Construction activities may create a problem of noise and vibration generated by construction equipment, truck traffic, work vessels and other similar sources. The principle source of noise along the project area is expected from vehicles. Since there is smooth movement without any obstruction and proper dispersion, there will not be any increase in the noise levels. However, noise levels on the existing roads are expected to be reduced considerably.

Mitigation measures

- All construction equipment shall be fitted with exhaust silencer. Damaged silencer to be promptly replaced by the contractor.
- Ready mix concrete will be used for any concrete work so that batching plant etc. will not be installed at site. This will totally reduce noise generation from this important construction machinery.
- Proper maintenance of equipment shall be undertaken with the provision of enclosures and intake silencers.
- DG sets, if used, shall adhere to the noise and air emission standards of MoEF & CC.
- Construction works will be undertaken in the agreed working hours. Construction activities involving generation of high noise shall be avoided between 10 pm and 6 am in the residential and sensitive areas.
- Personal protective equipments will be given to the workers working in the areas of high noise generation.
- Continuous Noise monitoring will be carried out during operational phase to collect comparative data.
- Transmission of noise and vibration are limited by the distance from their sources. Noise could be considerably reduced by adoption of low noise equipment or installation of sound insulation fences. Green belt shall also attenuate the noise levels.
- For the pilling activity hydraulic or manual pilling equipment will be put to use.

Water Quality

- Sea water quality will be affected during the construction, dredging & reclaimation activities.

- Disposal of dredge spoils removed during capital dredging.
- Site specific pollution of sea water due to seepage of oil/grease from construction equipments.
- Heavy metal pollution due to entry of various metal pieces emanating from equipments/ chemical etc. during construction phase.

Mitigation measures

- All diesel and motor oil stored on the site will be stored in bunded area away from the shore. Signs will be posted indicating the storage of these substances.
- Navigational aids such as channel beacons, buoys will be provided during construction phase.
- Adequate Storm water drainage will be provided all along the project area. Oil Traps will be installed in the Storm water drainage in order to remove any oil & grease matter caused due to run-off. The surface run off shall be made to pass through Oil and grease cum sedimentation chamber.
- Intercepting drains around the site of construction and designated places for the machines where refueling and change of lubricants shall be carried out.
- It is proposed to use pre-cast material wherever possible so as to avoid any mixing of run-off water with storm water drain during construction.
- Proper Construction methodologies will be adopted in order to avoid turbidity problem.
- Use of environmentally friendly construction technologies and international standard construction methods.
- To reduce impact of dredging on aquatic life silt curtain shall be erected during dredging process to control the suspension of sediment in limited area.
- Sewage generated from labour camp will be conventionally treated in Sewage Treatment Plant and therefore avoid mixing with surrounding environment.
- Water used for hydraulic testing will not be allowed to mix with any of the water body and will be disposed off through drainage system of local authority.

- No impacts on ground water quality and its availability are anticipated as a result of the proposed actions at the sites.
- Disposal of capital dredge taking into account bathymetry, hydrodynamics and model studies to ensure maximum dispersal of sediments.

Solid Waste Generation

The solid wastes will be generated in the form of during construction activities. There will be certain impacts on land environment due to the proposed developmental activity. During Construction phase slight impact on the land environment is envisaged because of constructional activity and dredge material disposal. Baseline sediment analysis for the project has been incorporated in the reports, which shows adequate amounts of nutrients in the soil quality. Greenbelt development within and around the project site will help in improving the ecology and aesthetic value of the area. The planted trees will absorb specific air pollutants, reduce noise pollution, control soil temperature, help for holding moisture in soil, attract more birds and will help in maintaining the overall environmental quality.

Mitigation measures

- Proper solid waste disposal management will be done during the construction phase. Color coded containers will be provided for solid waste collection.
- All the construction waste debris will be collected and segregated at site before disposal. The re-usable construction solid waste shall be reused for land filling and reclaiming activities, rest shall be disposed off as per existing local norms.
- Oil and fuel will be stored in separate bunkers.
- Use of environmental friendly degreasers and detergents will be promoted.
- Proper maintenance of the construction equipments will be done to avoid spillage or leakage of oil or other greasy material on the land environment.

Reserved Forest and Fauna

The project area does not fall under any of the reserved forest area. There are no endangered species of flora and fauna within the project area, neither any endemic

flora nor fauna species are found in the adjoining area. Wild life activity is also absent in the area.

Mitigation Measures

During construction all the care will be taken to preserve existing flora and fauna. The loss, if any of these will be compensated as per the standards norms and practice.

Land Use

The proposed terminals are within the Mumbai Port.

Mitigation Measures

- Development of any land will be in accordance with the government regulations.
- Proper navigational channels will be provided during construction phase and dredging activity.
- Reclamation bunds shall be constructed. Approach bund and temporary road will be constructed for reclamation purpose.
- Adequate ground treatment is proposed after reclaimed land.

Marine Environment and Biodiversity

The marine environment and biodiversity includes the fishes, benthic flora-fauna and other marine organisms which may be affected by the proposed project activity. Apart from turbidity marine water quality can be affected due to aqueous discharges (oily wastes, sanitary wastes) from the dredgers, barges and workboats involved in the activity.

Mitigation Measures

- During construction period, suitable barrier will be used to protect the adjoining water bodies from the falling earth materials and dust raised to avoid sedimentation.

- Removal of contaminated sediments, capping, as well as periodical beach nourishment could be effective measures against adverse effects on water quality.
- It is proposed to use pre-cast material wherever possible so as to avoid any mixing of run-off water with storm water drain during construction
- Proper construction methodologies will be adopted in order to avoid turbidity problem.
- Care will be taken to avoid layer of suspended solids in the air during transportation.
- No run-off water during construction phase will go outside the project area and will not enter in to the surrounding environment.
- Construction technologies to be adopted in such away to minimize effect on terrestrial ecosystem.
- Care will be taken to minimize any damage to terrestrial flora – fauna, if any and to restore ecological balance.
- Pollution prevention plans will be evolved and submitted to responsible authorities to minimized/prevent pollution at source.

4.5 IMPACTS DURING OPERATION PHASE

4.5.1 Positive impacts of the proposed project

Any developmental project related to infrastructure and community development would have its own impacts on various parameters. These impacts, besides moderate on environment during construction phase, may also have sustainable positive impacts. The said project certainly has certain positive impacts on socio- economy as well as environmental factors. The social impact will include additional employment and commercialization of the area. The project will also enhance aesthetic look of the area.

4.5.2 Air Quality

The proposed Terminal as it is will not involve any major impact on air quality of the area; however various activities related to proposed facility may cause negligible impact on air quality of the area. Ambient air quality at all the sites was monitored for baseline conditions. The results are already presented in the report. The proposed

project operations such as loading, unloading of ships, ship fuelling activities and ancillary activities shall generate dust emissions. Particulate emissions will be generated from loading and unloading operations. Moving of vehicles shall also emit small amount of vehicular pollution.

Impacts

- Fugitive emissions from the cargo loading and unloading activities.
- The major activities envisaged in the proposed project emitting dust emission from cargo loading and unloading activity.
- Emissions from DG sets, navigational equipment/machinery.
- During operation phase over-fueling or careless fuelling practices may also lead to impacts on air quality of the project site area.

Mitigation measures

- Regular monitoring of air quality as mentioned in EMP will enable to keep the air quality below permissible standards
- Proper sprinkling of water during loading unloading activity as well as proper house-keeping shall be maintained so as to reduce the fugitive dust emission.
- The exhaust emissions from ships/vessels shall be controlled on the ship itself by providing long stacks with air emission control option such as sea-water scrubbing and fuel substitution.
- Day to day management and maintenance of the facility and the ancillary structures.
- D.G Set will be used only in case of emergency with proper enclosures in order to reduce the impact of air emissions.
- Control and check will be kept on careless fuelling of the ships if any and excessive trafficking of the ships at one place.
- This vegetation cover will also act as a barrier for any penetration of air quality and odour in the nearby area.
- Approach roads will be covered with green belt on both the sides to avoid any air quality problems to the nearby residents.

- Road Furniture / Signboards will be put along the approach roads and at project building requesting motorists to avoid idling or/and stoppage of the vehicles at non-designated places.

4.5.3 Noise quality

Major noise generating sources are loading and unloading of materials abrasive blasting, chemical paint removals, mechanical paint removals, metal working, equipment handling and vehicular movement.

Mitigation measures

- Noise barriers in terms of vegetation cover wherever required will be used for attenuation of noise.
- Signboards will be put along the approach roads and at terminal building requesting motorists to avoid unusual use of horns and also for avoiding idling noise.
- Major Noise generating equipment will be designed such that the cumulative noise at a distance of 1 m remains 80 dB(A), by giving proper acoustic enclosures to such noise generating equipments and machinery.
- The occupational noise exposures to the workers in the form of 8 hourly time weighted average will be maintained within the prescribed OSHA standard limits.
- Workers exposed to excessive noise will be given appropriate PPE including ear plugs, muffs.
- Continuous noise monitoring will be carried out during operational phase to collect comparative data.
- Proper maintenance of the equipments in terms of noise generation will carried out.

4.5.4 Water Quality

- Impacts Due to Disposal of Maintenance Dredge Spoil.
- The proposed project activities such as ship washing and surface run-off during operation phase shall impact on the water quality of the area. The effluent ship parking facility and sewage water from the onsite support

facilities shall impact on the water quality of the area. Proper mitigative measures are proposed for the treatment and disposal of waste water generated from the ship multipurpose terminal and berthing facility.

Mitigation measures

- Bathymetry at site offers possibility to achieve attenuation levels in the shortest possible time.
- Discharge depths of approximately 7 m below the surface of the water reduces the impact of SS on the top water column.
- Locations identified also have low bio-productivity.
- Provision of a specially equipped pump truck to receive sewage from ships and transport it to the treatment plan.
- The sewage will be treated in the Sewage treatment plant and the treated water shall be utilized in landscaping and dust suppression, whereas the sludge shall be used as manure.
- The sewage from the ships will also be treated within the ships and hence there will be no risk of contamination of surface or groundwater as a result of the effluent or waste discharge from the ships when within the port area.
- Ships will not be allowed to release any oily bilge waste or ballast water in the sea within port limits and hence wastes from the ships will also not affect any surface or groundwater.
- Due care will be taken such that no water pollution/ contamination occurs during loading and unloading activities.
- Aqueous discharges from vessels such as dumping of ship waste (sullage/ sewage), oil contaminated bilge water, holding cleaning and tank cleaning residues (slop), and spillages upon re-fueling and lubricating oil changes and disposal of solid wastes will be regulated to minimize impact on marine water and sediment quantity.
- The runoff from the slipway, berths, etc will be collected and treated into the effluent treatment plant.

- Proper storm water drainage system with adequate oil & grease traps shall be incorporated in order to remove any spillage/ leakage/ runoff of oil and other wastes in sea water.
- During construction period, suitable barrier will be used to protect the adjoining water bodies from the falling earth materials and dust raised to avoid sedimentation.
- Removal of contaminated sediments, capping, as well as periodical beach nourishment could be effective measures against adverse effects on water quality.
- During Operation Phase, Water used for hydraulic testing will not be allowed to mix with the any of the water body and will be disposed off through drainage system of local authority.
- Alternatives for reducing waste include oil/water separation and waste oil reclamation.

4.5.5 Solid Waste Management

Since the proposed project deals with ship loading, unloading and berthing facility there will not be any major hazardous waste generation on site. The solid waste generated on the facility will be collected, segregated. Recycle bale waste will such as paper, glass metal etc. will be sold to authorise vendor and remaining waste will be disposed off to the local municipal authority. Major issues like oil spillage and refuelling will be handled carefully and proper solid/hazardous waste management will be implemented.

Mitigation measures

- Oil from oily wastes will be sent for recovery and reuse.
- Solid waste will be segregated into recyclable, degradable and non- degradable at site itself. The recyclable waste will be either reused or given to authorize end-users, whereas remaining solid waste shall be handled to local municipal authority.

4.5.6 Impacts on ecological resources and flora and fauna

There will be no significant impact of the project on flora and fauna and other vegetation due to any of the pollution parameters during operational phase of the project. No significant impact is envisaged on the local terrestrial flora fauna during the operational phase of the project. Since there are no reserved forests or forest area in the project area, there will not be any violation of flora and fauna.

Negligible Impact on marine flora fauna is envisaged due to navigation of ships, loading unloading of cargos, run-off water from the refueling stations etc. However, proper mitigative measures will be taken to minimize the impacts due to the operation of the proposed project. Green belt development will help in attenuating any pollution caused during the operation of the proposed facility.

Mitigation measures

- Criss-cross drainage system installed with oil & grease traps, channelized into main storm water drainage will be incorporated in so as to ensure all the run-off water is treated and channelized into storm water drainage leaving no run-off water going directly into sea.
- The route of navigation of ships will be designed such that the fishery in the nearby area will not be affected.
- The sewage from the ships will also be treated within the ships and hence there will be no risk of contamination of surface or groundwater as a result of the effluent or waste discharge from the ships when within the port area.
- Ships will not be allowed to release any oily bilge waste or ballast water in the sea within port limits and hence wastes from the ships will also not affect any surface or groundwater.
- Aqueous discharges from vessels such as dumping of ship waste (sullage/ sewage), Oil contaminated bilge water, holding cleaning and tank cleaning residues(slop), and spillages upon re-fueling and lubricating oil changes and disposal of solid wastes will be regulated to minimize impact on marine water and sediment quantity.

4.5.7 Land use

The land use pattern, which will undergo slight changes, will not be further altered during operational phase. The land use covered under green belt will be maintained during operational phase. The commercial activities may increase on the land ward side with proper layout and permission from local authorities. The aesthetic look of the said land use will be enhanced by green belt & other signage with proper architectural plans.

Mitigation measures

A green strip of local trees on both sides of the road will be planted on the approach roads.

4.6 IMPACTS ON QUALITY OF LIFE

4.6.1 Socio-Economic Condition

The socio-economic scenario in the region will certainly change with positive impact on the existing regional socio-economic pattern. There will be change in employment pattern with local residents will be given preference for jobs opportunities and/or self-employment. The economic growth will have positive impact; it will also help in increase in living standards of the local residents. Due to enhancement in infrastructure facilities and utilities in living condition will also improve. It will have positive effect on industrial sector in the area.

4.6.2 Accident hazards and safety

During operational phase accidents / hazards will be greatly minimized and ensure further safety of the local people. In addition, by adopting mitigation measures will ensure an efficient traffic system with high standards of safety. Onsite and Offsite Emergency Preparedness Plan as well as Disaster Management Plan have been sketched out and shall be implemented. The Risk analysis for the proposed project is also carried out.

Table 20 - Environmental Impacts and Mitigation Measures

Construction Phase

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
1	Physiography	Disturbance in relief feature	Mild	Will be achieved by systematic planning and designing of the project activities.	Promoter, Client, etc.
2	Land resources	No Change in land use	Neglisible	Will be achieved by systematic planning and implementation.	Promoter, Client, etc.
3	Human resources	No adverse impact	Negligible	Will be achieved by systematic planning and resources.	Promoter, Client, etc.
4	Ecological resources – Flora & Fauna	No impact anticipated to threatened or endangered plant species. Negligible impact on marine species which will be mitigated.	Negligible	Cutting of larger girth size trees are avoided by suitably adjusting the road alignment, if required. Greenbelt shall be developed around the site.	Promoter, Forestry Dept. involving NGOs and local people.

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
5	Environmental aesthetics values	No cutting of trees	Negligible	Loss of vegetation will be kept minimum as far as possible during site clearance. In case of any loss the same will be compensated by rehabilitation and restoration of the mangrove species that shall be affected.	Promoter through prospective contractor
6	Sub-surface hydrology	No wells and hand pumps are existing in project area	Negligible	Whenever possible, care is taken to avoid its relocation by judicious engineering road design. Temporary alternative water sources will be provided in case drinking water means are affected.	Promoter through prospective contractor
7	Religious places	-----	----	----------	-------
8	Geology	Not much affected	Negligible	Systematic planning and implementation during the construction and dredging. Reclamation will be done with systemic planning and least disturbance to the	Promoter through prospective contractor

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
				natural geology.	
9	Surface of water	Contamination from solid wastes and sewage generated due to construction labour camp Surface runoff from the ship terminal and parking facility.	Mild	Installations of lavatory for construction workers at a minimum distance of 200 m from water bodies and provision for composting the domestic refuse.	Promoter through prospective contractor
10	Air quality	Short-term deterioration of air quality due to generation of fugitive dust. Dredging activities and other	Moderate	Trucks carrying soil, sand, stone, will be covered to avoid spilling. Fugitive dust sources will be sprayed with water to suppress dust. Emissions from vehicles & machinery will be checked regularly & maintained properly to confirm to National and State Emission Standards Barriers during	Promoter through Prospective Contractor (PC)

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
		construction activities leading to fugitive emission.		construction activities such as dredging will be installed.	
11	Noise level	Increased noise levels due to project activities	Mild	All the equipments will be duly lubricated, maintained in good working condition to minimize noise levels. Stationary construction equipments will be placed as far as possible from dense habitation. Green belt barrier. Provision of protection devices (ear plugs) to be provided to the workers operating in the vicinity of high noise generating machineries.	Promoter through Prospective Contractor (PC)

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
12	Ecological resources – Flora & Fauna	No endangered species found	Negligible	Proper care will be taken to maintain eco-balance.	Promoter through Prospective Contractor (PC)
13	Land use	Land acquisition is as per the government regulation and policy. Mild impacts on local land use may be visualized	Mild	Proper management planning will be achieved.	Promoter through Prospective Contractor (PC)
14	Construction workers camp	Impacts on community health	Mild	Supply of safe drinking water to the construction camp. Provision of adequate drainage system to avoid undesirable water logging. Provision of hygienic facilities to	Promoter through prospective contractor (PC)

Sr. No.	Environmental Parameters	Impact Attributes	Degree of Impacts	Mitigation Measures	Implementing Organization
				construction workers camp is made. A system of regular disposal of domestic waste & sewage.	
15	Accident hazards and safety	Short term impacts from road accidents. Impacts from accidents during handling and use of Construction machinery.	Mild	Proper traffic diversion and management during construction. Construction Safety measures will be employed. Proper warning signs will be used at construction site.	Promoter through prospective contractor (PC)

Table 21 Environmental Impacts and Mitigation Measures

Operation Phase

Sr. No.	Project Related Issues	Actions to be Taken	Responsible Organisation
1	Prevention of Road side Squatters or indirect Urban Sprawls	Involve land use planning agencies like the Revenue Department at all levels during operation stage. Plan and control development activity. Removal, cleaning of squatter and temporary hutments of construction workers once construction activities has been completed.	P & C in consultation with the Municipal corporation.
2	Road Safety and Traffic Management	Adequate number of proper & legible signs will be installed along the road. Prepare and administer a monitoring system on road/ accidents.	P & C in consultation with Public Works Department and State Traffic Police.
3	Air Quality	Monitor periodically ambient air quality at selected sites. Confinement and absorption of the pollutants at source by creating vegetation along the length. Enforcing different control measures to check pollution (e.g. catalytic converters, unleaded petrol, proper serving etc.)	P & C in consultation with MPCB

Sr. No.	Project Related Issues	Actions to be Taken	Responsible Organisation
4	Noise level	Monitor periodically ambient noise level at selected sites. Minimization of use of horns near sensitive locations/ silence zones with the help of sign boards at proper places. Provide noise barriers with roadside plantation.	P & C in consultation with MPCB
5	Water Quality	Monitor periodically water quality for establishing the change of water quality, if any, and assessing its potentiality of surviving aquatic flora and fauna and for irrigation use.	P & C and MPCB
6	Soil Characteristics	Periodic monitoring of soil quality at specified distance for assessing contamination by vehicular emissions. Checking the overflow of spillage from the carriageway by promoting growth of vegetation cover along the road shoulders and preventing overflow to green belt.	P & C in consultation with Public Works Department (PWD).
7	Maintenance of Avenue trees	Plantation will be undertaken by the concession company on an aggressive note along the whole stretches on the both sides of the road. Restoration of Mangroves if any will be done adequately with proper technology.	P & C in consultation with authorities and State forest Department

Sr. No.	Project Related Issues	Actions to be Taken	Responsible Organisation
8	Human Health and Safety	Vulnerable stretches, which are prone to accidents, will be identified. Adopt Safety measures and other control measure during Operation of the facility. Installing proper road signs, marking along the whole stretch of the tolled highway in the form of cautioning, informatory and mandatory signs of gantry mounted overhead sizes. Installing fire safety measures, electrical safety measures, Personal protective Equipments and other work-safety measures. Incorporation of On-site Emergency Preparedness, Off-site Emergency Plan, Disaster Management Plan	P & C in consultation with MPCB. State Public Health Works Department (PWD)

4.7 EVALUATION OF IMPACTS

The Environment impact of the project has been discussed in this chapter and the potential of the impact is mainly under the construction and operational phase. The type and magnitude of the impact is entirely site specific.

Impacts from Project Location

Project location plays an important role in prevention of adverse impacts and to minimize the mitigative measures. These measures can be classified as follows: -

- Through Engineering Design
- Through project Scheduling
- Through Tree Planting
- Through Rehabilitation & Resettlement Planning or through property constructed and maintained labour camps.
- Through post construction by providing facilities other Government, Departments and agencies to watch, monitor, enforce environment standards. All this is presented in individuals tables

Evaluation of impacts

Evaluation of impacts has been done for following issues namely: -

- Environmental Impact due to project location
- Environmental Impact from construction camps
- Environmental Impact from road construction phase
- Environmental Impact from Operation phase.

These impacts have been evaluated and are enumerated in following Table 22

Table 22 - List of Possible Environment Impacts due to Proposed Project

Sr. No.	Attributes	Proposed Development Impact Marks	Mitigative measures
1	Project Location	0	Change in land use pattern
2	Soil Slippage	0	Proper soil engineering and foundation designs and structural protection
3	Rock Quarrying	0	Material will be obtained from authorized agencies
4	Air quality	2	Provision of adequate monitoring during operational phase, no changes in this line during construction period due to proper mitigative measures.
5	Water Quality	2	Sewage Treatment Plant (STP) and Effluent treatment plant (ETP) during operational phase.
6	Noise Quality	2	Proper noise control management plan during construction period, noise barriers in terms of thick vegetation proposed.
7	Land use	2	The land use pattern shall change after the proposed project.
8	Reduction in built habitation of structures	1	Not Applicable
9	Displacement of population	0	No encroachments or settlements.
10	Heritage / Archaeology	0	No such structures within the project area.
11	Socio - economy	1	Better employment opportunities, improved economic status, increase in commercial and business opportunity.
12	Loss of Environmental Aesthetics	0	Proposed project will cause changes in the existing Land use of the site, however, enhancement in the environmental settings envisaged.

Note: The total negative impact is only 14 where severe most could have been 5 x 12 = 60. So negative impact is 16% to positive impact is 84%.As the total positive impact of the project is 77, therefore, it can be concluded that this is an environment friendly project.

CHAPTER 05 - ENVIRONMENTAL MONITORING PROGRAMME

5.1 THE NEED

Monitoring is an essential component for sustainability of any developmental project. It is an integral part of any environmental assessment process. Any development project introduces complex inter-relationships in the project area between people, various natural resources, biota and the many developing Forces. Thus, a new environment is created. It is very difficult to predict with complete certainty the exact post-project environmental scenario; hence, monitoring of critical parameters is essential in the post-project phase.

Monitoring of environmental indicators signal potential problems and facilitate timely prompt implementation of effective remedial measures. It will also allow for validation of the assumptions and assessments made in the present study. Monitoring becomes essential to ensure that the mitigation measures planned for environmental protection function effectively during the entire period of projects Operation. The data so generated also serves as a data bank for prediction of post-project scenarios in similar projects.

Environmental monitoring during the construction phase shall comprise checking:
Appropriate permits, certificates, authorizations and
Compliance with the EMP and governmental regulations

This can be ensured through use of checklists for:

- Site Establishment.
- Monthly Audit.
- Site Closure.
- Environmental Management Plan implementation monitoring during the construction phase.

Monitoring Checklists are given below:

Table 23 - Project Start-Up Checklists

ENVIRONMENTAL ASPECTS	YES/NO	COMMENTS
Personnel on site are environmental aware of various issues of interest		
Telephone numbers of emergency services are available on site		
Solid waste management system has been established at both construction site and labor camp		
Wastewater management system has been establish at both construction site and labor camp		
Necessary fire fighting equipment is available and in good working order.		

Weekly Checklists

ENVIRONMENTAL ASPECTS	YES/NO	COMMENTS
Construction camp is neat and tidy and the laborers facilities are of the acceptable standard.		
Waste collection and removal system is being monitored.		
Sufficient fire fighting equipment is available at the construction site and is in good working order.		
All construction vehicles are in good working order and have a valid PUC certificates.		
Dust control measures (wherever necessary) are in place and are in working efficiently.		
Noise control measures (wherever necessary) are in place and are effective in controlling erosion.		
Erosion control measures (wherever necessary) are in place and effective in controlling erosion.		

Monthly Checklists:

ENVIRONMENTAL ASPECTS	YES/NO	COMMENTS
Environmental Management is reviewed in the monthly review project review meeting at site.		
All new personnel on site are imparted training on Environmental Awareness.		
Construction activities are undertaken according to the approved method statements.		
Fuel flammable material storage areas comply with general fire safety requirements.		
Public complaints have been recorded and dealt with the satisfactory manner.		

Site Closure Checklist:

ENVIRONMENTAL ASPECTS	YES/NO	COMMENTS
Contractor has cleared everything not forming the part of the permanent works.		
Re-vegetation has been satisfactorily completed.		
All areas disturbed during construction have been brought back to the near original condition in accordance with the conditions.		

5.2 AREAS OF CONCERN

From the monitoring point of view, the important parameters are ambient air quality, noise marine water quality, etc. An attempt is made to establish early warning system, which indicates the stress on the environment, suggested monitoring parameters and programmers are described in the subsequent sections.

5.3 WATER QUALITY

The chemical characteristics of the water quality should be monitored as well as the biological, parameters must be checked for its life sustainability. The parameters to be monitored are as follows:

Water Quality

- Physico-chemical parameters
- pH
- Salinity
- Conductivity
- TDS
- Turbidity
- D.O.
- BOD
- Phosphates
- Nitrates
- Sulphates
- Chlorides

Biological parameters

- Light penetration
- Chlorophyll
- Primary Productivity
- Phytoplanktons (No. of species and their density)
- Zooplanktons (No. of species and their density)

5.4 SOIL QUALITY

The soil sample is collected in and around the site to establish the baseline characteristics of the study area. Soil sample is collected using the auger from the depth of 60 cm from the project site. Soil sample collected from the project site is analyzed for the physical and chemical characteristics and is reported.

Sediments

- Physio-chemical parameters
- Texture
- pH
- Total Kjeldahl Nitrogen

- COD
- Sodium
- Potassium
- Phosphates
- Chlorides
- Sulphates

Biological Parameters

- Benthic Meio-fauna
- Benthic Macro-fauna

5.5 AMBIENT AIR QUALITY

Construction Phase

Ambient air quality monitoring is recommended to be monitored at three stations close to the construction sites. The monitoring can be conducted for one season. Monitoring can be conducted twice a week for 4 consecutive weeks. The parameters to be monitored are PM10, PM2.5, S02, NOx, CO, Pb, NH3, C6H6, O3, As, Ni, B- (a)-P.

Operation phase

Micrometeorology

An essential part of air quality monitoring would be to establish a small automatic Meteorological observation station to record daily continuous synoptic data. Arrangements for recording temperature, humidity, visibility, wind direction and speed, cloud cover, rainfall and meteorological phenomena like storms would be required to be established at the terminal site. The ambient air quality monitoring will have to be conducted at three locations; Air quality could be monitored for one season. High volume samplers can be used for this purpose. The frequency of monitoring shall be twice a week for 24 hours for four consecutive weeks. The parameters to be monitored are PM10, PM2.5, S02, NOx, CO, Pb, NH3, C6H6, O3, As, Ni, B-(a)-P. The ambient air quality monitoring during project operation phase can be carried out by project staff. Sufficient provision has been earmarked for

purchase of monitoring of: Ambient air quality and micro- meteorological instruments and equipments.

5.6 NOISE LEVELS

Personnel involved in the work areas, where high noise levels are likely to be observed during project construction and operation phases. For such in-plant personnel, audiometric examination should be arranged at least once per year Neighborhood (up to radius of 1 km)

It is recommended that during project operation phase, monitoring of sensitive areas like schools and Medicare centers be conducted within a distance of 1 km radius of the site to ascertain noise levels at receptors.

5.7 SUMMARY OF ENVIRONMENTAL MONITORING PROGRAMME

The summary of Environmental Monitoring Program for implementation during Project construction and operation phases is given in Table 24

Table 24 - The Summary of Environmental Monitoring Program for Implementation during Project Construction and Operation Phase

Sr. No.	Aspects	Parameters to be monitored	Frequency of monitoring	Location
1.	Water	pH, dissolved oxygen, phosphates, nitrates,	Once in three months	3 to 4 sites
2.	Ambient air quality	PM10, PM2.5, S02, NOx, CO, Pb, NH3, C6H6, O3, As, Ni, B-(a)-P.	Summer, post monsoon & winter seasons Twice a week for four consecutive weeks per season	Close to major construction sites
3.	Ambient Noise Quality.	Equivalent noise Levels	During peak construction activities	Construction sites
4.	Biological parameters	Phytoplankton Zooplankton	Once in a year	3 to 4 sites
5.	Greenbelt Development	Rate of Survival and Growth Of Various Species	Once per Month	Various sites

CHAPTER 06 - DISASTER MANAGEMENT PLAN

6.1 CONCEPT OF RISK ASSESSMENT

The concept of risk assessment and its engineering application has been well acclaimed since more than a decade. A variety of major accidents have focused attention on the dangers of risk exposure for human health and environment.

Risk analysis provides numerical measures of the risk that a particular facility posses to the public. It begins with identification of potential risk involving events and determination of the impact of each event. The consequences of each event are than calculated for numerous combinations of weather conditions and wind directions these consequence predictions are combined to provide numerical measure soft the risk for the entire facility.

Risk for a particular facility is based on the following variables:

- Multiple accident outcomes
- Population disturbance
- Site-specific meteorological data

—Risk analysis is a tool which helps to translate hindsight (accidents) into foresight (planning), showing ways and means (improved engineering, procedure and supervision) to prevent the calculated accident from happening.

Visualize failure scenarios for the structures, handling equipment and estimate distances safe from damage.

6.2 LEAKS AND SPILLAGES

It will be of prime importance to protect the marine and terrestrial ecosystem during the operations of crafts. The various possibilities of leakages and spillages of the fuel includes following probabilities

Spillages of fuel during re-fueling
Leakages of fuel during navigation
Leakage or spillage during ship washing process

6.2.1 Spillages of Fuel during Re- Fueling

The crafts will require fueling its operation depending upon its trips two end fro. There will be a possibility of spillage of fuel during re-fueling process. This may lead to disturbance in the shore ecosystem in a temporary manner. The spillage portion will be although small one, it may affect the natural marine life to save extent. To avoid such accidental spillage, following measures shall be adopted.

- Proper routine checks shall be performed on the pipeline used for re-fueling, various pumps, motor valves etc.
- Safety audit shall be performed yearly to account for the performance of the re- fueling system.
- Chemical foam system shall be kept accessible to spray on the spillage.
- Oil-water separator shall be installed.
- Area of spillage will be contented using HDP or PE pipes.
- Oil Spill Contingency Plan will be followed.

In case of such a spillage oil-water separator shall be immensely used to recover the oil spread on the surface of seawater. In case, of major spillage, containment technique shall be employed immediately and oil shall be recovered without further spreading it on the sandy shore. Therefore, the major methods to be adopted for the oil spill closed to the shore shall be;

i) Containment technique
ii) Scavenging
iii) Mechanical removal
iv) Dispersion technique
v) Use of absorbing material

Incase Oil spill accident, the below given mitigative measures shall be taken in order to preserve the mangroves from damage due to oil spill:

- Booming and skimming of oil on the water surface in mangrove creeks
- Pumping of bulk oil from the sediment surface, depressions and channels
- Water flushing of free oil from sediment surface and mangroves into areas where it may be collected
- Use of absorbent materials, with subsequent collection and disposal.

6.2.2 Leakages of Fuel during Navigation

There is a possibility of fuel being leaked from engine room because of various reasons. It may even happen during a collision and fuel/oil may spread on larger area in the sea. To prevent such accidental leakages, proper mechanical maintenance shall be carried out during the routine surveys as mentioned above. In case of such an event the craft shall be very well equipped with recovery system. These leakages can lead to water pollution leading to damage to mangroves, fishes, spawning problems, distraction of marine eco-culture, odor problem, and effect on marine bio-diversity.

Various mitigation measures will be enforced depending upon type of oil, quantity of spread, distance from the shore, etc. such as:

- Burning of the oil
- Scheming the surface with a suction device
- Absorbent technique
- Gelling method
- SilIking method
- Emulsification / dispersion

Leakage or spillage during ship washing process

Since the proposed terminal shall provide the berthing and parking facility of ships. There will be as such no major impact of oil spillage on the environment. However, there is a possibility of spillage during the ship washing process from the ship parking facility. These Spills can further contaminate the land as well as water, if it is not properly channelized. However, proper mitigative measures are taken in order to avoid any such spillage/ leakage during the workshop processes.

The intercepting drains passing from the area are installed with oil & grease trap to entrap the oil spills, the remaining water will then be sent to sedimentation tank where it will be further treated for oil spills and pollutants. The treated water will be reused within the plant.

Awareness amongst the workers regarding safe handling techniques and safety measures will be made through various workshops and seminars.
In case of any accidental spill, all processes will be brought to halt.

6.3 RISK ANALYSIS STUDY

- Identification of potential physical hazards which could trigger loss causing events such as fire and explosion, leakage of flammable materials etc. from the proposed facility.
- Identifying the Maximum Credible Loss Scenarios (MCLS) for the vulnerable areas in the storage areas in the facility for assessing the magnitude and severity of the impact of various failure scenarios in terms of damage to property and injury to personnel.
- Recommendations for risk reduction shall be made on the basis of the above for minimizing, if not eliminating various hazards and providing information on improvement of safety systems, where necessary.
- The major risk is envisaged from the storage yards, fueling of cargos. The leakage in piping, pumps and electrical fault can lead to hazardous event.
- A complete Risk Assessment will be done and the mitigative measures as well as safety measures will be proposed for the same.

6.4 ONSITE EMERGENCY PLAN

Assessing the adequacy of available resources to take care of emergencies as identified in the risk analysis study. Providing recommendations on the infrastructure, communication system and other facilities such as first aid, security, fire fighting etc. in view of effective handling of the emergencies identified. Specifying the roles and relationship amongst personnel from the facility and outside agencies for effective handling of the emergencies. Identification of assembly points and escapes routes

for evacuation. Preparation of an Onsite Emergency Response Plan Document is envisaged. The stages of On-site Emergency Plan include:

- Outline Emergency Response Team.
- Designated person in charge.
- Key responsibility of each individual.
- Telephone numbers for key people.

Risk Evaluation on preliminary hazards

Details regarding

- Location of Key-personals.
- Emergency Control room, if provided.
- Emergency Telephone numbers.
- First-aid Kit and Fire Extinguisher locations.
- Warning alarm, safety and security.
- Precautions during design and Engineering.
- Continuous surveillance.
- Details of Hospital and Fire-brigade facility.
- Procedures for notifying family members of injured employees.
- Procedure for reporting emergencies.

Awareness amongst workers for

- Use of fire-fighting equipment and first-aid.
- Mock-drill for Disasters.
- Use of personal protective equipment.
- Procedure for reporting emergency.
- Knowledge of alarm systems.
- Manuals for each Operating system.

Control Plans

Emergency Control plans.

- Safe time to resume work after an emergency.
- Control measures for any spillage, leakage, explosion, etc.

6.5 LIFE SAVING APPLIANCES AND ARRANGEMENTS

It is one of the important aspects towards the mitigative measures to be adopted on the craft. It is also recommended to have safety appliances and arrangements even at ship terminal facility, in case of emergency for the craft during its navigation. Various life saving arrangements/appliances shall be made available for such eventualities. The major issues to be tackled for the life saving or rescue operations will be during any eventualities arising out of collision or submergence of the craft. In case of such eventualities various life saving appliances such as embarkation ladder, float free launching pads, spaces for laundry emersion suit inflammable appliances shall be made available on the craft. In addition to this life saving appliances an effective ladder communication system shall be made available on the crafts.

In the event of noting such as event at the marine terminal control room to allowing life saving appliances kept ready.

- Rescue boat, which is design to rescue person in distress and to marshal survival craft.
- Retrieval rescue team for the safe recovery of the survivors and evacuation.
- Retro reflective material for detection of damaged craft in poor tight conditions
- Embarkation ladder to permit safe access at the survival craft.
- Live saving appliances such as thermal, protective aid emersion suit radio life saving appliances radiotelegraph installations in lifeboats shall be made available.
- Life buoys compiling with the requirement and regulation shall be kept ready and accessible during emergency life jackets etc shall be accessible
- Trained personnel with experience of rescue operation shall be provided on board on rescue boat.
- In case of addition to the rescue boat craft with all novel life saving appliances it is recommended that to take help of Coast Guards and also naval helicopters to search the exact site of accident.

6.6 OCCUPATIONAL HEALTH AND SAFETY

Specific occupational health and safety issues relevant to proposed project primarily include the following:

- Physical hazards
- Confined Spaces
- Exposure to Noise

The main sources of physical hazards at ports are associated with cargo handling and use of associated machinery and vehicles. However this shall be taken care of by applying all the terminal related norms and standards. The workers and vehicles passageway shall be kept separate. Avoiding entry of workers as far as possible in the area of ship loading and unloading activity and areas where grab is operational.

Noise pollution can cause due to one of the various activities at the terminal or parking facility. However, proper mitigative measures are out-lined for control of noise at the Facility. Onsite medical facility will be provided in case of any hazard or casualty during the operational phase. Fire safety measures shall be incorporated and implemented. Periodic health check-up of all the workers shall be carried out.

6.7DISASTER MANAGEMENT PLAN (DMP)

a) Objectives

The emergency DMP for onsite and offsite location will be inter-related. The overall objectives are

- To identify type of major disasters which may occur
- To localize the emergency and if possible eliminate
- To minimize the effect of accidents

Elimination of hazard will require equipment like fire fighting equipments, water sprays, emergency shutoff valves and purposeful construction. Minimizing the effect will be prompt action by operating and emergency staff, rescue, first aid, evacuation, rehabilitation and giving information promptly to people living / working nearby.

b) Types of Possible Emergencies

- Fire on berth / approach trestle / storage / buildings
- Spillage while bunkering by vessel
- Spillage due to collision in channel
- Natural calamities like cyclone / rough weather / earthquake/ Tsunami
- War situation/ Air strike.

c) Functions of Disaster Management

Controlling spread of accidental effects with minimum damage to men, material, machine and structures.

- To inform relevant agencies and request for help.
- To rescue victims and provide succor.
- To protect other and safely evacuate.
- To inform nearby inhabitations.
- To identify the affected persons and inform their relatives.
- To provide authentic information to news media and other.
- To preserve relevant records and equipment needed as evidence in any subsequent inquiry.
- To rehabilitate the affected areas and alot specific assignment to available manpower.

d) Classification of Accident

Level I : Operator Level

Level II : Local / Community Level

Level III : Regional level

Level IV : International level

e) Critical Targets

Disaster Management Plan is prepared after identifying the objects likely to be affected in the event of emergency. The target of fire includes personnel if emergency occurs at service platform during discharging of vessel and tank farm on shore.

f) Control Room (CR)

A control room will be established at a location away from likely spots of accidents and shall be easily accessible. Better location will be near the room from where all unloading operation are conducted and controlled.

6.8 EMERGENCY PLAN FOR BERTHS AND VESSEL

Terminal Emergency Plan

This plan will be drawn up in consultation with authority, fire brigade, coast guard and police etc. The plan will include:

- Specific initial action to be taken by those at the location of emergency (to notify time, position source and cause of spill) to control room and Coast guard.
- Immediate action to combat Oil –pollution.
- Evaluations of situation by on scene controller regarding threat posed by spill and identify threatened resources.
- Details of Communication system available siren code.
- An inventory including location details of emergency equipment.
- Sound alarm-terminal fire fighting staff to fight fire.
- Mobilize fire-fighting equipment.
- Electric power to switch off - emergency lighting to switch on.
- The ships calling at termianal will be advised of the terminal's emergency plan particularly the alarm signals and procedures to summon assistance in the event of an emergency, on board.

Rough Weather

The rough weather operations will be controlled in three stages

- Green Status - the operations of loading / unloading will be carried out as planned.
- Yellow status - This is an alert stage indicating possibility of rough weather, still operations can be continued with all emergency precautions.
- Red Status - Emergency situation or rough weather; operation will be suspended - Activities controlled by In charge of emergency operations. The

vessel / tanker is to be unearthed to safe anchorage or will be advised to proceed to sea.

First Aid & Fire Fighting Services

The proposed project will have full-fledged medical facilities as well as fire fighting facilities available in the area.

6.9 IDENTIFICATION OF MAJOR HAZARDS

The hazards occurring at Proposed Project can be broadly classified as natural and man-made hazards. Some of the hazards existing at site are as follows:

- Earthquakes
- Fire
- Tsunamis

Earth Quake

An earthquake is a sudden, rapid shaking of the Earth caused by the breaking and shifting of rock beneath the Earth's surface. This shaking can cause buildings, dams and bridges to collapse; disrupt gas, electric, and phone service; and sometimes trigger landslides, flash floods and fires; all these are possible scenarios at Proposed Project. Buildings with foundations resting on unstable soil and slopes are most at risk.

The greatest danger for people in an earthquake exists directly outside buildings, at exits, and alongside exterior walls. Ground movement during an earthquake is seldom the direct cause of death or injury. Most earthquake-related casualties result from collapsing walls, flying glass, and falling objects.

Fire

Fire is the most probable emergency scenario at proposed project. Fire can be caused in buildings and installations as electrical fire, fire in buildings. To handle this scenario an onsite emergency plan has been developed by the security team and is currently operational.

Tsunamis

Tsunamis are generated by large and rapid displacements of water, mainly from sudden and large scale changes in the configuration of the sea floor associated with fault displacement or gigantic underwater landslides, which could be mainly due to earthquakes.

Earthquakes generate tsunamis by vertical movement of the sea floor as in normal faulting or thrust faulting. If the sea floor movement is horizontal, tsunamis are not generated as in strike slip earthquake. Sometimes they are triggered by marine landslides into or under the water surface, also generated by volcanic activity and meteorite impacts, but such events are extremely rare. Tsunami hazard along a coastline is therefore a combination of all the potential sources of tsunamis that lie in the neighboring sea or ocean. Tsunami waves travel at a speed of approximately 700 km/ hr in 4000 m of water. In 10 m of water the velocity drops to about 36 km/hr.

6.10 EMERGENCY RESPONSE MEASURES FOR NATURAL HAZARDS

Natural Hazards cannot be prevented. However with mitigation measures the effects/damages could be reduced.

6.10.1 Response in case of Earthquake

Response procedures for worker

If indoors:

Take cover under a piece of heavy furniture or against an inside wall and hold on.

If indoors:

Stop the process and move away from machinery, equipment, etc.

Avoid moving around until the shaking stops.

If outdoors:

Move into the open, away from process area, office building and utility wires.

Gather at the Evacuation point mentioned during the safety drill and await instructions.

If in a moving vehicle:

Stop quickly and stay in the vehicle.

Once the shaking has stopped, proceed with caution.

After be prepared for aftershocks.

Although smaller than the main shock, aftershocks cause additional damage and may bring weakened structures down. Aftershocks can occur in the first hours, days, weeks, or even months after the quake.

Help injured or trapped co-workers.

Give first aid where appropriate. Do not move seriously injured persons unless they are in immediate danger of further injury. Call for help. Use Emergency numbers.

Stay out of damaged buildings, machinery, equipments.

Get your entire process area checked with Safety officers

Use the telephone only for emergency calls.

Emergency Response Procedure

- Initiate the Quick Response Team and First aid team for earthquake response
- Give a long siren for earthquake warning
- Inform the necessary authorities for aid
- Ensure no personnel or residents are stuck beneath any debris
- Ensure that all residents and personnel standing outside near the buildings are taken to open areas.
- Close the entry gate to restrict any incoming traffic
- Inform Electrical department to shut off the utilities.
- Ensure that the first aid ambulance and fire tender vehicles are mobilized
- Ensure that emergency telephone number is only used for this purpose
- Check the utilities and storage tanks for any damage.
- Inform structural engineers to check for any damage to the Dam structure

6.10.2 Response in case of Fire

On sighting a fire the first person should immediately inform the control room.

- If the fire is small engage in extinguishing the fire using the nearest fire extinguisher or retrieve the property that may be damaged.
- Guide staff to the emergency assembly point.
- The control room will immediately inform the first aid centre and the quick response team, which has trained fire-fighting personnel. Mobilize the fire tender.
- The Quick Response team will immediately move to the point of fire and take all necessary steps to stop the fire. If the fire is not controllable and spreads to the other area then immediately inform the security post who would in turn inform the district authorities and call for external help.
- First aid team will provide immediate relief to the injured personnel at the scene of incidence. The patients would then be evacuated on priority to the dispensary or hospital based on their condition.

Instructions for an Individual in case of fire

- Get out of buildings as quickly and as safely as possible. If outdoors stop all processes and inform the control room using emergency alarm system and Emergency numbers.
- Use the stairs to escape. When evacuating, stay low to the ground. Evacuate and assemble at the assembly point decided previously.
- If possible, cover mouth with a cloth to avoid inhaling smoke and gases.
- Close doors in each room after escaping to delay the spread of the fire.
- If in a room with a closed door.
- If smoke is pouring in around the bottom of the door or it feels hot, keep the door closed.
- Open a window to escape or for fresh air while awaiting rescue.
- If there is no smoke at the bottom or top and the door is not hot, then open the door slowly.
- If there is too much smoke or fire in the hall, slam the door shut.
- Call the security from the nearest phone

- Inform security gate if there are any persons trapped in the house or injured.
- Stay out of damaged buildings.
- Check that all wiring and utilities are safe.

6.10.3 Response in case of Tsunamis/Storm Surges

a) Structural measures:

Construction of cyclone shelters

- Plantation of mangroves and coastal forests along the coast line acting as Bioshields
- Development of a network of local knowledge centers (rural/urban) along the coast lines to provide necessary training and emergency communication during crisis time
- Construction of location specific sea walls and coral reefs in consultation with experts
- Development of well designed break waters along the coast to provide necessary cushion against cyclone and tsunami hazards
- Development of tsunami detection, forecasting and warning dissemination centres
- Development of a ―Bio-Shield‖ - a narrow strip of land along coastline. Permanent structures, if any in this zone with strict implementation of suggested norms. Bio-Shield can be developed as coastal zone disaster management sanctuary, which must have thick plantation and public spaces for public awareness, dissemination and demonstration.
- Increasing the river dike height, increasing the height of the coast by filling up of coastal sand of the same place.
- Identification of vulnerable structures and appropriate retrofitting for tsunami/cyclone resistance of all such buildings as well as appropriate planning, designing, construction of new facilities like
- Critical infrastructures e.g. power stations, warehouses, oil and other storage tanks etc. located along the coastline.
- All other infrastructure facilities located in the coastal areas.
- Public buildings and private houses.

- All marine structures.
- Construction and maintenance of national and state highways and other coastal roads.

b) Non-Structural Measures

- Coastal regulations Zone Act – Strict implementation.
- Aggressive capacity building requirements for the local people and the administration for facing the disasters in wake of tsunami and cyclone, ‗based on cutting edge level'
- Developing tools and techniques for risk transfer in highly vulnerable areas
- Conserving and developing Natural Bioshields (Mangroves) and shelterbelt plantations (Casuarina)
- Maintaining natural sand dunes.
- Maintaining and promoting beach development.
- Having diverse livelihood options.
- Launching a series of public awareness campaign throughout the coastal area by various means.
- Training of local administration in forecasting warning dissemination and evacuation techniques
- Awareness generation and training among the fishermen, coast guards, officials from fisheries department and port authorities and local district officials etc., in connection with evacuation and post tsunami storm surge management activities. Regular drills should be conducted to test the efficacy of the DM plans.

6.11 IDENTIFICATION AND ASSESSMENT OF HAZARDS

Storms, Floods and fires are potential disasters for such Projects. Their likelihood of occurrence and the resulting risk of damage should be incorporated into the design analysis of each project facility.

Table 25 - Summary of the Disasters Preparedness Plans

DISASTER	DESCRIPTION	RESPONSE PLAN	STAGES
Hurricanes and storms	Depending on the magnitude storms can damage the infrastructure of the project area to varying extent, thereby affecting its operation	Hurricane Preparedness Plan Secure insurance coverage.	Alert, Response Recovery
Earthquake	The proposed site is not in an earthquake prone area. Thus less chances of earthquake are envisaged.	Building and construction shall adhere to the earthquake efficiency norms.	Planning, Response, Damage Assessment and Recovery.
Fire	Fire outbreaks also vary in size and location and cause irreparable damage to the infrastructure.	Fire Prevention and Preparedness Plan Install fire fighting equipments Electrical work will be done by certified electrician. Provide proper Insurance coverage.	Response, Planning Fire Drills, Damage Assessment.
Oil Spills and leaks	Oil or fuel spill due to accidents or leakages pose a serious impact to the sensitive environment.	Spill Contingency Plan All petroleum products stored in bunded areas.	Report and Response, Recovery
Climate change	This natural occurring phenomena can pose a risk to the project if not adapted in time.	Contingency Plan	Alert, Response
Medical	Medical emergencies can occur at any moment and therefore requires a quick and coordinated effort to respond to the need.	Medical Emergency Plan First aid equipment and staff trained in CPR.	Response, Recovery

CHAPTER 07- PROJECT BENEFITS

The EXIM trade due to this facility will increase and creating employment opportunity, supply chain management, facilities for industries in the Mumbai region and Maharashtra.

CHAPTER 08 - ENVIRONMENTAL MANAGEMENT PLAN

8.1 INTRODUCTION

Impact assessment helps in identifying potentially damaging aspects of a proposed project. Based on the findings of the impact assessment, Environment Management Plan is devised to minimize adverse impacts and enumerated various steps to be taken for improvement of the environment.

However due to no major adverse impact on the Air, Water, Land, Biological Environment due to the project, the Environment Management Plan shall not be very complicated and only regular Monitoring of ambient air quality, marine water quality, noise level monitoring and soil quality monitoring shall be carried out as per the requirements.

All persons working near the noise generation equipments shall use Personal Protective Equipment such as ear-plugs, muffs and closely monitored for implementation. All workers should be made aware of adverse effect high noise levels through training program; this will ensure proper implementation of mitigation majors.

All sanitary facilities should have proper maintenance and all the wastes generated should be connected to a common septic tank followed by soak pits of adequate size. The supernatant should be regularly tested.

Environmental Management Plan (EMP) is the key to ensure a safe and clean environment. A project may have identified proper mitigation measures but without a management plan to execute it, the desired results may not be obtained. The present chapter on Environment Management Plan envisages proper implementation of mitigation measures to reduce the adverse impacts arising out of the project activities. The following issues have been addressed in this EMP:

- Mitigation measures for abatement of the undesirable impacts caused during construction and operation stages
- Details of management plans

- Institutional set up for implementation of the EMP
- Post project environmental monitoring programme to be undertaken after commissioning of the project
- Expenditures for environmental protection measures

The EMP is proactive in nature and should be upgraded if new facilities or modification of existing facilities, with environmental concerns, come up at a larger stage.
EMP included four major elements:

- Commitment and policy: The project will strive to provide and implement the Environmental Management Plan that incorporates all issues related to air, land and water
- Planning: This includes identification of environmental impacts, legal requirements, and setting environmental objectives. The various potential impacts are discussed under chapter 4
- Implementation: This comprises of resources available to the developers, accountability of contractors, training of operational staff associated with environmental control facilities and documentation of measures to be taken.
- Measurement and evaluation: This includes monitoring, corrective actions, and record keeping.

The EMP‘s that will be put into place consist of those during construction and operating stages of the project and includes the following elements:

- Hazardous and Solid Waste Management
- Air Pollution Control and Management
- Noise Control and Management
- Storm Water Management
- Plantation, Landscaping and Land Management
- Management of Soil Issues
- Occupational, Safety and Health Issues
- Best Management Practices
- Energy Conservation

- Environmental Monitoring
- Emergency Response Plans for Emergency Scenarios
- Environmental Management System

8.2 SUMMARY OF CRITICAL IMPACTS/ ISSUES

The major impacts due to different project activities and their mitigation measures have been identified in Chapter-5. These measures together constitute part of Environmental Management Plan (EMP). Environmental study carried out by the consultants has highlighted the following critical features of the project (Table-26).

Table 26 -Critical Impacts / Issues

Sr. No	Issues	Description
1.	Ambient Air Quality	During construction air and noise pollution may increase and would require mitigation.
2.	Ambient Noise Level	In the operation stage air quality and noise levels are generally expected to improve. However, measures would be taken to contain pollution due to increased vehicular traffic on the Thane side.
3.	Ecology	No terrestrial or marine ecosystem is involved at present in and around the project.
4.	Water Quality	Panvel Creek is nearby the project area
5.	Land Use Pattern	No additional land required, no alteration in the existing land use.
6.	Labour Camp	No labour camps on the site.

8.3. IDENTIFICATION OF IMPLEMENTING AUTHORITY

The responsibility for the implementation of the EMP will be with the Promoter and Contractor (P & C). An environmental management cell (EMC) will be established by the P & C for implementing the mitigative measures. To mobilize the appropriate expertise to design diverse type of mitigation measures. The P & C need to be collaborating with order institutions in the public and private sector. The EMC will ensure timely implementation of various mitigative measures at different stages of

the project i.e. during construction and operation stage and the completion of the project within scheduled time frame.

In addition, an environmental Officer will be appointed by project authorities for management of the project with the objective of reviewing and assessing the progress made by the concession company in implementing the suggested mitigative measures.

8.4 IMPLEMENTATION OF RECOMMENDED MITIGATION MEASURES

Different activities to be addressed in the management plan have been considered and discussed in detail in Table 28. This table presents an inventory of tasks to be performed for environmental management.

The mitigation measure for the impact is made a part of proposed activities. The major instruments of environmental management will be monitoring performance of the construction by the EMC. The conditions, which must fulfilled documents, are suggested below:

- All necessary measures and precautions will be cited so that the execution of the works and all associated operations on site or off- site are carried out in conformity with statutory and regulatory environmental requirements.
- Necessary measures and precautions to avoid nuisance or disturbance arising from the execution of the works will be included, preferably at the source it self.
- Wastes such as spoil or debris or silt from the sites will be immediately removed and the affected areas will be restored to their original state.

8.5 MONITORING PLAN

The regular components of monitoring and its frequency have been identified in Chapter 5. The air and noise quality monitoring will be performed at the same locations where baseline monitoring was carried out.

Environment Management Cell (EMC) Constituted by the P & C will be the prime agency for monitoring all activities. Project promoters will supervise all activities and accordingly advise the P & C to improve on areas where any shortcomings are observed. The EMC will provide all the monitoring results to project promoters who

will keep a record of all information and suggest suitable measures to be adopted by the Contractor if any aspect is found to be diverting from the anticipated values/ standards.

8.6 ENVIRONMENTAL TRAINING

The Environment Management Cell (EMC), in addition to implementing and monitoring different environmental attributes, will also be actively involved in imparting training and raising environmental awareness of Construction Engineers/ Contractors and other staff members/ workers so as to enable them take the environmental aspects into consideration as and when required. In the long run, the EMC can impart additional and specialized training in environmental management of the road and building construction system.

8.7 BUDGETS FOR ENVIRONMENTAL MANAGEMENT PLAN

The mitigative measures suggested in the preceding chapters forms costs related to measures incorporated into engineering design; project scheduling, site planning and preparation of tender documents. The cost on this account will be covered with the construction budget and should not be seen as items of cost for implementing Environmental Management Plan. The estimated environmental cost considered here includes:

a) During Construction phase

- Compensatory tree plantation for tree cutting, if any
- Provision of Sanitation at workers camp
- Provision of air, noise, and dust vegetative barrier/ special screens- both side of project area
- Dust suppression
- Erosion Control Measures, if any
- Solid barrier to check noise pollution for sensitive receptors like school etc
- Solid waste management due to construction activity

b) During Operation phase

- Air pollution monitoring
- Noise monitoring

- Water quality monitoring
- Soil quality monitoring
- Solid waste management

Provision of **Rs 1.0 Crore** in budget for implentation of Encironmental Managment Plan is earmarked. The environmental cost is consists of monetary value of the mitigative measures adopted to minimise the negative impact of project on environment. Environmental cost is divided into two categories, i.e. capital cost and operation and maintenance cost. Capital cost is the cost of all the structural measures proposed for environmental protection during construction phase while the operation an maintenance cost include the cost of monitoring air, noise, soil and water and maintaining the structural measures over project life.

The environmental cost is consists of monetary value of the mitigative measures adopted to minimise the negative impact of project on environment. Environmental cost is divided into two categories, i.e. capital cost and operation and maintenance cost. Capital cost is the cost of all the structural measures proposed for environmental protection during construction phase while the operation an maintenance cost include the cost of monitoring air, noise, soil and water and maintaining the structural measures over project life.

Table 27 - Reporting of The Major- Parameters And Responsible Organization- Construction Phase

Sr. No	Project Related Issue	Action to be Taken	Responsible Organisation
1	Provision of sanitation at workers colony	Workers Camp not on site. Prepare and administer land use control measure	Contractor in consultation with promoter

2	Utilities	Initiate and complete the process of shifting the utilities	Contractor and promoter in consultation with the local Electricity Department and Telecom Department
3	Reduction of vegetative cover	Mangrove area will be affected during Phase I of proposed aligment.	Promoter and Contractor in consultation with Forest Department
4	Road side Tree Plantation and compensatory afforestation	Select species and prepare location plan for roadside tree plantation.	Promoter and Contractor
5	Borrow pits and quarry sites development.	NA	NA
6	Sewerage disposal and solid waste at workers camp	Workers Camp not on site.	Promoter and Contractor
7	Traffic management	Secure assistance from local police for traffic control during construction phase. Safety measures will also be undertaken by installing road signs and making for safe and smooth movement of traffic	Promoter and Contractor in consultation with traffic department

8	Noise Level	Stationary equipment will be placed as far as possible form inhabited areas to minimise objectionable noise impacts Proper maintenance machinery and vehicles Construction activities strictly prohibiting between 10 pm and 6 pm near habitation Provision of using ear plugs by workers being exposed high noise levels Vegetative and solid barrier along inhabitant centre and sensitive receptor	P & C in consultation with MPCB
9	Air Quality	Vehicles and machineries will be regularly maintained so that emissions conform to National and State Standards. Vehicles carrying construction material will be covered by traps to avoid spilling and blowing by wind Asphalt mixing if	P & C in consultation with MPCB

		required will be over 500 m away for many communities. Mixing equipment will be seated and be equipped with dust removal device	
10	Water Quality	Monitor periodically water quality for establishing the change of water quality	P & C in consultation with MPCB
11	Soil Erosion	Strengthening of soil strata during open foundation	P & C

Table 28 Reporting of The Major- Parameters And Responsible Organization- Operation Phase

Sr. No.	Project Related Issues	Actions to be Taken	Responsible Organisation
1	Prevention of Road side Squatters or indirect Urban Sprawls	NA	P & C in consultation with the State Revenue and Planning.
2	Road Safety and Traffic Management	Adequate number of proper & legible signs will be installed along the road. Prepare and administer a monitoring system on road/ accidents.	P & C in consultation with CIDCO and State Traffic Police.
3	Air Quality	Monitor periodically ambient air quality at selected sites. Confinement and	P & C in consultation with MPCB

		absorption of the pollutants at source by creating vegetation along the length. Enforcing different control measures to check pollution (e.g. catalytic converters, EURO II engines, unleaded petrol, proper serving etc.)	
4	Noise level	Monitor periodically ambient noise level at selected sites. Minimization of use of horns near sensitive locations/ silence zones with the help of sign boards at proper places. Provide noise barriers with roadside plantation.	P & C in consultation with MPCB
5	Water Quality	Monitor periodically water quality for establishing the change of water quality, if any, and assessing its potentiality of surviving aquatic flora and fauna and for irrigation use.	MPCB and CIDCO
6	Soil Characteristics	Periodic monitoring of soil quality (mainly Pb) at specified distance for	P & C in consultation with MPCB

		assessing contamination by vehicular emissions. Checking the overflow of spillage from the carriageway.	
7	Maintenance of Avenue trees	Plantation will be undertaken by the concession company on an aggressive note along the whole stretches on the both sides of the road. Employment of local population for maintenance avenue plantation.	P & C in consultation with authorities and State forest Department
8	Human Health and Safety	Vulnerable stretches, which are prone to accidents, will be identified. Adopt Safety measures and other control measure for regulating traffic in such areas by: Installing proper road signs, marking along the whole stretch of the tolled highway in the form of cautioning, informatory and mandatory signs of gantry mounted	P & C in consultation with MPCB & CIDCO

		overhead sizes. Installing precautionary roads signs for controlling vehicle a speed in urban areas and curve stretches.	

8.8 DETAILS OF MANAGEMENT PLANS

8.8.1 Waste Water Management

- Reception Tank to receive Bilge water, Ballast water,sludge.
- Provision of a specially equipped pump truck to receive sewage from ships and transport it to the treatment plan.
- All Sewage will be connected to existing MCGM Sewage network

8.8.2 Solid and Hazardous Waste Management Plan

Construction Phase

All demolition work is completed. Minor Debris generated from development and costrctuion of building will be dumped at designated dumping place.

Mitigation Measures

- To avoid contamination due to leakage or spillage the chemicals such as bitumen, oil, diesel, paint etc shall be stored in earmarked place with proper platform with catch pits etc.
- Solid waste generated from labour camp will collected in dustbins.
- The waste will be segregated.
- The recyclable waste will be give to authorized vendor and the other waste will be disposed off to SWM facility.
- Indiscriminate settlement of workers should be avoided.
- The solid waste generated from labour camps will be disposed of in environmentally safe manner.
- The storage area will be covered during monsoon.
- Construction vehicles will be properly maintained to avoid leakage.

- The impact of increase in turbidity and siltation can be mitigated by providing coffer dams and silt trap around the foundation site before start of piling/drilling works

Operation Phase

All floating solids removed daily from the harbour water.

Waste generated from the operational phase of the project is likely to be restricted to small volumes associated with intermittent maintenance works and landscape upkeep. If appropriate waste handling, storage and disposal procedures are adopted, no significant impacts are considered likely.

8.8.3 PLANTATION, LANDSCAPING AND ECOLOGICAL MANAGEMENT:

On the basis of study, following management plans have been suggested:

OPERATION PHASE:

Creation of 12 Acre Botanical Garden adjacent to the Fishing harbour as a mitigation measure. One of the best maintained garden in Mumbai city. 2 Acre garden behind Taj Hotel, South Mumbai. 74 nos of small gardens, traffic islands developed and maintained by the port.

7500 Trees of average height more than 6m in Residential colonies, all along port roads, Avenues, Hospital, Jawahar Dweep. Revival of the Coastal Mangrove Wetland area under control of MbPT.

Declaration of MbPT‘s 5 km stretch at Sewree as ‗Sewree Mangrove Park‘ in 1996. Protection, classification, restoration as well as enhancement of Coastal Natural Resources.

CHAPTER 09 - EXECUTIVE SUMMARY

9.1 INTRODUCTION

The proposal relates to revival of Offshore Container Terminal (OCT) at Mumbai Port by re-biding the OCT with changed cargo profile to be handled at the terminal by including non-containerized cargo like steel and RoRO with a Right of First Refusal to the existing concessionaire M/s ICTPL. As a pre-requisite for the re-bidding process, Mumbai Port is desirous of having a Detailed Feasibility Report prepared for the re-configured project. The work of preparing the DFR has been entrusted to the Indian Ports Association.

9.2 TRAFFIC ANALYSIS

Based on the past traffic trends in Mumbai Port, steel and automobile are identified as the potential growth cargoes for OCT. Accordingly, the future prospects for these cargoes were analysed The assessment based on the present trends in the domestic and global scenario. The Study shows that in both steel and automobile cargo segments there will be appreciable growth in the near term, moderate growth in the medium term and subdued growth in the far term. Container traffic appears to have very limited scope for expansion. It may therefore be prudent to wait and watch for a few years how the traffic in the above cargoes behaves. Thereafter, the Port may take the decision for the construction of the third berth considering upon the behaviour of the cargo profile and the performance of the licensee. The expected future traffic of Steel, Ro-RO and containers for the whole of Mumbai

9.2.1 Port is as follows:

Traffic Projections for Steel (Million Tonnes)

Year	**17/18**	**18/19**	**19/20**	**20/21**	**21/22**	**22/23**	**23/24**	**24/25**	**25/30**	**30/35**	**35/40**
Mn Tons	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

Base: actual traffic of MbPT in 2015-16 @ 5.985 million tons

Traffic Projections for Automobiles (Lakh PCU units)

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Lakh Units	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

9.2.2 Traffic Projection for Containers

The scope for container are limited to about 40000 to 50000 TEUs assuming the entire present container traffic coming to Mumbai Port will be continued in future also.

Considering the above projections, the expected traffic at the proposed OCT is as follows.

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Steel Cargo											
Mn Tons	2.50	2.50*	2.63*	2.76*	2.89*	3.03*	3.03*	3.03*	3.03*	3.03*	3.03*
Automobiles											
Lakh Units	2.29	2.63	3.03	3.5	4.01	4.40*	4.40*	4.40*	4.40*	4.40*	4.40*
Containers											
Lakh TEU's	0.20	0.22	0.24	0.27	0.29	0.32	0.39	0.53*	0.53*	0.53*	0.53*

* Capped at Optimal Capacity

9.3 PLANNING PARAMETERS

The existing OCT jetty structure has an overall length of 700M designed to handle two container vessels upto 6,000 TEU capacity each ie. 75,000 DWT; 300 m LOA; 40 m Beam and 12.0 m Draft. With the revised traffic configuration i.e. with RoRo, Steel and containers through smaller ships, the design ship would be Panamax size general cargo vessels upto 75,000 DWT. Based on the past performance, the following conclusions are made.

Steel : The maximum ship size could be Panamax vessels of 225 m LOA but with draft limited to 12 m only. The beam will be 32 m.

RoRo: The largest pure car carrier of 8,000 unit capacity is only 30,000 DWT with 232 m LOA; 32 m Beam and 9.50 m draft.

Container vessels: With the present container traffic at around 50,000 TEUs and which is not expected to have any dramatic increase in future, the size of the veseels will be much smaller, Hence, it should be possible to berth 3 ships concurrently at berth.

Considering the size of the vessels of Ro-Ro and Steel calling Mumbai Port, besides Ro-Ro vessel all the time, it is consider to possible to handle 2 steel vessels part of time and one steel vessel all the time.

9.4 PROJECT DETAILS

The following facilities have been constructed by the Present Licensee, ICTPL, which are completed and are operational.

- Jetty (700 m x 58 m)
- Primary approach trestle (612 m x 18 m)
- Two forked extensions (518 m x 12m & 374 m x 12 m).

In addition they have constructed on the rear of the berth a sub-station, a fire- fighting pump room and a jetty office. However, these are incomplete and are to be equipped with the necessary machinery.

9.4.1 New Civil works to be executed for the revised cargo configuration;

The stackyard development:

Steel	:	26,600 sq.m of area of which 6,650 sq.m will be covered and 19,950 sq.m will be open area
RoRo	:	135,600 sq.m of paved area
Cotainers	:	20,160 sq.m of parking yard

For stacking steel and containers in the open a heavy duty pavement has to be prepared while for RoRo a yard developed with paver blocks will be required. Outof the total area of 18.23 Ha required, 14.65 Ha will be made available at Victoria Dock and the balance made available at Frere Basin.

In addition to the above, 23,900 sqm of railway yard area for handling steel cargo, 2552 sqm for Fire water platform including substation and an area of 6280 sqm for misc. operational buildings is required to be given. Hence the total area required to be given is 215092 sq m or say 21.51 Ha besides the Berth area and the trestle.

9.4.2 New Mechanical equipment required for the revised cargo configuration;

- 2 Nos. of 100 Ton Mobile Harbour Crane for Ship Shore Handling will be mounted on the crane rail tracks already provided at the berth.
- 27 Nos. of 20 Ton Tractor-Trailors for Shore at Stack yard.
- 7 Nos. of 30 Ton Mobile Cranes at Stack yard for Steel Cargo.
- 2 Nos. of Reach Stacker at Stack yard for Containers.

9.4.3 New Mechanical Equipment

Cargo handling equipment are needed for steel. For RoRo berth, the automobile units will be directly driven into the carrier through ramps by drivers. Containers will be handled through ships‘ own gear. If gearless vessels or barges are to be handled, the cranes that handle steel could be used by replacing hooks with spreaders.

For steel equipment are needed for ship-shore transfer, transfer from berth to the stackyard, handling at the yard and also for loading onto the railway wagons.

9.4.4 Ship-shore transfer

For ship-shore transfer, it is suggested to equip one berth with two Harbour Mobile Cranes for efficient handling of cargo. However, these HMCs should be mounted on the rail tracks laid for the container quayside gantry cranes. The deck slab will not be able to take the point loads from the crane leg pads. This crane rail track is of 30 m gauge. Hence the HMCs should be tailer made as rail mounted with this rail

span. The capacity should be to handle minimum 30 Te at 50 m radius. Market enquiry has already been made and a typical crane with technical details has been presented in the earlier section.

For the other berth, the cargo transfer will be through ships‘ own gear as is being done now.

9.4.5 Berth to Yard transfer:

It has been indicated earlier that the productivity of the HMC equipped berth will be 510 TPH while the ships‘ gear operated berth will have a productivity of 405 TPH. It has also been mentioned that the unit weight of steel coils will be 20 Te while that of the steel flats will be 10 Te.

For transfer of cargo between the quay and the storage, 10 truck-trailers will be required, with a distribution of 3 trailers on the each way, 2 trailers on the quay side and 2 trailers at the storage site for achieving the required productivity. Assuming 20 Te capacity trailer/trucks being used, the cycle time for each will be about 35 minutes:

5 minutes each at berth and at yard, 15 minutes travel time from berth to yard and 10 minutes travel time back to berth from the yard. Each trailer should be fitted with twist locks and other container securing arrangement as the same Truck-Trailers will be used for haulage for both steel and containers. Based on these, the berth will need 15 trailers/trucks and the other berth will need 12 trailers/trucks.

9.4.6 Handling equipment at the steel stack yard

When both the steel berths are being occupied and operated, 27 trailers/trucks will be arriving at the yard per hour. In order to unload the cargo, a minimum of 5 numbers of 30 Te mobile cranes will be required.. When the berths are not occupied or not in operation, these cranes will be used for loading the cargo onto evacuating trucks/trailers.

9.4.7 Handling equipment at the container yard

In addition to the tractor trailers, the containers yard will require two reachstacker and receipt/ delivery operations.

9.4.8 Handling equipment at the railway yard

For loading the cargo into railway wagons at the railway yard, at least 2 numbers of 30 Te mobile cranes will be required.

9.5 CAPITAL COST AND IMPLEMENTATION SCHEDULE

The total capital cost of the project as a replacement cost of the existing assets is estimated at Rs. 549.31 crores by M/s SBI Caps. An amount of Rs. 166.36 crores has been considered towards additional cap-ex for development of stack yards etc and procurement of cargo handling cranes without which Steel cargo cannot be handled. Thus total cost of the project is estimated at Rs. 715.67 crores. The entire project is expected to be completed in 18 months time from the date of grant of concession to the operator.

9.5.1 Operation And Maintenance Cost

The annual operation and maintenance cost of the Project based on the TAMP Guidelines 2008 is estimated at Rs. 112.79 Crores.

9.5.2 Revenue Earnining

The estimated annual revenue based on provisional tariff assessed as per the upfront tariff guidelines 2008 / Tariff orders is given below The estimated annual revenue from the project is given below:

Sr. no.	Particulars	2017-18
1.	Estimated Throughput	
	(a) RoRo (in lakh Per Car Units p.a)	4.40
	(b) Steel (in Lakh tons p.a.)	30.06
	(c) Containers (TEUs)	52560
2.	Handling RateRate (Rs)	

	(a) Automobiles (Per PCU)	1250
	(b) Steel(Per ton)	304
	(c)Container (PerTEU)	4344
3.	Annual Revenue Requirement (ARR) (Rs. In lakhs)	16929.34
	(a) RoRo	5497.96
	(b) Steel	9148.25
	(c)Conatiner	2283.13
4.	Estimated GRT (Lakh GRT hours)	4293.61
5.	Berth hire (Rs./ GRT hour)	1.35
6.	Revenue on Berth hire (Rs. In lakhs)	5800.73
7.	Total Estimated Income (Rs. lakhs)	22730.07

9.5.3 FINANCIAL VIABILIY ANALYSIS

The pre tax project IRR from the private operator‘s perspective is as follows:

Sr. No.	Pre-Tax Project IRR	IRR (%)	NPV @ 12% (in Rs. Cr)
1.	Base case	13.86%	93.61
2.	Capital Cost up by 10%	12.53%	28.91
3.	Revenue down by 10%	10.60%	(-) 67.21
4.	Annual O&M Cost up by 10%	12.14%	6.84
5.	Combined effect of Sl. no. 2, 3 & 4	7.64%	(-) 218.68

However, the viability of the project will further improve, if the operator achieves the productivity norms and become eligible for 15% increase in tariff over the notified tariff.

9.6 BASELINE ENVIRONMENT

9.6.1 Ambient Air Quality

In order to access the baseline air quality of the study area, ambient air quality monitoring was undertaken at 8 locations. The parameters for AAQM include Particulate Matter (PM10 AND PM2.5), Oxides of Sulphur (SO2), Oxides of Nitrogen (NOx), Carbon Monoxide (CO). All the parameters were monitored on 24 hourly basis except for CO, which was monitoring on 8 hourly basis.

The details of location are given in Table 2 and Fig. 7 above in chapter 3. The results of monitoring are represented in Table 8 & Figure 9 of Chapter 3.

9.6.2 Marin Water Quality

The surface and bottom marine water monitoring was carried out at 2 stations. The distance and direction of these stations from the project site are given in Table 9. The marine water monitoring locations are shown in the Figure 10 of Chapter 3. Niskin sampler was used for sampling the bottom water. The results are given in the Table 10 of Chapter 3

9.6.3 Sediment Quality

To know the present sediment quality near the project site, 2 monitoring stations were selected within 10km (aerial distance) from the project location. The sediment sampling was done using a grab sampler. The results for the sediment analysis are given in the Table 11 of Cahpter 3

9.6.4 Ambient Noise Levels

The main objective of noise monitoring in the study area is to establish the baseline noise levels, and assess the impact of the total noise that is expected to be generated by the proposed activities. Considering the site topography, proposed activities during construction and operation phase, noise monitoring has been undertaken for 24 hours at 8 locations. The noise monitoring locations are represented in Table 2 & Figure 5 above of Chapter 3. The noise levels in study area are represented in Table 13 of Chapter 3.

CHAPTER 10 - CONSULTANTS ENGAGED

We are one of the leading environmental and environmentally related infrastructure project consultants and completed no. of prestigious projects in the country. It includes Rs. 6,000 crore project of Sewri- Nhava Sheva sea link i.e. Mumbai Trans Harbour Link project, Rs. 500 crore of Inland Passenger Water Transport (IPWT) project. We have been also associated with major industrial houses in India, which includes ELDER PHARMA, DUPHAR GROUP, RUNWAL GROUP (ARIANE PHARMA), VINATI ORGANICS, ETC. along with FINOLEX group, CABOT India, NITCO group, ISPAT group, INDIAN OIL CORPORATION LTD., SAHARA group,
etc.

We have been also associated with various commercial development projects and companies. This includes RUNWAL GROUP, MATOSHREE REALTORS, FRAVASHI GROUP, SOLITUDE, PWD DEPT., etc. This project consultancy includes designing and commissioning of Sewage Recycling plants, Green Belt development, storm water design, etc.

We have our Lead Office in Mumbai at:
102, HIREN INDUSTRIAL ESTATE,
MOGUL LANE, MAHIM (W)
MUMBAI - 400 016.
MAHARASHTRA
To cater the needs of Industrial Corporate Houses in remaining part of the state, we have offices at Pune and Aurangabad. We also have our presence in the state of Goa and have office at Ponda. We also have our liasoning offices with our staff at Solapur, Kolhapur, Satara and Nagpur.

We have full-fledged office consisting of total of 5 Nos. of Computers operated by engineers themselves along with all the infrastructure facilities. The computers are loaded with different types of programs to cater various requirements.

The technical team is led by Dr. U. S. Kulkarni and is assisted by 5 nos. of Engineers. There are 3 nos. of field teams for carrying out various Surveys, Monitoring Programs, Operation and Maintenance. We also have resident engineers who are responsible for their regions. In all International Infrastructure Projects & Labs is fully self-sufficient, technically competent and therefore can handle any project irrespective of its size and capacity.

A.2. ABOUT OUR ENGINEERING/ARCHITECT OFFICES:

International Infrastructure Projects & Labs's entire Engineering and Architectural work is being carried out at 2 separate offices in Mulund, Mumbai and also at Pune. Both the offices have their locational advantage with respect to peripheral industrial developments and are very well connected by the State Highways and by Airports. These offices have their own Team Leaders and they independently handle the Engineering and Architectural drawings with the help of their strong technical team. These engineering offices carry out following jobs in general.

Preparation of all drawings viz. layout, hydraulic, GA, RCC etc.

Preparation of Summary Sheets for bill of quantities

Preparation of Comparative Statements

Design of columns, beams, slabs, etc.

We have facility to carry out Detailed Designing, Structural Designing, and Fabrication Drawings at our above offices.

A.3. ABOUT OUR LABORATORY:

This is our heart of the Consultancy Services as we carry out various feasibility and treatability studies before designing a particular project. The laboratory is fully sophisticated and computerized and moreover it is FDA approved laboratory. One can carry out any no. of chemical, physico-chemical analysis of water and wastewater using various analytical instruments. Such a facility of water, wastewater, air monitoring and noise measurements is also available in Pune.

CURRICULUM VITAE

OF

DR. UMESH S. KULKARNI

(M.Sc., Ph.D)

Name	DR. UMESH SHIVANAND KULKARNI
Contact details	102, Hiren Industrial Estate, Mogul Lane, Mahim, Mumbai – 400 016, Maharashtra, India. Resi: 2445 21 63 / 2444 58 36 Office: 2444 08 33 Fax : 2444 58 37 Mobile : 98200 63632 E-mail : druskul@gmail.com
Qualification	Ph.D. (Treatment of Industrial Wastewater) at Dept. of Chemical Technology (UDCT), University of Mumbai. M. Sc. (Physical Chemistry), University of Mumbai B. SC. (Chemistry), University of Mumbai
Nationality	Indian
Date of birth	12th March, 1964
Affiliations	• Associate Member – FICCI • American Solid Waste Association (ASWA) • Indian Water Works Association (IWWA) • Water Environment Federation (WEF) • Indian Institute of Environment Sciences(IIES) • International Eco – Tourism Society (IETS) • Associate member – Indo Brazilian Society • Member - Maharashtra Economic Development Corporation (MEDC)
Professional expertise	• Environmental Infrastructure Projects • Public Health and Bio-Medical Waste Sciences • Environment & Ecological Sciences • Hazardous Waste Management / Municipal Waste Management • Eco – Tourism • Development of Large Industrial Areas pertaining to Environmental Aspects • Environment Impact Assessment (EIA) • ISO 14001

	• Water Audit and Water Conservation • Lake Ecology and Restoration
Achievements	• Appointed as Expert member on Steering Committee of FICCI, Western Region • Appointed as Expert Member on Advisory Committee of National Hydro-Electric Power Corporation (NHPC) on the Environmental Aspects, Ministry of Power, New Delhi. • Nominated as Expert Member on Committee Formulation of Bio-Medical Rules – 1998 by Ministry of Environment and Forest, New Delhi, (MoEF) • Appointed as Technical Board Member of Maharashtra Pollution Control Board MPCB), Government of Maharashtra. • Nominated as an Expert member of Indian Institute of Rural Development (IIRD) (Joint Venture with Govt. of Rajasthan, WHO, UNDP etc.) • Member – Industrial and Environment Committee of Indian Merchant‘s Chambers (IMC). • Nominated as Expert Member on Environment Committee of Maharashtra Economic Development Corporation (MEDC) • Appointed as Hon. Secretary of Indo- Brazilian Society by The Ho. Consul Of Brazil
Associated Corporate	• Director – Ashok Alco-Chem Ltd. • Director – Ashok Alco Bio-Chem Ltd • Director – Vivid Chemicals Ltd. • CEO – International Infrastructure Projects and Labs. (IIPL)
Other achievements	• Associated with no. of Govt. & Departments such as NHAI, MSRDC, MIDC, MMRDA, MPCB, Dept. of Health, Water Resources etc., MoEF, MNES, NHPC, WAPCOS, NEERI • Also associated with no. of professional bodies such as IMC, BCCI, FCCI, CII, TERI, WHO, UNDP and also associated with no of Universities for their curriculum in the field of Environment and Ecology.

	• Recently developing Marine Bio – Technology Department in Ruparel College, Mumbai
Publications	• No. of research papers have been published in India and Foreign Journals including Indian Journal of Environment, American Chemical Society, Journal of Public Health etc. • Also delivered and presented no of research papers in person at various International level conferences at USA and Singapore • Recently presented research paper at 5TH European Conference on Environment and Ecology at Prague, Czech Republic, in Oct.-2001. • Attended no. of Conferences and Exhibitions at Singapore in the past.
Patents	• Involved in application of an Indian Patent for —Separation and Recovery of micro level Lignin from Bagasse (Pulp Effluents)॥ • Recently selected as qualified for Government of Maharashtra for carrying out Water and Energy audits as per World Bank standards.

Annexure 1

COASTAL ZONE MANAGEMENT PLAN
MAHARASHTRA

Sheet No: E 43 G 13 / NW

Projection :- UTM Datum :- WGS 1984 Map No. :MH 72

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Meters
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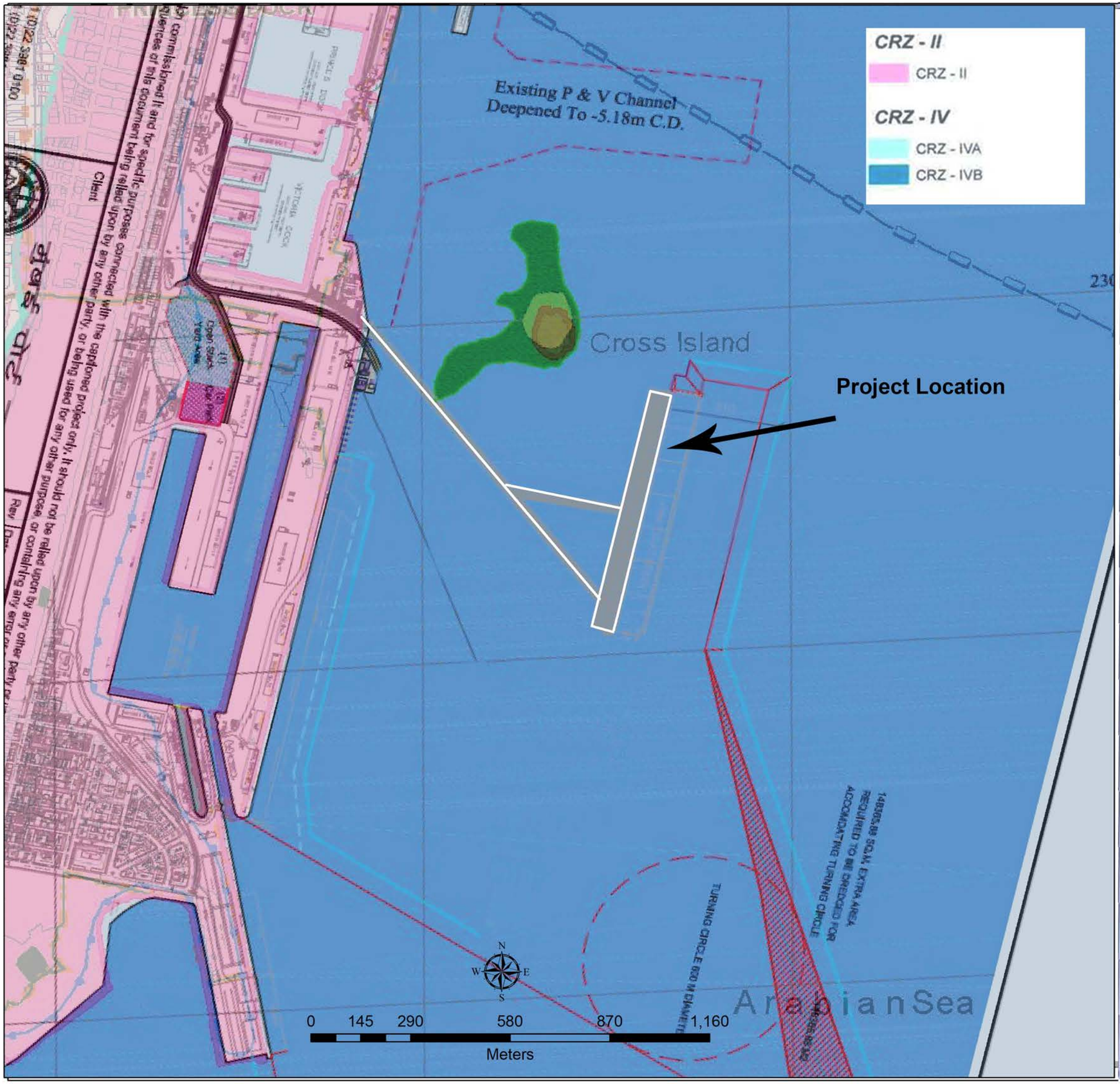

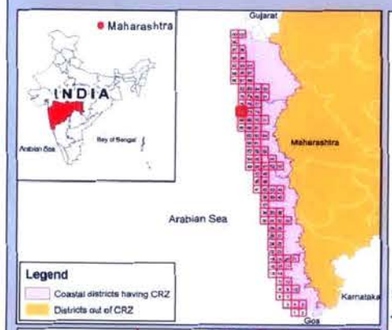

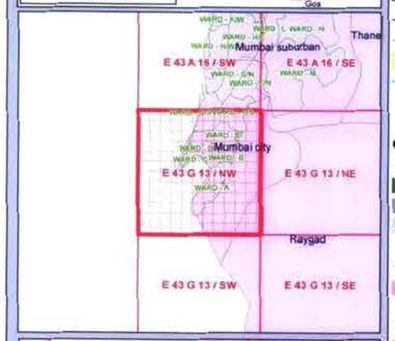

Legend

- Road
- Ferry
- Railway Line
- High Tide Line (HTL)
- Low Tide Line (LTL)
- Bund
- Seawall
- Ward Boundary
- Village Boundary
- 22 Survey Plots

CRZ Lines & Boundary

- Hazard Line
- 100 m Line in CRZ III Area
- 200 m CRZ Line - NDZ
- CVCA
- CRZ Boundary
  (500 m Line, 100 m for Bay, 100 m or width of the creek whichever is less along the tidal influenced water bodies)

CRZ CATEGORY

CRZ - I
- CRZ - IA
- 50 m Mangrove Buffer Zone (CRZ - IA)
- CRZ - IB

CRZ - II
- CRZ - II

CRZ - III
- No Development Zone
- 200 to 500 m from HTL

CRZ - IV
- CRZ - IVA
- CRZ - IVB

DATA SOURCE

I) National Centre for Sustainable Coastal Management
1) HTL, 2) CRZ - IA, 3) CVCA

II) Survey of India
1) Hazard Line

III) Maharashtra Coastal Zone Management Authority
1) Administrative Boundaries

IV) Institute of Remote Sensing
1) CRZ Categories and Lines (except CRZ - IA)
2) Infrastructure facilities such as Road, Ferry, etc

ABBREVIATIONS

CRZ : Coastal Regulation Zone
NDZ : No Development Zone
CVCA : Critically Vulnerable Coastal Area

Mapped During 2017-18

PREPARED AS PER COASTAL REGULATION ZONE NOTIFICATION 2011

Scrutinized by	Certified by	Approved by
Technical Scrutiny Committee National Centre for Sustainable Coastal Management	DIRECTOR National Centre for Sustainable Coastal Management Ministry of Environment, Forest & Climate Change	National Coastal Zone Management Authority Ministry of Environment, Forest & Climate Change Government of India

Prepared by

Institute of Remote Sensing
Anna University
Chennai - 600 025

State Government of Maharashtra

Annexure 2

CF 226(CRZ)/MB-38

MAHARASHTRA COASTAL ZONE MANAGEMENT AUTHORITY

4010437/4014701/4020781
4023516 / 4024068

Office of the :-
MAHARASHTRA POLLUTION CONTROL BOARD,
KALPATARU POINT, 3rd & 4th Floor,
Sion Maturga Scheme Road No.8,
Opp. Cine Planet Cinema, Near Sion Circle,
Sion (East), MUMBAI – 400 022.

Date: 9/5/05

MCZMA / 78

To,
The Secretary,
Ministry of Environment & Forests,
Govt. of India,
Paryavaran Bhavan, CGO Complex,
Lodi Road, New Delhi- 100 003.

	Office
	CF 226/CRZ
	PORT TRUST
	1120
	Dt:
	Date: 9/5/05

Sub: Application received from Mumbai Port Trust seeking CRZ clearance for two offshore berths and for construction of containers stack yard for new terminal.

Sir,

Mumbai Port Trust has made an application seeking CRZ clearance for two offshore berths and for construction of containers stack yard for new the terminal by its letter dt.28/3/2005. It has been stated that the traffic projections indicate that Mumbai Port Trust has got the potentional to increase its containers traffic from about 0.3 MTEUS in the year, 2001to about 0.73 MTEUS in 2006, 0.82 MTEUS in 2008 and 1.45 MTEUS by 2018 and therefore, they have proposed a new stack yard and offshore berths. The project proponent has carried out the Environment Impact Assessment study through NEERI.

The proposal was placed before 27th meeting of the Authority held on 5/4/2005 and discussed at length. The Authority has considered the above proposal from the point of view of traffic projections indicated by the Port Trust Authority and decided to recommend the said proposal to the Ministry of Environment & Forests, Govt.of India, having the cost of the project more than Rs.5 crores. The proposal is therefore recommended for grant of environment clearance. A copy of the an application received from the Mumbai Port Trust dt.28/3/2005 alongwith annexures thereto are enclosed for perusal and further doing needful in the matter.

Thanking you,

Yours faithfully,

D B Boralkar

(D. B. Boralkar)
Member Secretary

Encl.: As above

Copy submitted to –
The Chairman, MCZMA, office of the Principal Secreta
Mantralaya, 4th floor, Mumbai-400 032, for information plea

Copy to –
1. Deputy Secretary (Tech.), Env. Deptt., GOM, New Admn. Bldg., Mantralaya, Mumbai-400 032.
2. Sr. Law Officer(LW), MPCB, Sion, Mumbai – for information and necessary record please.
3. Regional Officer(P&P), MPCB, Sion, Mumbai – for information

CE(I/c) / Dy. CE (Dev.) - on leave
Sr EX Engr. Dredg.

AEP
9/5/05

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Annexure 3

104

Minutes of Environmental Public Hearing about proposed Project of Offshore container berths and development of container terminal on BOT basis by M/s. Mumbai Port Trust, at Mumbai Harbour. Held on 29/09/2005, at Durbar Hall, of the Asiatic Society of Mumjbai-23.

The Environmental Public Hearing about proposed project of "Offshore container berths and development of container terminal on BOT basis of M/s. Mumbai Port Trust was held on 29/09/2005 under the Chairmanship of Dr. D.B.Boralkar , Member Secretary, Maharashtra Pollution Control Board, at Durbar Hall, of the Asiatic Society of Mumbai-23. Convener of the public hearing panel welcomed the Chairman, panel members and the public participants present for public hearing.

Convener of the Public Hearing panel explained purpose and necessity of public hearing as per the Notification No. S.O.318(E), dtd. 10th April, 1997, under the Environment (Protection) Act 1986. Convener also placed facts about the procedure following by MPCB as per Notification.

As per the notice published in newspapers, written objections were called in the office of Maharashtra Pollution Control Board at Regional Office, Mumbai, from the Concerned persons. Also, copies of the executive summary each in English and Marathi about the proposed project were made available at Collector Office, Mumbai City, Department of Environment , Govt. of Maharashtra , Mumbai, Head Office of Maharashtra Pollution Control Board at Sion, Mumbai & Regional Office, Mumbai of Maharashtra Pollution Control Board at Shri Chhatrapati Shivaji Maharaj Mandai Municipal Bldg., 4th Floor, Mata Ramabai Ambedkar Road, Mumbai-400 001 to apprise the public about the proposed project.

Following are the panel members for the public hearing.

1)	Dr. D.B.Boralkar, Member Secretary, Maharashtra Pollution Control Board,	-Chairman
2)	District Collector or his Nominee,	-Member
3)	Shri. Surendra Jadhav, Dy. Secretary Env. Dept., Govt of Maharashtra, Mumbai.	-Member
4)	Representative of Urban Development & Sanitation, Govt. of Maharashtra, Mumbai.	-Member
5)	Shri. Father Dorkle Sr. Citizen.	-Member

6)	Smt. A.F.Remedios, Sr. Citizen.	-Member
7)	Shri. D.C Mane, Sr. Citizen.	-Member
8)	Smt. Kanoji Geetabai Chhe[illegible], Corporetor, M.C.G.B.	-Member
9)	Shri. Dosti Puran Mohanlal, Corporator, M.C.G.B.	-Member
10)	Shri. Dhulla Vijaya Kantilal, Corporator, M.C.G.B	-Member
11)	Shri. B.D.Wadde, S.R.O. Mumbai-I	-Convener

Convener of the Public Hearing panel declared that in spite of appeal in the newspapers, the office of Maharashtra Pollution Control Board have not received any written objection/suggestion in the register kept in the Regional Office-Mumbai.

Thereafter, the project proponents proceedings commenced with brief presentation on salient features of their proposed project. They covered description of site, location, details of construction, facilities provided, impacts on environment and related points.

Thereafter, Chairman [illegible]ed the participants/citizens to register the objections/points if any they have Chairman and Members have also raised certain points about the said proposed project to which project representative has given explanation. Details of Questions & Answers raised during the public hearing are as below:-

1) <u>Shri. Debu Goenka, (N.G.O.)</u>

Point No.1 :- Where is the CWPRS report?

Explanation:- The report is not enclosed and CWPRS have done siltation studies in the Mathematical Model. The Mathematical Model studies for the entire Mumbai harbour has been done. The study for the subject project has also been done and is a part of the report. The location and the orientation of the proposed berths have also been studied on the model and they have opined that the proposed berth is well aligned with the flow direction. They have also stated that since the proposed berth and its approaches to the shore are on piles the flow pattern is not affected. The quantum of siltation due to this project is expected to be about 2 million Cu. M. Thus the extra siltation is hardly 10-15% more than what we are doing now. The dumping of material due to capital dredging will be done at location where there is good dispersion and will not affect the mangrove locality. The nearest Mangrove from the proposed dumping ground is [illegible] km. He further explained the [illegible] siltation is [illegible] and [illegible] report will be [illegible]

2) Smt. Neera Punj, (N.G.O.)

Point No.1:- She asked abo . road management, space management, supporting infrastructure, mainland infrastructure, proper transport system. Why are we investing this money for when there is no proper infrastructure available to the general public?

Explanation:- Mumbai Port Trust owns a very good network of road and rail. MbPT owns roads from Docks to Wadala dedicated for the Port Traffic. At present hardly 2000 trucks ply on these roads owing to Port Traffic. We have handled 6 lakh containers with the existing road system in 1997-98 without Truck Terminus road at that time . Now, with the Truck Terminus Road, in place and also proposed Anik Panjrapole Link Road for which construction has already started, the Port Traffic will directly join the National Highways bypassing the city roads. Also when the proposed berthes are commissioned in 2009, the traffic at these berths for the first 5 years will not be more than 6 Lakh TEUs. The existing roads within the Port estate will be widened wherever required for smooth movement of Traffic.

As regards Railways, MbPT owns its own rail network from Docks to Wadala (55 km track length and 10 km route length). Mumbai Port is also proposing dedicated additional railway line from Wadala to Kurla for which the feasibility studies have been done and the pro is feasible. The final estimate and location survey have been completed and C al Railway is going to submit final eport within a week's time. With this dedic d link from Wadala to Kurla and also the 5th and 6th line laid by the Central Railway from Thane to Kurla (work almost completed) and Thane to Kalyan (work commenced) which is dedicated for the through trains and goods trains , there will be smooth sail of cargo movement through rail system.

Point No.2:- If feasibility study is already done, then it should be part of the report Then why this Public Hearing without the report?

Explanation:- In the Year 1996 we have handled 6 Lakh TEUs. Now i.e. in the year 2004-2005 we have handled only about 2-3 Lakh TEUs. Thus we have spare capacity available. To enhance the capacity of handling cargo it is necessary to construct offshore container berth.

3) **Shri. Father Berkle, Member, Environmental Public Hearing**

Point No.1 :- The project is necessary, but since the container traffic is shifted to JNPT, what is the benefits of the project and what is the ratio of evacuation by road and rail ?

Explanation:- 6:40 Ratio is c idered in feasibility report.

4) **Shri. A.O.Remedios, Member, Environmental Public Hearing**

Point No.1 :- What about MTHL project? If the project is delayed then the evacuation will be problem. The offshore project will be constructed by International firms and will complete the works well in time. Whereas the Bandra Worli link is delayed by the Indian company. Thus the evacuation will be problem.

Explanation: - Shri. Puran Doshi objected Mr. Remedios above point stating that the Indian companies are quite competent and comparable to any international firms. I have been in the standing committee MCGM for so long and I know the problems. The award of works to consultants is dictated by the World Bank and consultants are from abroad only. Don't blame Indian engineers and Companies alone for the delay.

Project proponent:- The evacuation of Port cargo is not at all dependent of the MTHL. What I want to say that it will only supplement the evacuation.

8) Shri. Debu Gonnka, (N.G.O.) :-

Point No.1:- You have assumed one-inch leakage. It may be more also at the site. What would be the follow up done by the Disaster Management Plan (DMP is minimum risk) while taking into consideration the minimum risk?

Explanation:- The Disaste nagement prepared by the Port is adequate to handle the condition. Accident and s, leakages (one or two inch) happen due to unloading operation, fire fighting is able to handle such situation and maximum precautions are taken. The hazardous chemicals come in the form of containers. There will be hardly few container carrying cylinders. The existing DMP of the Port is adequate to handle any risk.

Point No. 2:- Both the road and the rail network is congested/saturated and now. Traffic handled is proposed to be handled is not clear in the report. What is the waiting time? JNPT was created to decongest the city. Now what is the necessity of this project?

Explanation:- JNPT was set up as a part of Mumbai Port and subsequently it was developed as a satellite Port. We are not setting up as a competition to JNPT. The container traffic is increasing at a phenomenal rate and almost 60% of the Country's traffic is handled by Mumbai Region and this Project is the requirement of the trade and in the interest of the Nation. JNPT has already awarded for the construction of 3rd terminal. They are conducting feasibility studies for the 4th container terminal. Even after that there will be a substantial gap which is to be serviced by another Port so why not Mumbai Port. My other point is regarding the hinterland services Mumbai city itself will account for of the container traffic, which can be well served by the Mumbai Port with the new rminal. Mumbai Port contributes to Customs nearly 500-600 Crores worth goods which will be more than double if the project is envisaged. MbPT also contributes Rs. 350 Crores towads Octroi to MCGM, which will increase upto Rs. 800-900 Crores. Similarly Stamp Duty, Sales Tax will go up. We should gear up to handle huge container traffic projected in the next ten years and since 60% of Hinterland consumption is from the Mumbai Region, this project is necessary in the interest of Nation. Other thing is that it creates employment, other economical activities, labour contractors, service providers etc. These changes are positive aspects of this project. It is first of its kind in India and pride of Mumbai. Mumbai Port has studied all the issues comprehensively.

Point No.3:- He desired to know about the evacuation of cargo by road (roads already being congested)

Explanation: - There should be no apprehension on this count. On the road front, I divide the connectivity into 2 parts, one is internal roads within the Mumbai Port Estate, where we are planning to widen the roads, construct flyovers etc. We are investing about Rs. 90 Crores for this.

The second is the road improvements being carried out by MMRDA. The Anik Panjrapole Link Road. This is a [illegible] link and the work has already begun. Mumbai Port is also sensitive to the co[illegible]station issue. We are funding Rs. 35 Crores to ensure that the project is completed well in time before the offshore berths are commissioned. The work would be completed in 2 years time. From Panjrapul MMRDA is planning to develop two stretches, which directly connects to Eastern Express Highway. Eastern Express Highway is 10 lane road. With the above projects taken up by MMRDA, the entire Port traffic will bypass the City roads. I would like to tell you that Mumbai Prot handled 6 Lakh TEUs in 1996-1997 and evacuation was not at all a problem at that time even without Wadala Truck Terminus Road. This traffic of 1.2 Million TEUs will come only after 10 years by which time even MTHL will be completed.

About Rail connectivity, members / citizen's should appreciate that his traffic will put minimum constraints to the general public. The Wadala Kurla Link will come and it is being monitored by the Prime Minister Office. The ministry has given high priority for this Rail Link. Mumbai Port is funding this project and I am sure that this would be completed before commissioning of the berths. Even though I am very much confident that this rail link would be completed, assuming that this will not complete in time, there should be no concern to MbPt as MbPT is capable of handling 6 lakhs TEUs with the existing link itself. We have handled without any difficult previously. After all the traffic of [illegible] million TEUs will be achieved in the 10th Year.

I would like to state that MMRDA has submitted a report for construction of East island freeway from Museum to Ghatkopar. Mumbai Port Trust has given clearance for this link for aligning in Mumbai Port estate. This link will further ease the traffic on City roads.

6) Shri. Father Berkle, Member, Environmental Public Hearing

Point No – 1: - What comes to the common man? P. D'mello Road is fully blocked / encroached by the Slums. What about their rehabilitation? What profit will MbPT make out of this project?

Explanation: - I share your sentiment fully. MbPT has constructed several roads, which are given to public, and P.D' Mello road is one of them. This road is maintained by MCGM. Slums have developed not because of MbPT. This matter is being taken up with BMC / Chief Secretary, Government of Maharashtra. In fact for the Wadala Kurla rail link, about 2140 hutments will have to be resettled. We have made provision for their rehabilitation. MbPT have become victim of migration and hence the proliferation of slums. MbPT will see to that no further slums will develop. It is true that public needs to be [illegible] slums [illegible] redevelopment.

7) Smt. Neera Puni, (N.G.O.) & Dr. Boralkar, Chairman.

Point No.-1: - Whether any heritage buildings are to be demolished for this Project? No mention has been made in the report.

Prepare list of heritage structures under Mumbai Port and state if any of these heritage structures, are affected by this project, if yes, list out the heritage structures.

Explanation: - There is no heritage building in the Project area. All we are demolishing is the old sheds and warehouses.

8) Shri, Debu Goenka, (N.G.O.),

Point No.-1: - Whether any other land will be reclaimed? Whether mangroves will be affected due to additional dredging?

Explanation: - No land is being are claimed for this project. We are just filling up the old dock basin. About Mangroves, I should say this would not at all affected. The dumping area is about 25 km away from Mangroves. Mumbai Port has spent to dovelop and protect Mangrove at Sewree. Previously about 2000 Flamingos were there but of today there are about 14000 flamingos. This project will in no way affect this habitat. Nearly about 15 acres of area have been replanted. We spent Rs. 50 Lakhs to develop this area as sanctuary. The mangroves situated between Sewree-Wadala will be in no way affected.

9) Shri. Father Berkle, Mer . Environmental Public Hearing

Point No – 1 :- What about passenger water transport from Ferry Wharf. Whether this is affected by the project.

Explanation :- Not affected. In fact MbPT has given clearance to Ferry Wharf Belapur Cruise Terminal to the Maharashtra Government. They have appointed consultants for finalizing the terms.

10) Shri, Debu Goenka, (N.G.O).

Point No-1 :- Trucks are parked on the roads. Wadala Truck Terminus is more than 10 years old No use is made of it.

Suggestion by Remedios: - I suggest that MbPT think of beautifying the MbPT Roads so that general public can use the same and income is earned by MbPT. MbPT should create health centre, gymnasium and swimming pool at the recreation area -- Botanical Garden developed by them.

I should mention that Mumbai came up because of Mumbai Port Trust. There should be no objection for revival of the Port.

Suggestion by Dr. Bora : - I agree with you. In fact the entire MbPT should be properly planned to improve the environment of their estate, which includes proper drainage system for the slum etc. Proper environmental plan can be taken. There is an Institute for Environment Planning at Ahmedabad, which will give all details to finalize the plan.

Road within the MbPT estate should be paved.

The point is noted by project proponent.

Explanation :- Sheds will be demolished and the same will be used as filling material after processing Further excavated material and demolished structures due to construction activity in an around Mumbai will also be used. This in one way will solve the dumping of debris generated out of construction activity in Mumbai.

11) Shri. Doshi Puran Mohanlal, Member , Evvironmental Public Hearning

Point No- 1 :- MbPT should give NOC for providing drinking water, toilet blocks etc for the slums in their areas. Municipality is ready to provide. What about the water supply to be given to the hutments?

Explanation :- Requested all the members who attended the meeting not to delay the project. He also explained that these unauthorized hutments, which are post 1995, and are already having unauthorized water connections and MbPT is suffering at their cost. If water connection is given to them this will results in to huge proliferation of hutments and blame MbPT for the same and vice-versa. He again stated that already MbPT has constructed about 55 Sulabh Sanchalay mainly in Darukhana area.

1) **Suggestion by Father Berkie:-** MCGM and MbPT should co-ordinate with each other.

2) **Suggestion by Remedios:-** Stated that this query is related to MCGM.

3) **Suggestion by Doshi Puran :-** Stated that government has ordered that after 26th July torrential rains rehabilitate the slums outside the city limits.

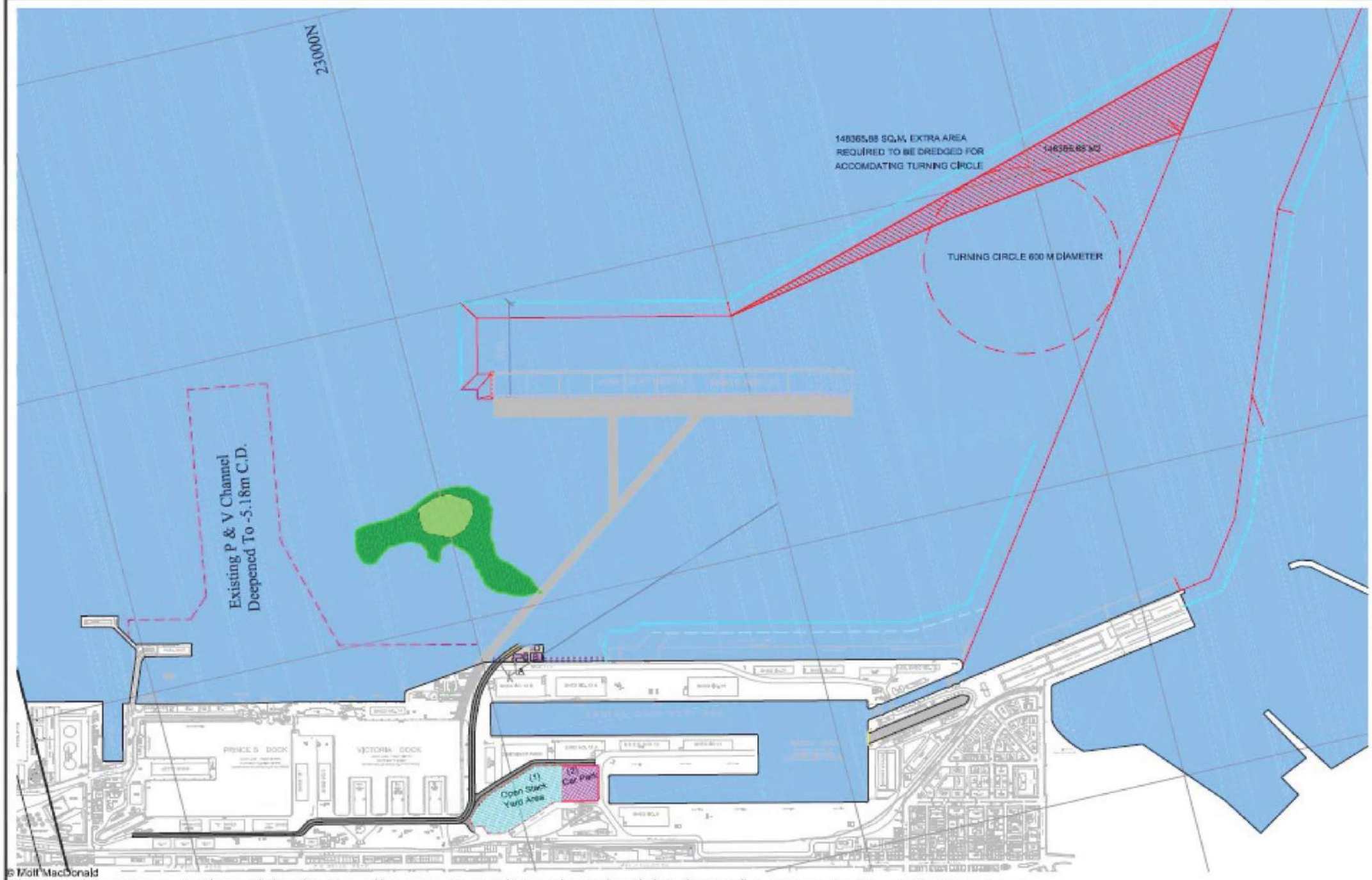
Lastly, Chairman and Convener of the public hearing panel again suggested the participants/public to ask questions/points/objections/doubts if still they have. However, nobody raised any questions/points/objections/doubts about the said.

Finally, the Public Hearing was concluded with vote of thanks to all those who were present by Chairman and convener of the Public Hearing panel.

(D.B.Boralkar)
Chairman
Environmental Public Hearing Panel

Member Secretary
Maharashtra Pollution Control Board

Annexure 4



Buildings :

A. Administrative Building	= 432 Sqm
B. Sub Station	= 875 Sqm
C. Water Tank	= 250 Sqm
D. Workshop	= 100 Sqm
1. Open Stack Yard Area	= 35888 Sqm
2. Car Park	= 12570 Sqm
3. Reclaimation Area	= 4875 Sqm

Legends :

Proposed Berth

Road ed Dredged Area

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Client

मुंबई पोर्ट ट्रस्ट

Mumbai Port Trust

MbPT ESTATE

Rev	Date	Drawn	Description	Ch'k'd	App'd

Drawn	K Patel
Checked	S Pallwal
Approved	U Phalke
Scale at A3	NTS
Status: PRE	Rev: P1

Annexure 5

ITEMS IDENTIFIED BY THE PUBLIC AND OTHER STAKEHOLDERS

The Public Hearing was held on 29/09/2005 under the Chairmanship of. Dr. D. B. Boralkar, Member Secretary, Maharashtra Pollution Control Board, at Durbar Hall, of the Asiatic Society of Mumbai-23. Convener of the Public Hearing panel explained purpose and necessity of the public hearing as per the Notification No. S.O.318(E), dtd. 10th April, 1997, and the Environment (Protection) Act, 1986. Convener also placed facts about the procedure followed by MPCB as per Notification.

As per the notice published in newspaper, written objections were called in the office of Maharashtra Pollution Control Board at Regional Office, Mumbai about the concerned proposed project. Also, copies of the executive summary in English And Marathi about the proposed project were made available at Collector Office, Mumbai city, Department of Environment, Govt. of Maharashtra, Mumbai, Head Office Maharashtra Pollution control Board at Sion, Mumbai & Regional Office, Mumbai of Maharashtra Pollution control board at Shri Chhatrapati Shivaji Maharaj Mandai Municipal bridge 4th floor Mata Ramabai Ambedkar Road, Mumbai-400 001 to appraise the public about the proposed project.

Convener of the Public Hearing panel declared that in spite of appeal in the newspaper, the office of Maharashtra Pollution Control Board has not received any objection/suggestion in the register kept in the Regional Office-Mumbai.

Thereafter, the project proponent's proceedings commenced with brief presentation on salient features of their proposed project. They covered description of site, location, details of construction, facilities provided, impact on environment and other related point.

The minutes of public hearing and action plan is given below in Table 1.

During deliberation of EAC meeting held on 25-27th May, 2017, the project proponent made a request for an exemption from Public Hearing as the same was conducted in 2005. In 18th meeting's MoM in point no. 4, the committee allowed the same (exemption from public hearing).

Table 1: Minutes of Public Hearing held on 29/09/2005

S.N	Name of the Person	Points Raised	Replies	Action Plan/Management plan
1.	Shri. Debu Goenkar (N.G.O.)	Where is the CWPRS report?	The report is not enclosed and CWPRS have gone siltation studies in the Mathematical Model. The Mathematical Model study for the entire Mumbai harbor is done. The Study for the subject project has also been done and is a part of the report. The location and the orientation of the proposed berths have also been studied on the model and they have opined that the proposed berth is well aligned with the flow direction. They have also stated that since the proposed birth and its approached to the shore are on piles the flow pattern is not affected the quantum of siltation due to this project expected to be about 2 millon Cu M. Thus the extra siltation is hardly 10 to 15 % more than what we are doing now. The dumping of material due to capital dredging will be dumped at location where there is go to dispersion and will not affected the mangrove nearby. The nearest mangrove from the proposed dumping ground is approx. 25 km. He further explain about the rate of siltation of ongoing and stated their record will be calculation	In view of the CWPRS study , no further action is deemed necessary.
2.	Smt. Neera Puni (N.G.O.)	She asked about road management, space management, supporting infrastructure, mainland infrastructure, and proper transport	Mumbai Port Trust owns a very good network of road and rail. MbPT owns roads from Docks to Wadala dedicated for the Port Traffic. At present hardly 2000 trucks ply on these roads owing to Port Traffic. We have handled 6 lakh containers with the existing road system in 1997-98 without: Truck Terminus road at that time. Now, with the Truck terminus Road, in place and also proposed Anik Panjrapole Link Road for which construction has already	Indian Railway will be entrusted the task of laying the dedicated Wadala Kurla railway line. The MbPT road evacuation network will be further improved by undertaking concreting of Roads Present Position: (1) The roads have been concreted wherever

		system. Why are we investing this money for when there is no proper infrastructure available to the general public?	started, the Port Traffic will directly join the National Highways bypassing the city roads. Also when the proposed berth is commissioned in 2009, the traffic. At these berths for the first 5 years will not be more than 6 Lakh TEUs. The existing roads within the Port estate will be widened wherever required for smooth movement of Traffic. As regards Railways, MbPT owns its own rail network from docks to Wadala 55 km track length and 10 km route length) Mumbai Port is also proposing dedicate additional railway line from Wadala to Kurla for which the feasibility studies have been done and the project is feasible. The final estimate and location survey has been completed and central railway is going to submit final report within the week time. With this dedicated link from Wadala to Kurla and also the 5th and 6th line laid by the Central Railway from Thane to Kalyan (work almost commenced) and Thane to Kalyan (work commenced) which is dedicated for the through trains and goods trains, there will be smooth sail of cargo movement through rail system.	necessary. (2) With the commissioning of Eastern Freeway in 2013, the traffic on MbPT's network has reduced thereby improving evacuation of cargo.
		If feasibility study is already done, then it should be part of this report then why this Public Hearing without the report?	In the Year 1996 we have handled 6 Lakh TEUs. Now i.e. in the year 2004-2005 we have handling only about 2-3 Lakh TEUs. Thus we have spare capacity available. To enhance the capacity of handling cargo it is necessary to construct offshore container berth.	Feasibility study is attached as **Annexure 7**
3.	Shri Fathe Borkle, Member, Environmental Public Hearing	The project is necessary, but since the container traffic is shifted to JNPT, what are the benefits of the	6:40 Ratio is considered in feasibility report.	Following are the project benefits: • Increase in EXIM trade due to this facility • Increase facilities for industries in the Mumbai

		project and what is the ratio of evacuation by road and rail?		region and Maharashtra, will also provide many economic benefits • Efficient and modern handling facilities for cargo and berthing of ships • Reduce dependence on transportation by other modes • Direct employment : skilled/unskilled and semiskilled laborers • Indirect employment: Local public in activities such as civil construction, mechanical erection, electrification, transportation, piping works and associated amenities. etc.
4.	Shri. A. O. Remediov: Member, Environmental Public Hearing	What about MTHL project? If the project is delayed then the evacuation will be problem. The offshore project will be constructed by international firms and will complete the works well in time. Whereas Bandra Worli link is delayed by Indian companies. Thus	Shri. Puran Doshi objected Mr. Remedios above point stating that the Indian companies are quite Competent and comparable to any International firms. I have been in the standing committee MCGM for so long and I know the problems. The award of works to consultants is dictated by the World Bank and consultants are from abroad only. Don't blame Indian engineers and Companies alone for the delay Project proponent replies that the evacuation of Port cargo is not at all dependent of the MTHL. What I want to say that it will only supplement the evacuation.	--

		excavation will be problem.		
5.	Shri.Debu Goenkar,(N.G.O.)	You have assumed one-inch leakage. It may be more also at the site. What would be the follow up done by the Disaster Management Plan (DMP is minimum risk) while taking into consideration the minimum risk?	The Disaster management prepared by port is adequate to handle the condition. Accidental analysis leakages (one or two inch) happen due to unloading operation, firefighting is able to handle such situation and maximum precautions are taken. The hazardous chemicals come in the form of containers. There will be hardly few container carrying cylinders. The existing DMP of the Port adequate to handle any like.	Detailed DMP study was conducted by MbPT in 2015 and brief of DMP given in EIA (Chapter 6)
		Both the road and the rail network is congested/saturated and Traffic handled is proposed to be handled is not clear in the report. What is the Waiting time? JNPT, was created to decongest the city. Now what is the necessity of this project?	JNPT was set up as a part of Mumbai Port and subsequently it was developed as a satellite Port. We are not setting up as a competition to JNP, The container traffic is increasing at a phenomenal rate and almost 60% of the Country's traffic is handled by Mumbai Region and this Project.is the requirement of the trade and in the interest of the Nation. JNPT has already awarded for the construction of 3rd terminal. They are conducting feasibility studies for the 4th container terminal. Even after that there will be a substantial gap which is to be substantial gap which is to be serviced by another Port why not Mumbai Port. My other point is regarding the hinterland services Mumbai City Itself will account for the container traffic, which can be well served by the Mumbai Port with the new terminal. Mumbai port contributes to customs nearly 500-600 crores worth goods which will be more than double if the project is envisaged. MbPT also contributes Re. 350 Crores towards Octroi to MCGM, which can be upon Rs. 800-900 Crores Similarly lamp Duty Sales Tax will go up. We should gear up to handle huge container traffic projected in the next ten years and since	In view of demand for container traffic at Mumbai port even after factoring JNPT into account, the proposed project will be implemented.

			60% Hinterland consumption is from the Mumbai Region, this project is necessary interest of Nation. Other thing is that it creates employment, other economic activities, labor contractors, service providers etc. these changes are positive aspects of this project. It is the first of kind in India and pride of Mumbai. Mumbai port has studied all the issues comprehensively.	
		He desired to know about the evacuation of cargo by (roads already being congested)	There should be no apprehension on this count. On the road front, I divide the connectivity into 2 parts, one is Internal roads within the Mumbai Port Estate, where we are planning to widen the roads, construct flyovers etc We are investing about Rs 90 Crore for this. The second is the Improvements being carried out by MMRDA. The Ankh Panjrapol link road. This is a link and the work has already begun. Mumbai Port is also sensitive to the congestion issue. We are funding Rs 35 Crores to ensure that the project is completed well in time before the offshore berths are commissioned. The work would be completed in 2 years' time. From Panjrapol MMRDA is planning to develop two stretches, which directly connects to Eastern Express Highway. Eastern Express Highway is 10 lane road. With the above projects taken up by MMRDA, the entire Port traffic will bypass the City road. I would like to tell you that Mumbai Port handled 6 Lakh TEUs in 1996-1997 and evacuation was not at all a problem at that time even without Wadala Truck Terminus road. This traffic of 1.2 Million TEUs will core only after 10 years by which time MTHL will be completed. About Ralf connectivity, members/citizens should appreciate that his traffic will put minimum constraints to the general public. The Wadala Kurla Link will come and it is being monitored by the Prime Minister Office. The ministry has given high priority for this rail Link. Mumbai	The MbPT road evacuation network will be further improved by undertaking concreting of Roads Present Position: (1) The roads have been concreted wherever necessary. (2) With the commissioning of Eastern Freeway in 2013, the traffic on MbPT's network has reduced thereby improving evacuation of cargo.

			Port is funding this project and i am sure that this would be completed before commissioning of the berths. Ever, though I am very much confident that this rail link would be completed, assuming that this will not completed In time, there should be no concern to MbPt as MbPT is capable of handling 6 lakhs TEUs with the existing link itself, We have handled without any difficult previously. After all the traffic 0.8 million TEU will be achieved in the 10^{TH} Year. I would like to state that MMRDA has submitted a report for construction of East island freeway from Museum to Ghatkopar. Mumbai Pont Trust has given Clearance for the link for aligning in Mumbai Port estate. This link will future ease the traffic on city road.	
6.	Sing, Father Berkie : Member , Environmental Public Hearing	What comes to the common man? P. D'mello Road is fully blocked/encroached by the Slums. What about their rehabilitation? What profit will MbPt make out of this project?	I share your sentiment fully, MbPT has constructed several roads, Which are given to public, and P.D Mello road is one of them. The road is maintained by MCGM. Slums have developed not because of MBPT. This matter is being taken up BMC/Chief Secretary, Govt. of Maharashtra. In fact for the wadala kulra rail link, 2148 hutments will have to be resettled. We have made provision for their rehabitation. MbPT have become victim of migration and hence the proliferation of slums. MBPT will see to that no further slums will develop its area. A firm policy needs to be adopted in terms of slum development.	2148 hutments affected by dedicated Wadala Kurla Line will be resettled.
7.	Smt. Neera Puni (N.G.O) & Dr. Boralkar, Chairman.	Whether any heritage building are to be demolished for this project? No mention has been made in the report. Prepare list of heritage structures	There is no heritage building in the Project area. All we are demolishing is the old shed and warehouses.	--

		under Mumbai Port and state if any of these heritage structures, are affected by this project, if yes, list out the heritage structures.		
8.	Shri Debu Goenka,(N.G.O.)	Weather any other land will be reclaimed? Whether mangroves will be affected due to additional dredging?	No land is being are claimed for this project. We are just filling up the old dock basin. About Mangroves, I should say this would not at all affected. The Mumbai Port has spent dumping area is about 25 km away from Mangroves. Mumbai port has developed and protected Mangrove at sewree Previously about 2000 Flamingos were there but of today there are about 14000 flamingos. This project will in no way affect habitat. Nearly about 15 acres of area as sanctuary has been replanted. We spent Rs. 50 lakhs to develop this area as sanctuary. The mangroves situated between Sewree wadala will be in no way affected.	Fill P&V docks after obtaining regulatory compliances. Present Position: Filling of P&V docks completed only yard development work pending.
9.	Shri. Father Berkle, Member	What about passenger water transport from Ferry what weather this will be affected by the project.	Not affected. In fact MbPT has given clearance to Ferry Wharf Belapur Cruise Terminal to the Maharashtra Government. They have appointed Consultants for finalizing the terms.	Navigation channel/routes of ferry are totally different to MBPT. Since No ferry activity is getting hampered due to proposed project. No
10.	Shri Debu Goenka, (N.G.O)	Trucks are parked on the roads. Wadala Truck Terminus is more than 10 years old No use is made of it.	Suggestion by Remedios: - I suggest that MbPT think of beautifying the MbPT roads and that general public can use the same and income is earned by MbPT. MbPT should create health center, gymnasium and swimming pool at the recreation area-Botanical Garden developed by them. I should mention that Mumbai came up because of	• MbPT has create its own parking plot for day to day operation, however for surplus number of trucks MbPT use Wadala truck terminal as per requirement • Environmental management

			Mumbai Port Trust. There should be no for revival of the Port. Suggestion by Dr. Bora.: I agree you in the fact the either MbPT should be properly claimed to improve environment of their estate which include proper drainage system for the slum etc. proper environment plan can be taken. There is one institute for environment planning at Ahmedabad which will give all support to finalize the plan. Roads within the MBPT estates should be paved. Project Proponent said that Sheds will be demolished and the same will be used as filling material after processing further excavated material and demolished structures due to construction activity in an around Mumbai will also be used. This in one way will believe the dumping of debris generated out of construction activity in Mumbai.	plan has been prepared and given in EIA report. • Roads within the MBPT estates are paved and regular maintenance of unpaved shoulder on paved road is practiced. • As mentioned in sr. no 2 above, further roads will be concreted wherever necessary. MbPT is doing number of project for the benefit of general public i. Construction of Domestic Cruise Terminal Building, New Dock Boundary Wall in Prince's Dock and Evacuation Plan. In this project, the MbPT is developing a coastal passenger transport complex to facilitate free movement of domestic cruise passengers and their vehicles. ii. Proposed repairs limited to reconstruction of existing Cruise Terminal Building (G+1) to G+3 of Mumbai Port Trust at Ballard Pier Extension, Indira Dock, Mumbai.

				The proposed building includes facilities like Business and commercial centres such as hyper market, food court and shopping outlets, travel agencies etc., for the convenience of departing and arriving passengers. iii. <u>Development of Tourism Projects on Kanhoji Angre Island, Dis. Raigad</u>.
11.	Shri, Doshi Puran Mohanlal, Member, Environmental Public Hearing	MbPT should give NOC for providing drinking water, toilet blocks etc. for the slums in their areas. Municipality is ready to provide. What about the water supply to be given to the hutments?	Requested all the members who attended the meeting not to delay the project. He also explained that these unauthorized hutments, which are post 1995, and are already having unauthorized water connections and MbPT is suffering at their cost. If water connection is given to them this will results in to huge proliferation of hutments and blame MBPT for the same and vice versa. He again stated that already MBPT has constructed about 55 sulabh sauchalaya mainly in darukhana area. 1. Suggestion by Father Berkie:- MCGM and MbPT should co-ordinate with each other, 2. Suggestion by Remedios: Stated that this query is related to MCGM. 3. Suggestion by Remedios :- Stated that this query is related to MCGM 4. Suggestion by Doshi Puran:- Stated that government has ordered that after 25th duty Torrential rains rehabilitate the slums outside the	MbPT will coordinate with MCGM to find solution to the problem.

			city limits.	

Management Plans

1. Dredging & disposal of dredge spoil

Construction Phase

The potential impacts on marine water quality may arise primarily due to dredging and disposal of capital dredge spoil at the reclamation area or at Site DS3. The dredging, piling and reclamation activities would also disperse the bed sediment in the water column thereby increasing the suspended solids in water.

Several types of miscellaneous construction vessels such as barges, transport vessels, barges equipped with grouting concrete forming equipment and work barges will be deployed during construction. Such vessels are a source of low level organic and petroleum pollution to local marine waters. The management plans to address these issues will focus on the following:

- The dredging and disposal of capital dredge (not used for reclamation) will be planned taking into account the bathymetry and hydrodynamics to ensure efficient dispersal of sediments. This will not result in accumulation of dumped sediment in a restricted area.
- The site will also be periodically monitored as a part of the marine environment management plan.
- The port vessels will be kept in proper working condition to minimize risks of accidents and avoid any kind of accidental spillages.
- The intertidal and near shore sub-tidal segments outside the project area will be restored to their original contours once the construction activities are completed.
- General clean-up of the site used for construction related activities, adjacent intertidal areas, creeks etc. will be undertaken and all the discarded materials will be removed from the site and aesthetic quality of the surroundings restored, once the construction operations are over.

#Note: dredging work (partly) and reclamation (completely) has been done as per earlier MOEFCC approval.

Operations Phase

a) Minimizing Impacts Due to Disposal of Maintenance Dredge Spoil

The disposal logistics of maintenance dredge spoil has been carefully planned based on economic considerations and meeting acceptable standards for its disposal. The following factors feature prominently as mitigation for impacts,

- The bathymetry at site locations offers the possibility to achieve attenuation levels in the shortest possible time.

- The discharge depths of approximately 7 m below the surface of the water reduce the impact of SS on the top water column.

b) Minimizing Impacts Due to Escapement of Cargo

As a policy towards maintenance of the health of the marine area, adequate technology and machinery will be selected normally employed in modern ports, to minimize escapement of cargo. It is acknowledged that even with a carefully chosen technology, cargo spillage sometimes occurs in quantities more than assessed, majority of which is due to human error or negligence.

- The well laid-out protocol for the movement of ships in the navigational channel, their berthing and un-berthing, cargo loading and unloading will be strictly followed to avoid accidents and spillages of cargo / fuel.
- The employees will be trained to inculcate a high level of competence not only in day to day operations, but also during emergency situations.
- Periodic refresher course will also be organized to maintain the level of their competence.
- Only ships having valid MARPOL Certificate will be permitted at the berths.
- It will be ensured that the ships are equipped with on board sewage treatment facility in operating condition.

2. Traffic management plan

As mentioned in the Traffic feasibility report, the sites (existing/ to be developed) are available around MBP under port trust for handling container cargo. Numbers of Link roads have been developed for the decongestion of the road traffic. MMRDA along with MBPT has done the development outside the port estate (Mumbai Trans harbour link, Eastern freeway, western freeway etc.) for the improvement of transportation system of Mumbai city. Number of roads within the MbPT premises has been developed. Following measures have been developed for effective Traffic Management Plan.

- Transportation vehicles will be properly and timely maintained and serviced regularly.
- It will be mandatory to use low sulphur diesel in transport vehicles, equipment and machinery where applicable.
- Movement of bulk materials will be preferred during non-peak hours and regulated during peak hours.
- Speed control of 20 km/h within the port premises as per the policy of MbPT will be maintained.
- The traffic diversion signs will be bold and clearly visible particularly at night.
- Since most of the port area is concreted / asphalted, dust generation will be relatively low. Nevertheless, there areas will be regularly swept as per the ongoing practice and collected dust will be adequately disposed in the designated areas in the port premises.

- Vehicles having Government license and certificate for transportation of construction will be commissioned.
- Record maintenance of vehicle coming within the port premises.
- Environmental awareness program training will be imparted to the key contractors' personnel and compliances will be enforced through in-house monitoring.

3. Point 7: Controlled blasting of rock and dredging

The weathered rock will be removed by cutter section dredger and hard rock by controlled blasting.

Ground vibration and underwater shock waves are two of the major impacts from underwater blasting. The ground vibrations can potentially cause damage to structures; while excessive underwater shock wave pressure, apart from causing damage to nearby structures, marine life may be impacted. Special precautions are taken since underwater blasting is frequently conducted in the vicinity of structures such as berths, breakwaters, port buildings etc. The magnitude of these vibrations is determined by factors such as quantity of explosive charge, characteristics of the rock and distance from the blast. Through controlled blasting these adverse effects can be effectively controlled and considerably minimized.

Controlled blasting generally adopts safe vibration level by developing appropriate Rock Blasting and Dredging Plan that takes into consideration attenuation relations with respect to propagation characteristics of blast vibrations, calculating safe explosive weights for different distances, design of blasting patterns and monitoring of blast vibrations during actual operations that enables modification in remaining blasting runs, if required. MbPT will contract known experts in the field such as CWPRS to examine the issue, provide their expertise while tendering the works and also critically evaluate the probable bids. The selected contractor will be bound to provide detailed plan for underwater controlled blasting and dredging which will be again examined by the expert and an agreed plan will be the included in the contract. The expert will be associated with the project in the supervisory capacity until the controlled blasting is completed.

From their previous experience of controlled blasting of rock and dredging MbPT is confident that the Elephanta Caves, a World Heritage Site located approx. 10 km from the site proposed for controlled blasting will not have any impact of ground vibrations and underwater shock waves. Underwater drilling and blasting of hard rock was successfully carried out for the construction of Liquid Chemical Berth within the Harbour and the safety of various structures was established by monitoring of blast vibrations on selected structures during actual blasting. Seismographs set-up at Elephanta Caves which was 2.6 km from the blast site did not record any vibrations due to blasting. The Elephanta Caves are about 7.9 km from the project site, hence, will not be affected.

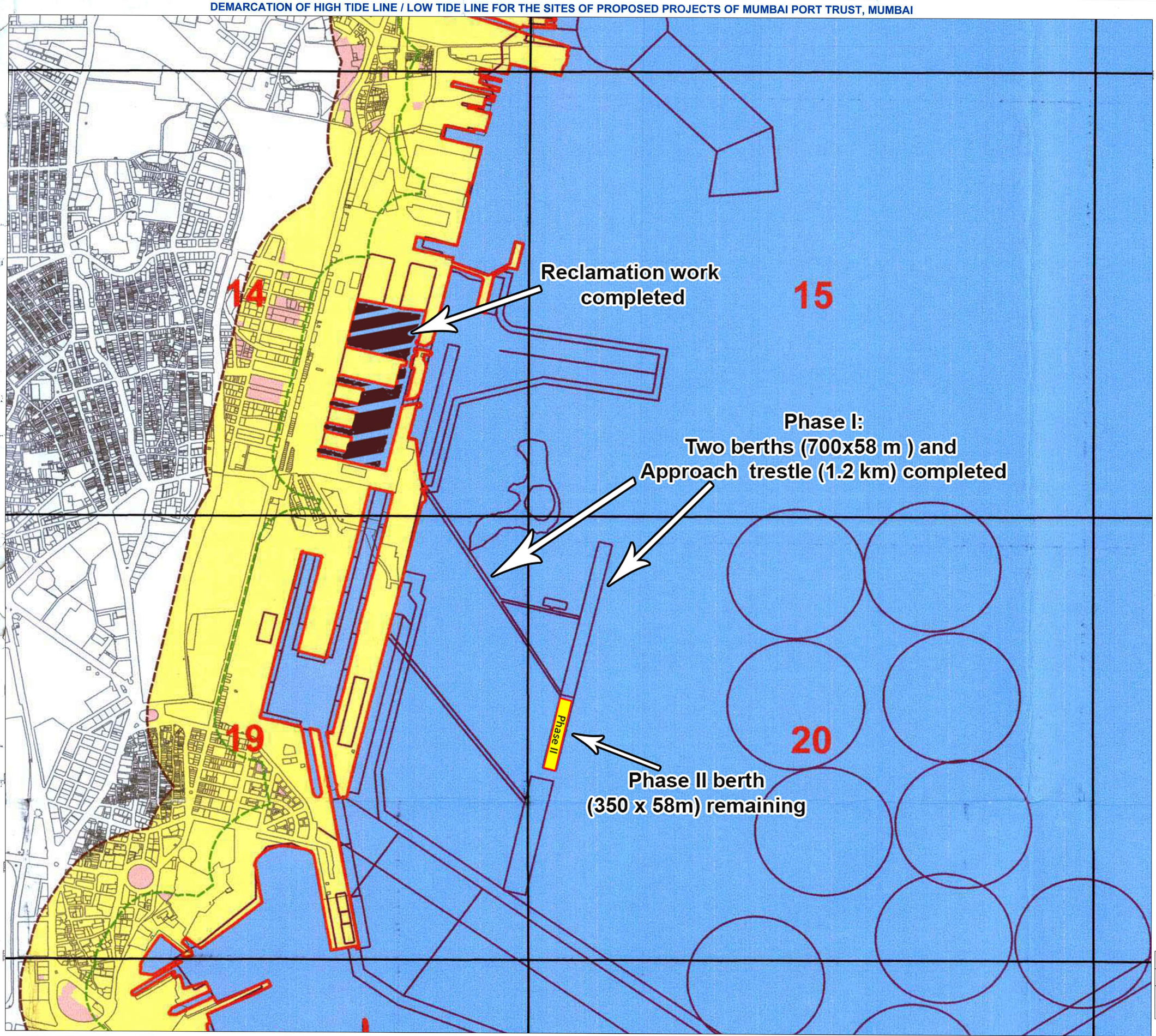
If recommended by experts the possibility of using an Air Bubble Curtain system will be examined and deployed if feasible. Air Bubble Curtain system an air bubble curtain is created by using heavily perforated hoses around the construction site as a ring and are driven into the seabed. The hose ends are connected to superchargers on ship deck and pump compressed air into the hoses. The air escapes through the holes according to a well-defined pattern. The result is a curtain of millions of small bubbles that rise to the water surface, forming a kind of whirlpool around the construction site. In The air bubbles change the density of the water and thus break down the sound waves thereby reducing the impacts of blast waves.

Underwater blasting operation

Generally drilling rigs mounted on pontoon or barge are stationed at the predetermined site to be dredged, drilling towers are positioned over the specified drill hole locations and drilling commences. In the Over Burden Drilling – the commonly used method, a casing pipe is driven upto the rock through the overburden, the inner drill rods are inserted through the casing pipe and the hole is drilled to the required depth. Usually a number of holes are drilled in a line. The barge is shifted to the next position for drilling and another row of holes are drilled. The operation is repeated until required number of holes are completed as planned. Each hole is loaded with pre-decided number of cartridges of explosives and primed with non-electric shock tube initiation system provided with delay detonators. Detonation is carried out through a lead-in line connected to an ignition device.

Annexure 6

DEMARCATION OF HIGH TIDE LINE / LOW TIDE LINE FOR THE SITES OF PROPOSED PROJECTS OF MUMBAI PORT TRUST, MUMBAI



LEGEND

- LOW TIDE LINE (LTL)
- HIGH TIDE LINE (HTL)
- 100m / CREEK WIDTH FROM HTL
- 200m FROM HTL
- 500m FROM HTL
- MANGROVES (CRZ - IA)
- 50m BUFFER FROM MANGROVES (CRZ - IA)
- MUD FLAT (CRZ - IA)
- CRZ - II
- CRZ - III
- CRZ - IVA
- CRZ - IVB
- PROPERTY BOUNDARY
- HTL REFERENCE POINTS

SOURCE : CLIENT

- PROJECT SITE

PROPOSED PROJECTS

1. FIFTH OIL BERTH
2. TRANSIT SHED AT 16/17 ID
3. TRANSIT SHED AT 4 ID
4. MULTIPURPOSE CARGO BERTH
5. DEVELOPMENT OF BERTHING FACILITIES FOR OFFSHORE SUPPLY VESSELS
6. NEW CRUISE TERMINAL AT GATEWAY OF INDIA

N

SCALE 1 : 25,000

0 312.5 625 1,250 1,875 2,500 M

	PREPARED BY INSTITUTE OF REMOTE SENSING ANNA UNIVERSITY CHENNAI - 600 025
	FOR MUMBAI PORT TRUST PORT HOUSE, SHOORJI VALLABHDAS MARG BALLORD ESTATE, MUMBAI - 400 001

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Annexure 7
Feasibility Report

SECTION 1
BACKGROUND

1.1 OFFSHORE CONTAINER TERMINAL (OCT) PROJECT

During 2006, Mumbai Port invited global tenders for development of an offshore container terminal on DBFOT basis. Based on this tendering process, the licence was awarded to Indira Container Terminal Private Limited (ICTPL), a special purpose vehicle promoted by Gammon India Ltd. & Gammon Infrastructure Projects Ltd, collectively called the Gammon Group (74%) and Dragados SPL, Spain (26%). The license was for the development and operation of OCT for a period of 30 years including construction period of 3 years at gross revenue share of 35.064%. The License also requires the company to operate and maintain the existing container operations of at the Ballard Pier Station (BPS) for the first five years of the License or 2 years from commencement of operations, whichever is earlier. The license was awarded during December 2007.

As per the concession agreement, the project was to be executed in two phases. During the first phase, the minimum port facilities will be as follows:

- Berth length of not less than 700 m and width of 58 m with adequate equipment such as ship to shore handling cranes, yard handling equipment, rail container yard handling facilities, reefer plug points and sufficient container stacking yard etc., commensurate with the traffic. The berth is expected to be designed to receive vessels of capacity up to 6000 TEUs

- A Y-Shape approach trestle to connect the berths with the onshore terminal facilities. For this an 18 m wide road will split into two 12 m wide roads reaching the berth and thus forming a Y-shape. Approximate length of the trestle will be 1000m.

- Minimum equipment to be commissioned at the berth:
    - Four QGC capable of handling 6000 TEU design vessels at the time of commissioning of berths and two more after achieving throughput of 600,000 TEUs in an operating year

- 12 RTGs at the time of commissioning of berths and six more after achieving throughput of 600,000 TEUs in an Operating year
- Two RMGs for rail container depot at the time of commissioning.

The project was designed to handle 1.2 million TEU of container traffic during the first phase. During the second phase, the berth length would be increased by 350 m, thereby increasing the handling capacity of the terminal to 2.0 million TEU.

Correspondingly, the Mumbai Port was to fill up of Princess and Victoria Docks and with the surrounding areas to create a stackyard space of 35 hectares; provide railway connectivity to create a railway container depot with 3 sidings; dredge the approach channel, turning basin and berth to handle container vessels of 13.5 m draft with tidal advantage.

1.2 PRESENT STATUS

Though the commercial operations at the offshore terminal should have been commissioned by December 2010, the project got delayed due to various reasons beyond the control of both the Concessionaire and the Licensee. The berths with the approach could be completed only by September 2013. During this period, the container traffic through Mumbai Port also did not grow as expected due to the influence of JNPT and it stagnated between 40,000 to 50,000 TEU only.

As of now, part of Project facilities like offshore jetty, approach trestle, and yard has been physically completed, however, the completion of the balance mechanical and other facilities of the Project have not been taken up. The civil works as existing now is shown in the figure hereunder.

As an interim measure to ensure utilization of assets created, MbPT decided to permit handling of RoRo at OCT for one year.

1.3 PROPOSAL FOR REVIVAL OF THE PROJECT

MbPT, desirous to revive the OCT Project, evaluated various options including the structure adopted by Kamarajar Port, whereat, Kamarajar Port had proposed to rebid the existing iron ore terminal to include coal cargo based on highest revenue share, with Right of First Refusal (ROFR) to the existing concessionaire and assessment of the existing iron ore terminal assets at Replacement cost. The same model is being followed by Kandla Port for Berth 13 and Berth 15. Accordingly, MbPTis contemplating to revise the cargo to be handled at OCT to include no-containerised cargo like steel and RoRo.

In view of this, ICTPL engaged KPMG as consultant for preparation of Feasibility Report of the Project for revised cargo configuration

MbPT, with the consent of ICTPL, has appointed SBI Capital Markets Limited as their Transaction Advisor for assessment of Replacement cost of existing facilities (viz. berths, approach trestle, fire water platform, etc), review of Feasibility Report of reconfigured Project, assist is the process of sanction and bidding etc. SBICAPS, with due approval from MbPT, has appointed Frischmann Prabhu (India) Pvt. Ltd. (FPI) as their technical sub-consultant.

1.4 DFR FOR THE RE-CONFIGURED OCT PROJECT

As a pre-requisite for the re-bidding process, MbPT is desirous of preparing the Detailed Feasibility Report for the re-configured project. This work of preparing the DFR has been entrusted to the Indian Ports Association. The project team of IPA visited Mumbai Port during 24th& 25th October, 2016 and held detailed discussions with the Chairman and other officials of the Port; related documents were also collected. The Scope of Work as defined by MbPT in their letter No CE.CF-226 (CTP) / 737 dated 28th October, 2016 is as follows:

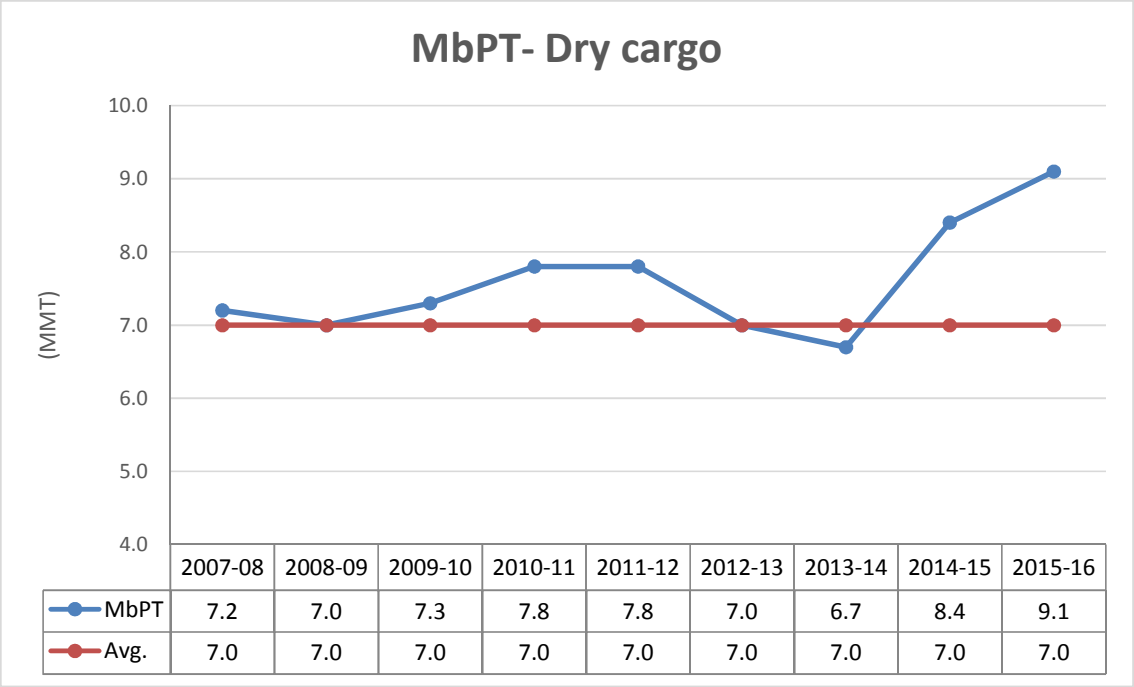
- To undertake Detailed Feasibility Report for OCT with the revised cargo mix viz. RoRo and steel cargo along with containers, any other cargo to make the project technically & financially viable.
- The traffic projections already made by McKinsey and KPMG will reviewed and supplemented, if necessary for remaining concession period ie. 27.5 years for steel, automobiles and containers.
- To look into the project scope with equipments for ship-shore transfer, jetty to stackyard and stackyard equipments
- To examine the requirement of stackyard area as required for the projected traffic
- The capital expenditure indicated by SBICAPS will be taken as frozen.
- In case, the 3rd berth is to be constructed based on other traffic projections, the capex thereof shall be worked out at current costs.
- To work out the technical and financial viability of the project considering civil cost, dredging cost and other costs.

SECTION 2

TRAFFIC PROJECTIONS

2.2 INTRODUCTION

Mumbai Port's dry cargo traffic (excluding stream-handled cargo) had been steady at around 7 million tons per annum for 7 years, from 2007-2008 to 2013-14. It spluttered to about 8.5 million tons in 2014-15 and over 9 million tons in 2015-16. Both the non-reducing and increasing performances are significant given the Port's limitation in handling deep-drafted and large vessels, which have been coming in increasing numbers in the last 15 years due to availability of abundant cargo and the economies of scale it brings.

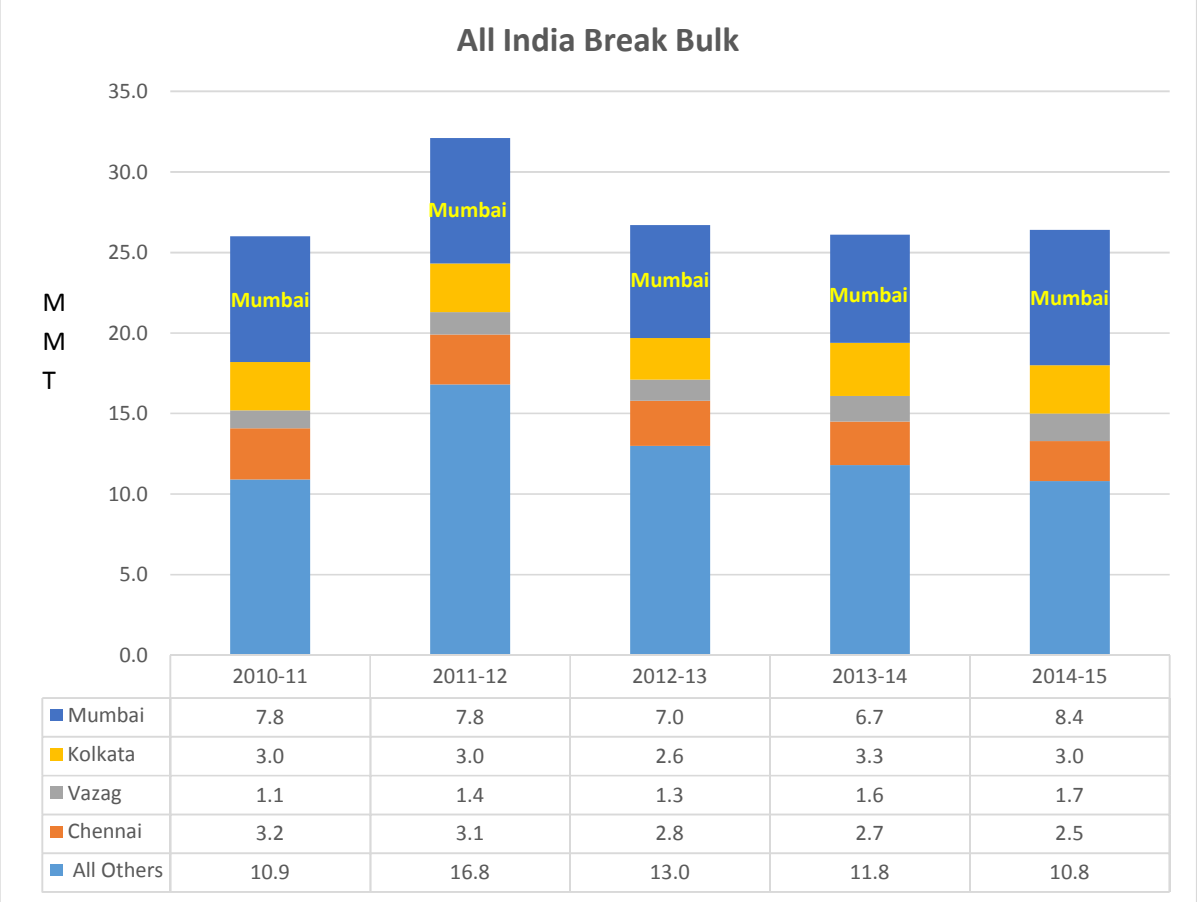


	2007-08	2008-09	2009-10	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16
MbPT	7.2	7.0	7.3	7.8	7.8	7.0	6.7	8.4	9.1
Avg.	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0

The reason for the Port's undiminished volume appears tolie in the kind of dry cargo it handles Break-bulk and automobiles are the two dry cargo items the Port predominantly handles. A good proportion (about 60%) of break-bulk cargo still comes to the Port in modest parcels. They are carried on small or medium-size vessels

(handysize/ handymax) of 150 metre to 205 length requiring 9.0 to 10.5 metre draft max. The Port handles smaller, lighter vessels with no difficulty. The medium-size vessels are a struggle as the Port has only one berth to cater to vessels beyond 9.5metre draft and 190 metre length.

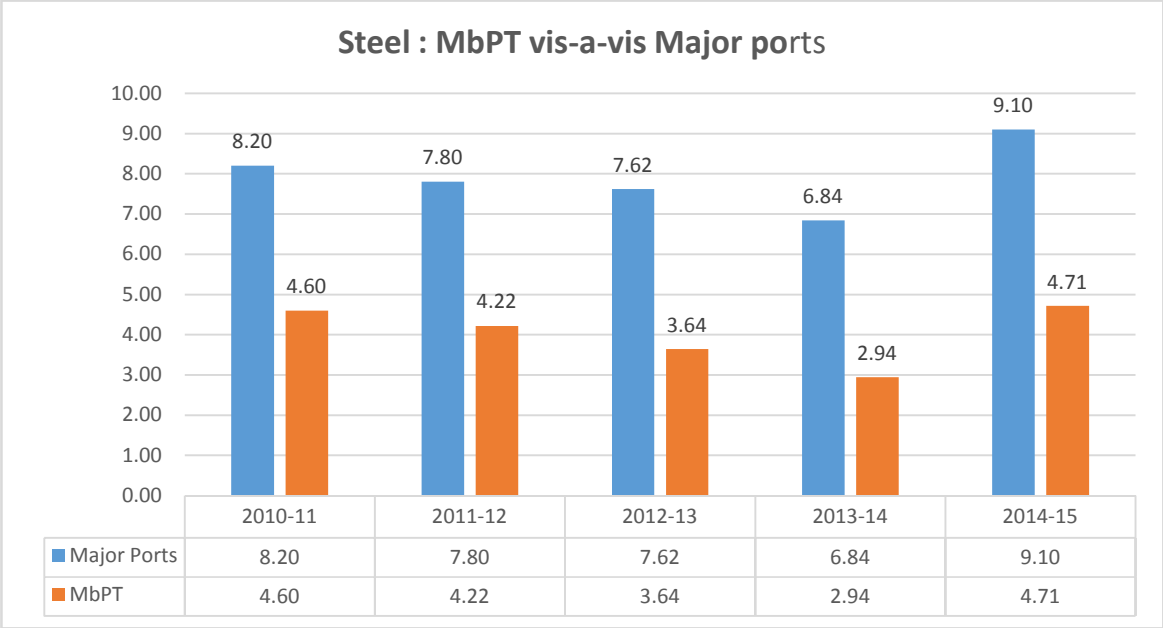
The above may mean one of two things. One, there is ample break bulk cargo for Mumbai Port and because of its limitation most of them are coming in smaller, lighter vessels. Or, some of the medium-size vessels and all the big vessels of the Supramax/ Panamax type are slipping through the fingers of the Port. Either way, it indicates the Port's position and potential in break bulk cargo. For many years now, Mumbai Port has been number one in break bulk traffic among the 12 major ports, handling 25-30% of the total traffic.



	2010-11	2011-12	2012-13	2013-14	2014-15
Mumbai	7.8	7.8	7.0	6.7	8.4
Kolkata	3.0	3.0	2.6	3.3	3.0
Vazag	1.1	1.4	1.3	1.6	1.7
Chennai	3.2	3.1	2.8	2.7	2.5
All Others	10.9	16.8	13.0	11.8	10.8

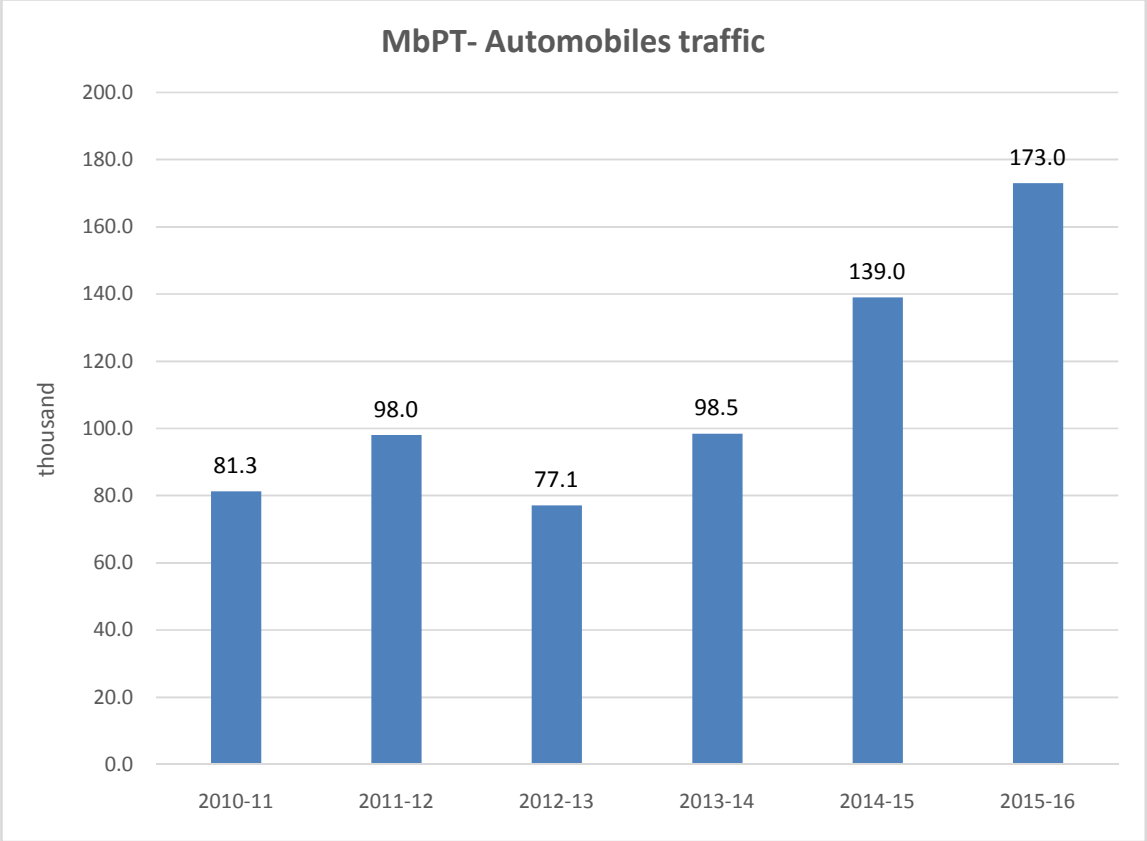
Another notable feature is steel is the major break bulk item the Port handles. Steel makes up more than 50% of its break bulk volume. In steel throughput also, Mumbai

Port occupies a preeminent position, handling more than 40% of the total steel traffic of the 12 major ports. In 2013-14 its share was 43% and in 2014-15, 52%.



	2010-11	2011-12	2012-13	2013-14	2014-15
Major Ports	8.20	7.80	7.62	6.84	9.10
MbPT	4.60	4.22	3.64	2.94	4.71

One more dry cargo the Port is handling in increasing quantity is automobiles. In 2015-16 it handled about 1.7 lakh vehicles. From April to September'16 it has handled above one lakh units and it is poised to clock more than 2 lakh units by the end of the financial year 2016-17. In automobile traffic, Mumbai Port now is number 1 in western region and number 3 at all India level. Soon it may become the 2nd largest among all the Indian ports including non-major ports.



These performances have come about despite the Port's handicap in two vital attributes: draft and berth size. Had the port been better endowed it would have attracted much more break bulk cargo and automobile as it has the geographical advantage and a rich and growing hinterland for its key cargoes, namely steel and automobiles.

That is where OCT (the terminal may be referred to as Offshore Cargo Terminal or ODT as it is now intended to handle all dry cargo and not just container) can make a difference. With 14.5metre depth at its berth pockets and a liner quay of 700 metre, extendable to another 350 metre, it is exactly what the doctor ordered for Mumbai Port's growing steel and automobile traffic.

Thus, from the above perspective, steel and automobiles have been identified as the most potential cargoes for OCT. Accordingly, their prospects for OCT have been analyzed and traffic projections thereof worked out, as detailed under.

2.1 Steel

2.1.1 State of the Indian steel industry

India is the third largest producer of crude steel and the largest producer of sponge iron or DRI (direct reduced iron) in the world. It accounts for 2% of India's GDP and 6% of industrial production.

It is one of the first core sectors de-licensed and de-regulated under the 1991 liberalization policy initiative. Since then, its capacity and production have grown in the range of CAGR 5 to 8%. The capacity increased from 20 million tons in 1991-92 to 110 million tons in 2014-15, production from 14.34 million tons to 91.6 million tons and consumption from 44.33 to 77.09million tons.

India's steel consumption currently is quite low, considering its population size of 1.29 billion. But the consumption is set to surge significantly over the coming 20 years. Upcoming big-ticket projects in infrastructure development, expansion and improvement, boom in construction and housing industry, rising activity in engineering and manufacturing sectors and an increasing appetite for personal consumption products such as automobiles and white goods are expected to drive the demand for steel to unprecedented levels.

Steel import-export scenario

2.1.2 Import prospects

Steel import has been on a steady rise in the past 5 years, reaching a peak in 2015-16. Primarily, 2 factors contribute to the increasing import. One: domestic price being higher than imports. Two: non-availability locally of some products like high quality, and special grade steel required by automotive and engineering goods industry. At present, they are not manufactured in the country.

Steel import is likely to increase tremendously for another reason: demand exceeding production. Per capita consumption of steel in India today is 65 kg compared with 510 kg in China and the world average at 235 kg. It is expected that India will in the coming two decades catch up at least with the world average. When that happens, there is a question mark whether domestic production will be ready for it.

An Exim Bank paper on India's prospects for steel export says two problems beset the steel industry in ramping up its production. One is that its expansion projects are moving slowly or stuck. A plethora of factors including high cost of funding, land acquisition delays, environmental issues, non-availability of reliable raw materials supplier and others have been identified as the chokers of capacity augmentation. Another problem is underutilization of capacity by about 25%. This level of underutilization may be tolerable and even prudent in the present climate of low consumption. But not when the demand shoots up in the foreseeable future.

Signs of falling production and increasing consumption were already seen in 2015-16. In that fiscal, production came down to 91.12 tons from the previous year's 91.46 million tons, a 1.1% drop. In the same period demand went up by 4.3%, from 76.96 million tons to 80.27 tons. Not coincidentally, in the same year import reached a new high of 11.71 million tons, a sharp increase of 25.6% over the previous year.

Another major challenge the steel industry faces is dumping of steel on India by China and Russia, to a lesser extent. On average, Chinese steel is cheaper by 15% than domestic steel and Russian by about 9%. So long as China's downturn continues and Russia's economy, battered by fall in oil prices, remains weak they will keep tweaking their currencies, which will further cheapen their prices.

The Government has taken a series of measures in the recent past to stem the flow of foreign steel into the country. These include anti-dumping duty, *ad valorem* safeguard tax and MIP or minimum import price. These are interim steps. They may not continue forever. Both MIP and safeguard duty have been challenged by a host of countries under WTO rules. Anti-dumping duty is the only one sustainable but it takes a long time to implement. Thus, these protective measures may work for a limited time. They are not a permanent solution for stanching the import.

The Hon'ble Finance Minister in a recent seminar on steel industry said as much. He remarked that the Government could do the hand-holding to the steel sector only for a limited extent and period. The industry has to stand on its own by staying competitive in world market.

In the backdrop of the above scenarios and trends the outlook for steel import has been examined and the key points bulleted as below:

2.1.3 Positives for import:

- Steel derives its demand from 3 broad sources of consumption: public, private and personal, roughly at45%, 35% and 20% respectively.

- Public consumption includes improvement, expansion and development of infrastructure like roads, rails, bridges and ports. As per the Working group on steel for the 12th five-year plan, spending on infrastructure will in the coming years go up to a trillion dollars. In the Union budget of 2016-17, Government has allocated an expenditure of Rs. 280000 crores for road and rail development and improvement.

- Other Government initiatives include housing for all by 2022, power for all by 2019, 100 smart cities by 2022 and Atal Mission for Rejuvenation of Urban Transportation or AMRUT.

- Private demand comprises manufacturing, engineering services, housing and other construction activities. Manufacturing sector is projected to grow from current 8 to 12%. Urban population is expected to increase to 600 million by 2030 from the present 400 million. All these will escalate the demand for steel appreciably.

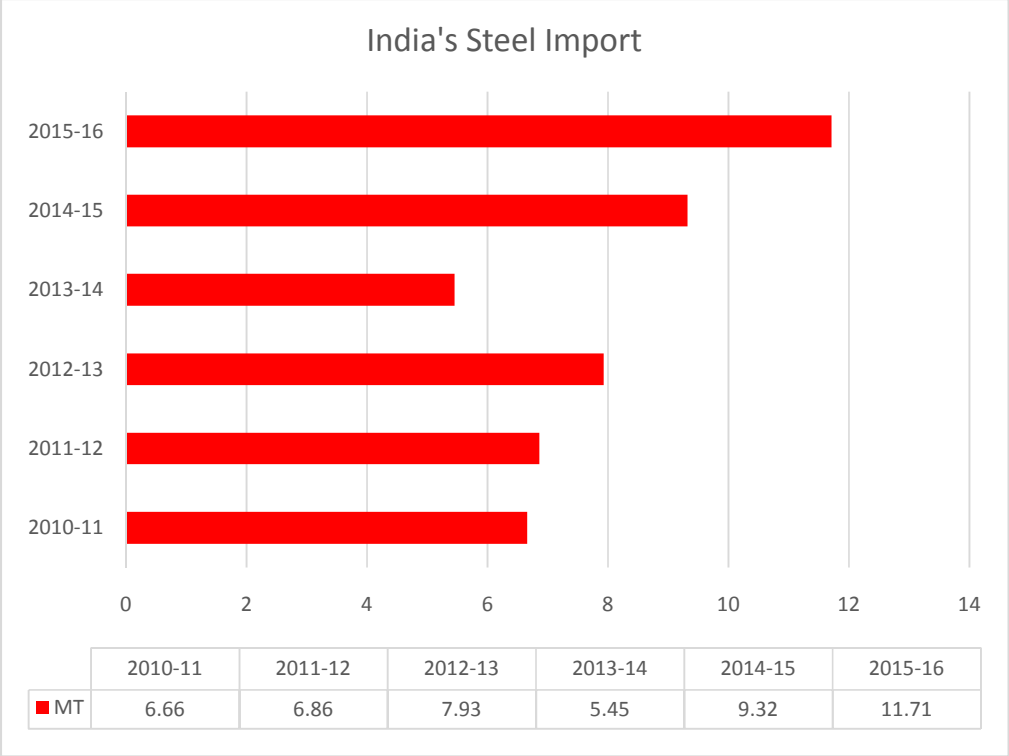
- On the personal front, due to rise in standard of living and disposable incomes, a steep increase is anticipated in the demand for automobiles and personal goods like television, washing machine, air conditioner and others, which all will need massive quantities of steel.

- As a result of these voracious demands, the country's steel consumption is expected to reach about 300 million tons by 2035, from the current 80 million tons.

- Meanwhile, China's steel juggernaut will roll on. Today China has an over capacity of 300 million tons. It will do all it can to dump them on other countries including India. At their price, it will be hard for us to resist it.

- South Korea and Japan are among the top steel producing countries in the world. We have free trade agreements with both. This means, duty-free or substantially reduced tariff on their steel imports.

- We have already seen the effect of the above factors: import on a continuous climb in the last 6 years, except for a drop in 2013-14.



	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16
MT	6.66	6.86	7.93	5.45	9.32	11.71

2.1.4 Negatives for import

- If the Government could hold on to the protective measures, at least in the near-term the import will come down

- If the local industry could control its cost and bring domestic prices on par with or below imports cost, it might retard import to an extent

- Once again, China factor: if China's economy revives and they go on an infrastructure bingeagain, they may not have enough steel to spare to others.

- These are all big 'ifs'.

2.1.5 On balance for import

- Steel import started falling from June-July 2016 as the Government's punitive measures took hold. However, import looked up again in Oct 2016

- The Government's interventions are likely to continue well into 2017. So, the imports may keep falling till around the end of calendar year2017.Nevertheless, it will not reduce drastically. It may bottom out at around 9 million tons.

- The reduction in import will most likely to be a temporary phenomenon. The demand is certainly set to surge sooner or later to about 300 million tons by 2035 from the current 80 million tons, as already indicated. When that happens, the import will override the restrictions or the Government may dilute or dissolve the curbs

- The present steel production looksfarmore than the consumption. But it is not. In 2015-16 the production was about 91 million tons, consumption 80 million tons and exportabout 4 million tons. Thus, the gap between production and consumption even at today's low demand is not much. And when the demand shoots up the gap will widen.

- In sum, regardless of the present reduction in import,caused by the curbs, the long-term prospects for steel import appears to be strong. It may scale to about 25-30 million tons around 2030-35

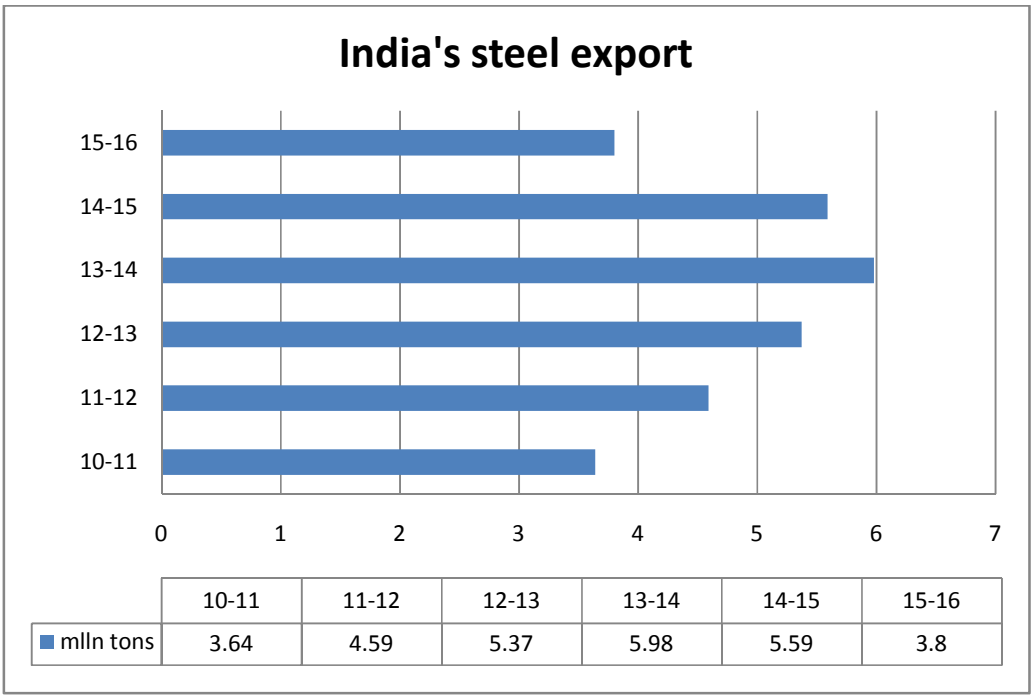
2.1.6 Export prospects

- India started exporting steel in 1964. Its real pace picked up after the liberalization in 1991.

- In 2013-14its export turned around when it briefly became a net exporter.

- Since then our export has been falling. It plunged to 3.8 million tons in 2015-16 against 5.59 tons in 2014-15, a precipitous fall of 32%.

- Even at the best of times our export was around 7 million tons. It is a fraction of the 110 million tons China is exporting.



- Our export market is well diversified with America leading the pack. America is our major steel buyer, taking almost 80% of our export. Today, it is the only market that has a strong demand outlook. Its steel use increased 6.7% increase last year. Robust growth in its automotive industry and energy sectors, and recovery in non-housing (their houses are wooden) construction – all point to a forthcoming massive steel demand in America. This augurs well for our steel export.

- Besides America, the rest of the world too is expected to recover from the present economic sluggishness, set off by China's downturn. With China's recovery expected by the end of 2017, global demand, too, for steel is expected to pick up. This offers hope for our exports.

2.1.7 Positives for steel export

- Like the shipping industry, the steel industry is cyclical and fiercely competitive. For the last few years, demand for steel and prices of steel in international market were bearish. This has struck a blow to our export. Last year's export was particularly bad at 3.8 million tons. However, the analysts

are predicting the cycle is turning and resurgence expected around 2018. If they are right prices, may perk up, which will give a fillip to our export.

- In his victory speech, Donald J Trump said infrastructure is his top priority. This is not surprising. America's infrastructure is fraying. Its roads, rail, subways and airports are cracking and crumbling. To repair and rebuild them America will need a humungous infusion of steel. And India may be a major source for their import with our long history of selling steel to them.

- It is expected that the Chinese economy will revive by the end of 2017 and it will revert to its infrastructure splurging days of the 2000s and early 2010s. If and when that turnaround happens, China might become a net importer of steel. This will be another opportunity for our export.

- On the domestic front, the Government has set a production target of 300 million tons by 2025. The steel industry is skeptical about it. However, if it does materialise, the industry may be able to offer its surplus for export at a competitive price.

2.1.8 Negatives for export

- China and Russia, to a lesser extent, remain a threat to our exports, with their predatory prices.

- America produces less steel than it needs. A few years ago, we took their place as the world's third largest producer of steel. The expected developmental boom in America's various sectors will further scale up its demand. Despite it, they have been talking of antidumping action against India, a regular and reliable supplier. Anti-dumping action is not easy to enforce, however. It requires investigation and proof that the unloading country is selling its products at prices less than the domestic prices. This may be the case with China, not with India. Yet, there is a lurking danger.

- If domestic demand rises, as anticipated, and international prices continue to remain weak there may be no incentive for export

2.1.9 Assessment of <u>overall</u> steel traffic

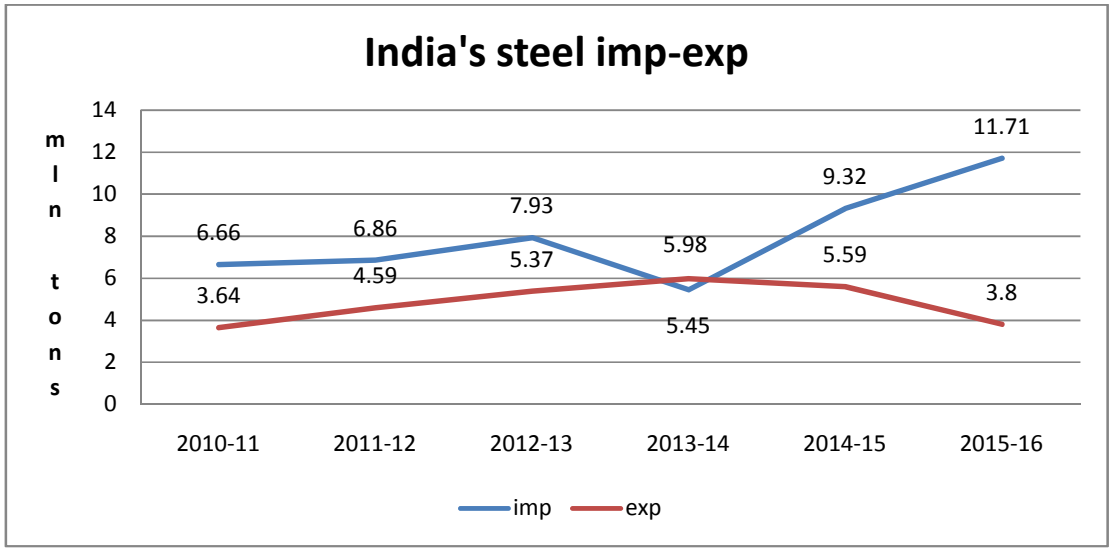
- Import prospects certainly look bright, notwithstanding the recently invoked counter measures. Over the long haul, demand is set for a huge spurt due to factors already mentioned. It is not certain production will be able to match it. Import is therefore most likely to climb.

- Export future does not look that cut and dried. It mostly depends on global factors. If major economies like Europe and China revive, that may push up demand and prices, giving some traction to our export.

- If the anticipated domestic demand does not materialize and production goes up, the surplus will go for export.

- In any case, the overall steel traffic will keep going up. If import falls export may pick up the slack or vice versa. It is interesting to note import and export seem to be inversely proportional. In the last 6 years whenever import fell export went up or vice versa, as the following graph shows.

- Last year, import constituted about 14% of the total steel consumed in India. At that rate, even if the consumption by 2035 increases to 250 million tons, if not 300 million tons astargeted, import alone will be around 35 million tons. At an optimistic scenario, export may grow to about 15 million tons by the same period.



2.1.10 Mumbai Port's place in steel traffic

Mumbai Port leads major ports in steel traffic. In 2014-15,it handled 4.7 million tons at a share of 43% and in 2015-16, 6.0 million tons at a share of 52%.The Port will continue its dominance as it caters to a rich and growing hinterland of both producers and consumers of steel.

The other ports in Mumbai region are Dharamtar and Dighi. They pose no threat to Mumbai Port. Dighi has a maximum draft of 8.5 meters and Dharamtar is a lighterage port. In any case, there will be more than enough steel cargo in the region that the ports do not have to poach each other's cargo.

2.1.11 Drivers for MbPT's steel traffic

- Mumbai Port's proximity to both major consumers and producers of steel and its good connectivity by road and rail and its commercial capital status give it a clear advantage in steel import and export

- Steel consumers in the region include automobile majors like Tata, Mahindra, Bajaj, Volkswagen and GM, white goods producers like Godrej, Voltas, Whirlpool, Blue Star, construction giants like L&T, Afcons and HCC, and engineering goods manufacturers such as Thermax, Sandvik and Atlas Copco

- Out of the top 10 steel manufacturers 5 have their plants at proximate distances from Mumbai Port
  - No 3: JSW Steel at Vasind, Kamleshwar, Tarapur and Dolvi
  - No 6:Bhushan Steel at Khopoli
  - No 7: Esssar Steel at Pune
  - No 8: Ferro Alloys at Nagpur
  - No 9: Mahindra Ugine atKhopoli
  - Besides the above, Posco and Uttam Galva have their plants at Pune and Khopoli for processing HR coil to CR coil for export

- Uttam Galva is a regular shipper of steel through Mumbai Port. They export about 2.5 lakh tons CR coils per year to US. They were also importing 30000

tons of HR Coils a month before the curbs. As per their representatives, if the Port could handle bigger vessels, like Panamax, their volume will increase.

- Maharashtra Seamless is another user of the Port for their pipes export. They now import about 7000 tons of billets for making pipes, which they export in equal quantity. Their projection is their import and export will increase to 10000 tons each per month.

- Posco exports to US about 25000 tons of coil every month. Their projection is 1 million tons a year starting next year

- Octroi was an irritant for using Mumbai Port. With the imminent introduction of GST, octroi will go. This may divert some more steel traffic to Mumbai Port

2.1.12 Steel traffic projection for Mumbai Port

For the past 6 years, Mumbai Port, on average, has been handling each year about 35% of the country's total steel traffic (including import and export). Based on this trend and the outlook visualized for the country's total steel import-export for the next 20-25 years, the forecast for Mumbai Port is:

- The Port's overall steel traffic including OCT's will grow at CAGR 5% until about 2025 and dip to CAGR 3% over the next 5 years, up to 2030. It is likely to drop to CAGR 2% over the next 10 years up to 2040 and then peter out to 1% by 2045 and plateau or retard from then on.

- The projection for OCT is done with its capacity limitation in view.

Projection for entire Mumbai Port

Base: actual traffic of MbPT in 2015-16 @ 5.985 million tons

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Mn tons	6.60	6.93	7.28	7.64	8.02	8.42	8.84	9.29	10.77	14.46	16.76

Projection exclusively for OCT

Year	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Mn tons	2.50	2.50*	2.63*	2.76*	2.89*	3.03*	3.03*	3.03*	3.03*	3.03*	3.03*

*capacity saturation of existing berth

2.2 Automobiles

2.2.1 State of the Indian Automobile industry

Liberated in 1991 from the cocoon of Ambassador-Premier era, Indian automotive industry has grown the fastest and fiercest, clocking a spectacular growth of ~ 10% in the last ten years. It is expected to grow even faster in the coming decade at a CGAR of ~ 15%

The industry contributes 7.1 to the GDP and accounts for 4.3% of country's exports. In the past decade, 2005-2016, it attracted $ 15 billion in FDI and created 19 million jobs.

Indian automotive industry is the largest in 3-wheelers, 2nd largest in 2-wheelers, 6th largest in passenger vehicles and 8th largest in commercial vehicles overall and 2nd largest in buses.

Drawn by the size of the market and the growing income level of the populace, almost every international automobile major has set up shop in the country. What is even more striking is most of them have made India as one of the manufacturing hubs for their export markets. As of 2015-16, 35% of the small cars sold in the world were made in India.

What pulls international auto makers to India to set up their production facilities for exports is availability of inexpensive labour, a large pool of blue-collared workers with previously acquired skills in auto industry, highly educated technical staff with expertise in design and engineering and a well-established spare parts network.

2.2.2 Indian automobile industry's export performance and prospects

During the decade 2005-15, overall automobile exports grew at a CGAR of 17.79%. Passenger vehicles grew at 7.8% and commercial vehicles at 15.1%. The growth in passenger vehicle segment was subdued in 2015-16 at ~ 5.2 % but the commercial vehicle segment clocked an impressive rise of 18.3% in that period.

Under the National Electric Mobility Mission Plan 2020 (MEMMP 2020), the Government has set an ambitious production target of 10 million passenger vehicles during 2015-20 at a CGAR of 25.6%.The industry expects the production of PVs will be at least 70% of the target. Presently, 20% of the cars manufactured are exported. At that proportion, by 2020 export will increase to 1.4 million units, from the present 0.65 million units.

As regards commercial vehicles, the export is expected to rise to about 3 lakh units by 2020, @ 25% CGAR

2.2.3 Positives for India's light vehicle (PV+CV) export

- There is a growing appetite for small cars (hatchbacks and compact sedans) from across developing and emerging economies including Latin American and some European countries. Most Indian car manufacturers are concentrating on this segment in order to tap into those fertile markets.

- Automobile is one market where we have no competition from China. China is two places behind us in automobile export. Because of quality issues China is likely to remain a laggard in auto export.

- After Latin America, Africa maybe a big market. Its potential has not been tapped so far. Our competitive price in small car segment will be a USP to them.

2.2.4 Negatives for auto export

- Thailand is a distinct threat to our passenger vehicle exports. They are the 15th largest exporter with a share of 1.4%. We are at the 20th position with a 0.8% share.

- Thailand is on par with us in cost, quality and productivity but better in logistics and supply chain efficiency. From India Suzuki has shifted its Celereio production to Thailand and Ford its Fiesta plant.

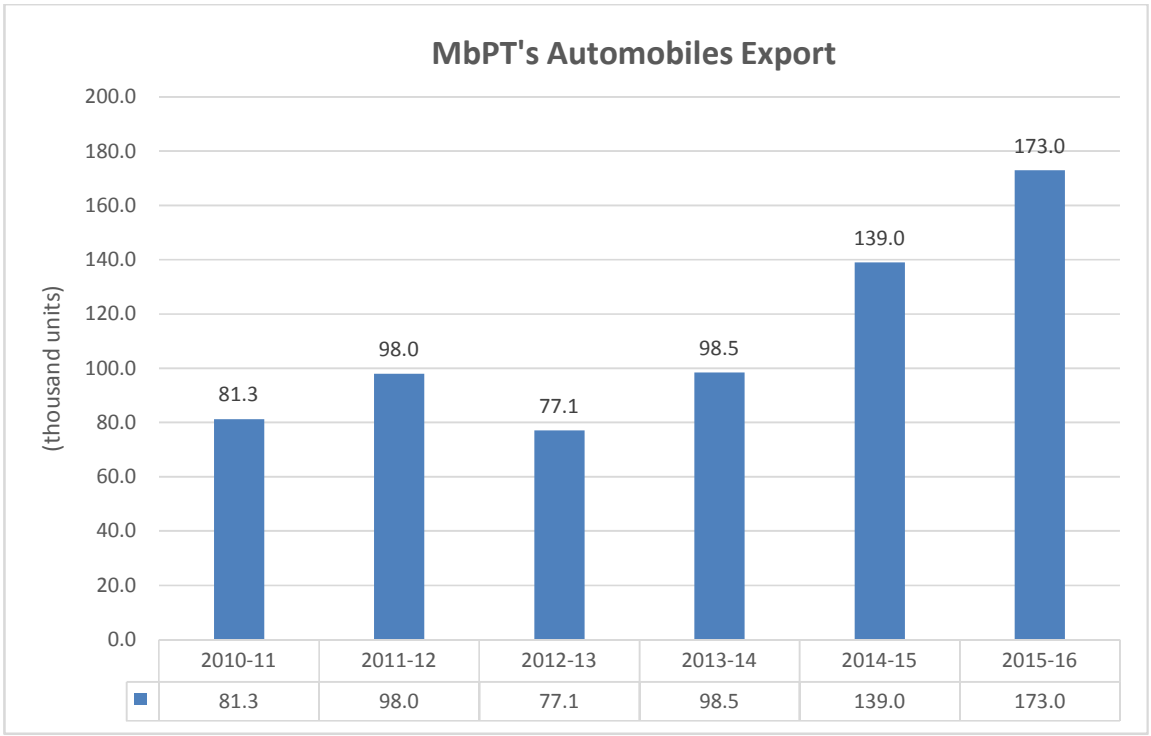
- Latin American and European markets may saturate in about 20 years. Unless we find alternative markets in Africa export may stagnate or fall.

2.2.5 Overall outlook for India's automobile export

- From all indications, it appears the country's light vehicle (PV and CV) export will boom in the near term (up to 2025), grow moderately in the medium term (up to 2035) and subside in the far term (after 2035)

2.2.6 Mumbai Port's place in automobile export

Mumbai Port's auto export traffic has been on the risesince the past 6 years except for a slide in 2012-13.Today, Mumbai Port is at the third position in passenger vehicles and commercial vehicles export with a share of 27%.



2.2.7 Growth drivers for Mumbai Port's auto export

- Mumbai Port is close to many indigenous and international automobile production facilities including that of Tata, Mahindra, GM, Volkswagen and Mercedes
- General Motors will be exportingout of Mumbai Port 70000 small cars (Beat) in 2016-17 and about 81000 units next year.
- GM expects the volume to substantially increase in the coming years.They have plans to invest $ 1 billion in India to boost exports
- Jaguar Land Rover plans to manufacture SUVs in Pune for local and overseas consumption
- Chrysler will invest Rs 3500 crores in Maharashtra for manufacturing its high-end Grand Cherokee model
- Tata Motors' representative stated they are presently shipping 16K to 20K commercial vehicles and they expect the volume to increase by 15% a year
- A representative of Mahindra and Mahindra, another commercial vehicle shipper from Mumbai Port, informed that in 16-17 they will be exporting about 36000 units and they expect the volume to grow by 10% every year
- It is expected that some volume of small cars exported to Europe from Thailand may divert to Mumbai Port due to shorter transit time from Mumbai. 22days is the transit time from Mumbai to European destinations whereas it is 28 days from Bangkok.
- As a policy, Mumbai Port does not handle dangerous cargo. Hence, an accident like the one that happened in Tianjin Port in China will not occur in Mumbai Port. On 12 August 2015, hazardous chemicals, Sodium nitrate and Potassium nitrate (IMCO class 5.1), stored in a warehouse near Tianjin Port caused huge explosion in which nearly 170 people and more than 10000 vehicles including about 8000 cars perished. This will never happen to the cars handled in Mumbai Port.

2.2.8 Estimated automobile export traffic for Mumbai Port and OCT

- Mumbai Port's auto export traffic has been continuously on the rise in the last four years. Besides Tata and Mahindra, who are the steadfast user of Mumbai Port, new players have come in including General Motors and Volkswagen. Signs are their exports will take off in a big way. This is evident from the auto traffic handled by the Port in the first half of the current fiscal. From April to September 2016 it has handled 1lakh units as against 1.7 lakh units in the whole of last year, 2015-16. Many more players are setting up shop in the Maharashtra cluster, including Jaguar, Chrysler and BMW.

- Mumbai Port currently handles about 27 % of country's total light vehicle exports. This proportion will certainly rise with the entry of new players and the expected increase in volumes of existing OEMs.

- In specific terms, Mumbai Port's automobile traffic will grow at 15% CAGR in the near term until 2020 and then come down to about 7% for the next 5 years, until 2025. Thereafter as our main export markets, South America and Europe, start maturing and other markets, mainly in Africa, open up, the growth may be at a sedate pace of CGAR 5% up to 2040. A subdued growth of 3% CGARcan be expected then on.

Automobile export projection
(for thewhole of MbPT)

Base: 1.73 lakh units in 2015-16

Year	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Lakh units	1.99	2.29	2.63	3.03	3.5	4.03	4.64	5.34	6.15	8.62	14.02	16.28

Projection exclusively for OCT

Year	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Mn tons	1.99	2.29	2.63	3.03	3.5	4.01	4.40*	4.40*	4.40*	4.40*	4.40*	4.40*

***capacity saturation**

2 Container Traffic

2.3. OCT appears to have limited scope for container traffic due to the following reasons:

- Mumbai Port on an average has been handling only 40000 teus per annum for the last 10 years
- These are mostly JNPT-bound or JNPT-origin containers moved on barge or by road. These are the only containers handled in the last 10 years. These containers carry import goods for local (Mumbai city's) consumption or export consignments shipped from city-based units. They may remain more or less at the same level.
- Feeder vessels up to the 2nd generation, which are light and small, could have come to Mumbai Port. They have not been coming.
- The present capacity of JNPT is 4.1 million teus. The 4th terminal is expected to go on stream in another 2 years. The capacity of JNPT will then reach nearly 9 million teus. This leaves little room for others in the region. In fact, JNPT will become a threat even to Mundra.
- Without quayside gantry cranes no mainline container vessel will call at the terminal

In view of above, the scope of OCT for container may be limited to about 40000 to 50000 assuming the entire present container traffic coming to Mumbai Port will shift to OCT. On this basis the capacity of OCT for container traffic is calculated as below

- The berth occupancy for the automobile berth of OCT has been assumed at 50%. So, there is a slack of 20% occupancy is available at the auto berth. This slack can be used for handling the above limited container traffic. On that basis, the capacity for container will be as follows:
- No. of moves per hour per crane: 12
- No. of cranes at a time : 2
- Ship output per hour: :24 moves
- Average no. of working hours per day: 20
- Ship output per day : 480 moves

- Proportion of 40’ to 20’: 50:50
- So, ship output per day: 720 teus
- Berth days available for container ships at auto berth: 365×0.2= 73 days
- So, **container capacity of OCT: 73×720= 52560 teus**

Conclusion on Traffic Analysis

Based on the past traffic trends in Mumbai Port, steel and automobile were identified as the potential growth cargoes for OCT. Accordingly, the future prospect for these cargoes for OCT was analyzed. The study shows that in both steel and automobile cargo segments there will be appreciable growth in the near term, moderate growth in the medium term and subdued growth in the far term. Container traffic appears to have very limited scope for expansion.

The above assessment is based on the present trends in the domestic and global scenario. It may therefore be prudent to wait and watch for a few years how the traffic in the above cargoes behaves. Thereafter, the Port and the prospective concessionaire, with mutual understanding, may take a call on the construction of the third berth

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SECTION 3

PLANNING PARAMETERS

3.1.1 GENERAL

The OCT berth as constructed has been designed to handle container vessels upto 6000 TEU capacity with dimensions as LOA : 300 m ; Beam 40 m; Draft 13.5 m. Considering the length of each berth as 350 m for accommodating such a vessel, 700 m length of jetty has been provided for accommodating two vessels at a time.

However, the expected container traffic did not materialise and the traffic in containers was limited to 45,000 ~ 50,000 TEU only. Consequently the ship sizes were also much smaller than the design vessel. With the present proposal for revival of this project with changed traffic configuration, in that RoRo, iron & steel and other general cargo have been added in addition to containers, possible revised design vessel is to be identified.

Accordingly, a detailed analysis of the port's past performance for 2015-16 & 2016-17 (*first 6 months*) has been carried out especially for larger vessels deployed for RoRo & Steel.

3.1.2 PAST PERFORMANCE

Steel : The detailed analysis of vessels carrying steel during 2015-16 & 2016-17 (*six months*) is presented in Annexure 3.1. It could be seen that bigger vessels are usually handled at the two berths BPS & BPX and hence these two berths are taken for reference.

The largest vessel handled was of 64,000 DWT with a LOA of 200 m. (*The average LOA was 189 m*)The maximum parcel size was 43,720 Te. The maximum draft on arrival was 11.0 m.

The average parcel size was 26,675 Te and the average working at time at berth was 70 hrs. This gives a productivity of 381 TPH.

RoRo : The detailed analysis of RoRo vessels during 2015-16 & 2016-17 (*six months*) is presented in Annexure 3.2. It could be seen that bigger vessels are usually handled at the two berths 0CT & BPX and hence these two berths are taken for reference.

The largest vessel handled was of 28,090 DWT with a LOA of 232 m. (*The average LOA was 204 m*). The maximum parcel size was 7,168 Te. The maximum draft on arrival was 9.3 m.

The average parcel size was 3,243 Te and the average working time at berth was 32 hours.

3.1.3 TRAFFIC PROJCTION

Projections for steel traffic for OCT

Year	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Mn tons	2.50	2.50	2.50*	2.63*	2.76*	2.89*	3.03*	3.03*	3.03*	3.03*	3.03*	3.03*

Projection for automobile traffic for OCT

Year	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25	25/30	30/35	35/40
Lakh units	1.99	2.29	2.63	3.03	3.5	4.01	4.40*	4.40*	4.40*	4.40*	4.40*	4.40*

Traffic Projection for container traffic

The container traffic at OCT may be mostly limited to 50000 teus for the entire period.

3.2.1 REVISED DESIGN SHIP

As indicated earlier, the existing jetty structure has been designed to handle container vessels upto 6,000 TEU capacity ie. 120,000 DWT; 300 m LOA; 40 m Beam and 13.5 m Draft.

With the revised traffic configuration i.e. with RoRo, Steel and containers through smaller ships, the design ship would be Panamax size general cargo vessels upto 75,000 DWT.

Based on the past performance, the following conclusions are made.

Steel :

The maximum ship size could be Panamax vessels of 225 m LOA but with draft limited to 12 m only. The beam will be 32 m.

RoRo :

The largest pure car carrier of 8,000 unit capacity is only 30,000 DWT with 232 m LOA; 32 m Beam and 9.50 m draft.

Container vessels:

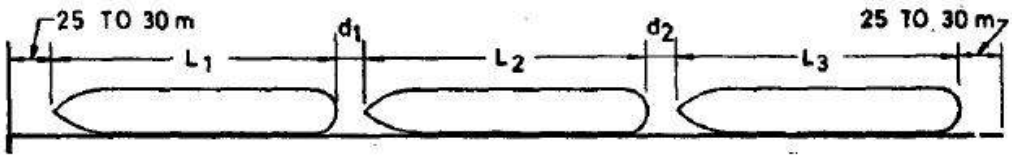
With the present container traffic at around 50,000 TEUs and which is not expected to have any dramatic increase in future, the size of the veseels will be much smaller.

3.2.2 NUMBER OF BERTHS

As has been brought out earlier, the existing jetty has an overall length of 700 m supposed to accommodate two berths, each for 6,000 TEU container vessels of length 300 m.

With the revised traffic configuration, the maximum size of vessels range from 232 m for RoRo and 225 m for steel cargo. Based on the past performance, the average LOA of RoRo vessels is 204 m and that of steel is 189 m. It is also noticed that the maximum LOA of vessels for steel cargo has been only 200 m. On detailed examination, it is noted that in case of RoRo vessels, the percentage of vessels exceeding 200 m LOA is only 12%.

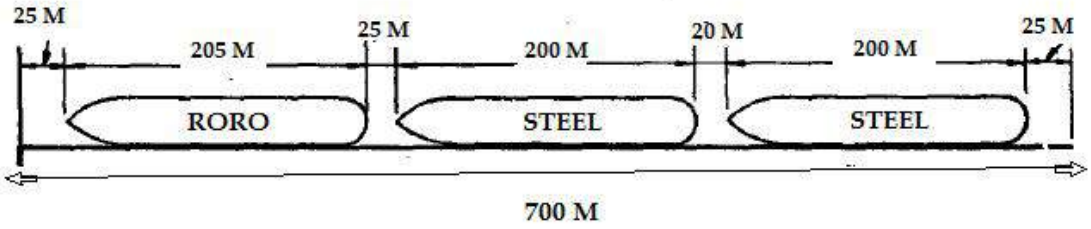
In view of these, it should be possible to berth 3 ships within the existing jetty length of 700 m. The stipulation for the spacing of the ships, as per IS: 4651 (Part V) – 1980 (Reaffirmed 1997), is reproduced hereunder.



$d_1 \leq \frac{L_1 + L_2}{20}$ SUBJECT TO A MINIMUM OF 15 m

$d_2 \leq \frac{L_2 + L_3}{20}$ SUBJECT TO A MINIMUM OF 15 m

Based on this, the berthing pattern for one RoRo vessel and two steel vessels will be as follows:



It has been indicated that the number of RoRo vessels with LOA exceeding 200 m is only 12%. Considering a still higher percentage of 15%, the middle berth could not be operated during these periods. Hence, besides a ro-ro vessel, it should be possible to handle 2 steel vessels part of the time and 1 steel vessel all the time.

3.3 CAPACITY OF BERTHS

I. Steel

Depending on the size of the automobile vessels and steel vessels it may be possible to work 2 steel vessels at a time at the steel berth part of the time and one vessel all the time, as indicated above. It is estimated that at the berth occupancy of 70% for the whole of steel berth, 60% of the time 2 steel vessels and 40% of the time 1 steel vessel can work there in a year. On that basis, the steel cargo capacity of OCT has been calculated as below.

Assumptions:

- Proportion of coil to flats: 7:3
- Average cycle time for coil: 6 minutes
- Average cycle time for flats: 9 minutes
- Average weight of coil load cycle: 20 tonnes
- Average weight of flat load cycle: 10 tonnes

On the above assumption, no. of cycles per hour:

- Coil: 10
- Flats: 7

On that basis productivity per hour per crane:

For a set of 7 coils and 3 flats the time taken is (7×6) + (3×9) = 69 minutes

That is, in 69 minutes, (7×20) + (3×10) = 170 tons achieved

So, the productivity per hour is (170÷69) × 60 = 147.83 tons, say 148 tons

Thus, **productivity per crane per hour is 148 tons**

Scenario 1: two steel vessels at a time

Total No. of cranes: 5 – 2 HMCs and 3 ship's cranes

Productivity per hour per crane: 148 tons (A above)

Productivity for 5 cranes: 148 × 5= 740 tons

Ship Productivity per day: 740 × 20 =14800 tons

No. of days for 2 vessels scenario at 70% berth occupancy: (365×0.7 × 0.6) = 153.30

So, capacity for 2 vessel scenario: 153.3×14800 = 2268840, say **2.27 million tons**

Scenario 2: one steel vessel at a time

The assumption in this scenario is vessels that are totally gearless may work with 2 HMCs. The ones that come with all ship's cranes working or partly working will use 3 cranes. Hence, an average of 2½ cranes is taken for this scenario

Total No. of cranes: 2½

Productivity per hour per crane: 148 tons (A above)

Productivity for 2½ cranes: 148 ×2½ =370 tons

Ship Productivity per day: 370 × 20 =7400 tons

Berth occupancy: 40% of 70%

No. of days for 1 vessel: (365×0.70) × 0.4 = 102.2

So, capacity for 1 vessel scenario: 102.2 × 7400 = 756280, say **0.76 million tons**

Thus, the total steel capacity of OCT: 2.27 + 0.76 = 3.03 million tons

It is understood that most steel vessel operators take ships on time charter and charter hire for gearless vessels is low. So, it is preferable to have shore cranes to attract more traffic. Since the berths have been designed for handling containers through quayside gantry cranes, the deck slab will not be able to take the load from pads of conventional Harbour Mobile Cranes. It is necessary to have custom-built rail mounted HMCs with a rail span of 30 m to suit the crane rails already provided on the deck.

It is possible to have such cranes and a typical rail mounted crane from Italgru is shown in Annexure 3.3. The basic technical detail sare furnished hereunder.

MAIN TECHNICAL DATA

1.1 CAPACITY AND OPERATING SPEEDS

• Crane Lifting Capacity	T	120/50/empty
• Hook Hoisting Speed	m/min	14/30/90
• Slewing Speed	rpm	1,45
• Boom luffing time: from max. to min. radius	s	55
• Traveling Speed	m/min	0 – 20

1.2 CRANE DIMENSIONS

1.2.1 Main dimensions

• Minimum Working Radius	m	11
• Maximum Working Radius	m	51
• Boom Length	m	51,50
• Max Hook Deep Below Ground	m	15
• Superstructure Slewing Radius	m	7,28
• Crane Height with Tower in Working Position	m	38,80
• Boom Pivot Point above Ground	m	23,70
• Operator's Eye Level	m	27,50

1.2.2 Portal on rail dimension

• Portal span	m	*30*
• Clearance under portal	m	*6,5*
• Number of bogies per corner	n	4
• Total number of wheels		32
• Wheels diameter	mm	500
• Rails size		A100

II. RoRo

Av. no. of units loaded per hour: 120

Net working hour per day: 20

Berth occupancy: 50%

Based on the above assumption

Average output per ship day: 2400 units

Berth capacity: 2400× (365/2) =4,38,000units or **0.44 MN units**

III. Container

The scope of OCT for container may be limited to about 40000 to 50000 assuming the entire present container traffic coming to Mumbai Port will shift to OCT. On that basis, the capacity of OCT for container traffic has been estimated as under:

The berth occupancy for the automobile berth of OCT has been assumed at 50%. Thus, there is a slack of 20% in the occupancy at the auto berth. This slack can be used for handling the above limited container traffic. On that basis, the capacity for container traffic works out to as follows:

- No. of moves per hour per crane : 12
- No. of cranes at a time : 2
- Ship output per hour: : 24 moves
- Average no. of working hours per day : 20
- Ship output per day : 480 moves
- Proportion of 40' to 20' : 50:50
- So, ship output per day : 720 teus
- Berth days available for container ships at auto berth : 365×0.2= 73 days
- So, **container capacity of OCT: 73×720= 52560 teus**

3.4 REQUIREMENT OF STORAGE AREA

I. Steel

- Weighted average stowage factor of coil and flats: 0.35
- Broken stowage allowance: 10%
- Stacking factor: 0.35×1.10 = 0.39
- Average Stacking height = 2 metre

Based on above factors one ton of steel will require 0.39÷2 = 0.20 m^2

Average dwell time for steel in MbPT is 16 days

On that basis, the area required for annual throughput of 3.03 million tons will be:

3030000 ÷ (365/16) ×0.2 = 26567m^2, say**2.66 hectares**

II. Automobiles

Average stowage factor for 1 ton of auto unit: 6 m^2

Average weight of 1 auto unit: 2 tons

So, average stowage factor of 1 unit of automobile: 12 m^2

Broken stowage allowance for automobile storage: 25%

So, stacking factor for automobile per unit: 12×1.25=15m^2

Thus, average area required for storage of one vehicle: 15m^2

Average dwell time: 7.5 days

Calculated throughput of automobile berth is 440000 units

So, total area required is: 440000 ÷ (365/7.5)×15 =135607 m2, 13.56 hectares

III. Containers

Assumptions:

Dwell time: 10 days

Type of yard equipment: Reach stacker

Calculation:

Space required for one teu ground slot at reach stacker yard: 35 m^2

Stacking height: 2½

So, area required for one teu: 35/2½ = 14 m^2

Total traffic: 52,560 teus/year

So, total area required for container traffic:

52560 ÷ (365/10) × 14 = 20160 m^2 ,say,**2 hectares**

Overall storage area required for estimated OCT capacity

Steel	**:**	**2.66 ha**
Auto	**:**	**13.56 ha**
Container	**:**	**2.01 ha**
Total	**:**	**18.23 ha**

ANNEXURE 3.1

PAST PORT PERFORMANCE

STEEL TRAFFIC

SL.No.	PARTICULARS		2015 - 16 BPS&BPX	2015 - 16 ID	2016 - 17* BPS&BPX	2016 - 17* ID
1	Number of shipcalls		150	162	50	201
2	Total volume handled		39,21,677	17,46,276	9,69,948	9,79,640
3	Imports		38,41,217	14,64,647	8,95,904	7,41,553
4	Exports		80,460	2,81,629	74,044	2,38,087
5	Deadweight Tonnage	Maximumum	64,000	63,562	63,478	58,098
		Minimum	22,289	3,482	23,500	2,048
		Average	51,325	31,879	47,653	12,076
6	Length Overall in M	Maximumum	200	200	200	200
		Minimum	154	87	161	75
		Average	189	163	188	108
7	Arrival Draft in M	Maximumum	11.00	10.50	10.50	9.20
		Minimum	5.90	4.90	6.20	2.20
		Average	9.40	7.40	8.90	4.90
8	Import Parcel Size in Te	Maximumum	43,720	32,469	40,912	26,159
		Minimum	1,084	48	1,707	427
		Average	26,675	10,388	20,352	4,418
9	Total Parcel Size in Te	Maximumum	43,720	32,469	40,912	35,155
		Minimum	1,084	239	21	54
		Average	26,145	10,779	19,391	4,674
10	Total time at berth in hrs	Maximumum	268	402	207	471
		Minimum	13	7	21	9
		Average	108	91	83	55
11	Total time worked at berth in hrs	Maximumum	159	189	145	317
		Minimum	5	1	3	2
		Average	70	54	50	29

2016-17 : April'16 to September'16

2015-16 : July'15 to March'16

ANNEXURE 3.2

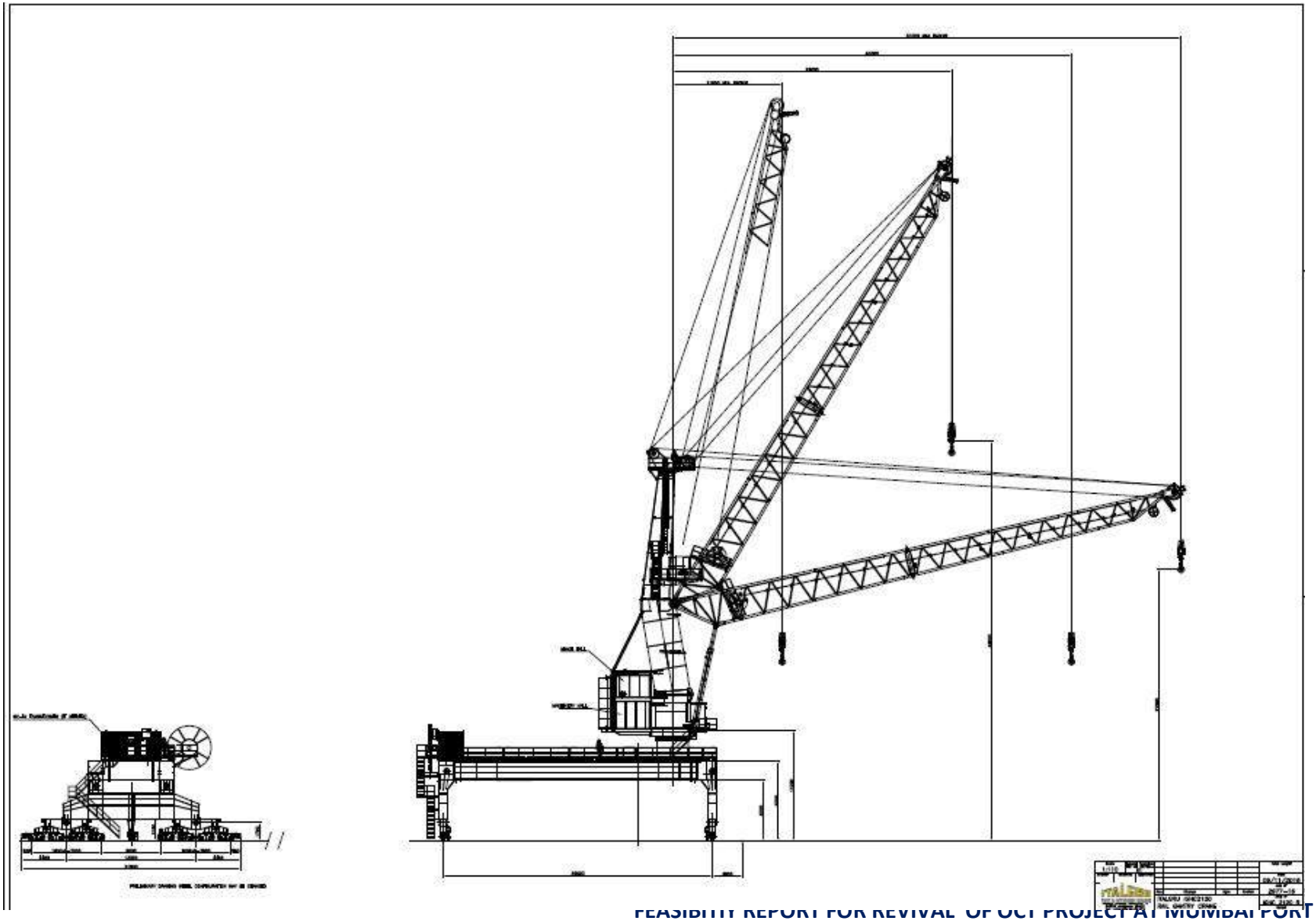
PAST PORT PERFORMANCE

RORO TRAFFIC

SL.No.	PARTICULARS		2015 - 16 OCT*	2015 - 16 ID	2015 - 16 BPX	2016 - 17* OCT	2016 - 17* ID
1	Number of shipcalls		50	41	35	47	7
2	Total volume handled		1,68,087	85,528	88,904	1,40,904	9,972
3	Imports		35,801	5,814	13,137	18,789	5,243
4	Exports		1,32,286	79,714	75,767	1,22,115	4,729
5	Deadweight Tonnage	Maximumum	28,092	21,421	27,564	28,075	22,675
		Minimum	12,229	5,271	10,019	12,229	3,629
		Average	20,630	15,673	19,131	20,100	15,340
6	Length Overall in M	Maximumum	232	215	232	232	215
		Minimum	173	162	180	180	130
		Average	204	181	204	206	178
7	Arrival Draft in M	Maximumum	8.70	8.60	9.30	9.00	8.40
		Minimum	6.90	5.30	6.50	6.70	5.80
		Average	7.80	7.30	8.00	7.90	7.50
8	Export Parcel Size in Te	Maximumum	6,605	5,323	7,168	6,990	2,935
		Minimum	675	116	91	332	431
		Average	3,243	1,993	2,228	2,598	1,182
9	Total Parcel Size in Te	Maximumum	6,901	6,065	7,169	7,660	3,183
		Minimum	809	219	692	426	732
		Average	3,362	2,086	2,540	2,998	1,425
10	Total time at berth in hrs	Maximumum	51	51	70	73	125
		Minimum	14	11	12	11	9
		Average	32	24	29	30	30
11	Total time worked at berth in hrs	Maximumum	49	48	53	67	104
		Minimum	10	4	5	7	4
		Average	24	14	20	23	22

2016-17 : April'16 to September'16

2015-16 : July'15 to March'16



SECTION 4

PROJECT DETAILS

4.1 GENERAL

In this section, the details of the revived OCT project are elaborated. The requirements for commissioning the revived project with a different cargo configuration such as RoRo and Steel are brought out. Due consideration has been given to the existing civil works and the additional works to be undertaken to commission this project are detailed. The re-configured equipments are also broadly indicated.

4.2 EXISTING CIVIL WORKS

The exisitng civil works are shown in the figure 4.1 hereunder.

Figure 4.1 : Existing civil works related to OCT

The Licensee, ICTPL have constructed the jetty (700 m x 58 m) ; primary approach trestle (612 m x 18 m) and two forked extensions (518 m x 12m & 374 m x 12 m). These are completed fully and are operational.

Some of the site photographs of the jetty and approaches are presented hereunder:

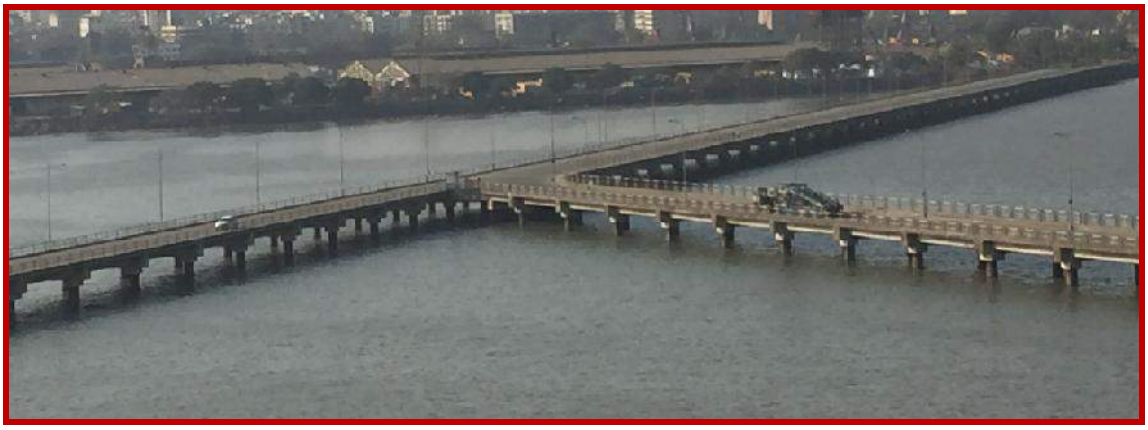


Figure 4.2 : Primary approach with forked extension

Figure 4.3 : Aerial view of the berths with parked car units

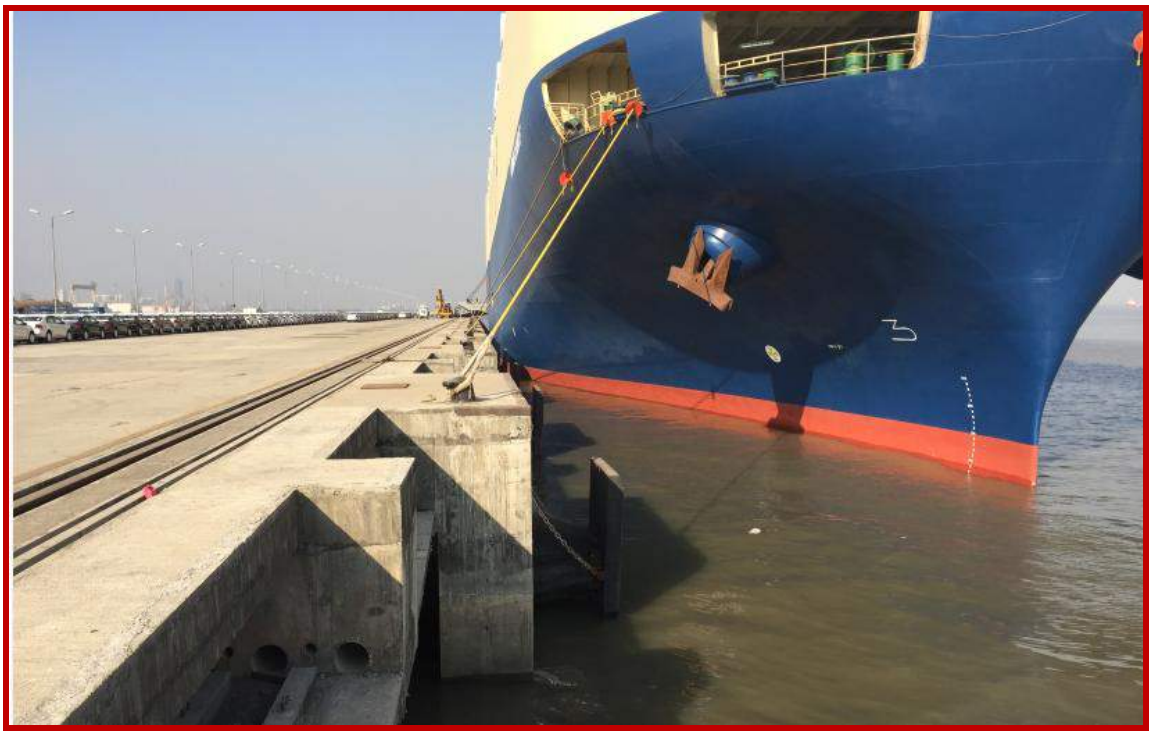


Figure 4.4 : Berth with a RoRo vessel berthed

In addition they have constructed on the rear of the berth a sub-station, a fire-fighting pump room and a jetty office. However, these are incomplete and are to be equipped with the necessary machinery.

A couple of pictures of these structures as existing are presented hereunder:

Figure 4.5 : Sub-station; Fire station and Office buildings

Figure 4.6 : Sub-station; Fire station and Office buildings

Mumbai Port, on their part, have filled up both Princess Dock and Victoria Dock and levelled the whole area shown in the Figure 4.1.

In addition, Mumbai Port has also developed a dedicated railway yard with 3 sidings each of 740 m length on the western side of P&V Docks as shown in Figure 4.7 hereunder. This railway yard is connected to MbPT railway network. Out of the 3 sidings, one is likely to be retained by the port for its own usage. Presently, the connectivity to the national network is available only around midnight and for a very limited time because of the suburbum traffic congestion. Hence the port has undertaken to lay a dedicated line (1 up + 1 down) from Wadala to Kurla. This is 4.41 km line which, when completed, is expected to facilitate smooth evacuation of MbPT cargo by rail. This project is being executed by Central Railway and MMRDA for the port. It is expected that this project will be completed within 2 years.

Meanwhile, MbPT has also facilitated receipt of export passenger car units by rail instead of by road by trailers reducing the trffiac congestion on roads.

A site photograph of the railway sidings is presented in Figure 4.8.

Figure 4.7 : Railway sidings west of P&V Docks

Figure 4.8 : View of the 3 railway sidings

4.4 NEW CIVIL WORKS TO BE EXECUTED

It has been brought out in the earlier section that the stackyard requirements will be as follows:

Steel ; 26,600 sq.m of area of which
6,650 sq.m will be covered and
19,950 sq.m will be open area

RoRo: 135,600 sq.m of paved area

Containers: 20,160 sq.m of parking yard

For stacking steel and containers in the open a heavy duty pavement has to be prepared while for RoRo a yard developed with paver blocks will be required. Out of the total area of 18.23 Ha required, 14.65 Ha will be made available at Victoria Dock and the balance made available at Frere Basin. These are shown in Figures 4.9 and 4.10 hereunder.

In addition to the above, 23,900 sqm of railway yard area for handling steel cargo, 2552 sqm for Fire water platform including substation and an area of 6280 sqm for misc. operational buildings is required to be given. Hence the total area required to be given is 215092 sq m or say 21.51 Ha besides the Berth area and the trestle.

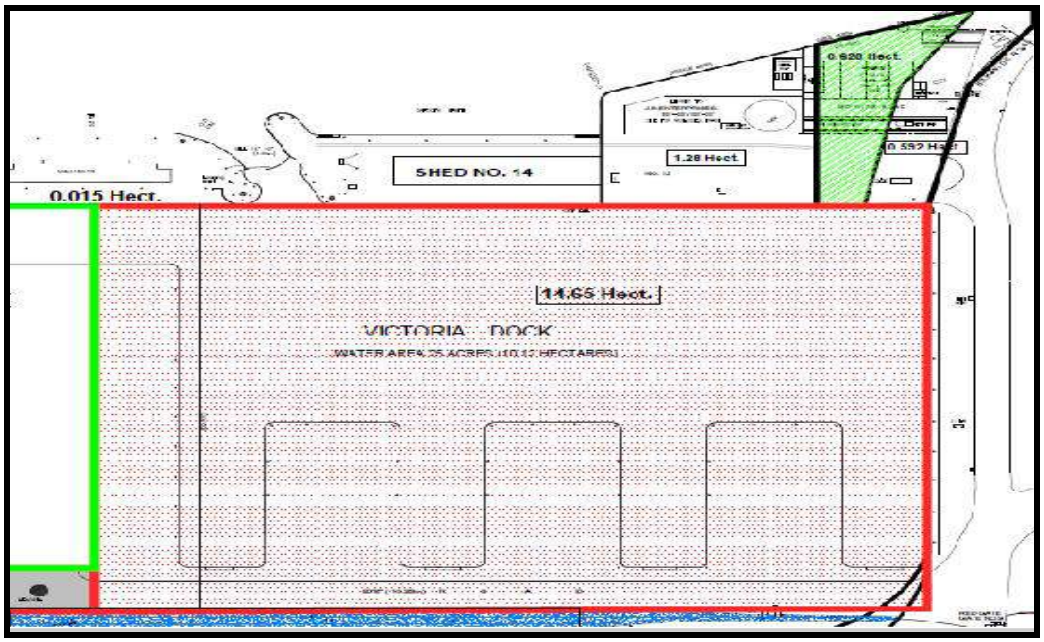


Figure 4.9 : Proposed area at Victoria Dock

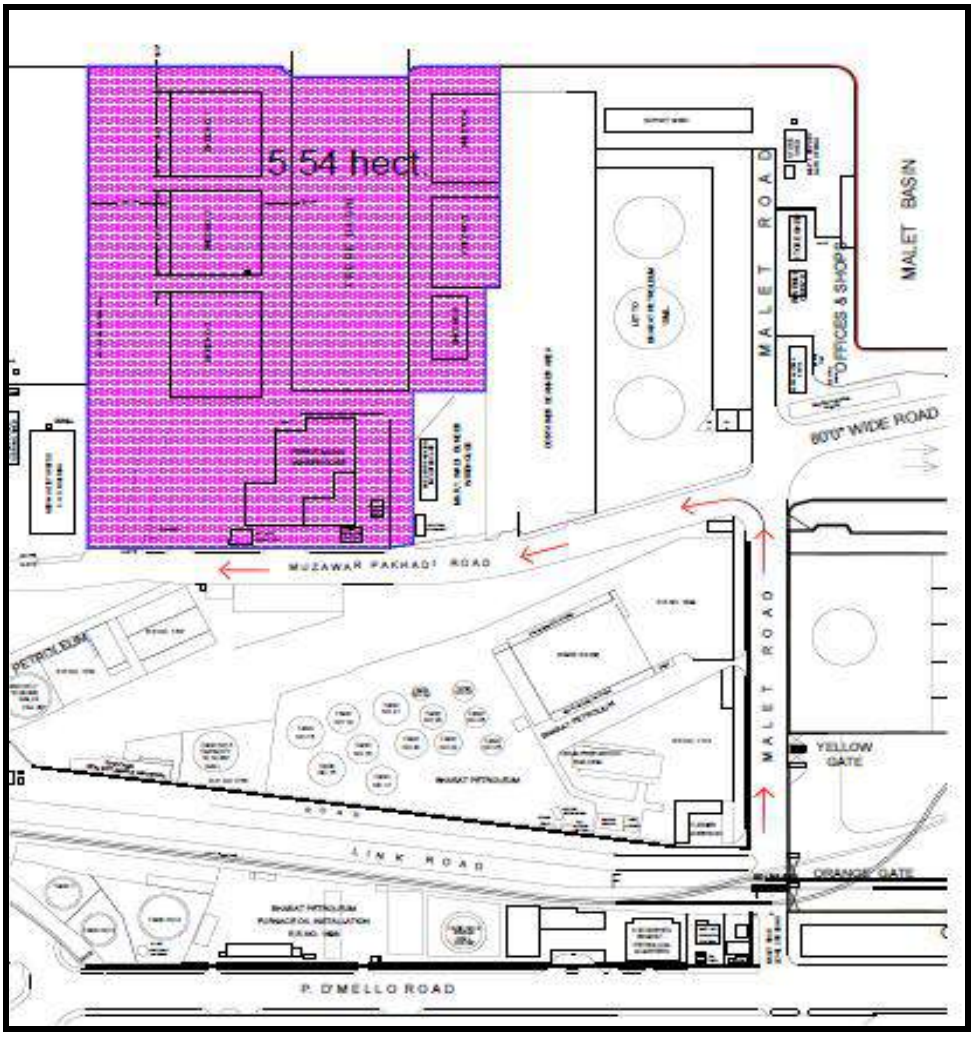


Figure 4.10 : Area available at Frere Basin

For railway loading of cargo, a 25 m wide strip of land has to be hardened along the railway siding along the railway siding of 760 m length accommodating a full rake.

In addition to all these, a maintenance workshop, operational building, compound wall and gate house, miscellaneous buildings such as dispensary, canteen etc. have to be provided. The partially constructed jetty-head building housing sub-station, fire station and jetty office has to be finished and completed properly.

4.4 NEW MECHANICAL EQUIPMENT

Cargo handling equipment are needed for steel. For RoRo berth, the automobile units will be directly driven into the carrier through ramps by drivers. Containers will be

handled through ships' own gear. If gearless vessels or barges are to be handled, the cranes that handle steel could be used by replacing hooks with spreaders.

For steel equipment are needed for ship-shore transfer, transfer from berth to the stackyard, handling at the yard and also for loading onto the railway wagons.

Ship-shore transfer:

For ship-shore transfer, it is suggested to equip one berth with two Harbour Mobile Cranes for efficient handling of cargo. However, these HMCs should be mounted on the rail tracks laid for the container quayside gantry cranes. The deck slab will not be able to take the point loads from the crane leg pads. This crane rail track is of 30 m gauge. Hence the HMCs should be tailer made as rail mounted with this rail span. The capacity should be to handle minimum 30 Te at 50 m radius. Market enquiry has already been made and a typical crane with technical details has been presented in the earlier section.

For the other berth, the cargo transfer will be through ships' own gear as is being done now.

Berth to Yard transfer:

It has been indicated earlier that the productivity of the HMC equipped berth will be 510 TPH while the ships' gear operated berth will have a productivity of 405 TPH. It has also been mentioned that the unit weight of steel coils will be 20 Te while that of the steel flats will be 10 Te.

For transfer of cargo between the quay and the storage, 10 truck-trailers will be required, with a distribution of 3 trailers on the each way, 2 trailers on the quay side and 2 trailers at the storage site for achieving the required productivity. Assuming 20 Te capacity trailer/trucks being used, the cycle time for each will be about 35 minutes: 5 minutes each at berth and at yard, 15 minutes travel time from berth to yard and 10 minutes travel time back to berth from the yard. Each trailer should be fitted with twist locks and other container securing arrangement as the same Truck-Trailers will

be used for haulage for both steel and containers. Based on these, the berth will need 15 trailers/trucks and the other berth will need 12 trailers/trucks.

Handling equipment at the steel stack yard

When both the steel berths are being occupied and operated, 27 trailers/trucks will be arriving at the yard per hour. In order to unload the cargo, a minimum of 5 numbers of 30 Te mobile cranes will be required.. When the berths are not occupied or not in operation, these cranes will be used for loading the cargo onto evacuating trucks/trailers.

Handling equipment at the container yard

In addition to the tractor trailers**,** the containers yard will require two reachstacker and receipt/ delivery operations.

Handling equipment at the railway yard

For loading the cargo into railway wagons at the railway yard, at least 2 numbers of 30 Te mobile cranes will be required.

SECTION 5

CAPITAL COST ESTIMATE & IMPLEMENTATION SCHEDULE

5.1 The total capital cost of the project estimated by SBI Caps as replacement cost of the existing assets for Rs. 549.31 cr has been considered as directed by Mumbai Port. However an amount of Rs. 166.36cr has been considered towards additional cap-ex for development of stack yards etc and procurement of cargo handling cranes without which Steel cargo cannot be handled. The detailed estimate is attached as ***Annexure 5.1.*** The summary breakup of the estimate is given as under:

Sl. No.	**Item of Work**	**Replacement cost** (Rs. Cr)	**Additional Cost** (Rs. Cr)	**Total Capital Cost** (Rs. Cr)
A	**Cost of Civil/Mech works**			
1	Civil Works	427.14	51.56	478.70
2	Mechanical Works		92.45	92.45
	Total - A	**427.14**	**144.01**	**571.15**
B	**Other Costs**			
1	Supervision Cost @ 2.5% of A	10.68	3.60	14.28
2	Design Cost @ 1% of A	4.27	1.44	5.71
3	Pre-operative exps @ 3% of A	12.81	4.32	17.13
4	Upfront fee paid to MbPT	25.00		25.00
	Total -B	**52.76**	**9.36**	**62.12**
C	**Interest During Construction**	**69.41**	**12.99**	**82.40**
	Grand Total (A+B+C)	**549.31**	**166.36**	**715.67**

Note : The above capital cost estimate relates to Private operator's investment only. Port has to incur capital expenditure on Capital dredging for the contracted depth.

5.2. IMPLEMENTATION SCHEDULE

The project is already commenced and is operational except for further development of stack yards and miscellaneous civil works and installation of Mechanical equipment which is expected to take 18 months from the date of award.

5.3. PHASING OF EXPENDITURE

The phasing of expenditure proposed is as under:

The new BOT operator is expected to take over the assets on “as is where is” basis and start operating from day one. In case the existing Licensee exercises their First Right of Refusal (FROR) the berths are already in their possession and they can continue to operate the berths. As stated above, in either case the civil works and procurement and installation of additional mechanical equipment is assumed to take 18 months.

Year – 1	Rs. 632.49 crores(UpfrontRs. 549.31cr + 50% of Addl cost)
Year – 2	Rs. 83.18 crores (Balance 50% of Addl cost)

5.4. REPLACEMENT OF OUTLIVED ASSETS

The civil works are assumed to last for more than 30 years as per the guidelines and hence no replacement of civil assets is required during the period of 27.5 yrs of Concession period. Since the life of Harbour Mobile cranes is only 20 years and the life of other cranes is 10 years, they are to be replaced at the end of 20 and 10 years respectively. The present day cost of the equipment is escalated at 3% to arrive at the cost of the respective cranes at the end of respective periods and considered as Cap-ex in the respective years.

ANNEXURE 5.1

CAPITAL COST ESTIMATE

Rs. Cr

S.No	Description	Qty	Replace-ment cost	Addl Cost	Total Cost
A	**Cost of Civil works**				
1	Site Establishment		6.60		6.60
2	Construction of Berth		274.74		274.74
3	Construction of Approach Trestle		100.12		100.12
4	Construction of Fire water platform		12.62		12.62
5	Stack yard / Container yard		23.44		23.44
6	Terminal Operational buildings		9.62		9.62
7	Dev of yard/terminal op bldg/ Misc			51.56	51.56
	Sub Total (A)		**427.14**	**51.56**	**478.70**
B	**Mechnical/Electrical works**				
1	Mobile Harbour Crane - (100T)	2		70.00	70.00
2	Tractor-Trailors (20 T)	27		6.75	6.75
3	Mobile Cranes (30 T)	7		10.50	10.50
4	Reach Stacker	2		5.20	5.20
	Sub Total (B)		**-**	**92.45**	**92.45**
C	**Total Cost (A+B)**		**427.14**	**144.01**	**571.15**
D	**Other Costs**				-
1	Supervision Cost - @% of C	2.50%	10.68	3.60	14.28
2	Design cost - @% of C	1%	4.27	1.44	5.71
3	Pre-operative expenses @% of C	3%	12.81	4.32	17.13
	(Financing and Head office cost)				-
4	Upfront fee paid to MbPT		25.00		25.00
5	Interest During constrution		69.41	12.99	82.40
	Sub Total (D)		**122.17**	**22.35**	**144.52**
	GRAND TOTAL		**549.31**	**166.36**	**715.67**

Summary of Cap-ex including apportioned other costs

			Cap-ex	Other Cost	Total
1	Civil Cost		478.70	121.13	599.83
2	Mech / Electrical Cost		92.45	23.39	115.84
3	**Total**		**571.15**	**144.52**	**715.67**

SECTION 6

ANNUAL OPERATON & MAINTENANCE COST

6.1 The annual operation and maintenance cost for the project has been estimated based on 2008 TAMP Guidelines for fixation of up-front tariff for PPP projects. The broad break-up of estimate is given in the table below.

S.No	Particulars	Basis		Amount (Rs.incrores)
1	Repairs & Maintenance Civil works	1% of Cap-ex	1% x Rs. 599.83crs.	6.00
2	Repairs & Maintenance Mech/Elec. works	5% of Cap-ex	5% x Rs. 115.84cr	5.79
3	Insurance	1% of Cap-ex	1% x Rs. 715.67 cr	7.16
4	Other expenses (towards salaries and overheads)	5% of Gross value of assets ofCargo handling Activity	5% x Rs. 441.66cr	22.08
5	Power Cost (Area= 305424 m2 less water front area of 28000 m2)	24 units per m2	24 units x Rs.8/unit x 277424 m^2	5.33
6	Fuel cost *			
	a) HMC -100 T-2 Nos	2 Nos. x 4000 hrs x 70 ltrs	560000 x Rs. 67/ ltr	3.75
	b) TT-20T – 27 Nos	27 Nos. x 4000 hrs x 10 ltrs	1080000 x Rs. 67/ ltr	7.24
	c) Mobile cr-30T-7 Nos	7 Nos. x 4000 hrs x 20 ltrs	560000 x Rs. 67/ ltr	3.75
	d) Reach Stacker-2 Nos	2 nos. x 4000 hrs x 20 ltrs	160000 x Rs. 67/ ltr	1.07
			Total	15.81
7	License Fee			9.03
	a) Berth Area	40600 m2	Rs 309.76 p.a	
	b) Approach Trestle	21732 m2	Rs 309.76 p.a	

S.No	Particulars	Basis		Amount (Rs.incrores)
	c) Covered area for steel	6,650 m2	Rs. 309.76 p.a	
	d) Open area for steel	19,950m2	Rs. 309.76 p.a	
	e) Paved area for Rly yard for steel	23,900 m2	Rs. 309.76 p.a	
	f) Open area for RoRO	135,600 m2	Rs. 309.76 p.a	
	g) Open area for Container yard	20,160 m2	Rs. 309.76 p.a	
	h) Area for fire water platform incl sub stn	2,552 m2	Rs. 309.76 p.a	
	i) Area for bldg. etc	6,280 m2	Rs. 309.76 p.a	
	j) Waterfront area (700 x 40)	28,000 m2	Rs. 154.88p.a	
	Total Area	**305,424 m2**		
8	Stevedoring Charges	440,000 PCUs	Rs. 250 per PCU as per market	11.00
9	Depreciation			
	a) Civil assets	3.17 % of CC	Rs. 599.83 cr	19.01
	b) Mech/Elec assets	10.00 % of CC	Rs. 115.84cr	11.58

** Note : Fuel cost is based on Optimal capacity of project. The cost based on the traffic is charged to the respective years.*

6.4 The operating expenditure is escalated at the following rates year on year.

a) Repairs & Maintenance - 5% p.a.
b) Insurance - 1% p.a.
c) Other Expenses - 3% p.a
d) Power - 3% p.a
e) Fuel - 5% p.a
f) Stevedoring charges - 5% p.a
g) License fee - 10% for every 5 years

6.5 The detailed Operating & Maintenance expenditure is given at **Annexure 6.1**

Annexure- 6.1

OFFSHORE CARGO TERMINAL (OCT) - MUMBAI PORT
OPERATING EXPENDITURE

Year	Fin Year	R&M Civil (Esc5%)	R&M Mech (Esc5%)	Insurance (Esc1%)	Other exps (Esc3%)	Power Cost Esc 3%	Fuel Chgs On Traffic (Esc5%)	Steve-doring (Esc5%)	L.Fee (Esc10% for every 5 yrs)	Total Op-ex Excl Depn
1	2017-18	6.00	5.79	7.16	22.08	5.33	12.67	11.00	9.03	**79.05**
2	2018-19	6.30	6.08	7.23	22.75	5.49	13.34	11.55	9.03	**81.76**
3	2019-20	6.61	6.39	7.30	23.43	5.65	14.74	12.13	9.03	**85.27**
4	2020-21	6.94	6.71	7.37	24.13	5.82	16.27	12.73	9.03	**89.01**
5	2021-22	7.29	7.04	7.45	24.86	6.00	17.98	13.37	9.93	**93.90**
6	2022-23	7.66	7.39	7.52	25.60	6.17	19.77	14.04	9.93	**98.08**
7	2023-24	8.04	7.76	7.60	26.37	6.36	20.85	14.74	9.93	**101.64**
8	2024-25	8.44	8.15	7.67	27.16	6.55	21.99	15.48	9.93	**105.37**
9	2025-26	8.86	8.56	7.75	27.97	6.75	23.21	16.25	10.92	**110.27**
10	2026-27	9.31	8.99	7.83	28.81	6.95	24.50	17.06	10.92	**114.37**
11	2027-28	9.77	9.43	7.91	29.68	7.16	25.89	17.92	10.92	**118.67**
12	2028-29	10.26	9.91	7.98	30.57	7.37	27.20	18.81	10.92	**123.03**
13	2029-30	10.77	10.40	8.06	31.49	7.59	28.56	19.75	12.02	**128.65**
14	2030-31	11.31	10.92	8.15	32.43	7.82	29.99	20.74	12.02	**133.38**
15	2031-32	11.88	11.47	8.23	33.40	8.06	31.49	21.78	12.02	**138.32**
16	2032-33	12.47	12.04	8.31	34.41	8.30	33.07	22.87	12.02	**143.47**
17	2033-34	13.09	12.64	8.39	35.44	8.55	34.72	24.01	13.22	**150.06**
18	2034-35	13.75	13.28	8.48	36.50	8.80	36.46	25.21	13.22	**155.69**
19	2035-36	14.44	13.94	8.56	37.60	9.07	38.28	26.47	13.22	**161.57**
20	2036-37	15.16	14.64	8.65	38.72	9..34	40.19	27.79	13.22	**167.71**
21	2037-38	15.92	15.37	8.73	39.89	9.62	42.20	29.18	14.54	**175.45**

Year	Fin Year	R&M Civil (Esc5%)	R&M Mech (Esc5%)	Insurance (Esc1%)	Other exps (Esc3%)	Power Cost Esc 3%	Fuel Chgs On Traffic (Esc5%)	Steve-doring (Esc5%)	L.Fee (Esc10% for every 5 yrs)	Total Op-ex Excl Depn
22	2038-39	16.71	16.14	8.82	41.08	9.91	44.31	30.64	14.54	**182.15**
23	2039-40	17.55	16.94	8.91	42.31	10.21	46.53	32.17	14.54	**189.16**
24	2040-41	18.42	17.79	9.00	43.58	10.51	48.86	33.78	14.54	**196.48**
25	2041-42	19.35	18.68	9.09	44.89	10.83	51.30	35.47	15.99	**205.59**
26	2042-43	20.31	19.61	9.18	46.24	11.15	53.86	37.25	15.99	**213.60**
27	2043-44	21.33	20.60	9.27	47.62	11.49	56.56	39.11	15.99	**221.96**
28	2044-45* (6 months)	11.20	10.81	4.68	24.53	5.92	31.69	20.53	15.99	**125.35**

SECTION 7

ANNUAL FINANCIAL REVENUE EARNINGS

7.1 The revenue earning from the project to the Operator under PPP / BOT is basically the Berth hire charges & Cargo handling charges. Hence the tariff shall be determined under Revised Reference Tariff guidelines 2013 or under Upfront Tariff guidelines 2008 in case no reference tariff is available for the given cargo profile in the port concerned or in any other Major Port. The said guidelines will also apply to Port's own Project. Although this project is said to be under 2005 guidelines and the operator has to submit a Tariff proposal for notification of the rates, there have to be rates and Annual Revenue permitted by TAMP. Accordingly, the financial analysis has been carried out considering the entire project is taken up through BOTand the upfront tariff permitted by the TAMP. The rates are escalated at 3% every year considering the average WPI of 5%.

7.2 The estimated annual revenue based on provisional tariff assessed as per the upfront tariff guidelines 2008 / Tariff orders is given below :

(Rs. In Lakhs)

S.No	Particulars	At Optimal Capacity
1.	**Estimated Throughput**	
	(a) RoRo (in lakh Per Car Units p.a)	4.40
	(b) Steel (in Lakh tons p.a.)	30.06
	(c) Containers (TEUs)	52560
2.	**Handling RateRate** (Rs)	
	(a) Automobiles (Per PCU)	1250
	(b) Steel(Per ton)	304
	(c) Container (PerTEU)	4344
3.	**Annual Revenue Requirement (ARR)** (Rs. In lakhs)	**16929.34**
	(a) RoRo	5497.96
	(b) Steel	9148.25
	(c) Container	2283.13
4.	**Estimated GRT** (Lakh GRT hours)	**4293.61**
5.	**Berth hire (Rs./ GRT hour)**	**1.35**
6.	**Revenue on Berth hire** (Rs. In lakhs)	**5800.73**
7.	**Total Estimated Income (Rs. lakhs)**	**22730.07**

7.3 The broad assumptions for the estimating the revenue are as follows.

7.3.1. The anticipated Handling charges and berth hire charges are worked out based on the preliminary calculations of annual revenue requirement and capacity as per the TAMP Guidelines for determination of upfront tariff (2008) / Tariff orders.

Although the guidelines of TAMP are followed for arriving at the Revenue requirement, the rates for specific cargoes are not worked out in view of the project being developed through BOT. The Tariff for the said cargoes has to be got notified by the BOT operator in accordance with 2005 / 2015 guidelines before commencing the operations.

TAMP vide their Guidelines of 2015 have stated that "Based on the Annual Revenue Requirements assessed, taking into account the traffic, Port will have the flexibility to determine the rates to respond to the market forces based on its commercial judgement and draw the SOR within the ceiling of indexed ARR". Hence, the BOT operator may have to take a call on the rates to be charged at OCT, based on the situation at the time of commissioning the operations and may approach TAMP for notifying Scale of Rates.

7.3.2. The anticipated Handling charges and berth hire charges are worked out based on the preliminary calculations of annual revenue requirement and capacity as per the TAMP Guidelines for determination of upfront tariff (2008.) / Tariff orders.

7.4. The port will earn revenue from Port Dues and Pilotage as per the General scale of rates, besides the Lease rental.

7.5 The detailed Traffic & Revenue workings are given at **Annexure 7.1.**

Annexure- 7.1

OFFSHORE CARGO TERMINAL (OCT) - MUMBAI PORT

OPERATING REVENUE

Year	Fin Year	RoRo		Steel		Containers		Total Revenue Rs Cr
		Traffic in PCUs	Revenue Rs Cr	Traffic in MTPA Esc 3%	Revenue Rs Cr	Traffic in TEUs	Revenue Rs Cr	
1	2017-18	229,000	50.94	2,500,000	100.92	20,000	33.53	**185.39**
2	2018-19	263,000	60.26	2,500,000	103.95	22,000	35.43	**199.64**
3	2019-20	303,000	71.51	2,625,000	112.42	24,200	38.82	**222.75**
4	2020-21	348,450	84.70	2,756,250	121.59	26,620	42.56	**248.85**
5	2021-22	400,718	100.33	2,894,063	131.50	29,282	46.68	**278.51**
6	2022-23	440,000	113.47	3,025,120	141.57	32,210	51.06	**306.11**
7	2023-24	440,000	116.87	3,025,120	145.82	35,431	54.27	**316.96**
8	2024-25	440,000	120.38	3,025,120	150.20	38,974	57.79	**328.36**
9	2025-26	440,000	123.99	3,025,120	154.70	42,872	61.67	**340.36**
10	2026-27	440,000	127.71	3,025,120	159.34	47,159	65.95	**353.00**
11	2027-28	440,000	131.54	3,025,120	164.12	51,875	70.68	**366.34**
12	2028-29	440,000	135.49	3,025,120	169.05	52,560	73.21	**377.75**
13	2029-30	440,000	139.55	3,025,120	174.12	52,560	75.41	**389.08**
14	2030-31	440,000	143.74	3,025,120	179.34	52,560	77.67	**400.75**
15	2031-32	440,000	148.05	3,025,120	184.72	52,560	80.00	**412.77**
16	2032-33	440,000	152.49	3,025,120	190.26	52,560	82.40	**425.16**
17	2033-34	440,000	157.07	3,025,120	195.97	52,560	84.87	**437.91**
18	2034-35	440,000	161.78	3,025,120	201.85	52,560	87.42	**451.05**
19	2035-36	440,000	166.63	3,025,120	207.91	52,560	90.04	**464.58**
20	2036-37	440,000	171.63	3,025,120	214.14	52,560	92.74	**478.52**
21	2037-38	440,000	176.78	3,025,120	220.57	52,560	95.52	**492.87**

Year	Fin Year	RoRo		Steel		Containers		Total Revenue Rs Cr
		Traffic in PCUs	Revenue Rs Cr	Traffic in MTPA Esc 3%	Revenue Rs Cr	Traffic in TEUs	Revenue Rs Cr	
22	2038-39	440,000	182.08	3,025,120	227.19	52,560	98.39	**507.66**
23	2039-40	440,000	187.55	3,025,120	234.00	52,560	101.34	**522.89**
24	2040-41	440,000	193.17	3,025,120	241.02	52,560	104.38	**538.57**
25	2041-42	440,000	198.97	3,025,120	248.25	52,560	107.51	**554.73**
26	2042-43	440,000	204.94	3,025,120	255.70	52,560	110.74	**571.37**
27	2043-44	440,000	211.09	3,025,120	263.37	52,560	114.06	**588.51**
28	2044-45* (6 months)	220,000	108.71	1,512,560	135.64	52,560	84.10	**328.44**

SECTION 8

FINANCIAL VIABILITY & SENSITIVITY ANALYSIS

8.1 The Financial viability of the project, considering the 27.5 years being the balance License period from the date of award of the restructured project works out to 13.86% even after payment of 35.064% revenue share to the port.

8.2 Sensitivity analysis has also been carried out to gauge the impact of increase in cost and reduction of revenue earnings on the viability of the proposal. The results of the analysis are presented below. The detailed Financial working are given at ***Annexure-8.1***

Sl. No.	Pre-Tax Project IRR	IRR (%)	NPV @ 12% (in Rs. cr)
1	Base case	13.86%	93.61
2	Capital Cost up by 10%	12.53%	28.91
3	Revenue down by 10%	10.60%	(-) 67.21
4	Annual O&M Cost up by 10%	12.14%	6.84
5	Combined effect of Sl. no. 2, 3 & 4	7.64%	(-) 218.68

From the above, it is evident that the FIRR of the Project at Base case is 13.86% and in the least case of sensitivity gives 7.64% (which is considered to be not probable in view of the fact that Major portion of the cap-ex has already been incurred) and hence the Project is Financially viable for taking up through BOT/PPP route since the Revenue share is going to be reserved at 35.064%. It is to be noted that the Revenue share payable by the operator is not considered as operating expenditure in accordance with TAMP Guidelines. In fact, the Project IRR at base case without revenue share works to 27.49% and in least case 21.71% and is financially a viable project.

8.3. The viability of the project will be further prospective, in the event the operator achieves the productivity norms and eligible for 15% productivity increase in tariff over the notified tariff.

Annexure-8.1

OFFSHORE CARGO TERMINAL (OCT) - MUMBAI PORT
PROJECT IRR (As per IPA Traffic)

Rs. in Crores

Yr	Fin Yr	Cap-ex	Gross Revenue	Revenue Share 35.064%	Nett Revenue	Op-ex Excl Depn+ Interest	Net Opn Cash Flows	Sensitivity Analysis			
								Cap-ex +10%	Revenue -10%	O&M +10%	Combined Effect
1	2017-18	549.31	185.39	65.01	120.39	79.05	(507.98)	(562.91)	(520.02)	(515.88)	(582.85)
2	2018-19	166.36	199.64	70.00	129.64	81.76	(118.48)	(135.11)	(131.44)	(126.65)	(156.25)
3	2019-20		222.75	78.11	144.65	85.27	59.38	59.38	44.91	50.85	36.39
4	2020-21		248.85	87.26	161.59	89.01	72.58	72.58	56.42	63.68	47.52
5	2021-22		278.50	97.65	180.85	93.90	86.95	86.95	68.86	77.55	59.47
6	2022-23		306.10	107.33	198.77	98.08	100.69	100.69	80.81	90.88	71.00
7	2023-24		316.96	111.14	205.82	101.64	104.18	104.18	83.60	94.01	73.43
8	2024-25		328.36	115.14	213.22	105.37	107.85	107.85	86.53	97.32	75.99
9	2025-26		340.36	119.34	221.01	110.27	110.74	110.74	88.64	99.71	77.61
10	2026-27		353.00	123.77	229.22	114.37	114.85	114.85	91.93	103.41	80.49
11	2027-28	31.68	366.34	128.45	237.89	118.67	87.53	84.37	63.75	75.67	48.71

Rs. in Crores

Yr	Fin Yr	Cap-ex	Gross Revenue	Revenue Share 35.064%	Nett Revenue	Op-ex Excl Depn+ Interest	Net Opn Cash Flows	Sensitivity Analysis			
								Cap-ex +10%	Revenue -10%	O&M +10%	Combined Effect
12	2028-29		377.74	132.45	245.29	123.03	122.26	122.26	97.73	109.96	85.43
13	2029-30		389.07	136.42	252.65	128.65	124.00	124.00	98.73	111.13	85.87
14	2030-31		400.75	140.52	260.23	133.38	126.85	126.85	100.83	113.51	87.49
15	2031-32		412.77	144.73	268.03	138.31	129.72	129.72	102.92	115.89	89.09
16	2032-33		425.15	149.07	276.08	143.47	132.60	132.60	105.00	118.26	90.65
17	2033-34		437.91	153.55	284.36	150.06	134.30	134.30	105.86	119.29	90.86
18	2034-35		451.04	158.15	292.89	155.69	137.20	137.20	107.91	121.63	92.35
19	2035-36		464.57	162.90	301.68	161.56	140.11	140.11	109.94	123.95	93.79
20	2036-37		478.51	167.79	310.73	167.71	143.02	143.02	111.95	126.25	95.18
21	2037-38	41.33	492.87	172.82	320.05	175.45	103.27	99.13	71.26	85.72	49.59
22	2038-39	132.75	507.65	178.00	329.65	182.15	14.75	1.47	(18.22)	(3.47)	(49.71)
23	2039-40		522.88	183.34	339.54	189.16	150.38	150.38	116.43	131.46	97.51
24	2040-41		538.57	188.84	349.72	196.48	153.24	153.24	118.27	133.59	98.62

Rs. in Crores

Yr	Fin Yr	Cap-ex	Gross Revenue	Revenue Share 35.064%	Nett Revenue	Op-ex Excl Depn+ Interest	Net Opn Cash Flows	Sensitivity Analysis			
								Cap-ex +10%	Revenue -10%	O&M +10%	Combined Effect
25	2041-42		554.73	194.51	360.22	205.59	154.62	154.62	118.60	134.06	98.04
26	2042-43		571.37	200.34	371.02	213.59	157.43	157.43	120.33	136.07	98.97
27	2043-44		588.51	206.35	382.15	221.96	160.19	160.19	121.98	138.00	99.78
28	2044-45		328.44	115.16	213.28	125.35	87.93	87.93	66.60	75.39	54.06
	(6 m)										
	Total	**921.43**	**11,088.77**	**3,888.16**	**7,200.60**	**3,888.99**	**2,390.17**	**2,298.03**	**1,670.11**	**2,001.27**	**1,189.07**
					PIRR	**=**	**13.86%**	**12.53%**	**10.60%**	**12.14%**	**7.64%**
					NPV@12%		**93.61**	**28.91**	**-67.22**	**6.84**	**-218.68**

Annexure 8

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मुंबई पोर्ट ट्रस्ट

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मुंबई पोर्ट ट्रस्ट
पोर्ट भवन
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CE.CF.226/ctp/ENV/194

Dated 15 JUN 2021

To,

Member Secretary,
Expert Appraisal Committee(Infra-2)
Ministry of Environment, Forest and Climate Change
Indira Paryavaran Bhavan, Jor Bagh
Aliganj, New Delhi – 110 003

[F.No.10-10/2017-IA.III], IA/MH/MIS/62185/2017]

Subject: Change of details in Project titled Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour

Dear Sir

With reference to the above subject proposed project titled Construction of two offshore container berths and development of container terminal on BOT basis in Mumbai Harbour received ToR on 12th July 2017 and proposal was considered in 32nd EAC meeting Infra – 2, 2-4 July 2018, however some of the project details has been revised and defer from EIA report submitted during application. Project details changed /updated are given in table below.

Hence we are hereby submitting revised project details as an Annexure to EIA report. We humbly request you to consider updated details for further evaluation of project.

Thanking you

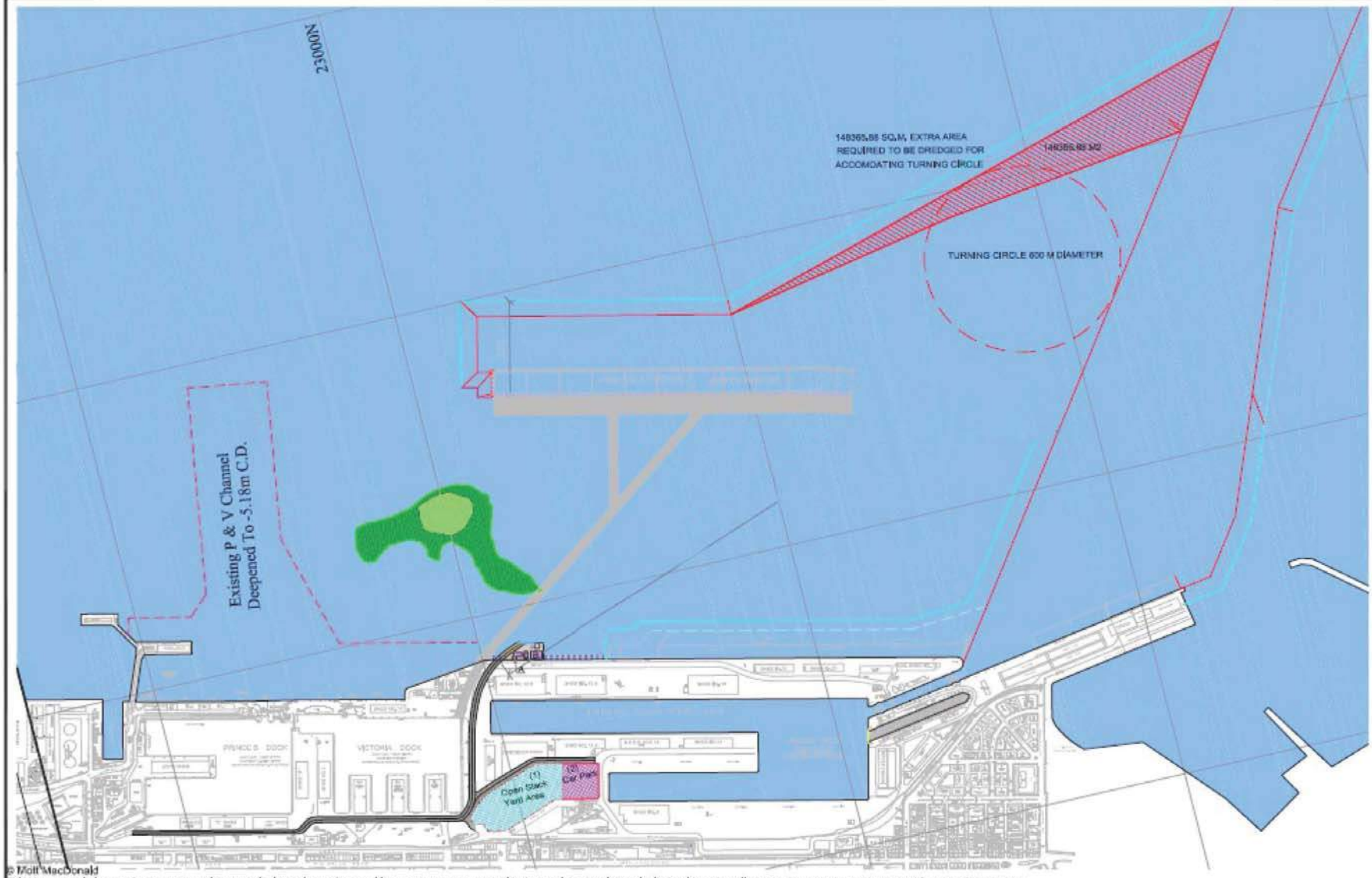
Yours faithfully,

N. G. Mirajkar
Chief Engineer(I/C)

Sr.No	Particulars	EIA report submitted to MoEFCC and discussed in 32ND EAC meeting (21.04.2016)	Updated EIA report submitted to MCZMA and Presented during 146th and 154th MCZMA meeting (04.09.2020)
1.	Capacity	**As Approved By MoEFCC :** 12 MTPA	Phase 1: As Approved By MoEFCC: 9.6 MTPA Now Proposed: 4.17 MTPA
2.	Cargo	**As Approved By MoEFCC:** New Cargo **Now Proposed:** Existing cargo handled at MbPT will be shifted to OCT.	**As Approved By MoEFCC:** New cargo(Containers only) **Now Proposed:** What is already been handled at the port will be shifted to OCT. Mainly RO-RO, Steel, any other clean cargo and occasionally passenger vessels
3.	Project cost	715.67 crores	1290 Crores
4.	Man power	Details was not incorporated	**Construction:** 115 **Operation:** 75
5.	Water requirements, sources (during construction and operation phases)	Details was not incorporated	The total water requirement, the per capita consumption for the in port consumption is taken as 45 litters per day. **Construction:**5.17 KLD **Operation:** 3.78 KLD
6.	Waste Management	Details was not incorporated	Total sewage generation **Construction:** 3.45 KLD **Operation:** 2.25 KLD solid waste to be generated **Construction:** 51.75 Kg/day **Operation:** 33.75 Kg/day

Annexure 1M

Copy of Final layout Plan



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Client
मुंबई पोर्ट ट्रस्ट
Mumbai Port Trust
MbPT ESTATE

Rev	Date	Drawn	Description	Ch'k'd	App'd

Drawn	K Patel
Checked	S Pallwal
Approved	U Phalake
Scale at A3	NTS
Status	Rev
PRE	P1

9. **Compliance: Plan for corporate Environment Responsibility (CER) as per specified under Ministry Office Memorandum vide F.No 22-65/2017-IA.III dated 1st May 2018 shall be submitted.**

Answer:

Ministry's Office Memorandum issued vide letter F. No. 22-65/2017-IA.III dated 01.05.2018 was supressed on 30th September, 2020.

With reference to the new Office Memorandum.following are the concerns raised during the public consultation and thereof the commitments made by the project proponent to address the same:

Public Hearing minutes along with action plan

S.N	Name of the Person	Points Raised	Replies
2.	Smt. Neera Puni (N.G.O.)	road management, space management, supporting infrastructure, mainland infrastructure, and proper transport system. Why are we investing this money for when there is no proper infrastructure available to the general public?	Mumbai Port Trust owns a very good network of road and rail. MbPT owns roads from Docks to Wadala dedicated for the Port Traffic. At present hardly 2000 trucks ply on these roads owing to Port Traffic. We have handled 6 lakh containers with the existing road system in 1997-98 without: Truck Terminus road at that time. Now, with the Truck terminus Road, in place and also proposed Anik Panjrapole Link Road for which construction has already started, the Port Traffic will directly join the National Highways bypassing the city roads. Also when the proposed berth is commissioned in 2009, the traffic. At these berths for the first 5 years will not be more than 6 Lakh TEUs. The existing roads within the Port estate will be widened wherever required for smooth movement of Traffic. As regards Railways, MbPT owns its own rail network from docks to Wadala 55 km track length and 10 km route length) Mumbai Port is also proposing dedicate additional railway line from Wadala to Kurla for which the feasibility studies have been done and the project is feasible. The final estimate and location survey has been completed and central railway is going to submit final report within the week time. With this dedicated link from Wadala to Kurla and also the 5th and 6th line laid by the

			Central Railway from Thane to Kalyan (work almost commenced) and Thane to Kalyan (work commenced) which is dedicated for the through trains and goods trains, there will be smooth sail of cargo movement through rail system.
		If feasibility study is already done, then it should be part of this report then why this Public Hearing without the report?	In the Year 1996 we have handled 6 Lakh TEUs. Now i.e. in the year 2004-2005 we have handling only about 2-3 Lakh TEUs. Thus we have spare capacity available. To enhance the capacity of handling cargo it is necessary to construct offshore container berth.
5.	Shri.Debu Goenka,(N.G.O.)	You have assumed one-inch leakage. It may be more also at the site. What would be the follow up done by the Disaster Management Plan (DMP is minimum risk) while taking into consideration the minimum risk?	The Disaster management prepared by port is adequate to handle the condition. Accidental analysis leakages (one or two inch) happen due to unloading operation, firefighting is able to handle such situation and maximum precautions are taken. The hazardous chemicals come in the form of containers. There will be hardly few container carrying cylinders. The existing DMP of the Port adequate to handle any like.
		Both the road and the rail network is congested/saturated and Traffic handled is proposed to be handled is not clear in the report. What is the Waiting time? JNPT, was created to decongest the city. Now what is the necessity of this project?	JNPT was set up as a part of Mumbai Port and subsequently it was developed as a satellite Port. We are not setting up as a competition to JNP, The container traffic is increasing at a phenomenal rate and almost 60% of the Country's traffic is handled by Mumbai Region and this Project.is the requirement of the trade and in the interest of the Nation. JNPT has already awarded for the construction of 3rd terminal. They are conducting feasibility studies for the 4th container terminal. Even after that there will be a substantial gap which is to be substantial gap which is to be serviced by another Port why not Mumbai Port. My other point is regarding the hinterland services Mumbai City Itself will account for the container traffic, which can be well served by the Mumbai Port with the new terminal. Mumbai port contributes to customs nearly

			500-600 crores worth goods which will be more than double if the project is envisaged. MbPT also contributes Re. 350 Crores towards Octroi to MCGM, which can be upon Rs. 800-900 Crores Similarly lamp Duty Sales Tax will go up. We should gear up to handle huge container traffic projected in the next ten years and since 60% Hinterland consumption is from the Mumbai Region, this project is necessary interest of Nation. Other thing is that it creates employment, other economic activities, labor contractors, service providers etc. these changes are positive aspects of this project. It is the first of kind in India and pride of Mumbai. Mumbai port has studied all the issues comprehensively.
		He desired to know about the evacuation of cargo by (roads already being congested)	There should be no apprehension on this count. On the road front, I divide the connectivity into 2 parts, one is Internal roads within the Mumbai Port Estate, where we are planning to widen the roads, construct flyovers etc We are investing about Rs 90 Crore for this. The second is the Improvements being carried out by MMRDA. The Ankh Panjrapol link road. This is a link and the work has already begun. Mumbai Port is also sensitive to the congestion issue. We are funding Rs 35 Crores to ensure that the project is completed well in time before the offshore berths are commissioned. The work would be completed in 2 years' time. From Panjrapol MMRDA is planning to develop two stretches, which directly connects to Eastern Express Highway. Eastern Express Highway is 10 lane road. With the above projects taken up by MMRDA, the entire Port traffic will bypass the City road. I would like to tell you that Mumbai Port handled 6 Lakh TEUs in 1996-1997 and evacuation was not at all a problem at that time even without Wadala Truck Terminus road. This traffic of 1.2 Million TEUs will core only after 10 years by which time MTHL will be completed.

			About Ralf connectivity, members/citizens should appreciate that his traffic will put minimum constraints to the general public. The Wadala Kurla Link will come and it is being monitored by the Prime Minister Office. The ministry has given high priority for this rail Link. Mumbai Port is funding this project and i am sure that this would be completed before commissioning of the berths. Ever, though I am very much confident that this rail link would be completed, assuming that this will not completed In time, there should be no concern to MbPt as MbPT is capable of handling 6 lakhs TEUs with the existing link itself, We have handled without any difficult previously. After all the traffic 0.8 million TEU will be achieved in the 10^{TH} Year. I would like to state that MMRDA has submitted a report for construction of East island freeway from Museum to Ghatkopar. Mumbai Pont Trust has given Clearance for the link for aligning in Mumbai Port estate. This link will future ease the traffic on city road.
6.	Sing, Father Berkie : Member , Environmental Public Hearing	What comes to the common man? P. D'mello Road is fully blocked/encroached by the Slums. What about their rehabilitation? What profit will MbPt make out of this project?	I share your sentiment fully, MbPT has constructed several roads, Which are given to public, and P.D Mello road is one of them. The road is maintained by MCGM. Slums have developed not because of MBPT. This matter is being taken up BMC/Chief Secretary, Govt. of Maharashtra. In fact for the wadala kulra rail link, 2148 hutments will have to be resettled. We have made provision for their rehabitation. MbPT have become victim of migration and hence the proliferation of slums. MBPT will see to that no further slums will develop its area. A firm policy needs to be adopted in terms of slum development.
8.	Shri Debu Goenka,(N.G.O.)	Weather any other land will be reclaimed? Whether mangroves will be affected due to additional dredging?	No land is being are claimed for this project. We are just filling up the old dock basin. About Mangroves, I should say this would not at all affected. The Mumbai Port has spent dumping area is about 25 km away from Mangroves. Mumbai port has developed and protected Mangrove at sewree Previously about 2000

			Flamingos were there but of today there are about 14000 flamingos. This project will in no way affect habitat. Nearly about 15 acres of area as sanctuary has been replanted. We spent Rs. 50 lakhs to develop this area as sanctuary. The mangroves situated between Sewree wadala will be in no way affected.
9.	Shri. Father Berkle, Member	What about passenger water transport from Ferry what weather this will be affected by the project.	Not affected. In fact MbPT has given clearance to Ferry Wharf Belapur Cruise Terminal to the Maharashtra Government. They have appointed Consultants for finalizing the terms.
10.	Shri Debu Goenka, (N.G.O)	Trucks are parked on the roads. Wadala Truck Terminus is more than 10 years old No use is made of it.	Suggestion by Remedios: - I suggest that MbPT think of beautifying the MbPT roads and that general public can use the same and income is earned by MbPT. MbPT should create health center, gymnasium and swimming pool at the recreation area-Botanical Garden developed by them. I should mention that Mumbai came up because of Mumbai Port Trust. There should be no for revival of the Port. Suggestion by Dr. Bora.: I agree you in the fact the either MbPT should be properly claimed to improve environment of their estate which include proper drainage system for the slum etc. proper environment plan can be taken. There is one institute for environment planning at Ahmedabad which will give all support to finalize the plan. Roads within the MBPT estates should be paved. Project Proponent said that Sheds will be demolished and the same will be used as filling material after processing further excavated material and demolished structures due to construction activity in an around Mumbai will also be used. This in one way will believe the dumping of debris generated out of construction activity in Mumbai.

11.	Shri, Doshi Puran Mohanlal, Member, Environmental Public Hearing	MbPT should give NOC for providing drinking water, toilet blocks etc. for the slums in their areas. Municipality is ready to provide. What about the water supply to be given to the hutments?	Requested all the members who attended the meeting not to delay the project. He also explained that these unauthorized hutments, which are post 1995, and are already having unauthorized water connections and MbPT is suffering at their cost. If water connection is given to them this will results in to huge proliferation of hutments and blame MBPT for the same and vice versa. He again stated that already MBPT has constructed about 55 sulabh sauchalaya mainly in darukhana area. 1. Suggestion by Father Berkie:- MCGM and MbPT should co-ordinate with each other, 2. Suggestion by Remedios: Stated that this query is related to MCGM. 3. Suggestion by Remedios :- Stated that this query is related to MCGM 4. Suggestion by Doshi Puran:- Stated that government has ordered that after 25th duty Torrential rains rehabilitate the slums outside the city limits.

MbPt is engaged in many CER activities such as Fish water management at Sassoon dock, a 60 KLD ETP (Effluent treatment plant) is also proposed at Sassoon dock. MbPT has also developed and protected mangroves at sewri. Previously about 2000 flamingoes were recorded in habitat but as of now there population is been increased to 14000. Nearly 15 Acres of area is replanted as flamingo sanctuary. Other CER activities will be undertaken as per stipulated norms with help of NGOS.

F.No.22-65/2017-IA.III
Government of India
Ministry of Environment, Forest and Climate Change
Impact Assessment Division
*****

Indira Paryavaran Bhawan
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New Delhi – 110003
e-mail: sharath.kr@gov.in

Dated: 30th September, 2020

Office Memorandum

Sub: Deliberation on the commitments made by project proponent and requirements to address the concerned raised during the public consultation and prescribe as specific condition(s) while recommending the proposal, for prior environment clearance, in physical terms in lieu of Corporate Environment Responsibility (CER) – regarding.

This is in supersession of the OM of even number dated 1st May, 2018, regarding guidelines in respect of Corporate Environment Responsibility. Ministry is in receipt of several representations regarding imposition of certain percentage of project cost or expansion cost as Corporate Environment Responsibility. Further, the said OM was also challenged before the Hon'ble High Court of Delhi in WP(C) No. 13252/2019 in the matter of CREDAI, NCR *Versus* Union of India and Ors.

The matter has since been examined in the Ministry and it is hereby decided that henceforth the Expert Appraisal Committee or State Level Expert Appraisal Committee shall deliberate on the commitments made by the project proponent to address the concerns raised during the public consultation and prescribe specific condition(s) in physical terms while recommending the proposal, for grant of prior environment clearance instead of allocation of funds under Corporate Environment Responsibility

Further, it is directed that all the activities proposed by the project proponent or prescribed by the Expert Appraisal Committee or State Level Expert Appraisal Committee, as the case may be, shall be part of the Environment Management Plan.

This issues with the approval of competent authority.

30/9/2020

(Sharath Kumar Pallerla)
Scientist 'F'/Director (IA-Policy)

To,

1. Chairman and Member Secretary of Central Pollution Control Board

2. Chairpersons and Members of all the Expert Appraisal Committees
3. Chairpersons and Member Secretaries of all the SEIAA/SEACs
4. Chairpersons and Member Secretaries of all SPCBs/UTPCCs
5. Member Secretaries of all the Expert Appraisal Committees
6. All the officers of IA Division

Copy for information to:

1. PS to Minister for Environment, Forest and Climate Change
2. PS to MoS for Environment, Forest and Climate Change
3. PPS to Secretary (EF&CC)
4. PPS to AS(RA) / AS(RSP)
5. PPS to JS(GM) / JS(AKN)/JS(SKB)
6. Website, MoEF&CC
7. Guard File.

30/9/2020

(Sharath Kumar Pallerla)
Scientist 'F'/Director (IA-Policy)